Facial Expression Recognition in Virtual Reality under Partial Occlusion by Head-Mounted Displays

by

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Declaration

The research presented in this thesis was carried out by the undersigned. No part of the research has been submitted in support of an application for another degree or qualification at this or another university.

Use of Generative AI

During the preparation of this thesis, ChatGPT by OpenAI was used as an assistive tool, including its reasoning models and deep research capability. It was used for language revision, grammar and style checking, improving concision and academic tone, structural feedback, argumentative brainstorming, literature-search support, and limited assistance with programming tasks, including plotting, debugging, and code refactoring. All AI-assisted suggestions were critically reviewed, verified, manually adapted, and integrated by the author. Generative AI was not used to generate research data, independently conduct empirical analyses, determine the reported results, create unverified citations, produce unsupported scientific claims, or replace the author's scholarly judgement. Where ChatGPT suggested academic sources or source-related claims, these were checked against the original publications, and all bibliographic entries were created and verified by the author. The author remains fully responsible for the content, accuracy, originality, analyses, figures, interpretations, and conclusions presented in this thesis. No confidential, restricted, commercially sensitive, personal, or participant-level research data were submitted to generative AI tools.

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Don't cry because it's over; smile because it happened.

— after Ludwig Jacobowski,
Leuchtende Tage

Abstract

Virtual reality (VR) has evolved into an established technology used across many research and application areas, including therapy, training, education, and entertainment. Across these settings, VR places users in controllable, virtual environments that offer highly immersive, interactive experiences and can elicit strong emotional responses. Recognising these responses is crucial for the evaluation and dynamic adaptation of VR experiences, as emotions influence human perception, decision-making, learning, and behaviour. Facial expression recognition (FER) is a widely used technique for automatic emotion recognition in affective computing. However, it is rarely applied to VR users because the head-mounted displays (HMDs) used to present VR experiences also occlude much of the wearer's upper face. This occlusion masks important visual cues and limits the applicability of conventional vision-based FER systems. Existing work has primarily examined this challenge by simulating HMD occlusion in images from established FER databases or by using custom HMD-integrated sensing setups, but lacks comparable benchmark evidence and data recorded during actual VR use.

To address this gap, this thesis develops and evaluates three artefacts in sequence. First, it presents EmojiHeroVR, a VR game designed to elicit and record posed facial expressions during HMD use. The game is then used in a user study with 37 participants to capture expressions of anger, disgust, fear, happiness, sadness, surprise, and neutral using external cameras and HMD-integrated face tracking. Second, the collected data are manually annotated, filtered, and structured into the EmojiHeroVR Database (EmoHeVRDB), a time-synchronised multimodal benchmark database for static and dynamic FER under natural HMD occlusion. Third, the thesis implements and evaluates baseline FER models on EmoHeVRDB. In these evaluations, face-tracking-based models outperform camera-based models, temporal modelling improves recognition, and multimodal fusion performs best. Overall, accuracy ranges from 69.84% for the static image-based baseline to 81.61% for the multimodal dynamic baseline, indicating that FER under HMD occlusion is challenging but feasible.

The source code for EmojiHeroVR, the baseline evaluation code, and the trained baseline models are made publicly available. EmoHeVRDB is available to researchers for non-commercial use through controlled access upon request.

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Finally, and most importantly, I want to thank my sister, my parents, and my friends. Even though time was often scarce and distances were long, I am truly thankful for all of the time we spend together.

List of Publications

This thesis revises, extends, synthesises, and integrates the work reported in the following publications into a single narrative:

P1: Thorben Ortmann, Qi Wang and Larissa Putzar (2023). 'Facial Emotion Recognition in Immersive Virtual Reality: A Systematic Literature Review'. In: *Proceedings of the 16th International Conference on PErvasive Technologies Related to Assistive Environments*. PETRA '23, pp. 77–82. DOI: 10.1145/3594806.3594861.

P2: Thorben Ortmann (2023). 'Towards Facial Expression Recognition in Immersive Virtual Reality with EmojiRain'. In: *2023 11th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*, pp. 1–5. DOI: 10.1109/ACIIW59127.2023.10388094.

P3: Thorben Ortmann, Qi Wang and Larissa Putzar (2024). 'EmojiHeroVR: A Study on Facial Expression Recognition Under Partial Occlusion from Head-Mounted Displays'. In: *2024 12th International Conference on Affective Computing and Intelligent Interaction (ACII)*, pp. 80–88. DOI: 10.1109/ACII63134.2024.00014. Best Student Paper Award.

P4: Thorben Ortmann, Mirco Hülsemann, Qi Wang and Larissa Putzar (2024). 'EmojiHeroVR: A Virtual Reality Game for Collecting Partially Occluded Emotional Faces'. In: *2024 12th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*, pp. 119–121. DOI: 10.1109/ACIIW63320.2024.00024.

P5: Thorben Ortmann, Qi Wang and Larissa Putzar (2025). 'Unimodal and Multimodal Static Facial Expression Recognition for Virtual Reality Users with EmoHeVRDB'. in: *2025 IEEE International Conference on Artificial Intelligence and eXtended and Virtual Reality (AIxVR)*, pp. 252–256. DOI: 10.1109/AIxVR63409.2025.00048. Best Short Paper Award.

P6: Thorben Ortmann, Qi Wang and Larissa Putzar (2026). 'Dynamic Facial Expression Recognition under Partial Occlusion by Head-Mounted Displays on EmoHeVRDB'. Invited submission to the Special Issue 'Best of ACII 2024' of IEEE Transactions on Affective Computing. Under review.

P1 provides the foundation for Section 2.7, the core section of the literature review in Chapter 2, and informed the initial formulation of the research gaps synthesised in Section 2.9.

P2 outlines the initial plan for the PhD project, which was later refined into the research design presented in Chapter 3.

P3 forms the central empirical basis for Chapters 4–6 and Sections 7.1.1–7.1.2 by introducing EmojiHeroVR and reporting on the user study, the construction of EmoHeVRDB, and the initial evaluation of image-based FER under natural HMD occlusion.

P4 complements P3 by presenting EmojiHeroVR as a VR game for collecting HMD-occluded emotional faces. Its evaluation of game experience and VR sickness is expanded in Section 5.6.

P5 provides the main basis for Sections 7.1.3–7.1.5 by evaluating static FER with HMD-derived facial expression activations and combining them with static facial images for multimodal baseline evaluations.

P6 establishes the dynamic FER baselines presented in Section 7.2. Its comparative discussion was revised and extended into Section 7.3.

Data, Code, and Model Availability

Demographic and questionnaire data. Chapter 5 reports the user study conducted with 37 participants for this thesis. Aggregated and anonymised demographic information is reported in Section 5.5. Aggregated and anonymised Game Experience Questionnaire and Virtual Reality Sickness Questionnaire results are reported in Section 5.6, with complete item-level response distributions provided in Appendix A. Participant-level demographic records and questionnaire responses are not made publicly available due to data protection and ethical considerations.

EmoHeVRDB. The EmojiHeroVR Database (EmoHeVRDB), constructed in Chapter 6 from the data recorded during the user study reported in Chapter 5, contains HMD-occluded facial images and HMD-derived face-tracking data. Because these data are derived from human participants, EmoHeVRDB is not distributed as a fully open public dataset. Instead, it is available to researchers for non-commercial use under controlled access and an end-user licence agreement. Instructions for requesting access are provided at `https://github.com/thorbenortmann/emoji-hero-vr-database`.

EmojiHeroVR. The source code for EmojiHeroVR, introduced in Chapter 4, is available at `https://github.com/thorbenortmann/EmojiHeroVR`. A demo video showing level 3 of the game version used in the user study is available at `https://youtu.be/TnJrGYOjKJc`.

Code and models. The code and models associated with this thesis are made publicly available through the central thesis repository at `https://github.com/thorbenortmann/phd-thesis`. It is organised according to the structure of the thesis and serves as the canonical access point for all related resources. Where resources are already maintained in repositories associated with the publications listed above (P1–P6), the central thesis repository links to those repositories rather than duplicating their contents.

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Abbreviations

API	Application Programming Interface
AU	Action Unit
CC BY-SA	Creative Commons Attribution-ShareAlike
CNN	Convolutional Neural Network
DSR	Design Science Research
ECG	Electrocardiogram
EDA	Electrodermal Activity
EEG	Electroencephalogram
EMG	Electromyography
EmoHeVRDB	EmojiHeroVR Database
EmoHeVRDB-DFEA	EmoHeVRDB Dynamic Facial Expression Activation Subset
EmoHeVRDB-DI	EmoHeVRDB Dynamic Image Subset
EmoHeVRDB-SFEA	EmoHeVRDB Static Facial Expression Activation Subset
EmoHeVRDB-SI	EmoHeVRDB Static Image Subset
FACS	Facial Action Coding System
FEA	Facial Expression Activation
fEMG	Facial Electromyography
FER	Facial Expression Recognition
FPS	Frames per Second
GDPR	General Data Protection Regulation
GEQ	Game Experience Questionnaire
GUI	Graphical User Interface

HAW	Hamburg University of Applied Sciences
HMD	Head-Mounted Display
HRV	Heart Rate Variability
IMU	Inertial Measurement Unit
IR	Infrared
KDEF	Karolinska Directed Emotional Faces
LOSO-CV	Leave-One-Subject-Out Cross-Validation
LSTM	Long Short-Term Memory
MLP	Multilayer Perceptron
PAD	Pleasure-Arousal-Dominance
PhD	Doctor of Philosophy
pp	Percentage Points
PPG	Photoplethysmography
RaFD	Radboud Faces Database
RAF-DB	Real-world Affective Faces Database
ReLU	Rectified Linear Unit
RG	Research Gap
RGB	Red–Green–Blue
RGB-D	Red–Green–Blue–Depth
RNN	Recurrent Neural Network
RO	Research Objective
RQ	Research Question
SDK	Software Development Kit
SSQ	Simulator Sickness Questionnaire
SVM	Support Vector Machine
UWS	University of the West of Scotland
VR	Virtual Reality
VRSQ	Virtual Reality Sickness Questionnaire
XR	Extended Reality

Chapter 1

Introduction

This thesis investigates automatic emotion recognition for virtual reality (VR) users. Specifically, it uses machine-learning models to recognise facial expressions associated with emotions such as anger, disgust, or happiness. Section 1.1 provides the background and motivation, Section 1.2 states the research aim, questions, and objectives, Section 1.3 summarises the main contributions, and Section 1.4 outlines the structure of the thesis.

1.1 Background and Motivation

Since the emergence of modern head-mounted displays (HMDs) around 2014, VR has developed into an established medium across diverse domains, including therapy, training, education, and entertainment (Cipresso et al. 2018). In emergency response training, VR can simulate high-stakes scenarios to practise procedures and decision-making under stress without real-world risk (Alshowair et al. 2024). In classroom settings, VR can provide interactive simulations or virtual field trips that promote engagement and active learning (Marougkas et al. 2024). Across these settings, users can experience and interact with virtual environments as if they were real (Jerald 2015, p. 9). At the same time, VR allows precise control over scenarios, detailed observation of behaviour, and repeatable presentation of situations, often

at lower cost and risk than comparable real-world settings (Marín-Morales et al. 2020).

In many VR applications, emotions play an essential role because they guide decisions, colour perceptions, influence learning, shape relationships, and affect well-being (Barrett, Lewis et al. 2016). Therefore, understanding users' emotions is crucial for evaluating VR experiences. Researchers rely on explicit and implicit methods to assess users' emotional responses (Halbig and Latoschik 2021). Explicit methods—such as questionnaires and interviews—require users to self-report their experiences. Consequently, they can be time-consuming and susceptible to memory or social desirability biases (Mauss and M. D. Robinson 2009). In contrast, implicit methods are based on observations—such as physiological or behavioural data—and infer emotions without requiring users' active participation. Recognising emotions automatically further enables the dynamic adaptation of VR content to users' emotional states. For example, systems may adjust exposure intensity in therapy, adapt learning experiences to frustration levels, or tailor entertainment content based on user engagement.

Automatic emotion recognition is a core challenge in the field of affective computing, which lies at the intersection of emotion science and computing (Picard 1997). Among the available input modalities, facial expressions are widely used because they are a natural and expressive way to convey emotions (Ekman 2006). Accordingly, facial expression recognition (FER) is a mature research area with established databases and evaluation protocols, achieving strong performance on many benchmarks (S. Li and W. Deng 2022). Most FER pipelines are camera-based and operate on visual facial recordings. In this context, static FER analyses the spatial information in single frames, whereas dynamic FER analyses sequences of frames to also leverage temporal information.

However, despite its prevalence in other settings, FER is rarely applied to VR users because the HMD occludes much of the wearer's upper face, particularly the eye and eyebrow regions (Luong et al. 2022; Marín-Morales et al. 2020). This partial occlusion masks important visual cues that conventional image-based FER

systems rely on and thereby substantially degrades their performance (Georgescu and Ionescu 2019; Houshmand and Khan 2020). While several approaches have been proposed to mitigate HMD-induced occlusion, reliable FER for VR users remains challenging in practice and is the central challenge addressed by this thesis.

1.2 Research Aim

The overarching aim of this thesis is to enhance emotion recognition for VR users by enabling effective FER despite partial occlusion caused by HMDs.

1.2.1 Research Questions

To operationalise this aim, the thesis addresses the following research questions:

RQ1: What are the key challenges and limitations that hinder effective FER for VR users?

RQ2: How can facial expressions of VR users be collected to construct a time-synchronised multimodal benchmark database that supports static and dynamic FER under HMD-induced occlusion?

RQ3: How do unimodal baselines and multimodal fusion perform for static and dynamic FER under HMD-induced occlusion?

RQ4: What interpretations, trade-offs, and design implications can be derived from these baseline results to advance FER research and applications in VR settings?

1.2.2 Research Objectives

To achieve the research aim and answer the research questions, this thesis pursues the following research objectives:

RO1: To review the state-of-the-art in FER for VR users, critically synthesise its limitations, and derive key research gaps.

RO2: To design and execute a user study that captures emotional facial expressions of VR users under HMD-induced occlusion.

RO3: To construct a time-synchronised multimodal database that combines external-camera recordings with HMD-derived signals to support benchmarking static and dynamic FER under HMD-induced occlusion.

RO4: To implement and evaluate unimodal baselines and multimodal fusion for static FER under HMD-induced occlusion.

RO5: To implement and evaluate unimodal baselines and multimodal fusion for dynamic FER under HMD-induced occlusion.

RO6: To compare the evaluation results of unimodal baselines and multimodal fusion across static and dynamic FER under HMD-induced occlusion, identifying relative strengths, synergies, and limitations.

RO7: To derive interpretations, trade-offs, and design implications from the baseline results to advance FER research and applications in VR settings.

1.3 Thesis Contributions

In addressing the research aim, questions, and objectives, this thesis makes three main contributions that form an evidence chain from data acquisition to benchmark construction and FER evaluation.

1. **EmojiHeroVR as a data-acquisition system.** EmojiHeroVR combines game-based elicitation of posed facial expressions with synchronised recording of naturally HMD-occluded facial images from external cameras and HMD-derived face-tracking signals represented as facial expression activations (FEAs). Developed in Chapter 4 and evaluated in Chapter 5, it provides a

repeatable procedure for collecting multimodal facial-expression data during actual VR use.

2. **EmoHeVRDB as a benchmark database.** The EmojiHeroVR Database (EmoHeVRDB), constructed in Chapter 6, provides manually labelled, time-synchronised multimodal data under natural HMD occlusion. Its participant-independent splits and task-specific subsets support comparable evaluations of static, dynamic, unimodal, and multimodal FER under a shared benchmark protocol.

3. **Baseline FER models and comparative benchmark evidence.** The evaluations in Chapter 7 establish reference results for image-based and FEA-based FER, temporal modelling, and multimodal fusion. They identify modality strengths, complementarity, and trade-offs, and examine the transferability of results from simulated to natural HMD occlusion, providing an empirical basis for further FER research in VR.

Together, these contributions establish an integrated and reproducible foundation for studying FER for VR users under HMD-induced occlusion. Their relationship to the research gaps outlined in Section 2.9.3 and the evidence generated throughout the thesis is examined in detail in Sections 8.1 and 8.2.

1.4 Thesis Structure

The structure of this thesis is designed to achieve the research aim and objectives and to answer the research questions. Its alignment with the research questions is visualised in Figure 1.1. The remainder of the thesis is organised as follows:

Chapter 2 reviews the literature relevant to this thesis. It begins with the theoretical foundations of emotion, emotion recognition, and FER, narrows its focus to the specific challenge of FER under HMD-induced occlusion, and concludes by identifying the research gaps addressed in the remainder of the thesis.

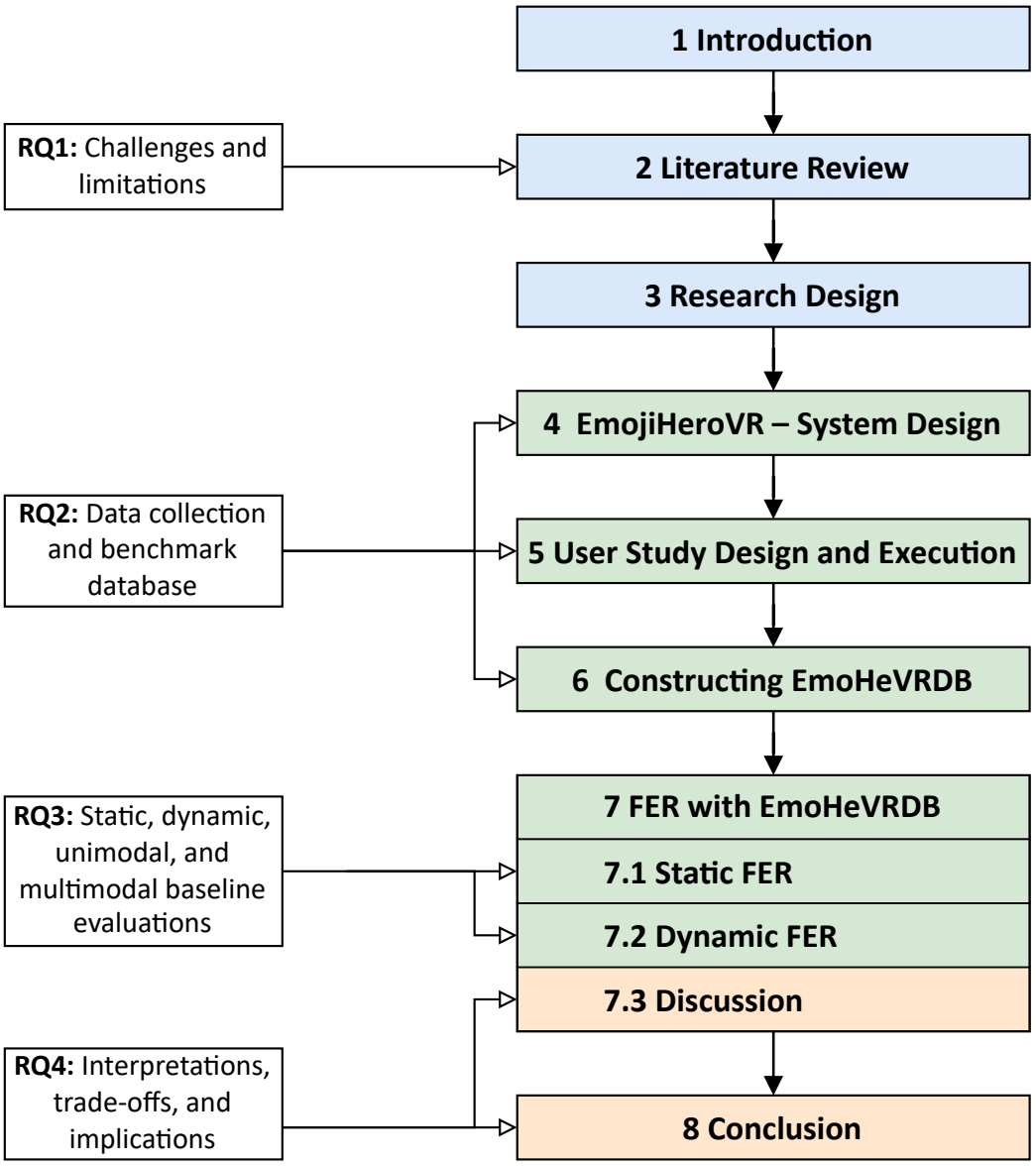


FIGURE 1.1: Structure of the thesis and its alignment with the research questions.

Chapter 3 presents the research design of this thesis. It describes the research philosophy and strategy, derives the evidence requirements from the research gaps, and outlines the design decisions and research plan used to address the research aim, questions, and objectives.

Chapter 4 details the system design of EmojiHeroVR, the VR game used in the user study in Chapter 5 to elicit and record facial expressions. It describes the game mechanics, the FER subsystem, and the multimodal recording pipeline.

Chapter 5 presents the user study conducted with EmojiHeroVR to collect facial-expression data from VR users wearing HMDs. It details the experimental setup, procedure, recruitment, ethical considerations, realised sample, and collected data. It also evaluates EmojiHeroVR as a study instrument based on questionnaire responses.

Chapter 6 explains how the data collected in Chapter 5 are transformed into EmoHeVRDB, a time-synchronised multimodal benchmark database for FER under HMD-induced occlusion. It describes the data annotation and filtering process, the construction of participant-independent training, validation, and test sets, and the definition of benchmark subsets for static and dynamic FER.

Chapter 7 presents the FER experiments conducted with EmoHeVRDB. It evaluates unimodal and multimodal approaches for static and dynamic FER under HMD-induced occlusion and concludes by discussing their performance, modality trade-offs, class-wise error patterns, and interpretive boundaries.

Chapter 8 concludes the thesis by synthesising the contributions in relation to the research gaps and answering the research questions. It then discusses the limitations and conditions of interpretation of the presented evidence before outlining directions for future work.

Chapter 2

Literature Review

To address the research aim of this thesis—*to enhance emotion recognition for virtual reality (VR) users by enabling effective facial expression recognition (FER) despite partial occlusion caused by head-mounted displays (HMDs)*—it is essential to understand the theoretical foundations and prior research in emotion recognition and VR. This chapter gradually narrows its focus from general theoretical concepts to the specific challenges addressed in this thesis. It proceeds as follows:

First, Section 2.1 introduces the concept of 'emotion' from the perspective of affective computing. Then, Section 2.2 discusses discrete and dimensional emotion theories, which are the most influential in affective computing and FER. Section 2.3 surveys approaches and modalities for emotion recognition, including central challenges around task definition and ground truth. Section 2.4 examines FER as a specific emotion recognition technique, covering static versus dynamic FER, multimodal FER, and common evaluation practices. Section 2.5 reviews major FER databases with emphasis on their historical development, annotation practices, and inter-rater reliability. Section 2.6 explores VR as an emotion elicitation and sensing context, and surveys related research and application areas. Section 2.7 reviews the state-of-the-art of FER in VR settings, contrasting external-camera approaches under natural HMD occlusion with headset-integrated sensing approaches. Section 2.8 outlines ethical and regulatory considerations for FER in VR settings.

Finally, Section 2.9 summarises the chapter's main findings and synthesises the key research gaps that motivate the research design and contributions of this thesis.

2.1 Emotions in Affective Computing

This section introduces the concept of 'emotion' from the perspective of affective computing, as this PhD thesis is situated in that research area. The term *'affective computing'* was introduced by Rosalind Picard in 1997, who defined it as *'. . . computing that relates to, arises from, or deliberately influences emotions'* (Picard 1997, p. 3).

Emotions are an integral part of human life (Damasio 1994). They guide our decisions, colour our perceptions, influence how we learn, shape our relationships, and affect our well-being (Barrett, Lewis et al. 2016). We experience emotions continuously and build an intuitive idea of what they are (Scherer 2005). However, there is still no precise definition of 'emotion' that researchers from different disciplines agree on (Izard 2010; Y. Wang, Song et al. 2022).

From a psychological perspective, emotions often involve an interplay of subjective experience, physiological changes, and expressive behaviours (Scherer 2005), with researchers debating whether emotions are discrete (Ekman 1992) or dimensional (Russell 1980). Neuroscience emphasises the role of brain structures in generating and regulating emotions (Barrett and Satpute 2019; LeDoux 1996). Social scientists stress the impact of cultural norms, social contexts, and power structures on the way individuals express and manage their emotions (Hochschild 1983; Lutz 1988; Wetherell 2015). Philosophical accounts of emotions focus on their nature as evaluative judgements and bodily perceptions, while also exploring their intentionality, consciousness, and ethical implications (Goldie 2002; Nussbaum 2001; Tappolet 2022).

Rosalind Picard proposed to think of 'emotion' as *'. . . an umbrella, under which a variety of processes cluster'* (Picard 1997, p. 225). She highlights that the human

emotion system is composed of various physical and cognitive components. Not all emotions engage these components equally, and different emotion theories highlight different components. When designing affective systems, whether for recognition or synthesis, researchers should focus on the models and components most relevant to the task and the emotions of interest. This pragmatic approach has been successfully applied in affective computing by many researchers (Calvo and D'Mello 2010).

2.2 Emotion Theories

Emotion theories are a theoretical foundation for affective computing applications. Although many variants and classifications exist, this section focuses on discrete (Subsection 2.2.1) and dimensional (Subsection 2.2.2) emotion theories because they are the most influential in affective computing and FER (S. Li and W. Deng 2022; Y. Wang, Song et al. 2022). For a broader perspective on emotion theories, see *'Emotion Theory: The Routledge Comprehensive Guide: Volume I: History, Contemporary Theories, and Key Elements'* (Scarantino 2024).

2.2.1 Discrete Emotion Theories

Discrete emotion theories posit that emotions—such as anger, fear, or happiness—are distinct categories with unique features, including specific facial expressions, measurable physiological markers (e.g. heart rate or hormone levels), and dedicated neural processes. Basic emotion theories are an essential subset of discrete emotion theories. They add that certain discrete emotions are innate, evolutionarily conserved, and universal across human cultures (Shiota 2024).

The following subsections first present a historical review of key contributions to discrete and basic emotion theories (2.2.1.1), then discuss critiques (2.2.1.2) and resulting advances (2.2.1.3), and finally conclude with remarks on the implications of discrete and basic emotion theories for affective computing and FER (2.2.1.4).

2.2.1.1 Historical Background and Key Contributions

The roots of discrete emotion theories can be traced back to the works of *Charles Darwin*. Darwin compared human expressions with those of other species. He found that certain emotional expressions—such as baring teeth in anger or wrinkling the nose in disgust—can be observed across species. In his work *'The Expression of the Emotions in Man and Animals'* (Darwin 1872), Darwin concluded that certain emotional expressions result from evolutionary processes and evolved from once-functional actions (e.g. biting or rejecting noxious substances).

Building on Darwin's evolutionary perspective, *Silvan Tomkins* explored the biological basis of emotional expressions. Through photographic analysis and observational studies, he identified consistent sets of facial muscle movements corresponding to distinct emotional states. Tomkins introduced the concept of discrete *'affects'*—such as interest, surprise, or fear—arguing they collectively form a primary motivational system that can override or amplify physiological drives like hunger or thirst (Tomkins 1962; Tomkins 1963). He proposed that these affects are biologically programmed responses, manifesting in characteristic facial expressions and bodily feedback that shape perception and behaviour across diverse contexts. By systematically cataloguing these expressions, Tomkins provided an early empirical framework suggesting that discrete emotions have a biological rather than a purely cultural basis.

Carroll Izard expanded on Tomkins' work by focusing on the developmental and cross-cultural dimensions of discrete emotions (Izard 1971; Izard 1977). Using observational and experimental methods, often involving children and infants, Izard provided empirical support for the idea that discrete emotions manifest before extensive socialisation and remain remarkably consistent across different societies. He proposed a set of ten fundamental emotions, including interest, joy, anger, and fear, each marked by distinctive facial expressions and bodily reactions.

Paul Ekman's work on universal basic emotions is arguably one of the most influential in FER and affective computing (Calvo, D'Mello et al. 2015; Y. Wang,

Song et al. 2022). Beginning in the 1960s, Ekman conducted cross-cultural studies, including research in remote communities in Papua New Guinea, to investigate whether people across diverse backgrounds could reliably identify specific emotional expressions (Ekman and Friesen 1971; Ekman, Friesen and Ellsworth 1972). Participants were shown photographs of faces displaying posed emotions—such as happiness, anger, and surprise—and then asked to select the most appropriate emotional label from a short list of choices. Due to the high recognition accuracy across study groups, Ekman concluded that certain emotions are universally recognised across cultures and universal in their facial expressions. Ekman proposed the six basic emotions: *anger*, *disgust*, *fear*, *happiness*, *sadness*, and *surprise* (see Figure 2.1). Following further research, he added *contempt* to that list (Ekman and Heider 1988). Based on his findings, Ekman and *Wallace Friesen* developed the *Facial Action Coding System* (FACS; Ekman and Friesen 1976). FACS is a



(A) Anger (B) Disgust (C) Fear

(D) Happiness (E) Sadness (F) Surprise

FIGURE 2.1: Images *AF01ANS, AF01DIS, AF01AFS, AF01HAS, AF01SAS, AF01SUS* from the KDEF database (Lundqvist et al. 1998), showing posed expressions of Ekman's six basic emotions.

comprehensive system for describing facial muscle movements, or *'Action Units'* (AUs), and mapping them to emotional expressions. For example, AU 12 represents the contraction of the zygomatic major muscle, or *'Lip Corner Puller'*, which is characteristic of smiling or happiness (Ekman, Friesen and Hager 2002).

According to Ekman and *Daniel Cordaro*:

> '*Emotions are discrete, automatic responses to universally shared, culture-specific and individual-specific events. The emotion terms, such as anger, fear, etcetera, denote a family of related states sharing at least 12 characteristics, which distinguish one emotion family from another, as well as from other affective states. These affective responses are preprogrammed and involuntary, but are also shaped by life experiences.*'
>
> (Ekman and Cordaro 2011, p. 364)

Ekman and Cordaro attribute the following characteristics to (nearly all) basic emotions:

> 1. *'Distinctive universal signals.*
> 2. *Distinctive physiology.*
> 3. *Automatic appraisal.*
> 4. *Distinctive universals in antecedent events.*
> 5. *Presence in other primates.*
> 6. *Capable of quick onset.*
> 7. *Can be of brief duration.*
> 8. *Unbidden occurrence.*
> 9. *Distinctive thoughts, memories, and images.*
> 10. *Distinctive subjective experience.*
> 11. *Refractory period filters information available to what supports the emotion.*
> 12. *Target of emotion unconstrained.*
> 13. *The emotion can be enacted in either a constructive or destructive fashion.'*
>
> (Ekman and Cordaro 2011, p. 365)

Further influential contributions to discrete emotion theories include the works of *Robert Plutchik* and *Jaak Panksepp*. Plutchik proposed eight primary emotions (anger, anticipation, disgust, fear, joy, sadness, surprise, and trust) based on psychoevolutionary functions and organised them in a colour-coded emotion wheel, shown in Figure 2.2, that illustrates opposing pairs, intensity variations, and potential compound combinations (Plutchik 1980). From a neuroscientific perspective, Panksepp identified seven primary emotional systems (care, fear, lust,

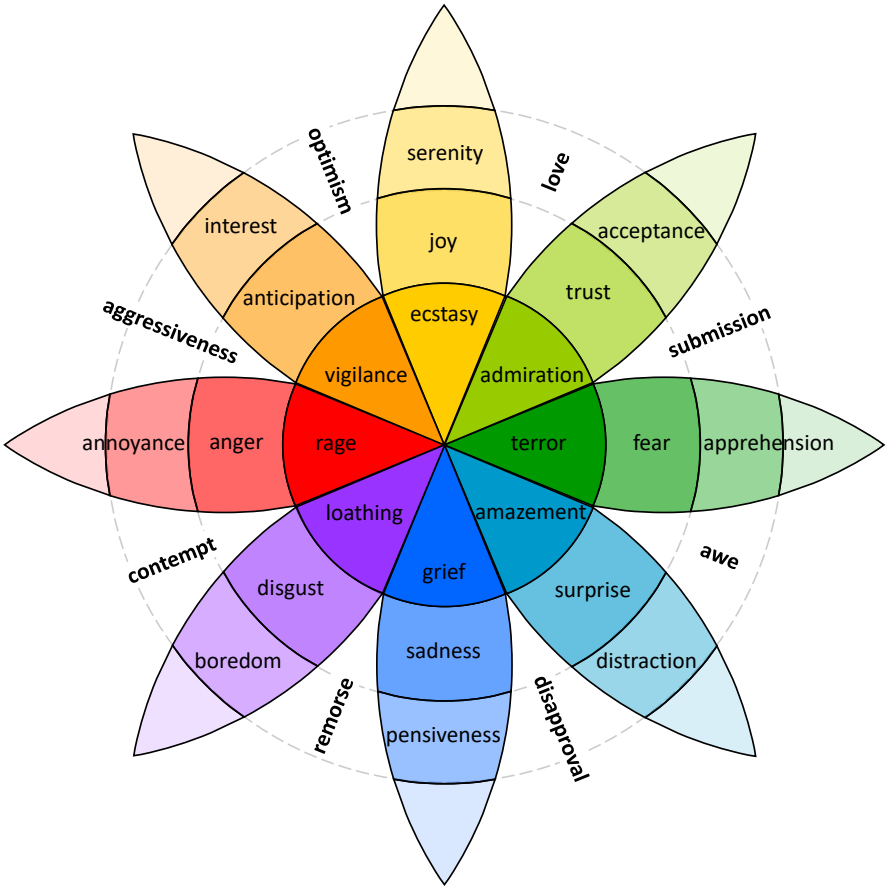


FIGURE 2.2: Plutchik's Wheel of Emotion; recreated from (Plutchik 2001, p. 349 Figure 6).

panic, play, rage, and seeking), each presumably rooted in distinct neural circuits (Panksepp 1998). Reviews of popular models of basic emotions and recent advances in basic emotion theory are provided by Tracy and Randles (2011) and Keltner et al. (2019), respectively.

2.2.1.2 Critiques

While discrete and basic emotion theories are often used in affective computing, significant critiques exist. Prominently, *James Russell* sparked an ongoing debate by questioning the universality of emotion recognition from facial expressions because of methodological flaws in cross-cultural studies (Russell 1994). He argued that using static pictures of posed facial expressions in combination with a forced-choice response format artificially inflated recognition rates. Existing research

indicated that recognition rates dropped substantially when participants freely labelled facial expressions instead of selecting from predetermined options.

Later cross-cultural research further challenged universality by demonstrating that East Asian participants represent emotions with different facial movement patterns than Western participants, revealing cultural variations in expression and perception (Jack et al. 2012).

Building on Russell's critique, Gendron, Roberson et al. (2014) highlighted that only six experiments prior to their work—most conducted nearly 40 years ago—employed the rigorous *'two-culture'* approach needed to test universality claims. In their study, participants from the US and the Himba ethnic group (a remote Namibian community) sorted images of posed facial expressions by emotion type. Without cues to emotion concepts, Himba participants did not produce the presumed universal basic emotion categories. When emotion words were introduced, both groups' sorting patterns converged slightly towards the universal model, yet cultural differences persisted.

Drawing from these findings, *Lisa Feldman Barrett's* theory of constructed emotion posits that emotions are not biologically hardwired but psychological constructs shaped by more basic neurophysiological processes, conceptual knowledge, and contextual factors (Barrett 2017). Barrett argues that emotion categories—such as anger or sadness—are not natural kinds but socially constructed folk concepts (Barrett 2006).

Neuroscientific evidence supports this view: meta-analyses of neuroimaging studies reveal no consistent neural 'signatures' for basic emotions (Kober et al. 2008; Lindquist et al. 2012). Instead, instances of the same emotion (e.g. fear or happiness) can recruit different brain circuits in different contexts, and the same brain region (e.g. the insula or amygdala) can participate in multiple emotions (Barrett and Satpute 2013).

Regarding facial expressions, critics emphasise that the link between facial movements and emotional experience is far from one-to-one. Classic experiments by

Carroll and Russell (1996) demonstrated that identical facial expressions can be perceived as different emotions depending on the situational context. Aviezer et al. (2012) found that intense positive/negative emotions, when winning/losing, share nearly identical facial expressions, with observers relying on body cues rather than faces for accurate valence discrimination. Gendron, Crivelli et al. (2018) revealed considerable cultural variability in the interpretation of facial expressions, suggesting that the same expression may be understood differently across cultural contexts. In their analysis of scientific evidence, Barrett, Adolphs et al. (2019) argued that inferring specific emotions solely from facial movements lacks reliability, as emotions can occur without accompanying facial expressions, and identical facial cues may be linked to various internal experiences. Thus, the emotional meaning of any set of facial movements is highly variable and context-dependent.

2.2.1.3 Responses and Advancements

In response to these critiques, many researchers have defended discrete and basic emotion theories, adapting and advancing them over time (Celeghin et al. 2017; Ekman and Cordaro 2011; Hutto et al. 2018; Keltner et al. 2019; Lench et al. 2013). Modern theorists rarely claim that emotions are rigid 'natural kinds'. Instead, they acknowledge contextual influences while maintaining that basic emotions remain fundamentally distinct categories with evolutionary origins (Shiota 2024). Many theorists have adopted more nuanced frameworks integrating biological universality and cultural specificity. For example, Elfenbein's dialect theory posits a universal emotional 'grammar' with cultural 'accents' in expression and recognition (Elfenbein et al. 2007). Moreover, while Ekman's early work identified prototypical facial expressions for basic emotions, his later research clarifies that these core categories often manifest as families of related states that vary with context (Ekman and Cordaro 2011). He suggests that although the evidence is particularly robust for a core subset of emotions, additional categories are likely to exist. A comprehensive summary of the evolution of basic and discrete emotion theories, with a focus on Ekman's work, is provided by Michelle Shiota's recent review (Shiota 2024).

2.2.1.4 Conclusion

As described in Section 2.1, the definition of emotions and related concepts remains a subject of ongoing debate. Despite the critiques regarding the universality of basic emotion, discrete emotion theories—particularly Ekman's theory of six basic emotions—continue to be the predominant approach in automatic FER. Their historical prevalence, intuitive categorisation, and computational simplicity have significantly facilitated the development of datasets and recognition models for affective computing researchers (S. Li and W. Deng 2022; X. Tan et al. 2022).

2.2.2 Dimensional Emotion Theories

Dimensional emotion theories describe emotions along a small set of continuous dimensions, most commonly *valence* and *arousal* (Colombetti and Kuppens 2024). Valence reflects how pleasant or unpleasant an experience feels, whereas arousal indexes the level of experienced or physiological activation. For instance, a coordinate such as $(+0.2, -0.5)$—slightly pleasant, moderately deactivated—corresponds to a relaxed feeling. In dimensional frameworks, emotion terms—such as anger or disgust—are fuzzy regions that blend into one another rather than rigid, mutually exclusive categories (Russell 1980).

The following subsections first review historical key contributions to dimensional emotion theories (2.2.2.1), then summarise relevant critiques (2.2.2.2), and finally conclude with practical implications of dimensional theories for automatic emotion recognition (2.2.2.3).

2.2.2.1 Historical Background and Key Contributions

The idea of describing emotions along continuous dimensions goes back to *Wilhelm Wundt*, who is often regarded as the founder of modern experimental psychology (Wassmann 2008). In the late 19th century, Wundt systematically employed introspection to measure subjective feelings in his research (Wundt 1897). He found

that emotions or *'affect'* vary continuously in quality and intensity. Thus, Wundt proposed to describe emotions along three bipolar dimensions of pleasantness–unpleasantness (valence), arousing–subduing (arousal), and strain–relaxation (intensity) rather than by rigid, discrete categories, such as anger or happiness (Barrett and Bliss-Moreau 2009).

In the mid-20th century, researchers began to revive and build on Wundt's ideas. In his studies of facial expression perception, *Harold Schlosberg* observed that subjects often confused similar expressions when forced to choose among discrete labels (e.g. surprise and disgust) (Schlosberg 1954). Consequently, Schlosberg investigated the use of multiple continuous dimensions and argued that emotions should be rated along three dimensions: pleasantness–unpleasantness, attention–rejection (serving as an approach–withdrawal metric), and activation level.

Around the same time, *Charles Osgood* introduced the semantic differential method to quantify the connotative meaning of words (Osgood 1952). In his cross-cultural studies, subjects rated a variety of 100 abstract and concrete terms—including *'House'*, *'Respect'*, *'Wealth'*, and *'Anger'*—on bipolar adjective scales such as good–bad and active–passive (Osgood 1964). Analysis of the ratings showed that the connotative meanings clustered along three dimensions: Evaluation (represented by scales such as good–bad and pleasant–unpleasant), Potency (represented by scales such as strong–weak and hard–soft), and Activity (represented by scales such as fast–slow and active–passive). Osgood concluded that, despite the complexity of subjective meaning, these three dimensions can capture how people interpret affective meaning in language.

Building on the work of Osgood and others, Mehrabian and Russell (1974) proposed the PAD model (see Figure 2.3) to measure emotional responses in environmental psychology systematically. Pleasure (P) is analogous to Osgood's Evaluation dimension, capturing positive or negative affect. Arousal (A) corresponds to the Activity dimension, representing the state of physical or mental activation. Dominance (D) is similar to Osgood's Potency dimension but focuses on feelings of control or power versus submissiveness or powerlessness. In their experiments,

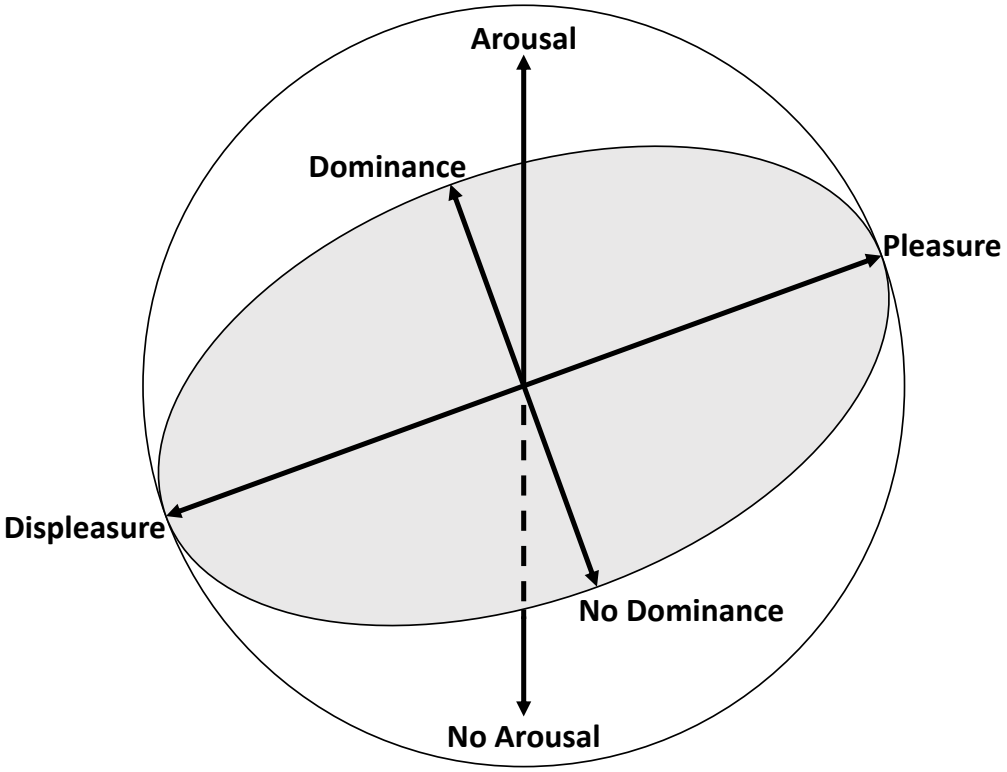


FIGURE 2.3: Visualisation of the Pleasure-Arousal-Dominance (PAD) model; adapted from (Bakker et al. 2014, Figure 4).

Mehrabian and Russell (1974) presented subjects with written descriptions of various environmental situations, such as sitting alone on a beach or attending a crowded hotel reception. Subsequently, subjects rated their feelings on several bipolar adjective scales (e.g. pleased–annoyed, aroused–unaroused, dominant–submissive). Factor analyses of these ratings showed that emotional responses, like the broad connotative meanings described by Osgood, reliably formed a three-dimensional continuum, supporting the validity of the PAD model.

Perhaps the most influential dimensional model in contemporary emotion research is Russell's Circumplex Model of Affect (Russell 1980), visualised in Figure 2.4. In a series of studies—including tasks in which subjects sorted emotion-denoting adjectives into categories and arranged these categories in a circular order—Russell investigated whether affect is best represented as a continuous, circular structure. Extensive analysis of the resulting data (including scaling, regression, and principal component analysis) consistently confirmed that nearly all the variance in emotional responses is captured by two key dimensions: pleasure–displeasure (later called

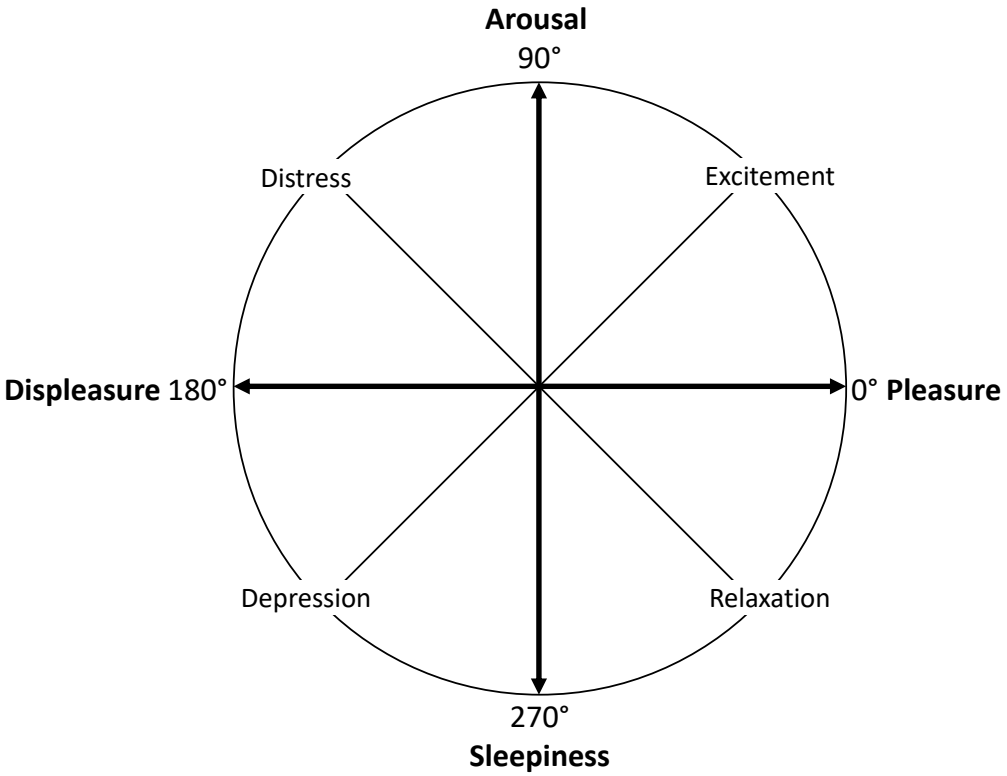
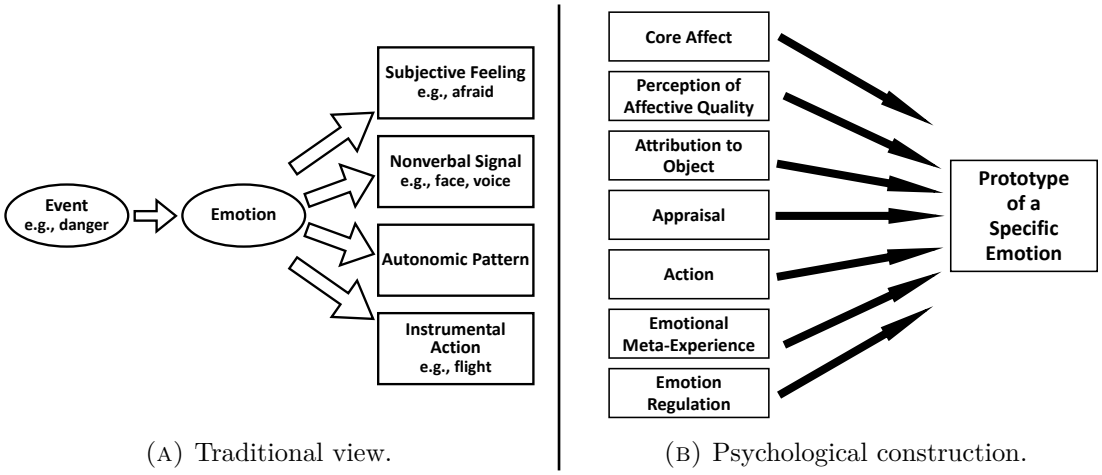


FIGURE 2.4: Russell's Circumplex Model of Affect; adapted from (Russell 1980, Figure 1).

'valence') and degree-of-arousal. Thus, Russell's model posits a structured affective space, which describes affect as a combination of these two dimensions. Additionally, the evidence suggested that affective concepts are best organised in a circumplex model—with pleasure, displeasure, arousal, and sleepiness positioned at 0°, 180°, 90°, and 270°, respectively. While Russell's model is fundamentally continuous, discrete emotion labels (e.g. angry, excited, and depressed) map onto specific regions of the circumplex, capturing gradual transitions and oppositions between feeling states. For example, excitement would be high in valence and arousal, whereas depression would be low in both.

Building on the circumplex model, Russell later articulated his theory of *'core affect'* to explain how specific emotions emerge (Russell 2003; Russell and Barrett 1999). Core affect refers to a continuously varying, elemental, neurophysiological state experienced as a combination of valence and arousal without attribution to a particular cause. Russell argued that each emotional episode begins with this core affect and that specific prototypical emotions (happiness, anger, fear, etc.)

emerge only when additional components—such as situational interpretations or appraisals—give context and meaning to it. This contrasted with the traditional view of emotional episodes, as visualised in Figure 2.5. Notably, dominance is also not part of core affect. Instead, it emerges during appraisal, reflecting how one perceives control in a given situation, which aligns with the PAD model's view. In this way, the continuous affective space depicted by the circumplex model is transformed by further components (appraisal, action, emotion regulation, etc.) into the complex array of emotions that we experience in everyday life. With this framework, Russell laid the foundation for further theories of the psychological construction of emotion, inspiring contemporary research such as Barrett's theory of constructed emotion (Barrett 2017).



(A) Traditional view. (B) Psychological construction.

FIGURE 2.5: Comparison of emotional episodes in (A) the traditional view and (B) in Russell's proposed psychological construction (including core affect); recreated from (Russell 2003) Figure 2 and Figure 3, respectively.

2.2.2.2 Critiques

Although dimensional emotion models have been highly influential in affective sciences, they have also faced several critiques. Critics argue that two or three dimensions cannot capture the rich variety of human emotions. For instance, anger and fear are high-arousal, negative-valence states and thus occupy a similar position in the valence-arousal space (Scherer 2005). However, they differ significantly in facial expression, action tendencies, and autonomic patterns: anger raises

skin temperature and readiness to confront, while fear triggers cooling and flight behaviours (Levenson 1992). Basic emotion theorists further emphasise that evolution-endowed emotions—such as anger, disgust, and joy—have distinct triggers and expressive signatures (Ekman 1992). Reducing each emotion to a single point in a low-dimensional space blurs these signatures, rendering category boundaries fuzzy. Consequently, mapping points back to discrete emotion categories is not trivial (Izard 2007).

Adding a dominance axis—as in the PAD model—can distinguish anger (high dominance) from fear (low dominance). However, an extensive semantic study with 531 students from Belgium, the United Kingdom, and Switzerland found that at least four dimensions (evaluation-pleasantness, potency-control, activation-arousal, and unpredictability) are required to capture the meaning of emotion-related words sufficiently (Fontaine et al. 2007).

Furthermore, Watson and Tellegen (1985) challenged the notion of positive and negative valence as opposites of the same bipolar dimension. In their reanalyses of several studies of self-reported mood, positive and negative affect consistently emerged as relatively independent dimensions. Subsequently, they developed the *Positive and Negative Affect Schedule* (PANAS), a 20-item self-report questionnaire, which has been used by many researchers (Watson, Clark et al. 1988). Larsen et al. (2001) expanded on this topic, providing evidence for mixed feelings where positive and negative valence co-occur (e.g. feeling bittersweet or nervous excitement).

Methodologically, most dimensional models stem from subjective self-reports or adjective ratings, reflecting subjects' ability to articulate feelings (Mauss and M. D. Robinson 2009). As a result, easily articulated dimensions—such as valence and arousal—are privileged over subtler physiological and neural markers. However, autonomic and neuroimaging research indicates that discrete emotions—such as anger, fear, joy, and disgust—have distinct response patterns beyond a general valence or arousal effect (Levenson 1992; Vytal and Hamann 2010). Underscoring the importance of language in shaping dimensional structures, cross-cultural work

revealed that certain Japanese emotion terms with no clear counterpart in English require an additional dimension of interpersonal engagement–disengagement, alongside valence and arousal, to capture their meaning sufficiently (Markus and Kitayama 1991).

In affective computing and facial expression research, practical limitations of dimensional emotion theories surface. For instance, assigning facial expressions to coordinates in a valence-arousal space generally feels less intuitive than choosing from a set of discrete labels, such as joy or anger. Consequently, most datasets designed for facial expression recognition rely on discrete labels (Mollahosseini et al. 2019). Conversely, a system that outputs only valence-arousal coordinates provides little actionable meaning without additional context. Thus, developers often need downstream logic to decide whether to treat a user as angry, afraid, or highly stressed, underscoring an interpretability gap (Pei et al. 2024).

2.2.2.3 Conclusion

Dimensional emotion theories have fundamentally shaped affective science by providing a framework to quantify the complexity of affect in continuous, low-dimensional spaces. Unlike discrete models, dimensional models—such as Russell's Circumplex Model of Affect (Russell 1980) and the PAD model (Mehrabian and Russell 1974)—are designed to capture the gradient nature of affect. Although the number and labels of the dimensions differ, the two axes of valence and arousal recur with remarkable consistency.

In practice, dimensional approaches excel when continuous physiological signals—such as heart-rate variability and electrodermal activity—are the primary input (Khare et al. 2024). In contrast, Ekman's six basic expressions (Ekman, Friesen and Ellsworth 1972) remain dominant in facial expression recognition, which is primarily vision-based (S. Li and W. Deng 2022).

However, contemporary research increasingly bridges the dimensional–discrete divide. Inspired by Russell's theory of core affect (Russell 2003) and Barrett's

theory of constructed emotion (Barrett 2017), models can predict valence–arousal coordinates and translate them with contextual cues into situation-dependent categories or action recommendations (Kosti et al. 2020). Furthermore, emotion recognition increasingly focuses on complementary techniques, such as multi-task learning (Kollias 2022) or the use of soft labels (Fard et al. 2024). Together, dimensional and discrete approaches aim to balance the elegant simplicity of low-dimensional spaces with the richer explanatory power of discrete emotion concepts.

2.3 Emotion Recognition

Emotion recognition is a cornerstone of affective computing, enabling computers to analyse and respond to human feelings (Picard 1997). However, it does not suggest that computers can directly recognise one's inner feelings. Instead, emotion recognition should be interpreted as *'. . . inferring an emotional state from observations of emotional expressions and behavior, and through reasoning about an emotion-generating situation.'* (Picard 1997, p. 167). Therefore, some researchers used and proposed alternative terms such as 'affect detection', 'emotion perception', or 'emotion inference' (Barrett, Adolphs et al. 2019; Calvo and D'Mello 2010). Nevertheless, 'emotion recognition' remains the most widely used term in affective computing practice and is thus used throughout this thesis.

This section provides a broader context for FER—the central topic of this thesis and a specific emotion recognition technique—before its in-depth discussion in Section 2.4. The following subsections define the emotion recognition task (2.3.1), discuss key challenges (2.3.2), and examine the issues of obtaining reliable ground truth (2.3.3).

2.3.1 Task Definition

Computationally, emotion recognition is framed as a pattern-recognition task of learning a function that maps observable input signals to a target emotion representation (D'mello and Kory 2015). The general inference process is visualised in Figure 2.6. Common input modalities are facial expressions, speech, text, and physiological signals (see Table 2.1 for an overview). The target emotion representation draws from discrete and dimensional emotion theories (see Section 2.2) and varies across systems and modalities. FER usually relies on Ekman's six basic emotions plus a neutral category, whereas systems relying on physiological signals—such as heart-rate or electrodermal activity—are more likely to use a continuous valence-arousal space (Khare et al. 2024).

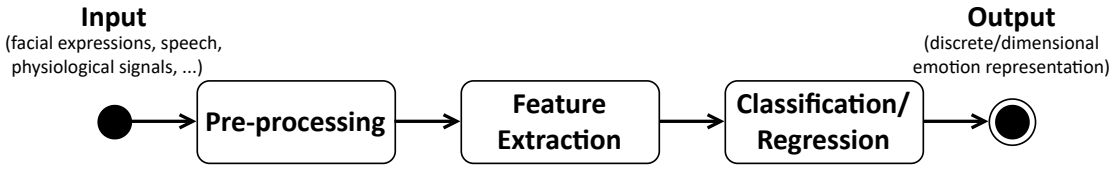


FIGURE 2.6: General emotion recognition process; based on (Cai et al. 2023).

Emotion recognition is typically formulated as a supervised learning problem (D'mello and Kory 2015). Accordingly, it becomes a multiclass classification task when the target space is categorical and a regression task when the target space is dimensional. Earlier work relied on classic machine learning models—such as support vector machine, logistic regression, or random forest—and hand-crafted features, such as geometric facial points, mel-frequency cepstral coefficients (MFCC) or skin-conductance statistics (Khare et al. 2024). Over the past decade, emotion recognition has undergone a deep-learning transition. Convolutional, recurrent, and Transformer networks learn feature representations directly from raw pixels, waveforms, text, or physiological signals, yielding significant accuracy gains (Khare et al. 2024). In addition, multimodal fusion architectures that combine several channels in end-to-end pipelines become increasingly important (Ahmed et al. 2023; D'mello and Kory 2015; S. Zhang et al. 2024).

TABLE 2.1: Input modalities for automatic emotion recognition.

Modality / Signal	**Sensor / Data Source**	**Raw Data**	**Hand-crafted Features**	**Reference Surveys**
Facial expressions (visual)	RGB/RGB-D/infrared/ thermal cameras	Pixel values of images or video frames, depth maps	Appearance (e.g. LBP, HOG), geometry (e.g. landmark positions, AU activations, optical flow)	Fasel and Luettin 2003, Corneanu et al. 2016, S. Li and W. Deng 2022
Facial expressions (muscle activity)	Facial EMG electrodes	EMG waveforms	Mean absolute value, variance, zero crossing, median frequency	M. Jiang et al. 2015 Wingenbach 2023
Speech / voice	Microphones	Audio waveforms	Prosody (e.g. rhythm, intonation), spectral features (e.g. MFCC), voice quality (e.g. shimmer)	El Ayadi et al. 2011, Wani et al. 2021
Text	Messages, posts, transcripts, documents	Sequences of text tokens	Bag-of-Words, part of speech tagging, n-grams, TF-IDF, lexicon sentiments	Nandwani and R. Verma 2021, Maruf et al. 2024
Body gesture and posture (visual)	RGB/RGB-D cameras, motion capture suits with visual markers	Pixel values of images or video frames, depth maps, marker coordinates	Skeletal keypoint positions, joint angles, distances, acceleration and velocity	Kleinsmith and Bianchi-Berthouze 2013, Noroozi et al. 2021
Eye-tracking	Infrared eye-trackers (remote or head-mounted), RGB cameras	Gaze coordinates, pupil diameter, blinks	Fixations, saccades, blink rate, pupil dilation	Lim et al. 2020, Skaramagkas et al. 2023
Physiological signals	ECG/EDA/EMG/EEG electrodes, PPG sensors, respiration belts, thermistors	Physiological signal waveforms (ECG, EDA, EEG, EMG, PPG, respiration, skin temperature)	Heart-rate variability, skin-conductance level, EMG peaks and variance, EEG band power, respiration rate, skin-temperature change	Shu et al. 2018, Bota et al. 2019, Saganowski et al. 2023

Abbreviations: AU = Action Unit, ECG = Electrocardiogram, EDA = Electrodermal Activity, EEG = Electroencephalogram, EMG = Electromyography, HOG = Histogram of Oriented Gradients, LBP = Local Binary Patterns, MFCC = Mel-Frequency Cepstral Coefficient, PPG = Photoplethysmography, RGB = Red–Green–Blue, RGB-D = RGB-Depth, TF-IDF = Term Frequency–Inverse Document Frequency.
Note: This table provides an overview and does not claim completeness.

2.3.2 Challenges

Emotion recognition is an inherently challenging task (Calvo and D'Mello 2010). First, identical observable signals can indicate different emotions depending on the cultural and situational context (Barrett, Mesquita et al. 2011). For instance, Namba, Rychlowska et al. (2020) showed that images of the same smile were rated as significantly more genuine in happy contexts and less genuine in contexts that implied politeness, relative to seeing the face alone. Such contextual information is rarely captured in emotion recognition datasets (Kosti et al. 2020).

Second, input signals are unreliable. Modalities—such as facial expressions, vocal prosody, or word choice—can be expressed deliberately, potentially masking genuine emotions. Additionally, autonomic responses—such as elevated heart rate or skin conductance—may reflect ambient temperature, exercise, or caffeine intake rather than emotions (Barry et al. 2008; Kreibig 2010).

Third, the target representations are only approximations of actual emotions. Discrete labels or low-dimensional coordinates cannot capture all aspects of rich, dynamic, multi-component, individual emotional experiences.

As Calvo and D'Mello stated, *'Affect detection is [...] a very challenging problem because emotions are constructs (i.e. conceptual quantities that cannot be directly measured) with fuzzy boundaries and with substantial individual difference variations in expression and experience.'* (Calvo and D'Mello 2010, p. 18).

2.3.3 Ground Truth

Consequently, producing ground truths for emotion recognition is challenging as well (D'mello and Kory 2015). Because models require standardised input–output formats for evaluation and learning, labels are collected using fixed schemas. In discrete settings, raters select labels from a set of categories (Mollahosseini et al. 2019). In dimensional settings, instruments—such as the Self-Assessment Manikin

(SAM; Bradley and Lang 1994) and the Affective Slider (Betella and Verschure 2016)—record valence, arousal, and, less commonly, dominance scores.

Self-reports can capture individual emotional experiences with fine granularity but are labour-intensive and susceptible to memory and social-desirability bias (Mauss and M. D. Robinson 2009). Thus, large-scale datasets rely on third-person ratings (Kosti et al. 2020; Mollahosseini et al. 2019). However, the inter-rater agreement is often only moderate (e.g. $\approx 60\%$ on the AffectNet dataset for FER (Mollahosseini et al. 2019)), particularly for naturalistic and in-the-wild settings (Afzal and P. Robinson 2011).

Early ground-truth datasets were recorded under laboratory conditions with actors posing emotions (Mattioli and Cabitza 2024). Such data do not require subsequent labelling and avoid the practical and ethical challenges of eliciting genuine emotions. On the other hand, this approach does not scale well, and the posed emotions differ from the messy, context-dependent, multimodal emotional expressions of everyday life (Barrett, Mesquita et al. 2011; Zeng et al. 2007). Moreover, posing emotions is only possible for deliberate emotional expressions, excluding responses of the autonomic nervous system (Kret 2015). Thus, contemporary research shifts towards collecting in-the-wild data, scraping publicly available images, videos, or text, and labelling them at scale through crowd-sourcing (Kosti et al. 2020; Mollahosseini et al. 2019). Although this approach scales well, it introduces additional label noise and cultural bias (Mattioli and Cabitza 2024; Mollahosseini et al. 2019).

As a result, emotion recognition models achieve well-above-chance accuracy but remain bound to human agreement levels, which constitute their ground truth (D'mello and Kory 2015).

2.4 Facial Expression Recognition

FER is the central emotion recognition technique used in this thesis. By definition, FER operates on facial appearance and motion, leveraging the fact that humans

naturally express and convey complex emotional states through facial expressions (Picard 1997). Although alternative contact-based measures—such as facial electromyography (fEMG)—can directly record muscle activations (Wingenbach 2023), this section focuses strictly on visual FER, reflecting its predominance in the literature and relevance for the remainder of this thesis. Low-cost cameras enable contact-free and scalable data capture, while modern processing hardware allows real-time inference (Ko 2018). Additionally, numerous large-scale datasets are available for training and evaluating person-independent FER models (Karnati et al. 2023). Drawing from a rich historical background, combined with steady advances in machine learning and deep learning, FER is one of the most studied methods in automatic emotion recognition (Corneanu et al. 2016; S. Li and W. Deng 2022). Consequently, mature FER solutions already support various real-world systems, including in-cabin driver monitoring systems (Smart Eye 2025), automated ad-effectiveness testing (iMotions 2025), socially interactive customer-service robots (SoftBank Robotics America 2025), and depression detection smartphone apps (Nepal et al. 2024).

The following subsections define the general FER task (2.4.1), contrast image-based (static) with image-sequence-based (dynamic) FER (2.4.2), examine multimodal FER approaches (2.4.3), and conclude with evaluation practices for FER (2.4.4).

2.4.1 Task Definition

FER maps facial expressions—as single images or image sequences—to a compact representation of the emotionally relevant facial information, most commonly one of Ekman's six or seven prototypical expressions (S. Li and W. Deng 2022). Although FER systems usually predict discrete labels such as 'anger' or 'fear', they target the visual appearance, not the inner emotional state. Inferring the emotional state from facial expressions is an additional step that extends FER to facial *emotion* recognition (Fasel and Luettin 2003). Facial emotion recognition can incorporate additional context (e.g. situational cues to distinguish a polite from a genuine smile) or knowledge (e.g. FACS mapping rules (Ekman, Friesen and Hager 2002)

for recognised AU activations). However, in practice, many studies treat FER results as emotional states, blurring the boundary between the two tasks (Fasel and Luettin 2003). Figure 2.7 illustrates the general FER pipeline. The following paragraphs describe each processing step.

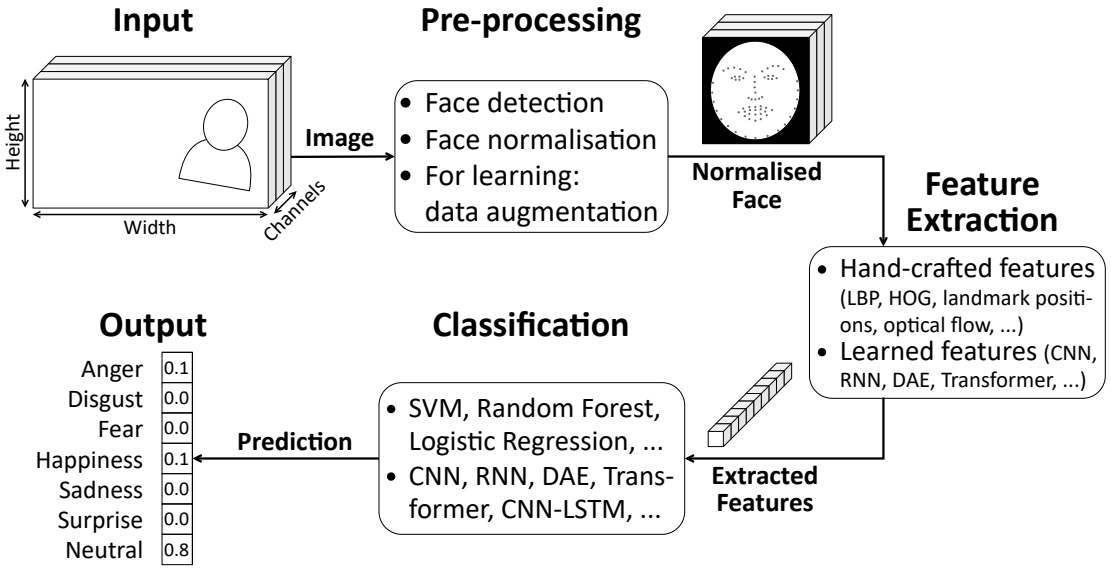


FIGURE 2.7: General facial expression recognition process; based on (Corneanu et al. 2016; Fasel and Luettin 2003; Karnati et al. 2023).

Abbreviations: CNN = Convolutional Neural Network, DAE = Deep Autoencoder, HOG = Histogram of Oriented Gradients, LBP = Local Binary Patterns, LSTM = Long Short-Term Memory, RNN = Recurrent Neural Network, SVM = Support Vector Machine.

Input. Visual FER operates on single images or temporal image sequences captured by camera (Y. Wang, Yan et al. 2024). Each frame is represented as a three-dimensional tensor of height×width×channels (Chollet 2021, p. 37). The number of channels depends on the sensor type: greyscale, infrared, or thermal cameras output one channel, RGB cameras provide three, and RGB-D systems add a depth channel for a total of four (Corneanu et al. 2016). RGB cameras are the most common choice because of their low cost, high availability, widespread use, and rich colour information (Ko 2018; S. Li and W. Deng 2022).

Pre-processing. Camera recordings usually contain more than just a subject's face. Thus, pre-processing begins with face detection, performed by classic Haar

cascade classifiers (Viola and Jones 2001) or modern deep-learning detectors, such as RetinaFace (J. Deng et al. 2019) and BlazeFace (Bazarevsky et al. 2019). Then, the detected facial region is cropped, geometrically aligned using facial landmarks, and resized to a fixed resolution (e.g. 224×224×3 for RGB input) (Corneanu et al. 2016). Additionally, photometric normalisation (e.g. histogram equalisation or colour-channel standardisation) reduces illumination differences (Karnati et al. 2023). In the learning phase, data augmentation introduces random rotations, zooms, noise, and colour perturbations to increase sample diversity and reduce overfitting (S. Li and W. Deng 2022). Note that this description is not exhaustive, and that most pre-processing steps are optional (although face detection and resizing are almost always applied). In particular, modern deep-learning approaches can learn many invariances themselves (Karnati et al. 2023).

Feature Extraction. Feature extraction in FER has evolved from interpretable, hand-crafted techniques to automated, data-driven representation learning (Ko 2018). Traditional approaches relied on explicit descriptors, such as local binary patterns for texture analysis, facial landmark coordinates for geometric relationships, and optical flow for motion dynamics (Corneanu et al. 2016). These hand-crafted features required domain expertise to design and often struggled with real-world variability (e.g. in lighting and pose) (Ko 2018). Modern approaches replace them with deep feature learning. Typically, convolutional neural networks (CNNs) extract spatial features from single frames, whereas recurrent units (e.g. long short-term memory (LSTM)) enable capturing temporal dynamics from sequences (S. Li and W. Deng 2022). Currently, Transformer-based architectures achieve state-of-the-art results by modelling long-range dependencies and global context through attention mechanisms (Karnati et al. 2023). The resulting deep embeddings provide discriminative input for downstream classification.

Classification. Classification maps the extracted features to discrete emotion labels. Traditional pipelines attached classic machine-learning classifiers—such as support vector machine or random forest—to hand-crafted features (Corneanu

et al. 2016). In contrast, modern deep-learning approaches integrate classification with feature extraction by appending trainable output layers, typically softmax-activated, to optimise the entire network through end-to-end learning, usually via cross-entropy loss (Karnati et al. 2023). Hybrid architectures bridge these paradigms by using deep networks purely as feature extractors, followed by an external classifier, most commonly a support vector machine, to combine deep representation learning with traditional discriminative power (S. Li and W. Deng 2022).

Output. FER systems predominantly output discrete labels or probability distributions aligned with Ekman's basic emotions anger, disgust, fear, happiness, sadness, and surprise, plus neutral (S. Li and W. Deng 2022). Compound emotions (e.g. happily surprised or angrily disgusted) extend this taxonomy but are addressed by fewer studies (Karnati et al. 2023). AU vectors that detail individual muscle activations are an alternative output format (Corneanu et al. 2016). AU coding offers a formalised pose-invariant description of facial configurations that can be mapped to prototypical emotions via FACS rules (Ekman, Friesen and Hager 2002). Dimensional models offer a continuous description of emotional states that can capture intensity variations, mixed states, and trajectories over time (Corneanu et al. 2016). In this setting, FER becomes a regression task, most commonly predicting continuous valence and arousal values based on Russell's Circumplex Model of Affect (Russell 1980). Nevertheless, dimensional FER remains the exception because datasets with continuous annotations are scarce, and dimensional values are often less intuitive for annotators and end-users than categorical labels (S. Li and W. Deng 2022).

2.4.2 Static and Dynamic Facial Expression Recognition

FER is either static or dynamic. Static FER (SFER) processes single images and extracts spatial features, such as texture and geometry. In contrast, dynamic FER (DFER) processes image sequences or videos, enabling the extraction of additional

temporal features across multiple frames (Y. Wang, Yan et al. 2024). Facial (macro) expressions usually progress through onset (initial muscle movement), apex (peak intensity), and offset (return to neutral) phases within about 0.5–4 seconds, as visualised in Figure 2.8 (Ben et al. 2022). DFER leverages this progression to disambiguate similar static frames (e.g. genuine vs. polite smiles), increase robustness (e.g. to occlusion or illumination changes), and capture temporal transitions that a single frame might miss.



FIGURE 2.8: An example of the temporal phases of a happy facial expression from onset, over apex to offset phase; from (Wu et al. 2014, Figure 7), © 2014 the authors, licensed under CC BY 3.0, no changes.

For spatiotemporal modelling, three model families dominate (S. Li and W. Deng 2022; Y. Wang, Yan et al. 2024). First, 3D CNNs perform convolutions across spatial and temporal dimensions simultaneously, capturing local motion patterns. Second, CNN–recurrent neural network (RNN) hybrids use a 2D CNN to extract spatial features from each frame, followed by an RNN (e.g. LSTM or gated recurrent unit (GRU)) to learn cross-frame temporal dynamics. Third, more recently, Transformers have emerged (Vaswani et al. 2017), applying self-attention to also capture global face dynamics and long-range temporal dependencies.

Many DFER systems output a probability vector of Ekman's six basic emotions plus neutral, analogous to SFER, for each sequence. However, DFER can also uniquely predict intensity levels and temporal phase transitions. When sequential data are available, DFER can outperform SFER, offering advantages through context awareness and increased robustness for emotion recognition, despite the higher computational costs. (Y. Wang, Yan et al. 2024)

2.4.3 Multimodal Facial Expression Recognition

Information about the same phenomenon can be captured by multiple sensors. Each sensor output can be termed a 'modality' (Lahat et al. 2015). For instance, the same facial expression of a person can be observed via an RGB camera or by electrodes placed on the person's face via fEMG. Different modalities have different advantages and disadvantages. A single modality rarely provides complete information about the target phenomenon. Thus, leveraging complementary information from multiple modalities has become a central aspect in modern machine learning approaches (Baltrušaitis et al. 2019), particularly in recognising emotions, which are expressed through facial expressions, speech, text, body gestures, and various physiological signals (see Table 2.1) (Ezzameli and Mahersia 2023). In their review of 90 multimodal affect detection systems, D'Mello and Kory found that, on average, multimodal systems were 9.83% more accurate relative to their respective best unimodal counterpart (D'mello and Kory 2015).

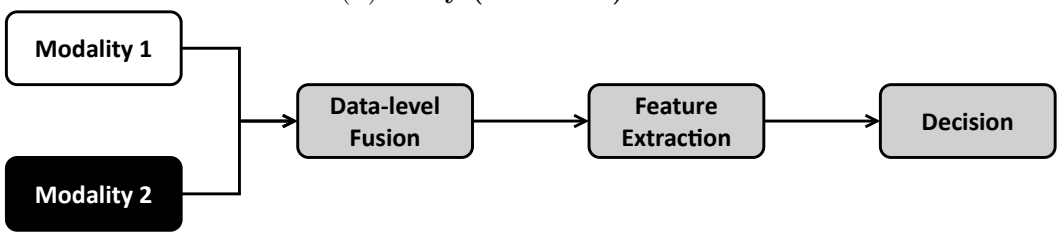
Likewise, multimodality can enrich FER (Corneanu et al. 2016). RGB images are widely available and provide rich visual information, but are sensitive to illumination and occlusion. Depth data and thermal images are largely invariant to ambient light and add information about 3D facial structure and changes in blood flow, respectively. Facial EMG measures muscle movements directly, but its signal can be noisy and requires electrodes to be placed directly on the skin (M. Jiang et al. 2015). For instance, by including depth maps alongside RGB data, Oyedotun et al. (2017) increased their system's recognition accuracy on the BU-3DFE database (Yin, Wei et al. 2006) from 82.92% to 89.31%. Similarly, S. Wang, He et al. (2014) fused RGB and thermal images from the NVIE database (S. Wang, Z. Liu et al. 2010), improving accuracy from 75.33% to 76.82%.

Multimodal fusion strategies can be characterised as *early* (data-level fusion), *intermediate* (feature-level fusion), *late* (decision-level fusion), or *hybrid* fusion (see Figure 2.9). In early fusion, raw or pre-processed data from different modalities are merged at the input stage into a joint representation, for instance, concatenating thermal images as a fourth channel to RGB images. Intermediate fusion merges

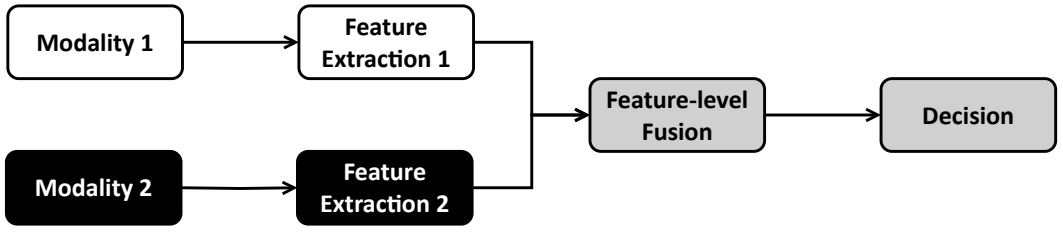
higher-level features extracted separately from each modality, for instance, concatenating CNN-extracted image features with hand-crafted fEMG features. Late fusion keeps modality-specific models separate and only merges their outputs at the decision level, for instance, averaging the probabilities of two prediction vectors for the same six basic emotion categories. Finally, hybrid fusion combines two or more of the above fusion levels within a single architecture. (F. Zhao et al. 2024)

Although recent works propose broader taxonomies for multimodal machine learning, challenging the intense focus on data fusion levels (Baltrušaitis et al. 2019; P. P. Liang et al. 2024; F. Zhao et al. 2024), these categories remain essential for the multimodal approaches presented later in this thesis (particularly Sections 7.1.5 and 7.2.4).

(A) **Early (data-level) fusion.**



(B) **Intermediate (feature-level) fusion.**



(C) **Late (decision-level) fusion.**

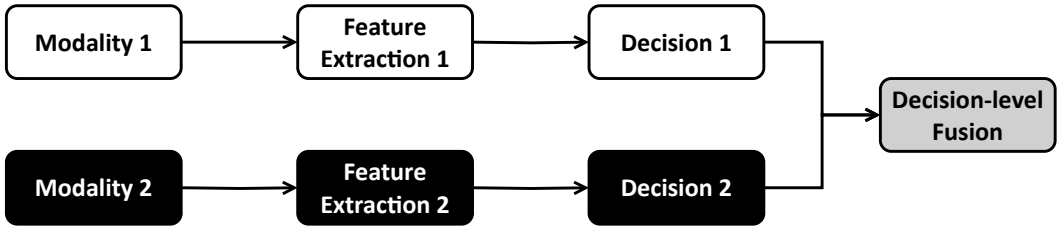


FIGURE 2.9: Classical taxonomy of multimodal fusion strategies; based on (Ramachandram and Taylor 2017; F. Zhao et al. 2024).

By fusing modalities, FER systems can capitalise on complementary information from heterogeneous sources, increasing accuracy and robustness to illumination shifts, sensor noise, and partial occlusion (Corneanu et al. 2016). However, these gains come at the price of additional complexity. Fusion techniques must be designed carefully to produce meaningful multimodal representations (Lahat et al. 2015). Data from different sensors must be aligned in terms of time, space, and semantics so that the corresponding information is associated correctly, despite varying sampling rates, coordinate systems, and representation scales (Baltrušaitis et al. 2019). Missing or degraded modalities demand fusion schemes that gracefully handle incomplete inputs or impute data. Moreover, larger data volumes and more sophisticated fusion pipelines require larger models and greater computational resources (F. Zhao et al. 2024). Despite—and partly because of—these challenges, multimodal learning remains a vibrant research area that is expected to drive continued advances in many machine learning tasks (P. P. Liang et al. 2024), such as FER.

2.4.4 Evaluation

Well-designed evaluation protocols and metrics are fundamental to advancing FER research because they ensure that results are reliable, comparable, and reproducible (Ko 2018). This section discusses essential evaluation protocols, performance metrics, and reproducibility practices in FER.

Protocols and Data Splits. Common evaluation protocols in FER include:

1. Subject-independent k-fold cross-validation (Hu et al. 2025; Ko 2018)
2. Leave-one-subject-out cross-validation (LOSO-CV) (Ab Razak and Sahran 2023)
3. Fixed database splits (S. Li, W. Deng and Du 2017; Mollahosseini et al. 2019)
4. Cross-database evaluation (Ramis et al. 2022)

Subject-independence in FER is essential to prevent models from learning identity-specific features that do not generalise beyond the training subjects. LOSO-CV is particularly effective for databases with few subjects. Fixed database splits require careful documentation and can provide maximum comparability. Cross-database evaluation is particularly relevant for testing model robustness to domain shifts caused by differences in recording conditions, demographics, and annotation standards. It typically yields lower accuracy but can provide a more realistic benchmark for a model's capabilities on unseen data. To prevent data leakage in dynamic FER, data must be split at the sequence level to avoid leaking temporal context between training and test sets (X. Jiang et al. 2020; Y. Wang, Y. Sun et al. 2022). Section 2.5 provides an overview of commonly used databases in FER research.

Performance Metrics. As the categorical approach is dominant in current FER research (S. Li and W. Deng 2022) and is followed throughout this thesis, this paragraph focuses on FER as a single-label multiclass classification task and does not include regression metrics.

Accuracy (i.e. micro-averaged across classes) remains one of the most widely reported evaluation metrics, providing an intuitive assessment of overall performance. However, it does not differentiate between classes and can be misleading for imbalanced test sets (Thabtah et al. 2020).

Definition 2.1 (Accuracy)**.** For N items and K classes with true labels $y^{(i)} \in \{1, \ldots, K\}$ and predictions $\hat{y}^{(i)}$, the accuracy Acc is defined as:

$$Acc = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left(\hat{y}^{(i)} = y^{(i)}\right) \tag{2.1}$$

$y^{(i)}$ is the true label of item i,

$\hat{y}^{(i)}$ is the predicted label for item i,

$\mathbf{1}$ is the indicator function (1 if true, 0 otherwise).

Precision, *Recall*, and *F1-score* offer more nuanced performance insights on a per-class basis, which is crucial in FER evaluation, as recognition performance often varies considerably across emotions (Barsoum et al. 2016; S. Li, W. Deng and Du 2017; Mollahosseini et al. 2019).

Definition 2.2 (Precision)**.** For K classes, the precision $Prec_k$ of class $k \in \{1, \ldots, K\}$ is defined as:

$$Prec_k = \frac{TP_k}{TP_k + FP_k} \tag{2.2}$$

TP_k is the number of True Positives for class k,
FP_k is the number of False Positives for class k.

Definition 2.3 (Recall)**.** For K classes, the recall Rec_k of class $k \in \{1, \ldots, K\}$ is defined as:

$$Rec_k = \frac{TP_k}{TP_k + FN_k} \tag{2.3}$$

TP_k is the number of True Positives for class k,
FN_k is the number of False Negatives for class k.

Definition 2.4 (F1-score)**.** For K classes, the F1-score $F1_k$ of class $k \in \{1, \ldots, K\}$ is the harmonic mean of precision $Prec_k$ and recall Rec_k:

$$F1_k = \frac{2 \cdot Prec_k \cdot Rec_k}{Prec_k + Rec_k} \tag{2.4}$$

$Prec_k$ is the precision for class k (Definition 2.2),
Rec_k is the recall for class k (Definition 2.3).

The *unweighted average recall* (UAR), also known as *balanced accuracy* or *macro-averaged accuracy*, calculates the average recall per class, regardless of the number of samples. Thus, it weights classes with fewer samples higher than the micro-averaged accuracy and is an important metric when test sets are imbalanced.

Definition 2.5 (Unweighted Average Recall)**.** For K classes with per-class recalls $Rec_1, \ldots, Rec_K$ (see Definition 2.3), the unweighted average recall UAR is defined

as:

$$\text{UAR} = \frac{1}{K} \sum_{k=1}^{K} Rec_k \tag{2.5}$$

Confusion matrices—constructed from per-class true positives, false positives, true negatives, and false negatives—are a standard tool for qualitative and quantitative error analysis in classification. In FER, commonly reported confusions on in-the-wild datasets include fear vs. surprise and anger vs. disgust (e.g. on FER+ (Barsoum et al. 2016), RAF-DB (S. Li, W. Deng and Du 2017), and AffectNet (Mollahosseini et al. 2019)), reflecting visual overlap in prototypical facial configurations and annotation uncertainty (Sebe et al. 2005, p. 196).

Reproducibility Practices. Modern reproducibility practices in FER research mirror broader trends in machine learning towards transparency and reproducibility (Pineau et al. 2021). Researchers are increasingly expected to release source code, environment configurations, trained models, and data splits, ideally via public repositories, to facilitate replication (NeurIPS 2025b; S. Zhou et al. 2024). Documenting random seeds, sharing data pre-processing scripts, and reporting training configurations all contribute to evaluations that can be independently verified. Major conferences have introduced reproducibility checklists (NeurIPS 2025b) and guidelines (NeurIPS 2025a) for authors, asking whether code and data are shared and whether results include error bars across multiple runs. Embracing these practices not only ensures the scientific integrity of FER results but also enables other researchers to confidently reuse and build upon FER models in downstream applications.

2.5 Facial Expression Recognition Databases

FER databases are essential for training and evaluating FER models. Their quality, diversity, and scale directly influence the capabilities and limitations of FER systems, making database creation a critical priority that underpins FER research. Table 2.2 offers a compact comparison of frequently used FER databases.

The following subsections present impactful FER databases in chronological order (2.5.1), discuss the challenges of obtaining ground truth (2.5.2), and introduce metrics to quantify inter-rater reliability (2.5.3).

2.5.1 Historical Development

Early FER databases captured posed expressions under laboratory-controlled conditions to ensure clarity and consistency. The *Japanese Female Facial Expression* (JAFFE) database (Lyons et al. 1998) comprises 213 greyscale images of 10 Japanese women posing the six basic facial expressions proposed by Ekman and Friesen (1975), plus a neutral expression. With amateur actors posing the same seven expressions, the *Karolinska Directed Emotional Faces* (KDEF) database (Lundqvist et al. 1998) contains 4,900 RGB images of 35 females and 35 males, captured from five angles.

The *Cohn-Kanade* (CK) database (Kanade et al. 2000) and its extension *CK+* (Lucey et al. 2010) marked a shift from static to dynamic analysis, capturing 593 video sequences of 123 subjects transitioning from neutral to peak facial expressions. Both databases also include AU annotations, which enable detailed analysis of facial expressions.

In the early 2000s, additional modalities emerged. The *Binghamton University 3D Facial Expression* (BU-3DFE) database (Yin, Wei et al. 2006) comprises 2,500 3D images of 100 subjects, each posing six basic emotional expressions at four different intensities plus a neutral expression. *BU-4DFE* (Yin, Xiaochen Chen et al. 2008) extended this approach by providing 3D videos. Furthermore, the *NVIE* (S. Wang, Z. Liu et al. 2010) and *Oulu-CASIA* (G. Zhao et al. 2011) databases introduced thermal and near-infrared (in addition to visible light) videos, respectively.

The limited subject pool, controlled recording conditions, and deliberate nature of laboratory corpora soon exposed a reality gap between posed in-lab and spontaneous in-the-wild expressions encountered in everyday life (Cohn and Schmidt 2004; Mollahosseini et al. 2019; Namba, Makihara et al. 2017). Collecting substantially

TABLE 2.2: Overview of frequently used FER databases.

Database	Samples	Subj.	Environ.	Elicit.	Occl.	Annotation Scheme
KDEF (Lundqvist et al. 1998)	4,900 RGB images from five angles in two sessions	70	Lab	P	N	6 BE + neutral
BU-4DFE (Yin, Xiaochen Chen et al. 2008)	606 3D model sequences paired with 2D texture videos	101	Lab	P	N	6 BE (+ neutral onset)
CK+ (Lucey et al. 2010)	593 greyscale / RGB image sequences	123	Lab	P(&S)	N	6 BE and contempt (+ neutral onset) + FACS coded apex frame
Oulu-CASIA (G. Zhao et al. 2011)	2,880 visible-light and near-infrared videos for 3 illumination levels	80	Lab	P	N	6 BE (+ neutral onset)
FER-2013 (Goodfellow et al. 2013)	35,887 greyscale images based on keyword search via Google image search	–	Web	P/S	Y	6 BE + neutral
RAF-DB (S. Li, W. Deng and Du 2017)	29,672 images based on keyword search via Flickr image search	–	Web	P/S	Y	6 BE + neutral + 12 compound expressions
AffectNet (Mollahosseini et al. 2019)	291,651 emotion-labelled images available (450,000 manually annotated images) based on keyword search via Google, Bing and Yahoo image search	–	Web	P/S	Y	6 BE and contempt + neutral + valence and arousal
DFEW (X. Jiang et al. 2020)	16,372 videos (movie clips)	–	Movie	P	Y	6 BE + neutral
FERV39k (Y. Wang, Y. Sun et al. 2022)	38,935 videos with 22 scene contexts based on keyword search via various search/video engines	–	Movie, TV show, Web	P/S	Y	6 BE + neutral

Heading Abbreviations: Environ. = Environment, Elicit. = Elicitation Method, Occl. = Occlusion present?, Subj. = Number of Subjects.
Cell Abbreviations: BE = Basic Expressions, N = No, P = Posed, S = Spontaneous, Y = Yes.

more data in a laboratory is time-consuming and expensive, particularly in terms of demographic and contextual diversity. Consequently, the field pivoted towards in-the-wild data gathered from the Internet and movie footage (X. Jiang et al. 2020; S. Li, W. Deng and Du 2017; Mollahosseini et al. 2019; Y. Wang, Y. Sun et al. 2022).

The first widely used landmark was the *FER-2013* database (Goodfellow et al. 2013), with 35,887 face crops of 48×48 pixels scraped using the Google image search API with 184 emotion-related keywords. Each image is annotated with one of Ekman's six basic expressions plus neutral.

S. Li, W. Deng and Du (2017) searched and found 29,672 facial images with Flickr's image search API to create the *Real-world Affective Faces Database* (RAF-DB). To ensure high data quality, each image was labelled by about 40 independent labellers. This also enabled RAF-DB to provide labels for 11 compound expressions in addition to the usual seven classes.

Pushing the scale an order of magnitude further, the *AffectNet* database (Mollahosseini et al. 2019) comprises 291,651 manually annotated emotional faces found through Google, Bing, and Yahoo image searches. The cropped and resized images have RGB colour and a resolution of 224×224 pixels. Besides eight discrete classes (six basic expressions and contempt plus neutral), AffectNet also includes continuous valence-arousal ratings, establishing a dual categorical-dimensional benchmark for modern FER models.

Although the previous three corpora revolutionised static FER, they do not contain dynamic information. To fill this gap, the *Dynamic Facial Expression in the Wild* (DFEW) database (X. Jiang et al. 2020) provides 16,372 short movie clips captured under challenging conditions (e.g. variations in pose, illumination, and occlusion). Each clip was rated by 10 labellers for Ekman's six basic expressions plus neutral. However, since the clips originate from movies, the expressions are posed (professionally) and not spontaneous.

The *FERV39k* database (Y. Wang, Y. Sun et al. 2022) comprises 38,935 clips from a more diverse range of sources, including online videos, TV shows, and movies. In addition to the usual seven expression categories, each clip is associated with one of 22 everyday scenes (e.g. talk show, street, official event), allowing for a systematic study of the scene context in expression perception.

Collectively, the progression from early databases—such as JAFFE and KDEF—to contemporary ones—such as AffectNet and FERV39k—illustrates a shift in key database properties:

- Small homogeneous cohorts → large diverse populations
- Laboratory-controlled conditions → in-the-wild conditions
- Posed expressions → spontaneous expressions
- Static FER → dynamic FER
- Unimodal FER → multimodal FER

2.5.2 Ground Truth

As for emotion recognition in general (see Sections 2.3.2 and 2.3.3), obtaining ground truth in FER is challenging. However, labels or annotations that indicate the true emotional state, perceived emotional expression, or observed facial configuration present in each data sample are essential for learning algorithms and evaluation procedures (S. Li and W. Deng 2022). In FER research, the ground truth is expressed in one of three annotation schemes (Corneanu et al. 2016):

- Discrete categories (anger, disgust, fear, happiness, sadness, surprise, optionally contempt, and neutral) based on Ekman's theory of basic emotions (Ekman 1992) (see Section 2.2.1)
- Continuous valence-arousal ratings (optionally plus dominance) based on Russell's Circumplex Model of Affect (see Figure 2.4) and Pleasure–Arousal–Dominance model (see Figure 2.3)

- AU activations based on FACS developed by Ekman and Friesen (Ekman, Friesen and Hager 2002)

The choice of scheme affects both the complexity of labelling and the metrics used for evaluation. Discrete categories are the dominant annotation scheme in current FER research (S. Li and W. Deng 2022).

Early FER datasets—such as JAFFE and KDEF—relied on posed facial expressions performed by subjects following explicit instructions. Thus, the instructed emotion served directly as the label (Lundqvist et al. 1998; Lyons et al. 1998).

To obtain richer structural labels, CK and CK+ were annotated by certified FACS experts, providing AU codes for peak frames in addition to emotion categories (Kanade et al. 2000; Lucey et al. 2010). Although these expert annotations are highly trusted, FACS coding requires specialised training and is immensely time-consuming (Ekman, Friesen and Hager 2002). Consequently, the resulting databases remain modest in size.

The gap between posed laboratory expressions and spontaneous in-the-wild expressions is another major challenge for ground truth in FER databases. Laboratory conditions do not accurately reflect the variability of real-world data (Mollahosseini et al. 2019). Spontaneous expressions can differ from posed expressions in terms of AU combinations, onset phases, apex phases, and intensity, and are difficult to elicit reliably (Cohn and Schmidt 2004; Namba, Makihara et al. 2017).

To address these limitations, modern corpora—such as RAF-DB and AffectNet—have replaced expert annotation of in-lab data with cost-effective crowd-sourcing of expressions scraped from the Internet.

S. Li, W. Deng and Du (2017) employed 315 students and staff members from universities to assign one of seven emotion labels to RAF-DB's images. Each annotator received a one-hour tutorial in emotion theory and online expression annotation. A custom website, which displayed each image with the exclusive label options, served as the labelling tool. RAF-DB's 29,672 images were distributed

equally and randomly among the annotators. Consequently, each image received labels from about 40 independent annotators. The resulting multi-annotator labels were then processed using an expectation-maximisation framework to assess the reliability of each annotator and filter out noisy labels. As a result, the labels of 285 annotators remained with a Cronbach's alpha score of 0.966, indicating excellent inter-rater agreement.

AffectNet (Mollahosseini et al. 2019) employed 12 trained annotators to label 450,000 images with eight categories and valence-arousal coordinates. Except for a subset of 36,000 images, each image was labelled by only one annotator. The 36,000 images labelled by two annotators revealed an agreement of 60.7% for the categorical labels, demonstrating the difficulty of FER for in-the-wild images.

An earlier validation study using the KDEF database showed that 272 female students from the University of Ghent identified the correct target emotion with an overall accuracy of 71.87%, ranging from 43.03% for fear to 92.65% for happiness (Goeleven et al. 2008). These results were comparable to those of other validation studies, exhibiting the general challenge of obtaining reliable emotion labels for facial expressions.

Another challenge in using Web content is the unequal class distribution. Online searches yield substantially more results for happy faces than for fearful or disgusted ones. Consequently, RAF-DB's single-label subset contains 5,957 happiness labels but only 867 for anger, 877 for disgust, and 355 for fear (the label distribution is not reported by S. Li, W. Deng and Du (2017) but based on direct inspection of the dataset). Likewise, AffectNet contains 143,991 happy and 78,780 neutral labels but only 27,793 for anger, 5,225 for disgust, and 8,136 for fear (Mollahosseini et al. 2019).

2.5.3 Inter-rater Reliability

Researchers use several metrics to quantify the agreement between raters or annotators (Hallgren 2012). Arguably, the simplest metric to compute is the *percent*

agreement.

Definition 2.6 (Percent Agreement)**.** For two raters, A and B, labelling the same N items into K classes, the percent agreement P_o is defined as follows:

$$P_o = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}(y_A^{(i)} = y_B^{(i)}) \tag{2.6}$$

$y_A^{(i)}$ is the label from rater A for item i,
$y_B^{(i)}$ is the label from rater B for item i,
$\mathbf{1}$ is the indicator function (1 if true, 0 otherwise).

However, the percent agreement does not account for coincidental agreements or the number of classes. Thus, the same percent agreement value can imply very different things depending on whether 2 or 20 classes are involved. *Cohen's kappa* (Cohen 1960) addresses this limitation by relating the observed agreement to the expected agreement by chance.

Definition 2.7 (Cohen's kappa)**.** For two raters, A and B, labelling the same N items into K classes, Cohen's kappa κ is defined as follows:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{2.7}$$

P_o is the observed agreement (see Definition 2.6),
P_e is the expected agreement by chance, defined as:

$$P_e = \sum_{k=1}^{K} p_A(k) \cdot p_B(k)$$

$p_A(k)$ is the proportion of items that rater A assigned to class k,
$p_B(k)$ is the proportion of items that rater B assigned to class k.

Cohen's kappa can only be computed for two raters. *Fleiss' kappa* (Fleiss 1971) extends its approach to more than two raters. Additionally, not all raters need to label all items. Instead, only the number of labels per item must be equal.

Definition 2.8 (Fleiss' kappa)**.** For L labels assigned to each of the same N items divided into K classes, Fleiss' kappa κ is defined as follows:

$$\kappa = \frac{P_o - P_e}{1 - P_e} \tag{2.8}$$

P_o is the observed agreement, defined as:

$$P_o = \frac{1}{NL(L-1)} \sum_{i=1}^{N} \sum_{k=1}^{K} n_{ik}(n_{ik} - 1)$$

n_{ik} is the number of labels of category k assigned to item i.

P_e is the expected agreement by chance, defined as:

$$P_e = \sum_{k=1}^{K} p_k^2$$

p_k is the overall proportion of all assignments made to category k.

The kappa family is most appropriate when labels are nominal, complete, and class-balanced. For interval or metric annotations—such as continuous valence-arousal ratings—the *intraclass correlation coefficient* (ICC; Shrout and Fleiss 1979) provides a better estimate of absolute agreement or consistency across multiple raters. When databases contain heterogeneous measurement scales or missing labels, *Krippendorff's alpha* (Krippendorff 2011) is preferable because it generalises to nominal, ordinal, interval, and ratio data and remains unbiased even when items receive different numbers of annotations.

2.6 Virtual Reality and Emotions

VR refers to computer-generated digital environments that users can experience and interact with as if they were real (Jerald 2015, p. 9). Historically, display devices used to simulate such environments have progressed from basic single-screen setups to large surrounding displays and projections, and ultimately to modern HMDs (Marín-Morales et al. 2020). Although the concept of HMDs

dates back to Ivan Sutherland's pioneering work in 1968 (Sutherland 1968), it was not until the emergence of high-quality, low-cost consumer devices—such as the Oculus Rift, HTC Vive, and Sony PlayStation VR—around 2014 that HMDs became the predominant hardware platform for VR (Cipresso et al. 2018). The advanced capabilities of these modern HMDs have since sparked a renewed wave of interest and investment in VR-related research (see Figure 2.10), technologies, and applications (Wohlgenannt et al. 2020). Given the predominance of HMDs in current VR research, this thesis exclusively focuses on VR delivered via HMDs.

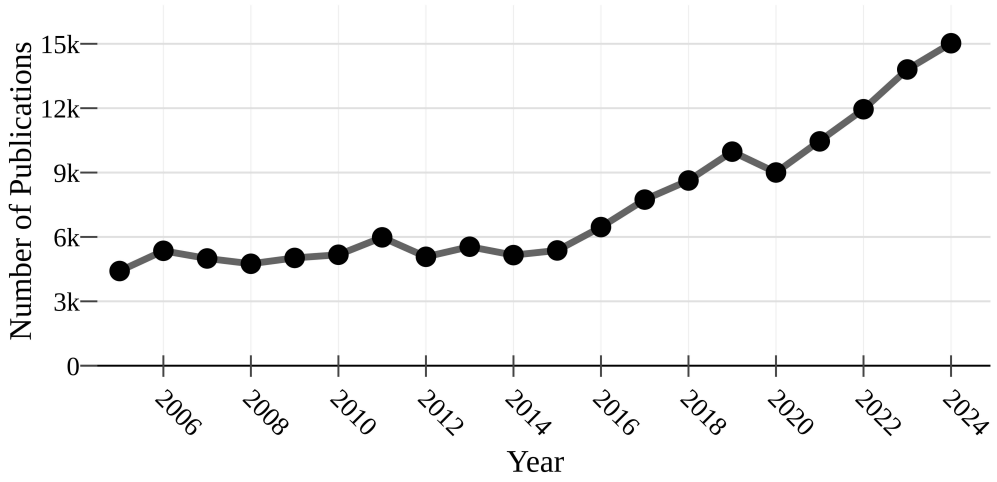


FIGURE 2.10: Number of publications related to VR since 2005, queried from the Scopus database on 6 August 2025. Query: *TITLE-ABS-KEY("virtual reality") AND (LIMIT-TO(DOCTYPE,"cp") OR LIMIT-TO(DOCTYPE,"ar"))*.

To provide essential context and motivation for the research aim of this thesis, this section explores the intersection between VR and emotion research. The following subsections examine the use of VR for emotion elicitation (2.6.1), review approaches to emotion recognition in VR (2.6.2), and survey key application areas that benefit from emotion-aware VR (2.6.3).

2.6.1 Emotion Elicitation

VR systems possess several properties that distinguish them from traditional media and make them especially well-suited for eliciting emotions.

Immersion. Immersion is the objective capability of a VR system to deliver a believable illusion of reality to its users' senses (Slater and Wilbur 1997). This capability results from the system's technological configuration, including display resolution, field of view, spatial audio fidelity, and tracking accuracy and latency (Slater 2009). Consequently, modern HMD-based systems are typically considered *immersive*, whereas basic single-screen setups are regarded as *non-immersive* (Slater, Banakou et al. 2022).

Presence. Presence is the subjective feeling of 'being there' in the virtual environment. It emerges from the successful integration of immersive technology with its user's perceptual processing (Slater and Sanchez-Vives 2016). Slater, Banakou et al. (2022) further decompose presence into *Place Illusion*—the feeling of actually being in the virtual place—and *Plausibility Illusion*—the feeling that events depicted in the virtual environment are actually happening.

Embodiment. Embodiment is a third illusion, which shifts the focus from the virtual environment to the virtual self (Slater, Banakou et al. 2022). When a virtual avatar moves in tight synchrony with the user's real body through head and body tracking, users can experience a sense of *agency*, *body ownership*, and *self-location* for that avatar—collectively referred to as embodiment (Guy et al. 2023; Kilteni et al. 2012).

Interactivity. Interactivity is another key property of VR. Unlike conventional image or film viewing, VR can allow users to move around, make choices, and modify the virtual environment (M. Balcerak Jackson and B. Balcerak Jackson 2024). With low latency and appropriate action-effect mapping, interactive environments increase plausibility and turn users from passive observers into active participants (Slater, Banakou et al. 2022).

VR Sickness. VR sickness, also known as *cybersickness*, refers to a cluster of symptoms—including nausea, eye fatigue, and disorientation—that can arise

during HMD use (Chang et al. 2020). Although individual susceptibility varies significantly, VR sickness is commonly attributed to a visual-vestibular sensory conflict, i.e. the user sees (self-)motion through the HMD without sensing actual physical movement (N. Tian et al. 2022). Hardware and content factors—such as high latency, low frame rate, intense optic flow, and continuous locomotion—can exacerbate symptoms (Biswas et al. 2024). Importantly, VR sickness disrupts effective emotion elicitation because it compromises the VR illusion (Weech et al. 2019) and elicits unwanted negative feelings. Consequently, VR sickness can shift emotional responses towards negative valence and high arousal, potentially even leading to premature termination of exposure. To account for VR sickness as a key confound, studies often measure it using established questionnaires (Biswas et al. 2024), such as the Simulator Sickness Questionnaire (SSQ; Kennedy et al. 1993) or the Virtual Reality Sickness Questionnaire (VRSQ; H. K. Kim et al. 2018).

Together, *immersion*, *presence*, *embodiment*, and *interactivity* enable VR to elicit emotions more reliably and intensely than non-immersive media (Gall et al. 2021; M. Li et al. 2022; Makransky and Lilleholt 2018; Yung et al. 2021). Leveraging this ability, researchers can conduct experiments in tightly controlled virtual environments—from relaxing landscapes (Anderson et al. 2017) to horror games (Lin 2017)—to elicit user responses similar to those observed in analogous real-world settings (Marín-Morales et al. 2020). Compared to real environments, VR can offer better control (over timing, intensity, and context), isolation (holding extraneous variables constant across trials), observability (capture of the VR scene, synchronised logs of behaviour, gaze, and physiology), reproducibility (standardised, shareable 'stimuli-as-code'), safety (high-arousal scenarios without real hazard, with instant abort), and cost efficiency (reusable scenes, minimal physical setup) (Aguilar et al. 2024; Bell, Nicholas et al. 2020; Marín-Morales et al. 2020; Somarathna et al. 2023). Consequently, VR has become an increasingly important medium for eliciting and studying emotions (Somarathna et al. 2023). This trend also extends to automatic emotion recognition in VR (Marín-Morales et al. 2020), which is discussed in the following subsection.

2.6.2 Emotion Recognition

Given VR's exceptional ability to elicit emotions (see Section 2.6.1), research interest in recognising the emotions of VR users has grown correspondingly (Luong et al. 2022; Marín-Morales et al. 2020).

Researchers rely on explicit and implicit methods to assess users' emotional responses to VR content (Marín-Morales et al. 2020). Explicit methods—such as questionnaires (e.g. SAM (Bradley and Lang 1994) or PANAS (Watson, Clark et al. 1988)) or interviews—require users to express their experiences actively. In contrast, implicit methods are based on observations—such as physiology (e.g. ECG or EDA) or behaviour—and do not require users' active participation (Halbig and Latoschik 2021). Many VR studies employ a combination of explicit and implicit evaluation methods and are conducted under laboratory conditions with approximately 10 to 60 subjects (Halbig and Latoschik 2021; Luong et al. 2022; Magalhães et al. 2024; Marín-Morales et al. 2020). Explicit self-reports can serve as ground truth and be used to assess the performance of implicit emotion recognition models (Luong et al. 2022). Conversely, implicit emotion recognition may compensate for the memory and social desirability biases inherent in subjective self-reports (Halbig and Latoschik 2021). Crucially, automatic emotion recognition enables VR applications to monitor users' emotional states in real time and adapt the environment accordingly (Luong et al. 2022).

Emotion recognition in VR reflects concepts and trends in general emotion recognition research (see Section 2.3), specifically the mapping of diverse input modalities to discrete or dimensional representations, which increasingly relies on machine learning, deep learning, and multimodal fusion (Indrasiri et al. 2024; Marín-Morales et al. 2020; Polo et al. 2025; Romero-Ramos et al. 2024). However, VR settings also present unique opportunities and challenges for emotion recognition. Modern HMDs integrate a multitude of sensors, including IMUs for motion sensing, inside-out cameras for tracking, microphones, and increasingly eye trackers (Adhanom et al. 2023; Garrido et al. 2024). Although these sensors' primary purpose is immersion, they also provide suitable signals for emotion recognition (Halbig and



FIGURE 2.11: Attachable mask paddings with EMG electrodes and a forehead PPG sensor for the Pico G2 4K model (emteqPRO); from (Gnacek, Quintero et al. 2024, Figure 3), © 2024 the authors, licensed under CC BY 4.0, no changes.

Latoschik 2021). Hand controllers add inertial and button-event streams that capture gestures and interaction patterns (Garrido et al. 2024). Beyond these built-ins, researchers often attach additional sensors—such as electrodes for ECG, EDA, or EEG—either embedded in HMDs (e.g. see Figure 2.11) or worn externally (Halbig and Latoschik 2021).

The most common modalities for emotion recognition in VR include ECG, PPG, EDA, EEG, eye tracking, and respiration, with many studies relying on time-synchronised multimodal data (Halbig and Latoschik 2021; Luong et al. 2022; Magalhães et al. 2024; Marín-Morales et al. 2020). Notably, facial expressions, a central modality for emotion recognition outside VR, are far less common in HMD settings. HMDs partially occlude the user's upper face, rendering conventional vision-based FER approaches less effective (Georgescu and Ionescu 2019; Houshmand and Khan 2020). Section 2.7 discusses this challenge in detail.

A unique advantage of VR for emotion recognition is that VR environments are fully controllable and observable (Marín-Morales et al. 2020). Consequently, contextual factors—such as environmental states, user interactions, or scene events—can be logged and incorporated alongside other modalities, enabling more sophisticated context-aware emotion recognition (Gupta et al. 2024).

There is no consensus on target emotion representations in VR research (Luong

et al. 2022; Marín-Morales et al. 2020). Some studies adopt discrete categories (see Section 2.2.1), usually selecting a subset of Ekman's basic emotions. Others favour dimensional models (see Section 2.2.2), most commonly valence and arousal, following Russell's Circumplex Model of Affect. Notably, arousal and arousal-related emotions—such as anxiety or stress—are examined more frequently than valence and valence-related emotions—such as happiness or sadness (Halbig and Latoschik 2021; Luong et al. 2022; Marín-Morales et al. 2020).

VR's context-aware multimodal data streams encourage sequence models—such as LSTMs or Transformers—that can operate online and provide robust data fusion (Gupta et al. 2024; Indrasiri et al. 2024). Leveraging implicit, real-time, multimodal, context-aware emotion recognition allows VR to benefit various research and application areas, as surveyed in the following subsection.

2.6.3 Research and Application Areas

VR can elicit strong emotional responses while immersing users in virtual environments with high ecological validity, control, and observability (see Section 2.6.1). At the same time, emotion recognition enables the assessment of these emotional responses for evaluating VR content (see Section 2.6.2). Consequently, automatic emotion recognition allows VR applications to adapt their environment based on users' emotional states in real time. Thus, as visualised in Figure 2.12, VR applications can implement an affective feedback loop (Novak et al. 2014), which

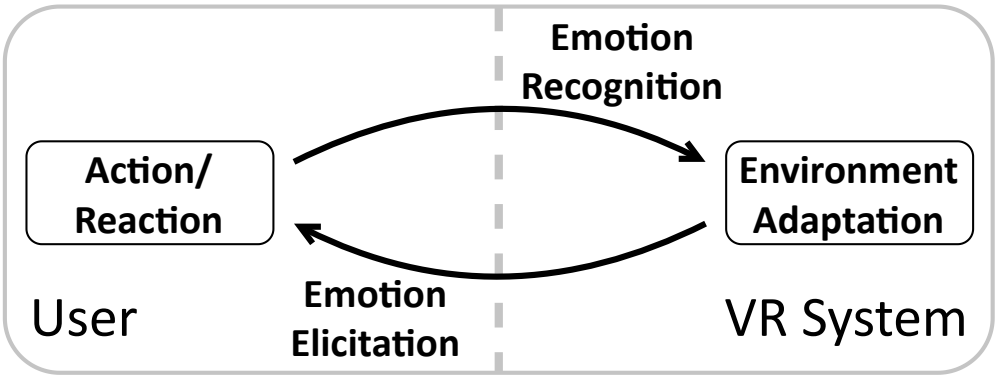


FIGURE 2.12: Affective loop in VR; based on (Andreoletti et al. 2021; Novak et al. 2014).

continuously adapts the VR environment based on the user's reaction to it, creating emotion-aware VR experiences. The potential of combining emotion recognition with VR's capabilities for emotion elicitation has led to a substantial increase in research interest since the emergence of modern HMDs around 2014. According to the Scopus database, the number of publications related to VR and emotions increased more than sevenfold (from 135 to 977) between 2014 and 2024 (see Figure 2.13). In the same timeframe, the number of VR-related publications approximately tripled from 5,153 to 15,027 (see Figure 2.10).

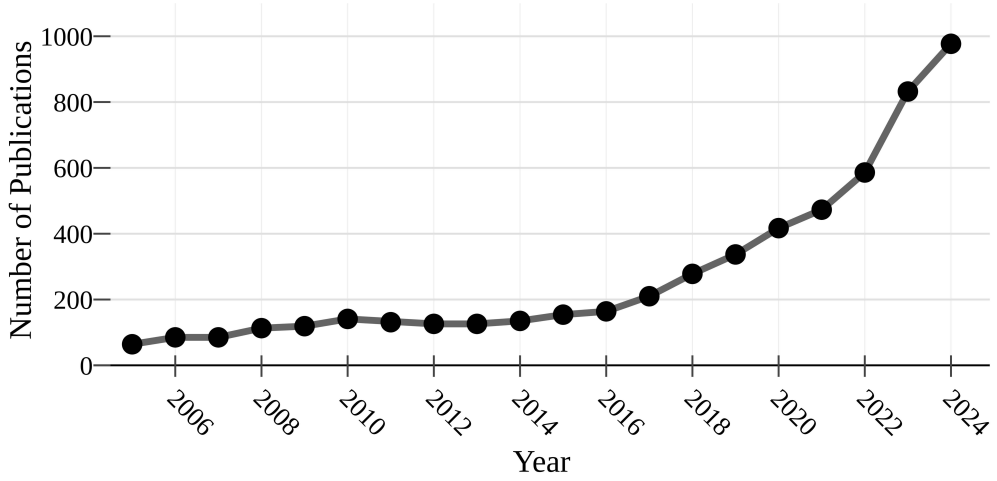


FIGURE 2.13: Number of publications related to VR and emotions since 2005, queried from the Scopus database on 1 September 2025. Query: *TITLE-ABS-KEY("virtual reality" AND "emotion*") AND (LIMIT-TO(DOCTYPE,"cp") OR LIMIT-TO(DOCTYPE,"ar"))*.

The remainder of this section presents the main research and application areas of emotion-aware VR, including mental health, therapy, education, training, and entertainment. However, it is not a comprehensive survey. Instead, it serves to motivate the research aim of this thesis, which is to enhance emotion recognition in VR by enabling effective FER despite HMD occlusion.

Mental Health and Therapy. VR has emerged as a powerful tool in mental health (Bell, Pot-Kolder et al. 2024), particularly for exposure therapy (Carl et al. 2019), relaxation applications (Riches, Azevedo et al. 2021; Riches, Jeyarajaguru et al. 2023), and pain management (Moreau et al. 2024; Teh et al. 2024).

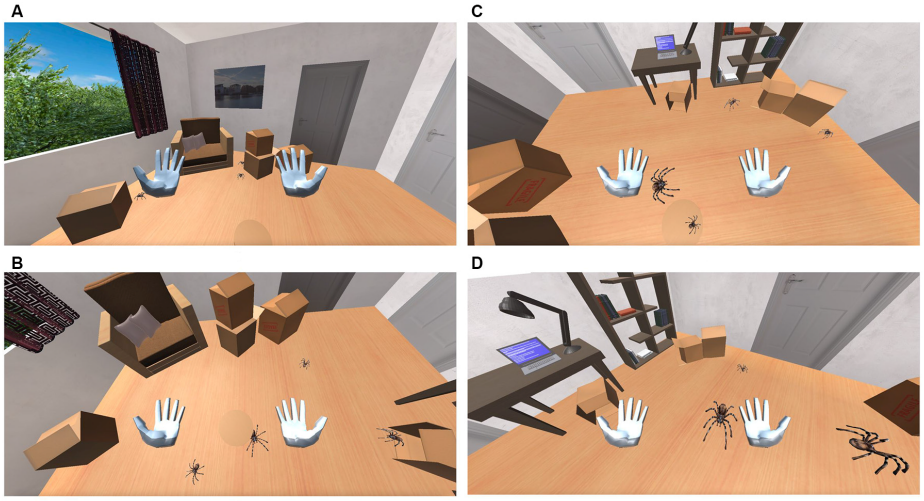


FIGURE 2.14: Variations of spider behaviour and sizes; from (Kritikos et al. 2021, Figure 4), © 2021 the authors, licensed under CC BY 4.0, no changes.

VR exposure therapy helps patients to overcome their phobias, anxieties, and similar mental disorders by safely exposing them to the feared object, activity, or situation in controlled virtual environments (Carl et al. 2019; Fradette et al. 2025). For instance, Kritikos et al. (2021) proposed a VR application to treat arachnophobia, which dynamically adjusts spiders' behaviour and appearance (see Figure 2.14) based on EDA measurements to keep users within a desired level of anxiety that is optimal for exposure therapy.

In relaxation applications, users can escape from their current situation into pleasant and calming virtual environments (Riches, Azevedo et al. 2021; Riches, Jeyarajaguru et al. 2023). These environments are typically nature-themed—such



FIGURE 2.15: Changes in fog density in response to fluctuations in heart rate variability. Figure 2 from Xu et al. (2025): *'Brief Interactive Virtual Reality Mindfulness Training with Real-Time Biofeedback for Anxiety Reduction: A Pilot Study'* in *Applied Psychophysiology and Biofeedback*, DOI: *10.1007/s10484-025-09718-w*. Reproduced with permission from Springer Nature.

as forests, mountains, or lakes—and designed to promote relaxation and reduce stress. Some applications actively engage users, for example, through mindfulness practices or guided meditations (Bell, Pot-Kolder et al. 2024). For instance, Xu et al. (2025) developed a mindfulness training application, in which dense fog in a forest gradually clears based on users' heart-rate variability (HRV) (see Figure 2.15).

Beyond escaping everyday stress, VR relaxation applications can also serve to distract users from painful medical procedures (Teh et al. 2024) and conditions (Moreau et al. 2024). For instance, Ghobadi et al. (2024) significantly reduced the experienced pain of dental implant surgery by immersing patients in a soothing nature-based VR environment during the procedure (see Figure 2.16). Psychophysiological assessments based on EMG, ECG, and EDA signals served as objective criteria for patients' emotional states, confirming questionnaire results.



FIGURE 2.16: Patient and VR setup for dental implant surgery; from (Ghobadi et al. 2024, Figure 2), © 2024 the authors, licensed under CC BY 4.0, no changes.

Training and Education. In training and education, VR provides safe, high-fidelity, interactive environments that monitor and modulate learners' engagement, cognitive load, and stress (Marougkas et al. 2024; Xie et al. 2021). Emotion awareness is crucial for these applications because emotions substantially influence human perception, attention, learning, memory, reasoning, and problem solving (Tyng et al. 2017). According to the Cognitive Affective Model of Immersive

Learning, particularly VR's affordances of presence and agency shape cognitive and affective factors—such as interest, motivation, self-efficacy, and self-regulation—that mediate learning outcomes (Makransky and Petersen 2021).

VR simulator training for various domains—including medical procedures (Elendu et al. 2024), emergency response (Alshowair et al. 2024), and safety training (Scorgie et al. 2024)—reproduces high-stakes conditions that elicit task-relevant arousal while allowing continuous monitoring through physiological and behavioural measures. For instance, Michela et al. (2022) developed a VR simulation to train police officers for shoot/don't shoot scenarios. The trainees' field of view widened with calm, controlled breathing and narrowed under dysregulated breathing (see Figure 2.17). Consequently, eight out of nine trainees improved their self-regulation of arousal under pressure, showing better breathing control and higher HRV.

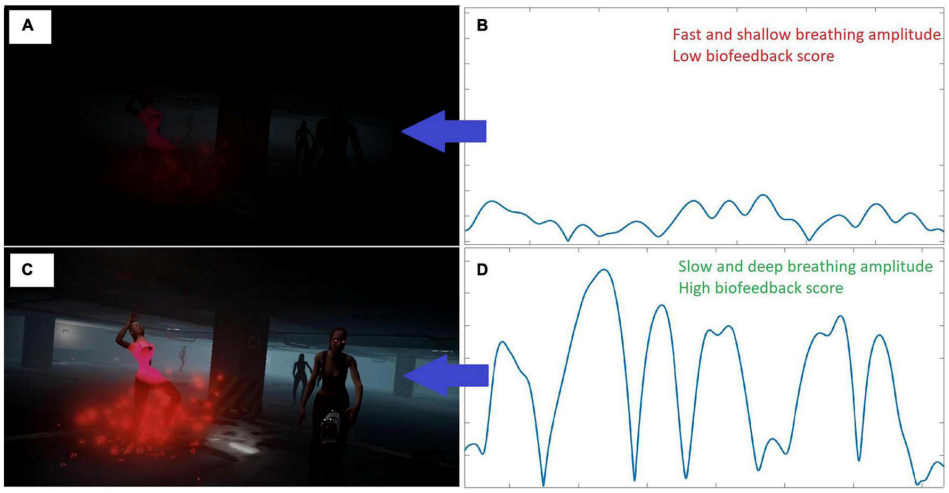


FIGURE 2.17: Implementation of biofeedback for peripheral vision modulation; from (Michela et al. 2022, Figure 1), © 2022 the authors, licensed under CC BY 4.0, no changes.

In classroom settings, VR can offer high-fidelity, interactive lessons across many disciplines—including scientific simulations, concept visualisations, historical reconstructions, and virtual field trips—that improve comprehension and knowledge retention by increasing learner engagement and fostering active learning (Mazhar and Al Rifaee 2023; Pellas et al. 2021). By tracking behavioural and physiological signals (Halbig and Latoschik 2021), as well as cognitive and affective states (Luong et al. 2022), VR lessons can monitor learners, adapt the learning experience, and

support teachers (Marougkas et al. 2024). Eye tracking enables attention cues that redirect gaze to relevant content (R. Liu et al. 2022), while estimates of cognitive load or frustration can trigger adjustments of task difficulty or visual complexity (Chiossi et al. 2025). For instance, Drey et al. (2020) implemented an educational VR game on social engineering and proposed monitoring learners' progress through head, hand, and foot movements, as well as gaze behaviour. When the game detected that a player was stuck, it would automatically display hints to guide the player on how to proceed.

Entertainment. Entertainment applications—such as video games, VR films, and social platforms—are among the most appealing VR activities for consumers (YouGov 2023). Combining embodied movement (head and hand tracking), interactive gameplay, decision making, and compelling storytelling, VR games can leverage VR's capabilities particularly well and amplify emotional responses. Reer et al. (2022) conducted a study with 133 participants, comparing the VR and non-VR versions of the popular game *The Elder Scrolls V: Skyrim*. They found that VR's realistic and natural interface (headset and motion controllers) significantly increased enjoyment by satisfying autonomy and competence needs. Likewise, Pallavicini et al. (2019) observed that the VR version of *Smash Hit* elicited higher levels of happiness, surprise, and presence than the non-VR version. Similar amplification effects were also reported for negative emotions. Lemmens et al. (2022) conducted a study where 31 participants played the VR version of the horror game *Resident Evil 7*, while 28 played on a TV screen. The VR condition produced stronger presence, higher fear, and lower HRV, with presence mediating VR's effect on fear. Beyond elicitation, de Lima et al. (2022) employed a machine-learning model to identify fear-inducing elements in their custom VR horror game and adjusted the game's pacing to maintain fear within a target range, enhancing the player experience.

Cinematic VR places viewers inside pre-recorded scenes with little or no agency, emphasising viewpoint control and presence over interaction. Compared to conventional video, HMD-delivered 360-degree films can strengthen emotional responses

(Bujić et al. 2023; Ding et al. 2018; Pavic et al. 2023), particularly in terms of arousal (F. Tian et al. 2021). For instance, Kuhne et al. (2023) exposed 79 participants to identical 360-degree nature, confrontation, and neutral clips in VR and on a 2D screen. VR yielded higher presence scores and more intense appraisal ratings: confrontation clips felt more tense, while beach and forest clips felt more relaxing. Physiologically, heart rate and skin conductance remained elevated across all VR scenes, indicating a baseline increase in arousal due to immersion itself.

Social VR platforms—such as *Meta Horizon Worlds* or *VRChat*—enable experiences in which users interact with each other through embodied avatars in shared virtual environments. Communication via spatial proximity, gesture, and voice fosters spatial and social presence, which are associated with higher enjoyment and relatedness (Barreda-Ángeles and Hartmann 2022). In a survey of 1,231 social-VR users, social presence and self-presence were found to significantly predict perceived social support, which, in turn, was positively associated with subjective well-being (van Brakel et al. 2023). Consistent with these findings, Deighan et al. (2023) conducted a qualitative study of VRChat users and reported that, during the COVID-19 pandemic, VRChat was used as a mental health tool to socialise in a safe environment, alleviating loneliness and feelings of isolation.

Further Areas. Beyond the areas discussed above, several adjacent strands underpin the value of combining emotion and VR research. These include perspective-taking and empathy (Trevena et al. 2024), emotion regulation (Montana et al. 2020), exergaming and physical training (Peng et al. 2024), as well as spectatorship of virtual live events (Park et al. 2024). Finally, methodological work uses emotional responses to evaluate VR properties and leverages VR to elicit and study emotions (Halbig and Latoschik 2021; Markowitz and Bailenson 2023; Somarathna et al. 2023).

2.7 Facial Expression Recognition in Virtual Reality

Outside VR settings, visual (camera-based) FER is one of the dominant approaches to automatic emotion recognition, reflecting the central role of facial expressions in human affective communication (S. Li and W. Deng 2022). In contrast, VR emotion recognition research predominantly relies on physiological measures, such as ECG, EDA, and EEG. Marín-Morales et al. (2020) report that only 5 of the 42 studies included in their review on emotion recognition in VR assessed facial expressions. All used EMG to capture facial muscle activity rather than visual FER. Similarly, Luong et al. (2022) find that, among 63 affective and cognitive VR studies, affect is predominantly inferred from psychophysiological measures. They mention only a few VR studies that attempt to sense or reconstruct facial expressions, explicitly noting that FER in VR settings can be more challenging due to HMD-induced occlusion of the face. In a related systematic review focused on multisensory VR, Magalhães et al. (2024) observed that objective emotion measures are dominated by autonomic signals, such as EDA and heart rate. Against this background, Ortmann, Q. Wang et al. (2023) conducted the first systematic literature review dedicated specifically to FER in VR settings. Following the PRISMA protocol (Liberati et al. 2009), they identified 256 records by searching the Scopus database and included 21 studies for in-depth analysis. EMG was the most common facial measure, while a smaller set of works explored visual FER using external cameras or (near-)infrared cameras integrated in HMDs.

This section reviews the current state of research on FER in VR settings by modality and lays the foundation for the key research gaps addressed in this thesis, which are synthesised in Section 2.9. First, Subsection 2.7.1 examines visual FER using external cameras and how existing studies address HMD-induced facial occlusion. Subsequently, Subsection 2.7.2 explores HMD-integrated sensing approaches, including HMD-mounted cameras (2.7.2.1), EMG (2.7.2.2), and other experimental techniques (2.7.2.3).

2.7.1 External-Camera-Based Approaches

External-camera-based approaches for FER in VR settings apply processing pipelines similar to those used in conventional visual FER (face detection, normalisation, CNN-based classification; see Figure 2.7). To address HMD occlusion, prior studies trained FER models with artificially occluded images from established FER databases.

Georgescu and Ionescu (2019) experimented with the AffectNet (Mollahosseini et al. 2019) and FER+ (Barsoum et al. 2016) databases. For the FER model, they used the VGG-Face network (Parkhi et al. 2015), a variant of the VGG-16 architecture (Simonyan and Zisserman 2015) extensively pre-trained for face recognition on the VGG-Face dataset. First, they trained and evaluated FER models on each unoccluded database. Their models achieved 59.03% accuracy on the 8-class subset of AffectNet and 84.79% accuracy on FER+. Then, they created artificially occluded versions of the databases by blacking out the entire upper half of each image. When evaluated against the occluded test set, the models' accuracies dropped from 59.03% to 37.70% on AffectNet and from 84.79% to 68.89% on FER+. However, when the FER models were trained on occluded data, their accuracy only decreased to 49.23% on AffectNet and 82.28% on FER+.

Houshmand and Khan (2020) applied a more sophisticated HMD simulation. They detected facial landmarks to accurately place a black rectangle on each image. Their experiments with the VGG-Face and ResNet50 architectures on the AffectNet, FER+, and RAF-DB databases yield results similar to those of Georgescu and Ionescu (2019).

Following a similar occlusion technique (see Figure 2.18), Gotsman et al. (2021) employed valence and arousal as their emotion model. Based on the continuous valence and arousal labels in AffectNet, which range from -1 to 1, they report root mean squared errors (RMSEs) of 0.434 and 0.402 for valence and arousal, respectively.



FIGURE 2.18: Artificial occlusion to simulate VR headsets; recreated from (Gotsman et al. 2021, Figure 2), © 2021 IEEE.

Yong et al. (2019) experimented with the 8,040 images of the *Radboud Faces Dataset* (RaFD; Langner et al. 2010) and simulated HMD occlusion by overlaying the eyes and eyebrows with a black horizontal bar at a fixed height. A DenseNet (Huang et al. 2017) achieved 95.9% accuracy on the eight emotion classes of RaFD. However, the images were randomly split rather than subject-independently, so that a substantial number of images of the same persons reenacting the same emotions were shared across the training, validation, and test sets.

Dehghani and Zaman (2023) added black horizontal bars to images from the FER-2013 database (Goodfellow et al. 2013) to occlude the eyes and eyebrows. They report 68.59% accuracy using a VGGNet model. In a small 'real-world validation' with five participants, they report that their model successfully detected happy and neutral emotions while playing *Astro Bot Rescue Mission*, and fear, neutral, and surprise while playing *Paranormal Activity*. However, they do not provide a formal frame-level evaluation.

Overall, prior studies have relied on established FER databases and artificial occlusion to simulate HMDs, typically by blacking out the eye and eyebrow regions, and then retraining or fine-tuning standard CNN-based FER pipelines. Table 2.3 summarises the reported results. Across datasets, simulated HMD occlusion consistently reduces recognition accuracy, yet performance remains clearly above chance. When training and testing on artificially occluded images, the largest decrease was reported on AffectNet (59.03% to 49.23%). In contrast, performance

TABLE 2.3: Overview of external-camera-based FER studies that train and test models on artificially HMD-occluded facial images.

Study	Database	Model	Label Space	Result	Unoccluded Baseline
Georgescu and Ionescu (2019)	AffectNet	VGG-Face	8 classes	Acc. 49.23%	59.03% [1]
	FER+	VGG-Face	8 classes	Acc. 82.28%	84.79% [1]
Houshmand and Khan (2020)	AffectNet	VGG-Face	8 classes	Acc. 50.13%	63% [2]
	FER+	VGG-Face	8 classes	Acc. 79.98%	84.99% [3]
	RAF-DB	ResNet-50	7 classes	Acc. 74.76%	77.28% [4]
Yong et al. (2019)	RaFD	DenseNet	8 classes	Acc. 95.90%	99.17% [5]
Dehghani and Zaman (2023)	FER-2013	VGGNet	7 classes	Acc. 68.59%	73.28% [6]
Gotsman et al. (2021)	AffectNet	EmoFAN-VR	Valence, Arousal	$RMSE_V$ 0.434, $RMSE_A$ 0.402	0.33,[7] 0.30 [7]
	AFEW-VA	EmoFAN-VR	Valence, Arousal	$RMSE_V$ 0.268, $RMSE_A$ 0.303	0.23,[7] 0.22 [7]

Abbreviations: Acc. = Accuracy, RMSE = Root Mean Squared Error.
Baseline References: [1]Georgescu and Ionescu (2019), [2]Mollahosseini et al. (2019), [3]Barsoum et al. (2016), [4]S. Li, W. Deng and Du (2017), [5]N. Sun et al. (2019), [6]Dehghani and Zaman (2023), [7]Toisoul et al. (2021).
Note: If the *Unoccluded Baseline* value comes from a different study than the *Result* value, the two values are only loosely comparable because dataset subsets, models, and evaluation protocols may differ.

on the less challenging FER+ database decreased only slightly (84.79% to 82.28%). Taken together, these findings imply that the lower face retains substantial affective information and that visual FER using external cameras can contribute to emotion recognition in VR settings despite HMD occlusion.

2.7.2 Headset-Integrated Sensing

In contrast to external-camera setups, HMD-integrated sensing places sensors on and in the headset itself to capture facial information from a headset-fixed viewpoint. Because the sensing is part of the HMD, no additional environmental hardware—such as external cameras—is required, keeping the VR setup compact and portable. Modern HMDs already integrate a wide range of sensors (e.g. tracking cameras, IMUs, microphones, and increasingly eye tracking) that can be repurposed for emotion recognition (see Section 2.6.2). In practice, however, many

FER research prototypes augment consumer HMDs with additional on-device sensing, such as inward-facing cameras (see Section 2.7.2.1) or electrodes (see Section 2.7.2.2) integrated into the facial interface. Unlike external cameras, HMD-integrated sensors can be positioned to 'look inside' the HMD to capture facial information that is not visible from the outside. Finally, because HMD-mounted sensors remain in a fixed position relative to the wearer's head, HMD-integrated sensing is less sensitive to characteristic VR challenges (head motion, body motion, occlusion, etc.), thus providing improved geometric and temporal stability.

At the same time, these benefits come with notable limitations. HMD-integrated facial sensing is hardware-dependent and poorly standardised: sensor availability, placement, and output formats vary markedly across HMDs, and more advanced devices are typically more expensive and less widely available. Public datasets and benchmarks remain scarce. Signal quality can be sensitive to headset fit, slippage, and contact conditions. Finally, since the HMD sits on the upper face, capturing the lower face can be challenging for embedded sensors, such as cameras or electrodes.

The following subsections illustrate the advantages and limitations outlined above and review visual FER using HMD-mounted cameras (2.7.2.1), EMG-based FER (2.7.2.2), and other experimental approaches (2.7.2.3).

2.7.2.1 Headset-Mounted Camera-Based Approaches

A direct way to make visual FER compatible with VR is to mount the camera(s) onto or into the HMD, resulting in a sensor pose constant relative to the wearer. H. Li et al. (2015) focused on continuous facial motion capture for driving 3D blendshapes and avatar animation. Their prototype combined an HMD-mounted RGB-D camera aimed at the lower face with strain gauges embedded in the HMD's foam liner to sense deformations in the occluded upper face (see Figure 2.19).

In a similar camera setup, Miranda and Orvalho (2016) used an RGB camera to capture the visible lower face. They proposed reconstructing a trackable 'full' face

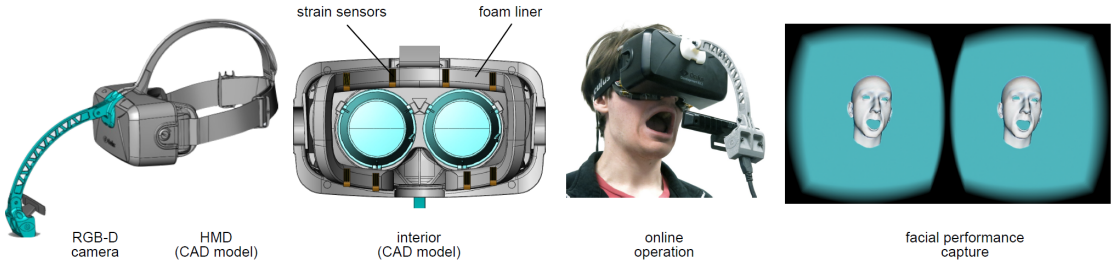


FIGURE 2.19: HMD-mounted facial sensing prototype; from (H. Li et al. 2015, Figure 1). Used with permission of ACM, from H. Li et al., Facial performance sensing head-mounted display, ACM Transactions on Graphics, 34.4, 2015; permission conveyed through Copyright Clearance Center, Inc.

by overlaying the upper face from a pre-recorded neutral calibration frame, and then performing FER from lower-face face-tracking features using a random forest classifier. They simulated this approach with the CK+ database (Lucey et al. 2010) and reported 64.80% accuracy using 10-fold cross-validation (six basic emotions plus neutral).

Rather than observing the mouth region, Hickson et al. (2019) repurposed the inward-facing near-infrared eye-tracking cameras integrated into commercial HMDs and performed FER directly on periocular image pairs using a CNN (see Figure 2.20). They collected a dedicated VR dataset with two separate HMDs and 46 participants who posed target expressions following exemplar videos. Using a subject-independent 5-fold cross-validation, they reported an accuracy of 66.6%

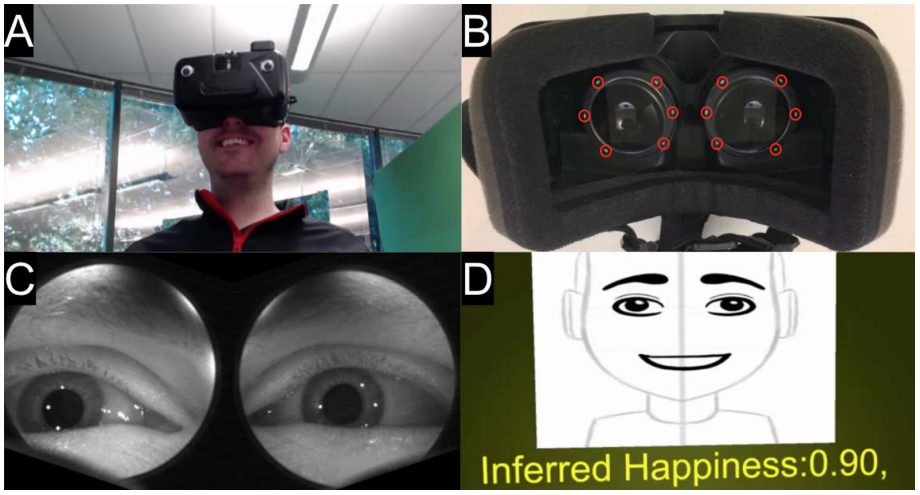


FIGURE 2.20: Eyemotion visual schematic; from (Hickson et al. 2019, Figure 1), © 2019 IEEE.

on five classes (anger, happiness, neutral, surprise, and closed eyes) with an InceptionV3 network (Szegedy et al. 2016). A lightweight one-shot personalisation step based on subtracting a per-session mean neutral image increased accuracy to 73.7%.

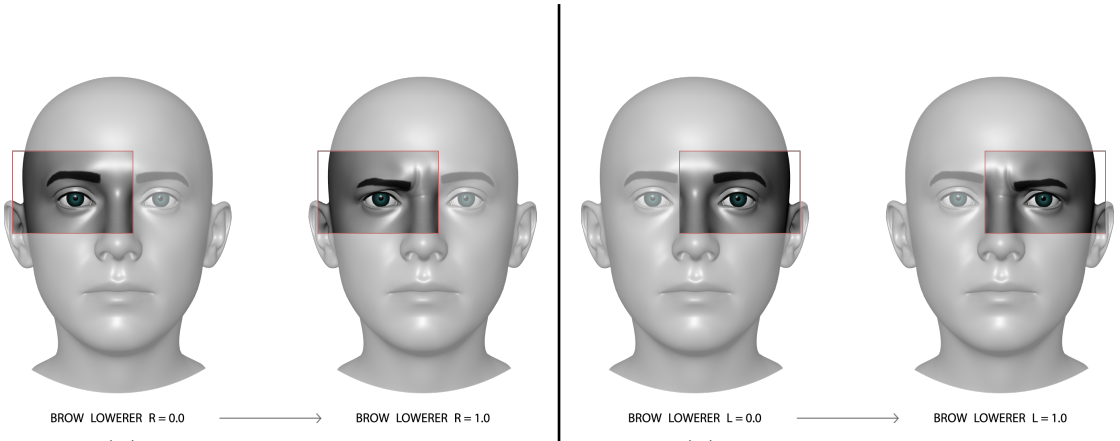
Xinrun Chen and H. Chen (2023) built a custom multi-camera setup by integrating two inward-facing near-infrared cameras into a Vive Focus Plus to record both eye regions and adding an external camera to capture the lower-face region. With this setup and a guidance application that presented CK+ exemplar sequences for imitation, they collected 60 expression sequences for each of anger, contempt, disgust, fear, happiness, sadness, and surprise. Using 10-fold cross-validation, they reported up to 93.91% accuracy with a multi-region deep model that fuses left-eye, right-eye, and external-region sequences. However, these results should be interpreted with caution. The reported cross-validation was not strictly subject-independent, which may have inflated the reported accuracy (Xinrun Chen 2026). Additionally, the authors indicated that the original dataset and code are no longer available for verification (Xinrun Chen 2026).

Ružickỳ et al. (2022) used a purely commercial sensing stack, combining an HTC Vive Pro Eye with the Vive Facial Tracker to record headset-native facial signals during nanoarthroscopy. They operationalised emotion tracking as coarse positive, neutral, and negative states to compare the stress-related effects of VR sessions with a 2D-monitor condition. Results derived from the facial-tracker stream suggested higher positive and lower negative values in VR.

Zhihui Zhang et al. (2023) leveraged the Meta Quest Pro's built-in face-tracking to mirror users' facial expressions onto a virtual avatar and then performed FER on rendered avatar images. Using the Unity engine, they captured frontal avatar face textures, detected and cropped the face, converted images to greyscale, and inferred seven emotion classes (six basic emotions plus neutral). For FER, a MobileNetV2 (Sandler et al. 2018) was trained on FER-2013 (Goodfellow et al. 2013) and synthetic virtual-character images. In a feasibility study with 15 participants who posed the seven target expressions, recognition accuracy was high for neutral, happiness,

sadness, and surprise (≈ 89–98%) but low for anger (24.84%) and fear (16.48%), which were largely confused with neutral.

While the number of studies remains limited, the available evidence suggests that HMD-mounted camera approaches can enable FER in VR by relocating the sensing viewpoint onto or into the HMD itself. Both lower-face recordings (Miranda and Orvalho 2016) and periocular recordings (Hickson et al. 2019) have been shown to contain affect-relevant cues. More recent work indicates that fusing complementary regions can further improve recognition performance (Xinrun Chen and H. Chen 2023; Zhihui Zhang et al. 2023). A practical strength of HMD-integrated FER is that such pipelines can leverage sensors increasingly present in commercial headsets. Eye tracking, in particular, is becoming more common in modern headsets (Adhanom et al. 2023), and some devices offer full-face tracking, either as an integrated feature (e.g. Meta Quest Pro (Meta 2025), Pico 4 Pro (Pico 2026)) or via dedicated add-on hardware (e.g. HTC Vive Full Face Tracker (HTC 2026b)). However, access to these sensors is usually mediated by manufacturer SDKs that expose tracking outputs in different formats (HTC 2026a; Meta 2025; Pico 2026). These interfaces typically provide blendshape coefficients, which are intended for avatar animation (see Figure 2.21), but can also be repurposed for FER (Zhihui Zhang et al. 2023). In its SDK documentation (Meta 2025), Meta explicitly frames its expression outputs as grounded in FACS (Ekman, Friesen



(A) BROW_LOWERER_R (B) BROW_LOWERER_L

FIGURE 2.21: Blendshapes for the *Brow Lowerer* Action Unit from Meta's Face Tracking API documentation (Meta 2025).

and Hager 2002), offering an interpretable, semantically structured representation. While concrete output formats remain provider-specific, industry efforts such as the OpenXR initiative (Group 2026) may help to reduce this fragmentation over time and enable more device-agnostic access.

2.7.2.2 Electromyography-Based Approaches

EMG records muscle activity via skin-contact electrodes. In VR settings, fEMG mitigates HMD occlusion because electrodes can be embedded into the HMD's facial interface and do not depend on the visibility of the face. As HMD contact is concentrated on the upper face, EMG-based FER typically relies on periocular, brow, and cheek muscles. Model outputs range from discrete expression classes to AU activations to continuous blendshape coefficients.

A major line of work centers on the *Faceteq* prototype and its evolution into the productised *emteqPRO*. Specifically, Mavridou, McGhee et al. (2017) presented Faceteq, a facial interface with eight dry electrodes (alongside other sensors) embedded in an HMD insert foam, targeting the frontalis, corrugator, orbicularis oculi, and zygomaticus major. Following Faceteq's application in several studies on emotion recognition in VR (Lou et al. 2020; Mavridou, Seiss, Hamedi et al. 2018; Mavridou, Seiss, Kostoulas, Nduka et al. 2018; Reidy et al. 2020), Gnacek, Broulidakis et al. (2022) introduced the product version emteqPRO, which integrates seven pairs of dry electrodes plus PPG and IMU sensors (see Figure 2.22). For instance, Reidy et al. (2020) placed the Faceteq sensor foam inside an HTC Vive to recognise emotions during a VR game for cognitive training. After collecting data in a user study with 18 subjects, a kNN model achieved 68.8% accuracy when classifying signals into one of four valence-arousal quadrants. In a subsequent study, M. Gjoreski et al. (2022) used emteqPRO with 38 participants, who watched 25 affective VR videos, and found that fEMG signals varied systematically with valence and arousal. To further support scale, Gnacek, Quintero et al. (2024) released AVDOS-VR, a database collected with emteqPRO that includes fEMG, PPG, and IMU signals with continuous valence and arousal ratings from 37 participants

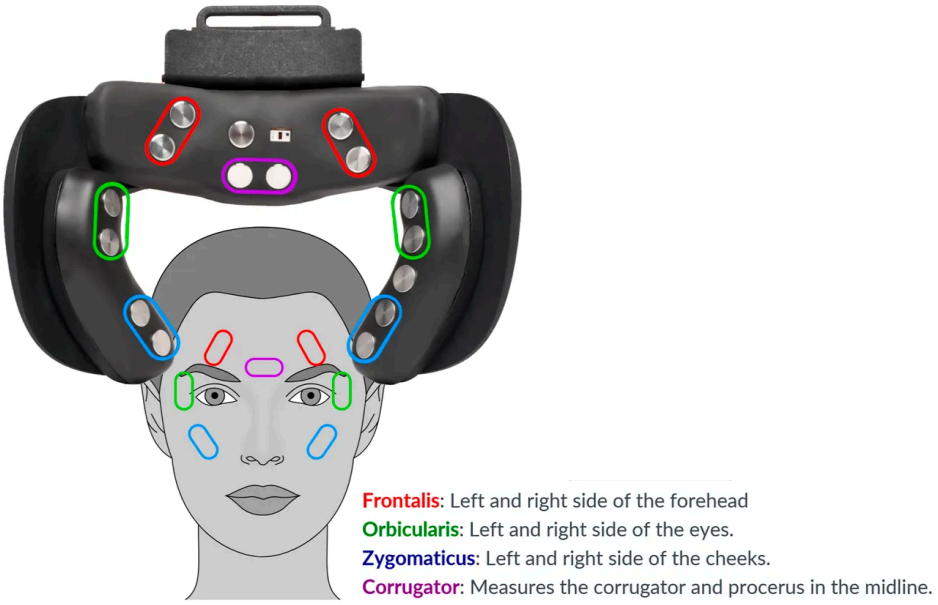


FIGURE 2.22: Mapping between emteqPRO sensors and facial muscles; adapted from (Gnacek, Broulidakis et al. 2022, Figure 5), © 2022 the authors, licensed under CC BY 4.0, changes: moved facial muscle captions from bottom to right.

viewing 30 VR videos. Following the LOSO-CV protocol, they reported fEMG amplitude as the most reliable modality. Their best-performing classifiers, solely based on fEMG, achieved mean F1-scores of 78% for 2-class arousal classification and 79% for 3-class valence classification. Moving towards increased ecological validity, Mavridou, Balaguer-Ballester et al. (2025) deployed EmteqVR (Mavridou, Seiss, Kostoulas, Hamedi et al. 2019) in a room-scale museum VR experience. During a six-week period at the Science Museum in London, they collected self-ratings, EMG, and PPG data from 291 participants who experienced neutral, positive, and negative environments under passive and active conditions. Their mixed-ANOVA analyses showed robust group and environment effects on fEMG root-mean-square amplitude, supporting the feasibility of affect sensing beyond seated, highly controlled VR settings.

Alongside the Faceteq/emteqPRO lineage, several other works explored periocular HMD-pad layouts using sticker electrodes. H.-S. Cha, Choi et al. (2020) evaluated different electrode configurations and achieved 85.01% accuracy for 11 expressions with Riemannian covariance features, enabling real-time avatar animation. Building on this, H.-S. Cha and Im (2022) improved robustness through cross-subject LDA

adaptation, reporting 82.02% accuracy in a user-independent setting. C. Kim et al. (2023) then combined periocular fEMG and EOG from ten HMD-mounted electrodes to estimate facial motion for real-time mirroring onto a 3D avatar face. Most recently, J.-H. Kim et al. (2025) aimed to reduce calibration effort by training participant-specific high/low valence classifiers from short, posed-expression prompts, achieving an F1-score of 88.20%. Using a similar custom electrode setup, Granato et al. (2020) introduced the RAGA database. It includes five fEMG channels from 33 participants playing racing games on a monitor and in VR. Regressing players' self-reported valence and arousal ratings from the fEMG signals, they report NRMSEs between 0.073 and 0.097 in a subject-dependent cross-validation.

2.7.2.3 Other Experimental Sensing Approaches

Beyond camera-based and facial EMG solutions, several other approaches have been explored for head-worn FER. The following overview is intended to highlight the range of these methods rather than provide an exhaustive survey. Many approaches remain at the prototype stage and are evaluated using small, non-standardised benchmarks. Additionally, several works target general facial sensing rather than VR-specific applications.

Non-imaging near-field optical sensing relies on IR emitter-receiver pairs (see Figure 2.23). These can be embedded into an HMD's foam liner to infer facial expressions from local reflectance changes (J. Cha et al. 2016; J. Kim et al. 2020; Murakami et al. 2019; Nakamura et al. 2023; Suzuki et al. 2017). Closely related, the *OCOsense* glasses apply distributed IR light sources and sensors to measure skin surface features for optomyography. Based on these features, the OCOsense system provides mappings to facial expressions, such as smile, frown, and brow raise. Currently, OCOsense is positioned primarily as a smart-glasses platform for augmented and extended reality interaction, but integration into HMD liners appears plausible in principle (Archer et al. 2023; H. Gjoreski et al. 2023; Kiprijanovska et al. 2023; Marinova et al. 2025).

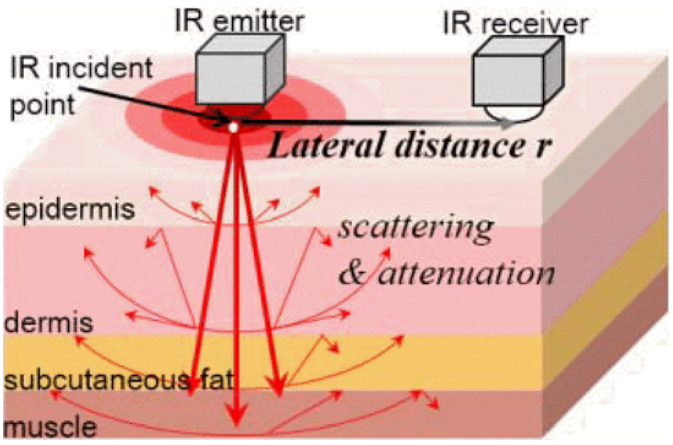


FIGURE 2.23: Illustration of radial propagation and attenuation of IR light vertically incident into the human skin due to physical action of scattering and absorbing mechanism; from (J. Cha et al. 2016, Figure 1), © 2016 IEEE.

Capacitive and electric-field sensing employ smart glasses, earables, or masks equipped with electrodes. They infer facial activity from deformation-dependent changes in capacitive coupling and near-field electric fields, offering a potentially occlusion-robust FER alternative for VR settings (Z. Guo and R.-H. Liang 2023; D. J. C. Matthies et al. 2017; D. J. Matthies et al. 2021; Rantanen et al. 2013).

Acoustic and vibration-based sensing leverage mechanically induced signals. Active systems, typically earables or glasses, emit inaudible probe signals and infer facial motion from deformation-dependent changes in echo (K. Guo et al. 2025; K. Li, R. Zhang, S. Chen et al. 2024; K. Li, R. Zhang, B. Liang et al. 2022). Passive contact microphones capture body-conducted muscle vibrations (Bello, B. Zhou et al. 2020) and can be combined with inertial sensing for facial activity recognition (Bello, Marin et al. 2023; S. Lee et al. 2019; D. Verma et al. 2021).

Finally, several works (Cudeiro et al. 2019; Fan et al. 2022; Karras et al. 2017; J. Lee et al. 2025; Richard et al. 2021; Y. Zhou et al. 2018) have explored speech-driven models that predict facial motion for avatar animation directly from audio (see Figure 2.24). However, these methods are primarily intended for mirroring speech-related motion onto 3D models rather than for recognising affective facial behaviour, and they remain blind to silent expressions.

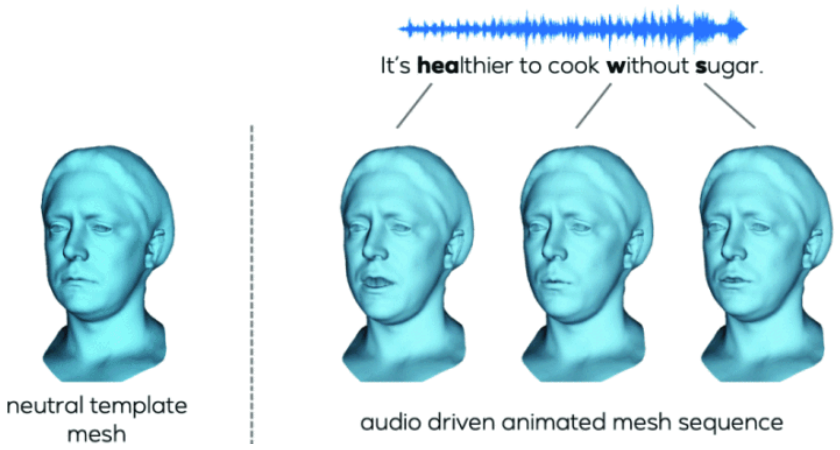


FIGURE 2.24: Given a neutral face mesh of a person and a speech signal as input, MeshTalk generates face animations; adapted from (Richard et al. 2021, Figure 1), © 2021 IEEE, changes: removed second and third row.

2.8 Ethical and Regulatory Considerations

FER in VR settings combines affect-related inference with sensor-rich interactions and immersive experiences. This combination makes the ethical context especially relevant because (1) affect inference relies on observable proxies and theory- and data-dependent models and assumptions (Mohammad 2022; Stark and Hoey 2021), (2) VR systems can capture detailed behavioural traces that increase identifiability and secondary-use risks (Miller et al. 2020), and (3) VR studies and applications may elicit strong emotional responses (Madary and Metzinger 2016; Slater, Gonzalez-Liencres et al. 2020).

The following subsections provide the ethical frame for this thesis. They discuss key ethical issues around emotion recognition in general (2.8.1), the specific challenges and risks of FER (2.8.2), the implications introduced by VR as an immersive and sensor-rich medium (2.8.3), ethical considerations when conducting human-participant studies (2.8.4), and a brief governance-oriented perspective linking these concerns to the relevant regulatory context (2.8.5).

2.8.1 Emotion Recognition

Emotion recognition is usually motivated by beneficial applications—including assistive technologies, healthcare, well-being, and education—as well as by more commercial 'neutral' goals, such as improving interaction and personalisation in media and entertainment (Mohammad 2022; Picard 1997). At the same time, emotion recognition can enable privacy-invasive monitoring and surveillance, particularly when sensing becomes ambient, continuous, or difficult to refuse in practice (Hernandez et al. 2021; Koskela 2000; Mohammad 2022). Hernandez et al. (2021) argue that the recent commercialisation and 'democratisation' of emotion recognition lower barriers for non-expert use and increase the likelihood of harmful deployments. These concerns are amplified in settings with clear power asymmetries (e.g. workplaces, education, public spaces), where individuals may have limited agency to opt out or contest the meaning of inferences (Hernandez et al. 2021; Mohammad 2022; Stark and Hoey 2021).

A central issue is that uncertainty in these inferences can translate into social harm. Even if a system is technically accurate with respect to its training labels, it may be unjustified to interpret outputs as reliable statements about a person's internal emotional state (Mohammad 2022; Stark and Hoey 2021). Such outputs can appear authoritative and may be used to legitimise decisions about people (e.g. performance, suitability, risk), shifting uncertainty away from the system and onto affected individuals (Mittelstadt et al. 2016; Stark and Hoey 2021). Therefore, risk-oriented guidance in affective computing recommends evaluating emotion recognition not only by predictive performance but also by foreseeable deployment contexts, harms, and misuse pathways (Hernandez et al. 2021). Further, it highlights clear communication of uncertainty and limits, meaningful opt-in consent, and careful attention to privacy and group-specific impacts (Hernandez et al. 2021; Mohammad 2022).

2.8.2 Facial Expression Recognition

FER is the central emotion recognition technique of this thesis, and thus, related ethical concerns are particularly relevant. The following paragraphs focus on validity and interpretation, data annotation and ground truth, bias and discrimination, and model opacity and accountability.

Validity and Interpretation. As for emotion recognition in general, a key issue in FER is that outputs are sometimes treated as evidence for internal emotional states. However, FER predicts patterns in facial appearance and motion. Interpreting these outputs as emotions is an additional inferential step that depends on theory and context. Barrett, Adolphs et al. (2019) conclude:

> '*The available scientific evidence suggests that people do sometimes smile when happy, frown when sad, scowl when angry, and so on, as proposed by the common view, more than what would be expected by chance. Yet how people communicate anger, disgust, fear, happiness, sadness, and surprise varies substantially across cultures, situations, and even across people within a single situation.*'
>
> (Barrett, Adolphs et al. 2019, p. 1)

Therefore, ethical analysis must distinguish between recognising facial behaviour and inferring internal emotion, and must make explicit which emotion theory and label space are being operationalised (Barrett, Adolphs et al. 2019; Martinez and Valstar 2016). Overstating what FER can infer risks legitimising high-stakes use where the underlying construct validity is weak.

Ground Truth and Annotation. As discussed in Section 2.5.2, obtaining ground truth in FER is challenging. FER datasets typically rely on posed expressions, self-reports, stimulus assumptions, or third-party annotation. Each approach introduces uncertainty and normative interpretation. Additionally, annotation processes can introduce systematic bias, reflecting annotators' personal biases (Y. Chen and Joo 2021). When labels are treated as objective ground truth, models may reproduce dataset assumptions and amplify them at scale. Consequently,

transparency about data provenance, annotation procedures, disagreement, intended use, and known limitations is not only a methodological concern but also a requirement for responsible communication (Gebru et al. 2021; Mitchell et al. 2019).

Bias and Discrimination. Bias and algorithmic discrimination are well-established risks in facial analysis and extend to affect recognition from faces. Buolamwini and Gebru (2018) report large intersectional performance disparities in commercial gender classification, with substantially higher error rates for darker-skinned women (up to 34.7%) than for lighter-skinned men (not more than 0.8%). In the FER domain, Pahl et al. (2022) identified demographic skews (particularly age and ethnicity) in data and algorithms, and demonstrated that model performance was best for people under 34 but dropped sharply for those over 44. Dominguez-Catena et al. (2025) further distinguished representational imbalance from stereotypical bias by showing that label-gender correlations in AffectNet (e.g. 'happy' about 60% female, 'angry' <30% female) can propagate into model predictions, primarily affecting the biased class. These findings connect to broader work showing that discriminatory outcomes can arise from seemingly 'neutral' data and design choices (Chander 2017; Mittelstadt et al. 2016). Consequently, ethical evaluation should treat subgroup performance as well as the distribution and meaning of errors as first-class considerations, rather than reporting only aggregate accuracy (Mitchell et al. 2019).

Model Opacity and Accountability. Machine learning intensifies these concerns. First, many FER models are difficult to interpret, complicating accountability, auditing, and meaningful explanation (Mittelstadt et al. 2016). Second, learning systems can exploit spurious correlations—such as identity cues, recording conditions, and demographic proxies—that inflate apparent performance while undermining robustness and fairness across contexts (Mohammad 2022). Third, when model outputs influence consequential decisions, responsibility gaps can arise, particularly if systems are treated as authoritative or autonomous rather than

fallible decision aids (Matthias 2004). For these reasons, the ethical assessment of FER should be understood as part of the broader ethics of socio-technical decision systems, not only as a question of facial modelling performance (Mittelstadt et al. 2016).

2.8.3 Virtual Reality

VR can elicit strong emotional responses (see Section 2.6.1). This increases potential benefits—such as therapy, training, and perspective-taking (see Section 2.6.3)—but also increases the risk of distress, coercion, and manipulation, as immersive environments can shape attention, behaviour, and affect in ways users may not fully anticipate (Cotton 2021; Madary and Metzinger 2016; Slater, Gonzalez-Liencres et al. 2020).

Moreover, as discussed in Section 2.6.1, VR can cause cybersickness, with symptoms similar to motion sickness, including nausea, headaches, and dizziness. Accordingly, cybersickness is both a user-welfare concern and a potential confound for affective measurements, since discomfort can influence reported and measured affect (Bayro and Jeong 2025; Rebenitsch and Owen 2016).

Furthermore, VR systems are sensor-rich and produce fine-grained behavioural traces. Even after explicit identifiers are removed, tracking data may still be personally identifying. Miller et al. (2020) demonstrated that motion tracking during VR tasks can identify users from large participant pools with high accuracy. Thus, data privacy, access control, retention, and secondary use are crucial ethical issues for VR studies and applications (Gebru et al. 2021; Mohammad 2022).

2.8.4 Human-Participant Studies

Studies on FER in VR typically collect sensitive data, including facial signals, behavioural tracking data, and potentially subjective or physiological measures (Bayro and Jeong 2025; Miller et al. 2020). Thus, informed consent and transparent,

purpose-limited data processing are essential (Hernandez et al. 2021; Mohammad 2022). Consent should also describe the study's goals and procedures in concrete terms, including tasks, stimuli, duration, foreseeable discomfort, and what will be recorded and inferred (Hernandez et al. 2021; Mohammad 2022). Where feasible, studies should support meaningful opt-in and practical withdrawal, particularly in contexts with power asymmetries (e.g. student or employee recruitment), where voluntariness may be harder to guarantee (Hernandez et al. 2021).

Participant welfare is a further meaningful consideration. If emotion elicitation involves negative or intense material, procedures should allow breaks and stopping without penalty and include appropriate debriefing (Madary and Metzinger 2016). Likewise, study protocols should account for VR sickness and also include exclusion criteria and monitoring during exposure (Rebenitsch and Owen 2016).

2.8.5 Regulatory Context

The ethical issues above intersect with governance and regulatory requirements. In the EU, facial recordings and VR tracking data can constitute personal data and therefore fall under the *General Data Protection Regulation* (GDPR; GDPR 2016). Accordingly, their processing requires a lawful basis, transparency, purpose limitation, data minimisation, storage limitation, and appropriate security measures (GDPR 2016). Depending on purpose and processing conditions, facial data may also fall under heightened rules as biometric data or special-category personal data (EDPB 2022). In addition, the *EU Artificial Intelligence Act* (EU AI Act; EU AI Act 2024) introduces a risk-based framework and restricts certain uses of emotion inference in power-asymmetric contexts, such as employment and education, while setting governance expectations for high-risk AI.

These legal anchors are not the primary basis of the argument here. Rather, they reinforce the academic literature's call for conservative claims, transparency about uncertainty and construct validity, and explicit attention to discriminatory impacts (Hernandez et al. 2021; Mitchell et al. 2019; Mohammad 2022). In

research practice, governance is supported by institutional ethics review and data-management procedures, as well as by structured artefacts—such as datasheets and model cards—that make the intended use, limitations, and evaluation boundaries explicit (Gebru et al. 2021; Mitchell et al. 2019).

2.9 Summary, Critical Synthesis, and Research Gaps

This section concludes Chapter 2 by synthesising the reviewed literature into the scope and research gaps addressed in the remainder of this thesis. First, Subsection 2.9.1 summarises the preceding sections of this chapter with a focus on research-gap-relevant findings. Then, Subsection 2.9.2 provides a critical synthesis of the main FER-in-VR approaches identified in the detailed review in Section 2.7. Finally, Subsection 2.9.3 formulates the research gaps that follow from this synthesis and guide the subsequent chapters.

2.9.1 Summary

Section 2.1: There is no single, universally agreed definition of 'emotion'. Thus, affective computing adopts a pragmatic approach by operationalising those emotion models and components that are most relevant for the task at hand.

Section 2.2: Discrete emotion theories—such as Ekman's six basic emotions (Ekman, Friesen and Ellsworth 1972)—and dimensional emotion theories—such as Russell's Circumplex Model of Affect (Russell 1980)—have been the most influential in affective computing, emotion recognition, and FER.

Section 2.3: Automatic emotion recognition is commonly framed as a supervised learning task that maps observable input signals to a target emotion representation.

Physiological modalities (e.g. HRV or EDA) are often paired with dimensional representations, whereas visual FER predominantly uses discrete categories. Across modalities, emotions are context-dependent, and their expression and interpretation can be ambiguous and unreliable. Ground truth is constrained by self-report biases and only moderate third-person agreement. Additionally, recognition targets remain only approximations of real, complex, emotional states. Consequently, emotion recognition results should not be interpreted as definitive evidence of a person's inner emotions. Furthermore, assumptions, uncertainties, limitations, the underlying data, and the employed emotion model should be communicated transparently (also see Section 2.8.1).

Section 2.4: FER is one of the most widely used approaches to emotion recognition and models facial expressions and movements. It typically relies on visual input (images or image sequences) and commonly operationalises discrete emotion categories aligned with Ekman's basic emotions. Importantly, FER does not directly infer inner affective states. Rather, it classifies observed facial behaviour as prototypical of the chosen target categories. Facial behaviour can also be described more granularly with FACS (Ekman, Friesen and Hager 2002), which encodes facial muscle activations as AUs. Static FER operates on single frames, whereas dynamic FER processes sequences, allowing it to leverage temporal information, potentially improving robustness and accuracy. Multimodal FER combines complementary signals from multiple modalities and fuses them at early, feature-level, late, or hybrid stages to mitigate modality-specific limitations and improve reliability. For evaluation, subject-independent splits are essential, and performance on balanced test sets is commonly reported using metrics such as Accuracy (Definition 2.1), Precision (Definition 2.2), Recall (Definition 2.3), and F1-score (Definition 2.4).

Section 2.5: Progress in FER is closely tied to the availability and properties of FER databases. Over time, datasets have evolved from small, posed laboratory corpora—such as KDEF (Lundqvist et al. 1998)—to large, in-the-wild collections—such as AffectNet (Mollahosseini et al. 2019)—and from unimodal static images to

dynamic and, increasingly, multimodal recordings. Yet, obtaining reliable ground truth remains challenging. In particular, third-party labelling often yields only moderate agreement (e.g. around 60.7% for AffectNet (Mollahosseini et al. 2019)). Furthermore, in-the-wild datasets tend to exhibit strong class imbalance, with more frequently observed expressions—such as happiness and neutral—dominating the label distribution.

Section 2.6: Through immersion, presence, embodiment, and interactivity, VR can elicit strong emotional responses while providing controllable and observable virtual environments. Reflecting the sensor-rich nature of VR systems, emotion recognition in VR often combines modalities and most commonly relies on physiological and behavioural signals. Across the reviewed literature, arousal and arousal-related emotions (e.g. anxiety or stress) are studied more frequently than valence and valence-related emotions (e.g. happiness or sadness). Notably, FER is applied significantly less often in VR settings than in general emotion recognition research because HMDs occlude the wearer's upper face. The research and application areas for emotion-aware VR are manifold and include exposure therapy, relaxation therapy, pain management, high-stakes simulator training, classroom settings, video games, films, social platforms, perspective-taking and emotion regulation.

Section 2.7: In VR settings, FER relies on external cameras or on HMD-integrated sensing. External-camera approaches place cameras outside the HMD and largely reuse conventional visual FER pipelines. Studies rely on established FER databases and simulate HMD occlusion by artificially masking the upper face. Although ecological validity is limited, promising results on static databases demonstrate that the visible lower face retains substantial affective information. Headset-integrated approaches capture facial information via sensors attached to or embedded in the HMD, including camera-based and fEMG-based solutions. HMD-mounted camera approaches relocate the sensing viewpoint into or onto the headset, keeping it constant relative to the wearer's face. Existing works

mount RGB or RGB-D cameras to capture the lower face or repurpose eye-tracking cameras to classify periocular images. Recent studies also leverage face-tracking capabilities available in some modern HMDs and add-on hardware, using the resulting blendshape coefficients for FER. Facial EMG approaches embed skin-contact electrodes into the HMD's facial interface to infer facial expressions from measured muscle activity. While independent of visual occlusion, they are typically tied to dedicated sensor integration and stable electrode-to-skin contact. Further experimental approaches include IR emitter-receiver sensing, electric-field sensing, acoustic sensing, and speech-driven facial-motion prediction for avatar animation.

Section 2.8: FER in VR settings combines affect-related inference with sensor-rich interactions and immersive experiences, making the ethical context especially relevant. Emotion recognition can enable beneficial applications but also privacy-invasive monitoring and harmful deployments, particularly under power asymmetries and when outputs are over-interpreted as evidence of inner emotional states. For FER specifically, key concerns include construct validity and interpretation, uncertainty in ground truth and annotation, bias and discrimination, and model opacity and accountability. VR further amplifies these issues through strong emotion elicitation, potential manipulation, cybersickness, and the identifiability and secondary-use risks of fine-grained behavioural traces. Therefore, human-participant studies require informed consent, concrete and transparent study descriptions, meaningful opt-in, practical withdrawal, and safeguards for participant welfare. These concerns also intersect with governance and EU regulations (e.g. GDPR (GDPR 2016) and EU AI Act (EU AI Act 2024)), reinforcing the need for transparent, purpose-limited data processing and clear documentation of intended use and limitations.

2.9.2 Critical Synthesis and Scope Decision

As reviewed in Section 2.7, FER in VR settings is primarily approached via external cameras observing the visible lower face region (see Section 2.7.1), via HMD-mounted cameras that record facial information from a headset-fixed viewpoint, including camera-derived face tracking outputs (see Section 2.7.2.1), and via HMD-integrated fEMG that measures facial muscle activity (see Section 2.7.2.2). The experimental approaches discussed in Section 2.7.2.3 are not considered for the following critical synthesis because they are largely exploratory prototypes and lack sufficiently established, comparable evidence. Speech-driven models, while more mature, are also excluded as they target speech-related facial motion rather than affective behaviour and remain blind to silent expressions.

External-Camera Approaches. External-camera approaches have adapted conventional FER pipelines to VR by simulating HMD occlusion through masking the eye and eyebrow regions. Because these works reuse established FER databases and pipelines, they can benefit from the maturity of conventional visual FER and provide controlled baseline comparisons. At the same time, external-camera approaches and prior studies face important limitations. First, HMD occlusion causes an irreducible loss of information about the occluded upper face region (eyes/eyebrows). Second, the ecological validity of experiments with established FER databases and artificial occlusion is limited. Challenges of real VR behaviour (head motion, body motion, self-occlusion, etc.) and settings (camera placement, HMD fit variation, face detection reliability, etc.) are largely absent. Third, prior work has focused exclusively on static image-based FER, despite VR's inherently dynamic, viewpoint- and motion-dependent nature. Despite these constraints, external-camera approaches remain worth exploring. Cameras are low-cost and widely available, visual FER has a mature ecosystem, and existing occlusion-aware results provide a strong baseline (see Table 2.3).

Headset-Mounted Camera Approaches. HMD-mounted camera approaches relocate the sensing viewpoint onto or into the HMD, keeping it fixed relative to the wearer's face. While the number of studies remains limited, the available evidence suggests that both lower-face recordings and periocular recordings contain affect-relevant cues. However, the ecological validity of current work is limited because recognition performance is predominantly reported for posed expressions elicited through imitation protocols. Moreover, dynamic FER remains rare, with most approaches operating on static frames rather than modelling temporally evolving expressions during real VR behaviour. Beyond these validity concerns, custom sensor setups and study-specific capture conditions limit comparability, leaving cross-study and cross-device generalisation under-tested. Sample sizes are mostly small, and recognition targets vary significantly, ranging from non-emotional expressions ('closed eyes' (Hickson et al. 2019)) and coarse affect proxies (neutral, positive, negative (Ružickỳ et al. 2022)) to basic-emotion categories (Xinrun Chen and H. Chen 2023; Miranda and Orvalho 2016; Zhihui Zhang et al. 2023). Nevertheless, HMD-mounted camera approaches are a promising and practically relevant direction. Inward-facing cameras can capture facial regions that external cameras cannot observe, and the headset-fixed viewpoint reduces sensitivity to characteristic VR challenges (head motion, body motion, self-occlusion, etc.). In addition, commercial HMDs and add-ons increasingly offer eye tracking and face tracking, which can be repurposed for FER, keeping the overall setup more standardised, compact, and portable. In particular, face-tracking outputs in the form of blendshape coefficients provide an interpretable, structured, full-face representation that, at least for some vendors (Meta 2025), is partially grounded in FACS (Ekman, Friesen and Hager 2002).

Facial-Electromyography-Based Approaches. HMD-integrated fEMG approaches infer facial expressions from facial muscle activity measured via skin-contact electrodes. Consequently, these approaches are independent of visual appearance and largely unaffected by HMD occlusion. Although evidence suggests that fEMG can support FER in VR settings in principle, several challenges

and limitations remain. Reliable measurement requires stable electrode-to-skin contact, making performance sensitive to fit, slippage, and impedance changes during use (Gnacek, Broulidakis et al. 2022; Gnacek, Quintero et al. 2024). Ecological VR behaviour can further introduce motion and speech artefacts (Mavridou, Balaguer-Ballester et al. 2025). Together, these factors contribute to substantial inter-user and cross-session variability, leading many systems to rely on per-user calibration to achieve robust performance (H.-S. Cha and Im 2022; Granato et al. 2020; J.-H. Kim et al. 2025). Subject-independent LOSO-CV evaluations report only moderate performance (e.g. F1-scores of 78% for 2-class arousal and 79% for 3-class valence (Gnacek, Quintero et al. 2024)). Furthermore, current work varies substantially in electrode layouts, recognition targets, and evaluation protocols, hindering standardisation and comparability. Capturing lower-face dynamics can also be challenging because HMD-compatible electrode layouts primarily target the upper face (C. Kim et al. 2023). Finally, fEMG typically requires dedicated sensing interfaces—such as Faceteq (Mavridou, McGhee et al. 2017) or emteqPRO (Gnacek, Broulidakis et al. 2022)—rather than leveraging sensors already integrated into mainstream consumer HMDs. Taken together, these considerations motivate treating fEMG as out of scope for the remainder of this thesis. A central objective of this thesis is to create a database for the reproducible evaluation of FER in VR. Meeting this objective with fEMG would be difficult because current work often relies on user- and session-specific calibration and requires specialised sensing hardware that is not widely available.

Scope Decision. Based on the critical synthesis above, the remainder of this thesis focuses on external-camera approaches and HMD-mounted camera approaches, including HMD-provided face tracking. This scope decision prioritises sensing that is already accessible in consumer hardware and can be replicated without specialised electrode interfaces. Consequently, the databases and baselines developed in this thesis are easier to reproduce and more likely to be adopted within the established visual FER ecosystem.

2.9.3 Research Gaps Addressed by This Thesis

The preceding critical synthesis and scope decision in Section 2.9.2 identified key limitations that currently hinder research progress and practical use of FER in VR settings. In line with this scope, this section formulates research gaps for external-camera approaches and HMD-mounted camera approaches. The following five research gaps guide the remainder of this thesis and align with the thesis aim of enabling effective FER in VR despite HMD occlusion.

RG1: Unvalidated Transfer from Artificial to Natural HMD Occlusion. Current external-camera approaches simulate HMD occlusion by masking the eye and eyebrow region in established FER databases. While this enables controlled baselines, it remains unclear how well results obtained under artificial occlusion transfer to natural HMD occlusion in real external-camera recordings. Consequently, current research lacks a benchmark database of naturally HMD-occluded emotional faces recorded by external cameras during actual VR use.

RG2: Limited Comparability and Reproducibility of HMD-Mounted Camera Approaches. Current HMD-mounted camera and face-tracking studies often rely on custom sensor setups, study-specific capture conditions, small sample sizes, heterogeneous recognition targets, and varying evaluation protocols. This makes reported results difficult to reproduce and prevents meaningful comparison across hardware configurations and sensing pipelines. As a result, current research lacks benchmark resources that enable reproducible, participant-independent assessment of HMD-mounted camera approaches using commercially available hardware without modification under a consistent evaluation protocol.

RG3: Insufficient Ecological Validity of Current Evidence. Across both external-camera and HMD-mounted camera approaches, the ecological validity of existing work is limited. External-camera studies reuse established FER databases that were recorded outside VR settings and therefore do not reflect VR-specific

behavioural and capture conditions. Similarly, HMD-mounted camera studies usually rely on imitation protocols rather than emotion-eliciting VR content, leaving it unclear how well reported results generalise to naturalistic VR use, which involves spontaneous expressions, interactive tasks, speech, and head and body motion. This gap motivates collecting facial recordings during actual emotion-eliciting VR sessions to enable evaluation under conditions representative of real use.

RG4: Limited Consideration of Dynamic FER. Current work on camera-based FER in VR settings focuses almost exclusively on static, frame-based recognition despite VR's inherently dynamic, interactive, and motion-dependent nature. Consequently, the potential of temporal modelling to improve FER performance and robustness remains insufficiently explored for both external-camera and HMD-mounted approaches. Addressing this gap requires a database that provides sequential recordings of HMD-occluded facial expressions, together with reproducible, participant-independent evaluation of dynamic FER models.

RG5: Lack of Comparable Multimodal Evaluation. Existing work on FER in VR settings largely evaluates external-camera and HMD-mounted camera streams in isolation and under heterogeneous protocols. As a result, reported findings are not directly comparable across modalities, and the potential of multimodal fusion to improve robustness under HMD occlusion remains unclear. Accordingly, current research lacks time-synchronised multimodal benchmark resources that combine external-camera recordings with HMD-mounted signals of VR users' emotional expressions. It further lacks corresponding benchmark evaluations under a reproducible, participant-independent protocol that compares unimodal baselines and multimodal fusion.

Table 2.4 summarises how the related work discussed in Sections 2.7.1 and 2.7.2.1 compares with this thesis regarding the characteristics underlying RG1–RG5.

TABLE 2.4: Comparison of related work and this thesis with respect to gap-relevant characteristics.

Study	FER Database or Sensing Setup	Natural HMD Occlusion	Actual VR Use	Posed Expressions	Categorical Target	Static FER	Dynamic FER	Multimodal FER	New Benchmark Database
Yong et al. (2019)	RaFD	–	–	✓	✓	✓	–	–	–
Georgescu and Ionescu (2019)	AffectNet, FER+	–	–	○	✓	✓	–	–	–
Houshmand and Khan (2020)	AffectNet, FER+, RAF-DB	–	–	○	✓	✓	–	–	–
Gotsman et al. (2021)	AffectNet, AFEW-VA	–	–	○	–	✓	–	–	–
Dehghani and Zaman (2023)	FER-2013 and a small informal VR-game validation	○	○	○	✓	✓	–	–	–
Miranda and Orvalho (2016)	Custom HMD-mounted external lower-face camera	✓	–	✓	✓	✓	–	–	–
Hickson et al. (2019)	Custom inward-facing eye cameras in commercial HMDs	✓	–	✓	✓	✓	–	–	–
Ružický et al. (2022)	HTC Vive Facial Tracker	✓	✓	–	✓	–	–	–	–
Xinrun Chen and H. Chen (2023)	Custom inward-facing eye cameras and external lower-face camera	✓	–	✓	✓	–	✓	✓	–
Zhihui Zhang et al. (2023)	Meta Quest Pro face tracking and rendered avatar images	✓	–	✓	✓	✓	–	–	–
This thesis	Meta Quest Pro face tracking and external cameras	✓	✓	✓	✓	✓	✓	✓	✓

Symbols: ✓= addressed, ○= partially addressed, –= not addressed.
Note: *Actual VR Use* denotes interactive or applied VR use, and excludes pure HMD-based imitation protocols.

Chapter 3

Research Design

This chapter translates the research aim and questions (see Section 1.2) and the identified research gaps (see Section 2.9.3) into a coherent research design that guides the remainder of this thesis. First, Section 3.1 positions the thesis in terms of research philosophy and research strategy to clarify the basis on which subsequent methodological choices are justified. Next, Section 3.2 derives the evidence requirements implied by the gaps (RG1–RG5) and specifies which artefacts and evaluation evidence are needed to address them. Then, Section 3.3 justifies the thesis-level design decisions adopted to translate these requirements into a concrete research route. Finally, Section 3.4 presents the resulting research plan and provides an overview of how the subsequent chapters implement it.

3.1 Research Philosophy and Strategy

Research Philosophy. This thesis adopts the research philosophy of *Pragmatism*. Pragmatism argues that the research problem and questions should be the primary determinants of the research approach, rather than prior commitment to a single philosophical position (J. W. Creswell and J. D. Creswell 2023; Saunders et al. 2023). Consequently, researchers are free to select and combine the methods, techniques, and procedures that are most suitable for their needs and purposes to

generate useful and defensible evidence within the problem context (J. W. Creswell and J. D. Creswell 2023). Pragmatism is appropriate for this thesis because it legitimises the integration of multiple complementary research activities to address the identified research gaps (RG1–RG5), rather than relying on a single canonical methodology.

Research Strategy. Building on this philosophical positioning, this thesis follows *Design Science Research (DSR)* as its research strategy. DSR is an established strategy in information systems research that generates knowledge about a problem domain and its potential solutions through the construction and rigorous evaluation of purposeful artefacts (Hevner and Chatterjee 2010; Hevner, March et al. 2004). This strategy is appropriate here because several research gaps (see Section 2.9.3) concern missing or weakly comparable benchmark resources and evaluations for studying FER under the target condition of VR use with natural HMD occlusion, rather than only deficiencies in existing model architectures. Accordingly, this thesis treats the creation of purposeful artefacts—including data-acquisition tools, benchmark resources, and baseline models—as core research contributions and justifies them through systematic evaluation. In this thesis, evaluation evidence is primarily quantitative and obtained using predefined metrics under established evaluation protocols to support comparable and reproducible conclusions.

3.2 Evidence Requirements

Building on the DSR positioning established in Section 3.1, this section derives which artefacts and evaluation evidence are required to address the identified research gaps (see Section 2.9.3) and to answer the formulated research questions (see Section 1.2.1). It focuses on what must exist and what must be shown in order to produce defensible evidence. Because the required artefacts are interdependent, the requirements are derived sequentially from Artefact 1 to Artefact 3.

3.2.1 Artefact 1: Baseline FER Models and Evaluations

Motivation. The research aim of this thesis is to enable effective FER for VR users despite HMD occlusion (see Section 1.2). The central empirical contribution of this thesis towards this aim is the production of benchmark evidence for that problem context. RQ3 asks how unimodal baselines and multimodal fusion perform for static and dynamic FER under HMD occlusion, while RQ4 asks what benchmarks, practical trade-offs, and design implications can be derived from these results. These questions are motivated primarily by RG4 and RG5, which identify the limited consideration of dynamic FER and the lack of directly comparable multimodal evaluation. In addition, RG1 and RG2 imply that these comparisons must be grounded in benchmark evidence that addresses both naturally HMD-occluded external-camera data and reproducible HMD-derived signals. Consequently, the first required artefact is a set of baseline FER models that enable these evaluations and comparisons.

Requirements. To fulfil this role, Artefact 1 must include unimodal models for the input modalities within scope (see Section 2.9.2) and multimodal models that combine synchronised modalities. It must also support both static and dynamic FER to examine the role of temporal information in VR settings. These baseline FER models are not intended as final state-of-the-art solutions. Rather, they serve as comparable reference points whose evaluation quantifies feasibility, reveals limitations, and shows whether temporal modelling and multimodal fusion yield meaningful gains under HMD-induced occlusion.

Evaluation Requirements. The evaluation of Artefact 1 must provide the central quantitative evidence for answering RQ3 and for deriving the benchmarks, practical trade-offs, and design implications sought in RQ4. It must therefore follow a reproducible benchmark protocol with participant-independent data splits, consistent task definitions for static and dynamic FER, and reporting based on established metrics. In line with RG4 and RG5, the evaluation must support

direct comparison across static and dynamic as well as unimodal and multimodal FER. To address RG1 specifically, it must also include an explicit comparison between results obtained under simulated HMD occlusion and results obtained under natural HMD occlusion.

3.2.2 Artefact 2: Benchmark Database

Motivation. Artefact 1 requires a benchmark resource with suitable properties for training and evaluating the baseline FER models. Existing FER databases do not provide naturally HMD-occluded emotional faces recorded during actual VR use for external-camera approaches (RG1, RG3). Additionally, HMD-mounted camera approaches lack benchmark resources that enable reproducible, comparable, and participant-independent evaluations (RG2). Across approaches, dynamic FER (RG4) and multimodal fusion (RG5) remain insufficiently explored. Therefore, a second required artefact is a benchmark database designed specifically for FER in VR under HMD occlusion.

Requirements. To address RG1, the database must provide external-camera recordings of naturally HMD-occluded faces, rather than only artificially occluded images. To address RG2, it must also provide HMD-derived facial signals obtained with unmodified, commercially available hardware, using a consistent, participant-independent protocol. Both modalities must be recorded during actual VR use (RG1, RG3) and support both static and dynamic FER (RG4) by including data that can be structured into single-instance and sequence-based benchmark tasks. To address RG5, the database must provide time-synchronised multimodal samples that align external-camera recordings with HMD-derived facial signals on matched instances. In addition, the database must define a target emotion representation suitable for the benchmark task and provide corresponding ground truth. It must further define clear subset definitions, transparent preprocessing and construction rules, and participant-independent data splits.

Evaluation Requirements. Because the validity of all later model evaluations depends on Artefact 2, the benchmark database itself requires two kinds of evidence. First, it requires evidence of intrinsic quality showing that its ground truth is defensible, its sample-selection procedure is transparent, and its cross-modal synchronisation is precise enough for sequence construction and multimodal fusion. Second, it must ultimately demonstrate its utility as a benchmark resource by enabling reproducible and comparable baseline evaluations for static and dynamic as well as unimodal and multimodal FER under one consistent protocol.

3.2.3 Artefact 3: Data-Acquisition System

Motivation. Artefact 2 requires a means to generate the data needed to construct such a benchmark database. Reusing existing studies and databases or scraping web data would not provide the required combination of naturally HMD-occluded external-camera recordings, actual VR use, and synchronised HMD-derived facial signals under one consistent protocol. Therefore, a data-acquisition system that serves both as an elicitation instrument and as a capture tool is the third required artefact. This requirement follows most directly from the insufficient ecological validity of current evidence (RG3) and addresses RQ2, which asks how facial expressions of VR users can be collected to enable the construction of a benchmark database such as Artefact 2.

Requirements. The system must support the repeated elicitation and capture of emotional facial expressions during actual VR use (RQ2, RG3). The resulting recordings must be structured in a way that supports the later construction of both static samples and dynamic sequences (RG4). At the same time, the system must record the required modalities under consistent conditions, including external visual facial recordings and HMD-derived facial signals (RG1, RG2). To support later multimodal benchmarking, all recordings must be logged in a way that allows reliable temporal alignment across modalities (RG5). Accordingly, Artefact 3 must

provide a reproducible means of eliciting, capturing, and synchronising multimodal facial-expression data for the construction of Artefact 2.

Evaluation Requirements. The evaluation of Artefact 3 must demonstrate that the data-acquisition system is feasible and fit for repeated use as a VR study instrument. Specifically, it must show that the system can be deployed with human participants without inducing substantial VR-related discomfort and with an acceptable user experience, while producing recordings of sufficient quantity, consistency, and technical quality for constructing Artefact 2. It must further show that timing, logging, and multimodal synchronisation are reliable enough to support the later construction of static and dynamic benchmark tasks as well as cross-modal comparison.

3.3 Design Decisions

Building on the evidence requirements derived in Section 3.2, this section justifies the key design decisions that shape how these requirements are translated into a concrete research route. These decisions operate at the thesis level and affect multiple artefacts and evaluations. More specific design choices concerning individual artefacts or evaluations are deferred to the subsequent chapters.

3.3.1 Categorical Emotion Representation

This thesis adopts a categorical emotion representation using the seven classes: anger, disgust, fear, happiness, sadness, surprise, and neutral. As discussed in Sections 2.2–2.3, emotion recognition can be formulated with either discrete or dimensional targets, whereas FER usually relies on Ekman's six basic emotions plus a neutral category (see Sections 2.4–2.5). Consequently, this choice provides maximal comparability with established FER benchmark datasets, models, and evaluations. At the same time, these class labels should not be understood as direct

evidence of participants' inner emotional states, but as facial-expression categories for classifying observed facial behaviour under HMD occlusion.

3.3.2 Posed Expressions

This thesis relies on posed facial expressions rather than spontaneous ones. As discussed in Section 3.2, Artefact 3 must generate the data needed to construct a benchmark database for FER under HMD occlusion by repeatedly eliciting and recording facial expressions during actual VR use. Spontaneous expressions are difficult to elicit reliably, and repeatedly inducing genuine anger, disgust, fear, or sadness would also raise substantial ethical concerns. In contrast, posed expressions make it easier to collect larger amounts of data, control class distributions, and establish clearer correspondence between prompts and labels. Nevertheless, the thesis acknowledges that relying on posed expressions introduces a reality gap with respect to spontaneous expressions (see Section 2.5.2) and therefore limits ecological validity.

3.3.3 Face Tracking

This thesis adopts HMD-provided face tracking as the HMD-integrated sensing modality. As established by the scope decision in Section 2.9.2, the thesis focuses on external-camera approaches and HMD-mounted camera approaches using sensing that is already accessible in consumer hardware and can be replicated without specialised electrode interfaces. As an HMD-mounted camera approach, HMD-provided face tracking is already available in several commercial HMDs and add-on devices (see Section 2.7.2.1). Manufacturer SDKs typically expose face-tracking outputs as blendshape coefficients intended for avatar animation. Although the exact output formats vary across providers, they are broadly similar in providing an interpretable, semantically structured description of facial expressions. Consequently, HMD-provided face tracking offers a practical, reproducible, and comparable signal for FER under HMD occlusion.

3.4 Research Plan

Based on the evidence requirements derived in Section 3.2 and the design decisions justified in Section 3.3, this section outlines the staged research plan for realising the required artefacts and evaluations. The plan proceeds from the design and implementation of a data-acquisition system (Artefact 3), to its deployment in a user study, to the construction of a benchmark database (Artefact 2), to the implementation and evaluation of baseline FER models (Artefact 1), and finally to the comparative synthesis of the resulting benchmark evidence. These steps reflect the dependency structure of the thesis and are realised in Chapters 4–7, which describe each step in detail.

Design and Implementation of the Data-Acquisition System. The research plan begins with the design and implementation of Artefact 3, the data-acquisition system. To realise this artefact, the thesis develops a custom VR game called *EmojiHero VR*, in which players reenact seven emotional facial expressions (see Section 3.3.1) symbolised by emojis to earn points. EmojiHeroVR combines this elicitation method with the capture of facial expressions via external cameras, HMD-provided face tracking, and synchronised multimodal logging during actual VR use. The full system design and implementation are presented in Chapter 4.

User Study Design and Execution. The next step is the design and execution of a user study, in which the data-acquisition system is deployed to generate the recordings required for constructing Artefact 2, the benchmark database. Accordingly, the thesis uses EmojiHeroVR in a controlled study with human participants to collect synchronised multimodal facial-expression data under consistent VR conditions. At the same time, the study empirically evaluates the suitability of EmojiHeroVR as a study instrument. In addition to collecting raw data, the study also assesses user experience and VR-related discomfort via questionnaires. Chapter 5 details the full study design and execution.

Construction of the Benchmark Database. The third step transforms the recordings collected in the user study into Artefact 2, the benchmark database. To do so, the raw samples are manually labelled with the seven classes chosen in Section 3.3.1, filtered based on labeller agreement, divided into participant-independent splits, temporally aligned across modalities, and structured into task-specific subsets. The resulting database, called *EmojiHeroVR Database (EmoHeVRDB)*, comprises benchmark subsets for static and dynamic FER on external-camera recordings and HMD-derived face-tracking signals. Multimodal FER is supported through temporal synchronisation across modalities. Chapter 6 describes the annotation procedure, database construction, and subset definitions in detail.

Implementation and Evaluation of Baseline FER Models. EmoHeVRDB is then used to implement and evaluate the baseline FER models (Artefact 1). This evaluation covers static and dynamic FER, unimodal and multimodal models, and an explicit comparison between natural and simulated HMD occlusion under a consistent participant-independent benchmark protocol. These static and dynamic experiments are reported in Sections 7.1 and 7.2, respectively.

Comparative Synthesis. Finally, the resulting benchmark evidence is comparatively synthesised across modelling settings and sensing modalities to derive interpretations, trade-offs, and design implications for FER in VR settings. This comparative synthesis is presented in Section 7.3.

Chapter 4

EmojiHeroVR – System Design

This chapter presents the design and implementation of EmojiHeroVR, the required data-acquisition system that realises Artefact 3 from the evidence requirements outlined in Section 3.2. First, Section 4.1 explains the system's role in the thesis and derives its architecture from the evidence requirements and design decisions established in Sections 3.2 and 3.3, respectively. Then, Section 4.2 describes EmojiHeroVR's game mechanism and its role as a facial-expression elicitation instrument. Next, Section 4.3 presents the FER subsystem of EmojiHeroVR, which rates whether a player's reenacted expression matches the emoji's target expression. Finally, Section 4.4 details the multimodal recording and synchronised logging capabilities that support the later construction of the benchmark database.

4.1 Design Rationale and Architecture Overview

Requirements. According to the requirements of Artefact 3 in Section 3.2.3 and the design decisions in Section 3.3, EmojiHeroVR must support: (1) the repeated elicitation of posed facial expressions from the seven target classes of anger, disgust, fear, happiness, sadness, surprise, and neutral, (2) external-camera recording of HMD-occluded faces, (3) HMD-provided face tracking, and (4) logging in a form suitable for later static and dynamic, unimodal and multimodal benchmarking. In

addition, EmojiHeroVR must be suitable for repeated deployment with human participants in VR, including an acceptable user experience and low VR-related discomfort.

A VR Game for Data Acquisition. The thesis adopts a game-based approach to satisfy these requirements. A VR game enables repeated elicitation within a structured interaction sequence while providing a more coherent and engaging experience than strict imitation protocols. At the same time, integrating elicitation, capture, and logging into a single system makes it possible to automatically associate the recorded data with essential game context.

Meta Quest Pro for Face Tracking. To realise the face-tracking requirement (see RG2, Artefact 3, and Section 3.3.3), the system employs the *Meta Quest Pro* as the concrete HMD. It is a commercially available consumer HMD that relies on five built-in infrared cameras for face tracking. The face-tracking outputs are accessible via the Face Tracking API (Meta 2025) of the Meta XR Core SDK (Meta 2023), which returns 63 blendshape coefficients, or facial expression activations, defined on the basis of FACS (Ekman, Friesen and Hager 2002). Accordingly, the system's face tracking depends on provider-specific hardware and software, but, unlike previous studies (see Section 2.7.2), it remains commercially available, reproducible, and usable without additional custom configurations or hardware.

Architecture Overview. At the highest level, EmojiHeroVR consists of two software components: a Unity-based VR game and an FER subsystem implemented as a Python web service. The Unity application is the system's central component. It provides the user interface, controls the game flow, integrates the connected sensing hardware, and handles logging. The FER web service receives facial images from the Unity application via HTTP requests and returns class probabilities for the seven target classes, which are used for recognition-based gameplay feedback. On the hardware side, the Unity application is connected to a central-view external camera, the Meta Quest Pro, and an optional 45-degree side-view external camera.

The central-view camera supplies the facial images used by the FER web service during runtime, whereas the side-view camera's purpose is only to provide a second viewpoint for the facial recordings. The Meta Quest Pro displays the game to players and provides the HMD-based face-tracking data via the Meta XR Core SDK. The overall system architecture and flow are visualised in Figure 4.1.

The user experiences the Unity application via the Meta Quest Pro and is prompted by the VR game to reenact facial expressions. During this process, central-view images are sent from the Unity application to the FER web service. Based on the response, the VR game determines whether a reenactment was successful. In parallel, the Unity application logs recordings from both cameras, the Meta Quest Pro's face-tracking data, and the relevant game context with timestamps, so that the resulting multimodal recordings can later be aligned.

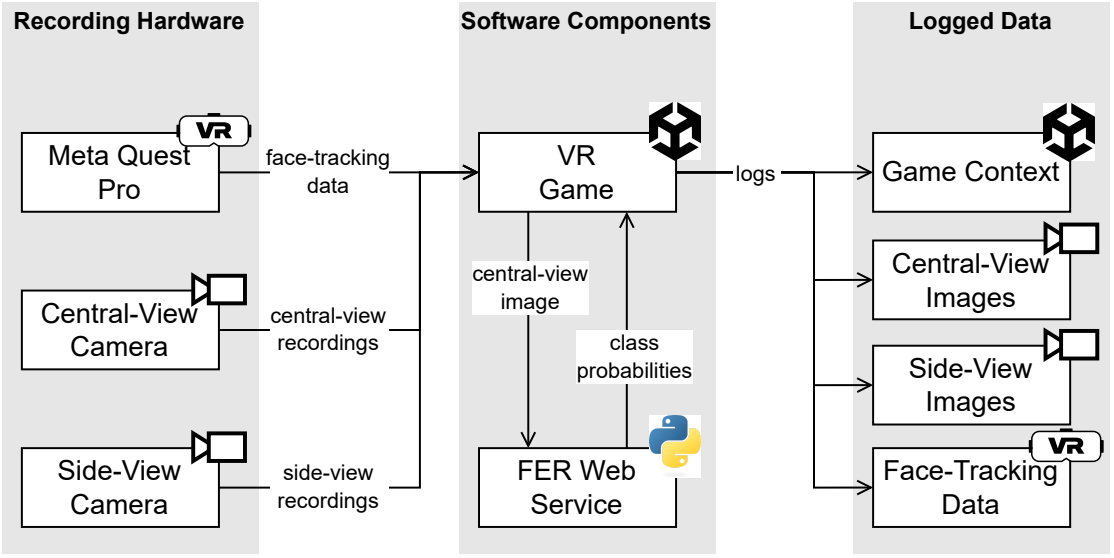


FIGURE 4.1: System architecture of EmojiHeroVR. The Python 'two snakes' logo is a trademark of the Python Software Foundation. The Unity 'cube' logo is a trademark of Unity Technologies or its subsidiaries. Both logos are used here for identification purposes only.

4.2 Facial Expression Elicitation

Requirements. The purpose of EmojiHeroVR is to collect data for the later construction of a benchmark database for FER under HMD occlusion. Accordingly, it must support the repeated elicitation of emotional facial expressions. In line with the design decision in Subsection 3.3.2, EmojiHeroVR relies on posed expressions to generate the data volume required for such a benchmark database. In line with the design decision in Subsection 3.3.1, these expressions correspond to the seven classes of anger, disgust, fear, happiness, sadness, surprise, and neutral. EmojiHeroVR realises this elicitation task through the following game concept.

Game Concept. EmojiHeroVR is a single-player high-score game, inspired by the popular *Guitar Hero* video game series. As depicted in Figure 4.2, emojis spawn at the back of a pinball-machine-like object and move towards the player in one of four lanes. When an emoji enters the red reenactment zone at the front of the lanes, the player has to reenact a facial expression associated with the emoji within about one second to earn points. Concretely, the game observes the player's facial expressions via an external camera and awards points if the FER subsystem recognises the prompted target expression.

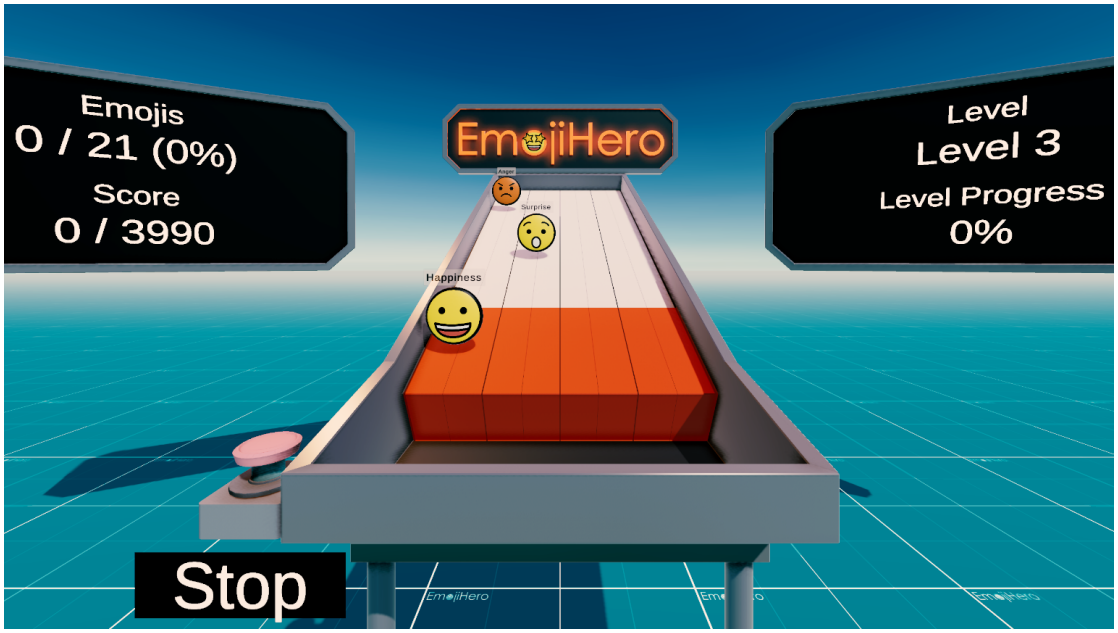


FIGURE 4.2: Screenshot of EmojiHeroVR's gameplay.

User Engagement. The number of awarded points depends on how quickly the game recognises the target expression, creating an engaging high-score mechanic. In addition, the game comprises multiple levels of varying difficulty determined by the number of emojis, their movement speed and their spawn rate.

Target Visualisation. Beyond timing and difficulty progression, the elicitation concept also depends on how the target classes are visually presented. For this purpose, EmojiHeroVR uses a dedicated set of emoji prompts. As shown in Figure 4.3, each expression category is associated with two different emoji textures. This is intended to encourage players not to mimic an emoji's specific facial expression but the emotion category it symbolises. Additionally, the category name is displayed directly above each emoji during gameplay to make the intended target category explicit.

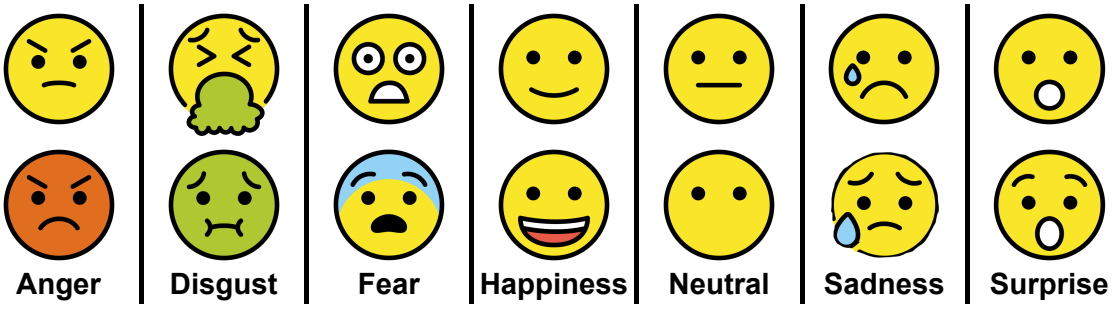


FIGURE 4.3: Emojis used in EmojiHeroVR; all emoji graphics are from the *OpenMoji* library (HfG Schwäbisch Gmünd 2023) and licensed under CC BY-SA 4.0.

Summary. Overall, this game concept enables repeated elicitation of the seven target classes within a temporally structured interaction loop, while the scoring and level progression help sustain player engagement. Integrating multimodal recording (see Section 4.4) with the game procedure further allows the system to create clear links between each reenactment and the corresponding recorded data.

4.3 Facial Expression Recognition Subsystem

As described in Section 4.2, EmojiHeroVR's game logic requires a mechanism to determine whether a player has successfully reenacted the prompted target expression. This mechanism is provided by the FER subsystem, which is implemented as a web service that exposes the underlying FER model to the Unity application.

4.3.1 Web Service

As shown in Figure 4.1, the FER subsystem is implemented as a Python web service using the FastAPI framework (Ramírez 2023). The service serves as a lightweight wrapper for the underlying FER model and exposes it to the Unity application through an HTTP-based interface.

During gameplay, the Unity application sends multiple asynchronous POST requests to the FER web service while an emoji is in the red reenactment zone depicted in Figure 4.2. Each request contains an image of the player's face recorded via the external central-view camera. For each request, the service first detects the face and facial keypoints using the MediaPipe package (Google 2023), with the maximum number of detected faces set to 1 and the minimum detection confidence set to 0.2. These settings are heuristics that help to detect the player's face despite HMD occlusion. Subsequently, the image is cropped based on the detected keypoints and passed to the FER model, which predicts a probability vector over the seven target classes. The FER web service returns this vector to the Unity application.

The Unity application then selects the class with the highest probability as the recognised class. If this class matches the target class of the current emoji for any one of the submitted images, the corresponding reenactment is rated as successful.

4.3.2 Facial Expression Recognition Model

The FER model wrapped by the FER web service must perform FER on HMD-occluded images. However, at the time of system development, no suitable model or benchmark database of naturally HMD-occluded facial expressions was available.

Consequently, this thesis followed the approach of related work (see Section 2.7.1) and trained the FER model on artificially occluded data from existing FER databases. For this purpose, EmojiHeroVR adopts the Poster++ architecture (Mao et al. 2023) due to its strong results on AffectNet (Mollahosseini et al. 2019) and RAF-DB (S. Li, W. Deng and Du 2017), its comparatively low computational complexity, and its open-source code. To match the seven target classes used in EmojiHeroVR, training data were combined from the AffectNet, EmotioNet (Benitez-Quiroz et al. 2016), ExpW (Zhanpeng Zhang et al. 2018), FER+ (Barsoum et al. 2016), and SFEW (Dhall et al. 2011) databases. The resulting images were processed with MediaPipe for face detection and facial-keypoint detection, and then uniformly cropped.

HMD occlusion was simulated by adding a black rectangle to each image based on the detected facial keypoints, following the same general approach as Houshmand and Khan (2020) and Gotsman et al. (2021). Figure 4.4 illustrates the applied artificial occlusion using example images from KDEF (Lundqvist et al. 1998). The resulting training set comprised about 360,000 artificially occluded images, together with a balanced validation set of 5,600 images, consisting of 800 randomly selected



FIGURE 4.4: Artificially occluded images *AF10HAHL, AF10HAS, AF10HAHR* from the KDEF database; from (Ortmann, Q. Wang et al. 2024, Figure 2), © 2024 IEEE.

images per class. Because the training set was highly imbalanced, especially due to AffectNet, the training process used sample selection probabilities inversely proportional to class frequency, combined with data augmentation and adjusted class weights in the categorical cross-entropy loss to favour the anger, fear, and sadness classes. To further validate the model under conditions closer to the intended application, it was evaluated on artificially occluded central-view and 45-degree side-view images from the KDEF database (see Figure 4.4). KDEF was chosen because it is balanced and was recorded under laboratory conditions similar to those of the later EmojiHeroVR study described in Chapter 5. In this cross-dataset evaluation, the model achieved an overall accuracy of 55%, with class recalls of only 10% and 7% for anger and fear, respectively.

To improve the model, training was resumed with a very low learning rate using artificially occluded images from 62 of the 70 KDEF subjects. This fine-tuning step increased accuracy on the remaining held-out KDEF subjects to 81%, providing sufficient confidence to deploy the model in EmojiHeroVR. Per-class precision, recall, and F1-score for the deployed FER model are listed in Table 4.1.

TABLE 4.1: Performance metrics in % for the fine-tuned FER model on the held-out artificially occluded KDEF test set.

Expression	**Precision**	**Recall**	**F1-Score**
Anger	60.87	58.33	59.57
Disgust	78.95	93.75	85.71
Fear	66.18	93.75	77.59
Happiness	95.92	97.92	96.91
Sadness	94.12	66.67	78.05
Surprise	100.00	70.83	82.93
Neutral	87.23	87.23	87.23
Average	83.33	81.21	81.14

4.4 Multimodal Recording and Logging

As outlined in Section 4.1, EmojiHeroVR must record the required modalities and log them in a form suitable for later database construction and benchmarking.

Recording Modalities. EmojiHeroVR incorporates three recording devices. First, a central-view external camera captures players' facial expressions from a frontal viewpoint under natural HMD occlusion. These recordings also provide the facial images used for the runtime FER described in Section 4.3. Second, a 45-degree side-view external camera records the same reenactment processes from an additional viewpoint. Third, the Meta Quest Pro provides HMD-based face-tracking outputs via the Face Tracking API (Meta 2025) of the Meta XR Core SDK (Meta 2023).

Recording Window. EmojiHeroVR defines a dedicated recording window for each emoji to directly associate all recorded data with the corresponding reenactment process (see Figure 4.5). Each recording window spans the time during which the emoji is present in the red reenactment zone depicted in Figure 4.2, plus a small preceding buffer. This buffer accounts for cases in which players begin reenacting an emoji before it enters the reenactment zone. Depending on the level, a recording window effectively covers about 1.5–2 seconds (including the buffer). All camera frames and face-tracking outputs recorded within a recording window are stored, whereas all data recorded outside of recording windows are discarded.

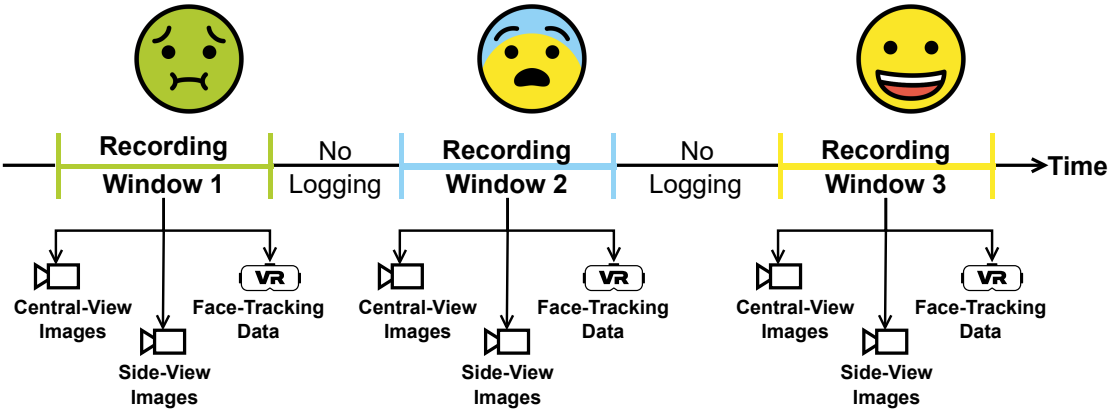


FIGURE 4.5: Association of recorded data with emojis during recording windows; all emoji graphics are from the *OpenMoji* library (HfG Schwäbisch Gmünd 2023) and licensed under CC BY-SA 4.0.

Logging Format. EmojiHeroVR stores camera recordings as separate PNG files at 30 FPS and a resolution of 1280×720 pixels. The Meta Quest Pro's 63 blendshape coefficients are appended to a CSV file once per iteration of the game loop. Consequently, the logging frequency depends on runtime performance. It averaged about 60 Hz during the user study described in Chapter 5. Each stored image and each face-tracking vector is associated with identifiers for the current session, level, and emoji, as well as a Unix timestamp at millisecond resolution. This logging format preserves the temporal information required to align data within and across modalities for each reenactment process.

Chapter 5

User Study Design and Execution

This chapter presents the design and execution of the user study in which EmojiHeroVR is deployed as the data-acquisition system (Artefact 3). It links the implementation of EmojiHeroVR in Chapter 4 to the construction of the benchmark database (Artefact 2) in Chapter 6 by generating the required raw multimodal recordings and evaluating EmojiHeroVR as a VR study instrument.

Section 5.1 describes the experimental setup. Section 5.2 details the experimental procedure. Section 5.3 outlines the sample selection and recruitment strategy. Section 5.4 presents the ethical considerations and approval. Section 5.5 reports the study execution and realised sample. Section 5.6 evaluates user experience, VR sickness, and gameplay performance. Section 5.7 details the collected raw data.

5.1 Experimental Setup

The experimental setup realises the system architecture introduced in Section 4.1 and is depicted in Figure 5.1. Two consumer Logitech webcams, a C930e and a C920e, capture participants' faces from a central view and a 45-degree side view at 30 FPS and a resolution of 1280×720 pixels. In addition, two softboxes set to a colour temperature of 5400 K provide consistent illumination. Participants are seated facing the cameras, with a white wall behind them. A Meta Quest Pro



FIGURE 5.1: Experimental setup with two cameras, two softboxes, a PC and a participant wearing the Meta Quest Pro VR headset; from (Ortmann, Q. Wang et al. 2024, Figure 3), © 2024 IEEE.

serves as both the VR display for EmojiHeroVR and the source of face-tracking data. Both cameras and the Meta Quest Pro are connected to a PC that runs EmojiHeroVR as a Unity application together with the FER web service described in Section 4.3. All recorded data are written to the PC's file system, with camera frames stored as PNG files and face-tracking outputs as CSV data, as detailed in Section 4.4. The PC runs Windows 11 version 22H2 and has an Intel i9-12900K CPU, 64 GB of DDR4 RAM, a 1 TB Samsung SSD 970 EVO Plus and an Nvidia GeForce RTX 3090 GPU.

5.2 Experimental Procedure

The study follows a fixed session protocol to ensure consistent elicitation and recording conditions across participants. The procedure is visualised in Figure 5.2.

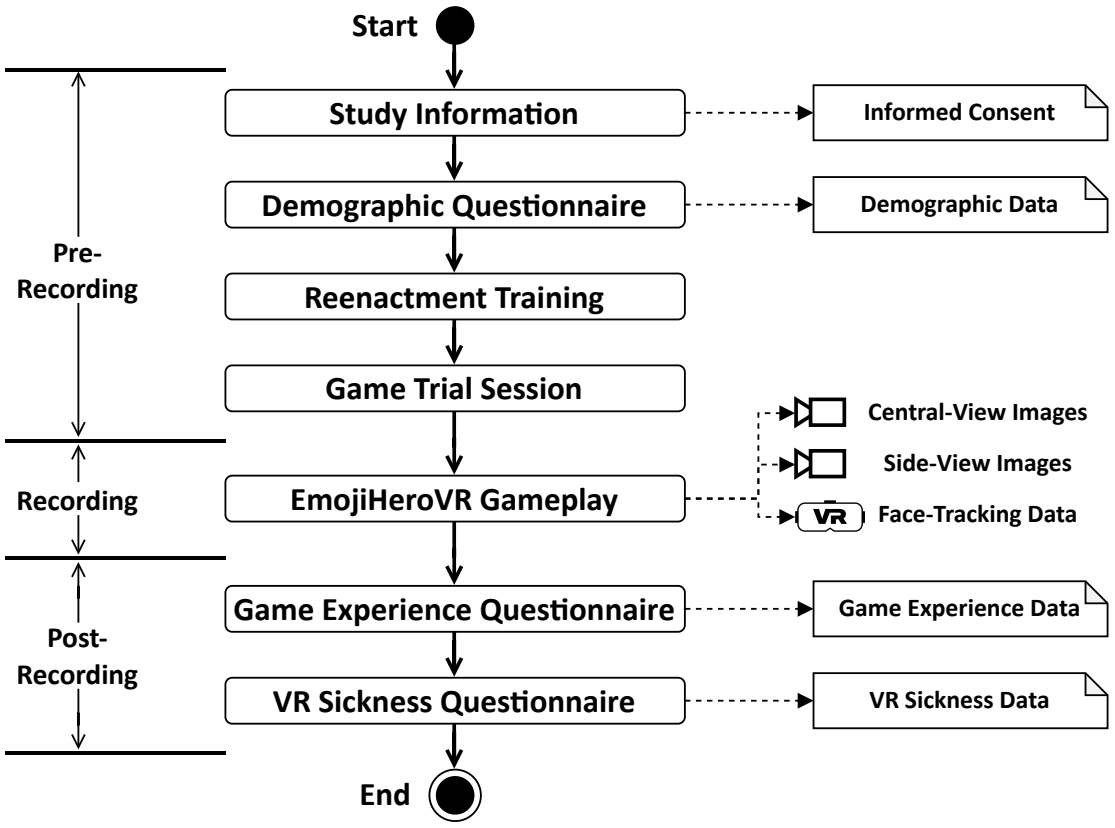


FIGURE 5.2: Overview of the experimental procedure.

First, participants are briefed on the study's purpose and procedure. They then receive information sheets and consent forms that describe the study design and data processing practices.

After providing informed consent, participants complete a short demographic questionnaire covering age, gender, ethnicity, and prior VR experience. The realised sample is reported in Section 5.5. Its characteristics provide context for the later interpretation of results with regard to possible demographic limitations.

In the next step, participants are introduced to FACS (Ekman, Friesen and Hager 2002) and trained for about 10 minutes in reenacting the seven target expressions used in EmojiHeroVR to improve the consistency of the posed expressions during gameplay.

Subsequently, the research team positions each participant at the recording station, adjusts the cameras, and sets up the Meta Quest Pro. A brief trial session allows

participants to familiarise themselves with the game before the recording session begins.

The recording session consists of four predefined levels of EmojiHeroVR separated by breaks of about 90 seconds. The levels present fixed emoji sequences and become more challenging as the number of emojis, their movement speed, and their spawn rate increase. Level 1 starts with 9 emojis in about 25 seconds, whereas Level 4 comprises 28 emojis in about 50 seconds. In total, each participant reenacts 70 emojis, corresponding to 10 reenactments per expression category. This repeated and structured elicitation procedure is designed to produce recordings of sufficient quantity and consistency for the later construction of the benchmark database.

After completing the gameplay session, participants answer the Game Experience Questionnaire (GEQ; IJsselsteijn et al. 2013) and the Virtual Reality Sickness Questionnaire (VRSQ; H. K. Kim et al. 2018). The GEQ is a widely used instrument for assessing game experience, while the VRSQ is specifically designed to capture VR-related sickness symptoms. Together, they cover the evaluation requirements concerning user experience and VR-related discomfort of EmojiHeroVR, defined in Section 3.2.3.

5.3 Recruitment and Sample Selection

The study relies on convenience sampling, as its purpose is not to obtain a representative sample of the population, but to collect data for the benchmark database targeted in this thesis (see Artefact 2). Accordingly, participants were recruited primarily from the environment of the Hamburg University of Applied Sciences (HAW Hamburg). Recruitment was promoted via e-mail using mailing lists, in lectures, and through posters and flyers placed around the Finkenau campus. Participation was voluntary and restricted to adults aged 18 years or above. Participants received €10 in cash as compensation for their contribution.

The study aimed for a minimum sample size of 30 participants. Relative to the established in-lab FER databases reviewed in Section 2.5.1, a participant count of 30 is comparatively small. However, the corresponding 2,100 expression reenactments are comparable in size and therefore expected to provide a sufficient basis for the later training and evaluation of FER models. At the same time, this target exceeds the sample sizes of many FER-in-VR studies reviewed in Section 2.7.2. Ortmann, Q. Wang et al. (2023) report an average of 21 participants across the user studies included in their review of FER in VR settings. Overall, the chosen target size represents a pragmatic compromise between dataset size, participant variation, and the practical effort of controlled multimodal VR data collection.

5.4 Ethical Considerations and Approval

As discussed in Section 2.8, human-participant studies on FER in VR settings raise several ethical considerations. These are directly relevant to the present study, as the setup in Section 5.1 captures multimodal facial data, while the procedure in Section 5.2 repeatedly elicits posed facial expressions through a VR game. The study addresses these concerns in several ways.

In line with Section 2.8.4, participants receive written study information and consent forms before taking part, with particular emphasis on the study procedure and data processing practices. Written informed consent is obtained before any recording begins.

As described in Section 5.3, participation is voluntary and restricted to adults. Recruitment takes place through open group-based channels such as mailing lists, lectures, posters, and flyers rather than through direct individual solicitation. This supports meaningful opt-in in the university context and reduces the risk of undue pressure. Practical withdrawal is possible at any time without negative consequences, for instance, in cases of discomfort during VR use.

Participant welfare during VR exposure is further supported by the study procedure. As discussed in Sections 2.8.3 and 2.8.4, the study uses a seated setup, a brief trial session, short level durations, and breaks between levels to reduce participant burden and mitigate potential VR-related discomfort. In addition, the thesis adopts posed expressions, as justified in Section 3.3.2, which allows controlled and repeated elicitation without relying on stronger emotion-induction procedures.

Finally, in line with Section 2.8.5, the associated data-processing practices are communicated transparently and designed to follow the principles of purpose limitation and data protection emphasised by the GDPR.

The recruitment process, the study conduct, and corresponding data processing were approved by the ethics committees of the University of the West of Scotland (application 21639) and the HAW Hamburg (application 2023-25).

5.5 Study Conduct and Realised Sample

The study was conducted on eight days between 15 November and 1 December 2023 at the Finkenau campus of the HAW Hamburg. In total, 37 participants took part in the study.

The realised sample consisted primarily of students and faculty members. Participants' ages ranged from 19 to 50 years, with a mean age of 27.19 years and a standard deviation of 7.66 years. Age statistics by reported gender are summarised in Table 5.1. Of the 37 participants, 15 identified as female, 20 as male, and 2 preferred not to disclose their gender. Thirty-one participants reported a European

TABLE 5.1: Age statistics of the realised sample by reported gender.

Statistic	**Female**	**Male**	**No disclosure**	**All**
n	15	20	2	37
Min	19	20	20	19
Max	50	36	23	50
Mean	29.20	26.25	21.50	27.19
SD	10.58	4.16	1.50	7.66

heritage. Regarding prior VR experience, 14 participants reported none, 16 rare use, 4 occasional use, and 3 frequent use. Overall, the realised sample reflects the university-based recruitment context described in Section 5.3 and is therefore not representative of the general population.

All 37 participants completed the full study procedure (see Figure 5.2), including all four levels of EmojiHeroVR and both post-study questionnaires. No participant withdrew during the study. However, due to a technical malfunction, face-tracking data were not recorded for one participant.

5.6 Evaluation of EmojiHeroVR

This section evaluates EmojiHeroVR as a study instrument using the post-study questionnaires introduced in Section 5.2, as well as recorded gameplay data. It thereby addresses the evaluation requirements for the data-acquisition tool derived in Section 3.2.3. Subsection 5.6.1 employs the core module of the GEQ to examine user experience, while Subsection 5.6.2 uses the VRSQ to assess VR-related discomfort. Subsection 5.6.3 analyses recognition success across repeated facial expression reenactments. One participant's questionnaire responses were excluded due to an incorrectly answered quality-assurance item. Accordingly, the questionnaire evaluations are based on responses from 36 participants, whereas the gameplay analysis includes all 37 participants.

5.6.1 Game Experience Questionnaire

The user study used the core module of the GEQ, which is designed to capture the player's immediate gameplay experience (IJsselsteijn et al. 2013). It comprises 33 items—for instance, 'I felt content', 'I felt skilful', and 'I forgot everything around me'—grouped into the 7 components of *Competence*, *Sensory and Imaginative Immersion*, *Flow*, *Challenge*, *Tension/Annoyance*, *Negative affect*, and *Positive*

affect. Each item is rated on a five-point Likert scale from 0 (*not at all*) to 4 (*extremely*). Each component score is computed as the mean of its item ratings.

Although the GEQ is a widely used instrument for assessing game experience, the validity of its seven-factor structure is not uncontroversial (Johnson et al. 2018; Law et al. 2018; Norman 2013). Therefore, the present study uses the GEQ primarily as a descriptive instrument rather than as a basis for psychometric claims.

Figure 5.3 visualises the GEQ component results for the conducted study. The complete response distribution for all items is provided in Appendix A.1. Overall, the results indicate that EmojiHeroVR provided an engaging and largely positive user experience. The highest mean component scores were observed for *Positive affect* ($\mu = 2.90$, $\sigma = 0.76$) and *Flow* ($\mu = 2.73$, $\sigma = 0.80$), suggesting that participants generally experienced the game as fairly enjoyable and absorbing. At the same time, *Competence* ($\mu = 2.14$, $\sigma = 0.87$), *Immersion* ($\mu = 1.92$, $\sigma = 0.76$), and *Challenge* ($\mu = 2.03$, $\sigma = 0.77$) were in a moderate range, which is consistent with a task that remains structured and demanding enough to support repeated elicitation. In contrast, *Tension/Annoyance* ($\mu = 0.91$, $\sigma = 0.80$) and *Negative affect* ($\mu = 0.54$, $\sigma = 0.59$) remained low. Taken together, these results suggest

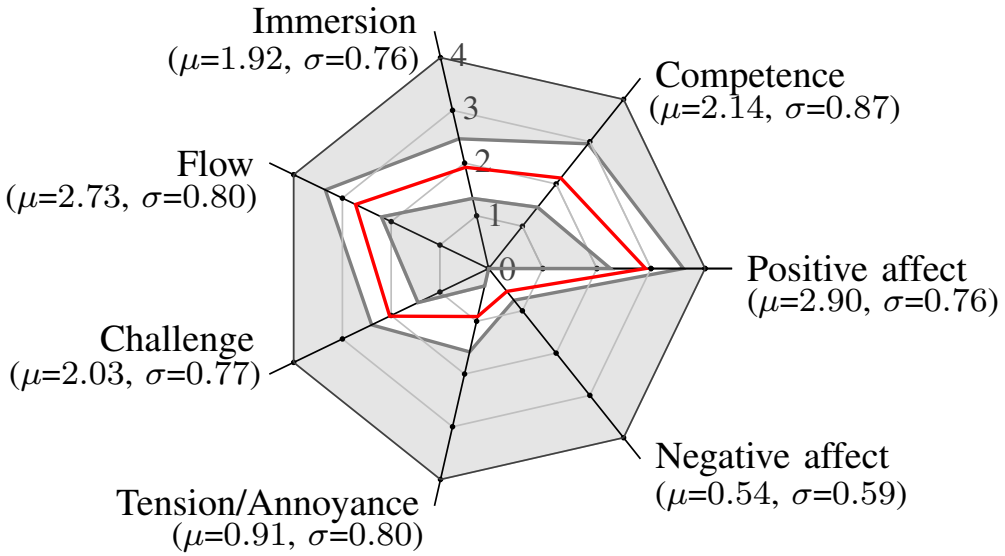


FIGURE 5.3: Radar chart showing the mean values (0=not-at-all, 1=slightly, 2=moderately, 3=fairly, 4=extremely) for the seven GEQ components in red. The first and third quartiles are visualised by the inner and outer grey lines; from (Ortmann, Hülsemann et al. 2024, Figure 2), © 2024 IEEE.

that EmojiHeroVR successfully balanced engagement and task difficulty without causing pronounced frustration or annoyance.

5.6.2 Virtual Reality Sickness Questionnaire

To assess VR-related discomfort, the study used the VRSQ (H. K. Kim et al. 2018). The VRSQ was developed specifically for immersive VR settings as a concise alternative to the Simulator Sickness Questionnaire (Kennedy et al. 1993). It comprises 9 items rated on a four-point Likert scale from 0 (*not at all*) to 3 (*very*) and groups them into the two components *Oculomotor* and *Disorientation*. The oculomotor component covers symptoms of general discomfort, fatigue, eyestrain, and difficulty focusing, whereas the disorientation component includes headache, fullness of the head, blurred vision, dizziness with closed eyes, and vertigo. Following the original scoring procedure, the item scores are summed per component and then upscaled to a range from 0 to 100. The total score is computed as the average of the two component scores. Table 5.2 summarises the mean item responses together with the resulting component and total scores. The complete response distributions for all VRSQ items are provided in Appendix A.2. Overall, the results indicate

TABLE 5.2: Mean VRSQ item responses (0=not-at-all, 1=slightly, 2=moderately, 3=very) and resulting component scores ($n = 36$).

VRSQ Symptom	**Oculomotor**	**Disorientation**
1. General discomfort	0.11	–
2. Fatigue	0.17	–
3. Eyestrain	0.58	–
4. Difficulty focusing	0.39	–
5. Headache	–	0.14
6. Fullness of head	–	0.50
7. Blurred vision	–	0.56
8. Dizzy (eyes closed)	–	0.08
9. Vertigo	–	0.08
Component Score	**10.42**	**9.07**
Total Score	**9.75**	

Oculomotor score = $(\sum$ Oculomotor items$/12) \times 100$;
Disorientation score = $(\sum$ Disorientation items$/15) \times 100$;
Total score = (Oculomotor score + Disorientation score)$/2$.

that playing EmojiHeroVR caused almost no symptoms of VR sickness. The mean total score was 9.75 ($\sigma = 7.97$), with similarly low scores for both *Oculomotor* ($\mu = 10.42$, $\sigma = 10.23$) and *Disorientation* ($\mu = 9.07$, $\sigma = 8.60$). The highest individual participant total score was 30. In conclusion, EmojiHeroVR caused very little VR-related discomfort under the present study conditions and was thus suitable for repeated facial-expression elicitation.

5.6.3 Gameplay Performance and Learning Effects

EmojiHeroVR rewards participants when the in-game FER model recognises the prompted expression. This feedback may encourage participants to favour facial configurations reliably recognised by the model, potentially leaving more difficult-to-recognise expression variants underrepresented or absent from the recordings. To examine whether recognition success increased during the recorded session, gameplay performance was analysed across levels and repeated reenactments. All 37 participants completed 70 emoji reenactments, comprising ten repetitions of each of the seven expression categories. A reenactment was considered successful if at least one prediction of the in-game FER model matched the prompted expression within its recording window.

Overall, 1,747 of the 2,590 reenactments (67.45%) were successfully recognised. Recognition success varied considerably between expression categories, ranging from 31.62% for anger to 97.03% for neutral. However, no consistent improvement across repeated reenactments was apparent, as illustrated in Figure 5.4. Comparing the first five with the last five reenactments of each category, the average success rate decreased by 1.78 percentage points (pp) from 68.34% to 66.56%. Small improvements were observed for sadness (+1.62 pp) and disgust (+0.54 pp), whereas success rates decreased for the remaining five categories, most notably anger (−5.95 pp).

At the participant level, recognition success remained relatively consistent for 26 of the 37 participants, with differences between the first and last five reenactments

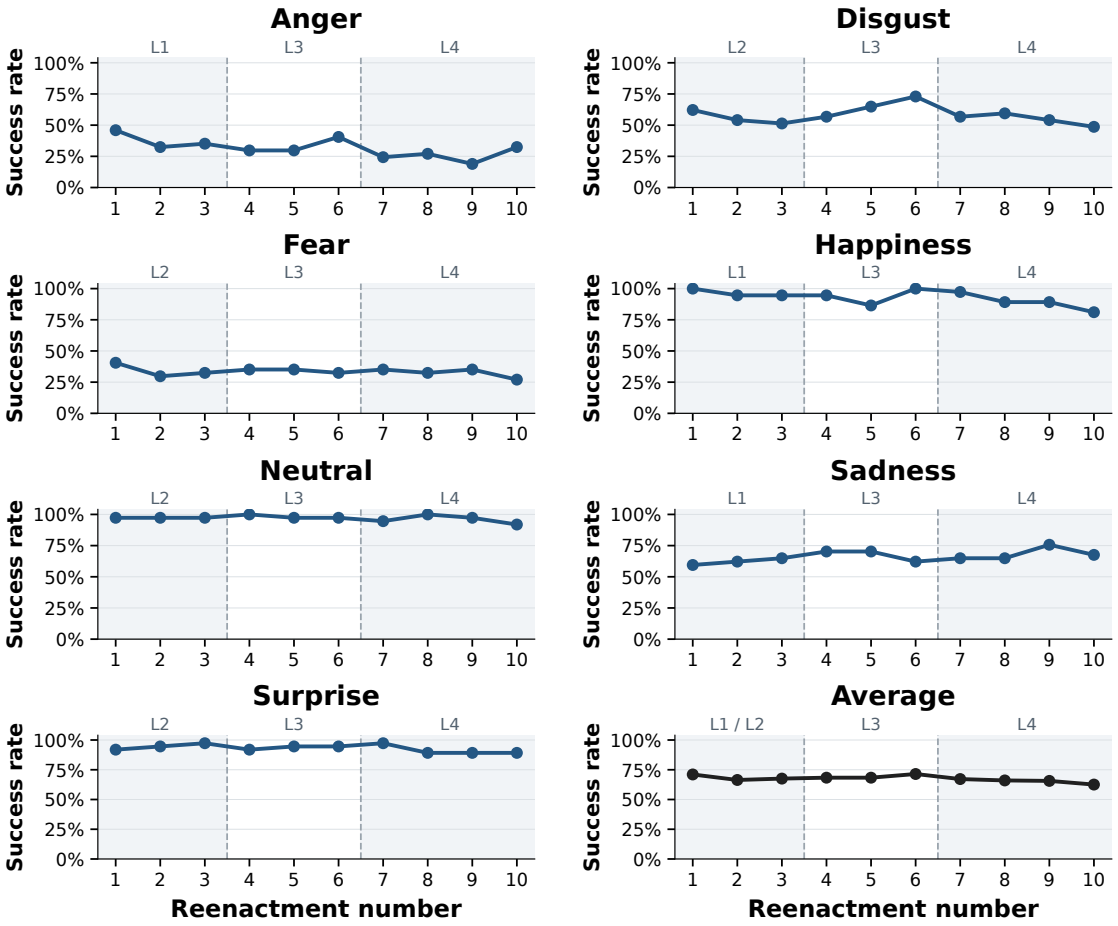


FIGURE 5.4: Recognition success across four levels (L1–L4) and ten reenactments per expression category ($n = 37$). L1 comprised only anger, happiness, and sadness. L2 comprised only disgust, fear, neutral and surprise.

ranging from −8.57 to +5.71 pp. Of the remaining 11 participants, five showed improvements ranging from 11.43 to 20.00 pp, whereas six exhibited decreases ranging from 11.43 to 31.43 pp.

Across the four levels, recognition success averaged 65.47%, 70.50%, 69.37%, and 65.35%, respectively, showing no consistent increase over the course of the gameplay session. However, all results must be interpreted in the context of changing gameplay conditions. Level 1 included only anger, happiness, and sadness, while Level 2 included only disgust, fear, neutral, and surprise. Level 3 introduced all seven expression categories at a higher spawn rate, while Level 4 additionally increased emoji movement speed. Consequently, the mean recording-window duration decreased from 2.14 s in Levels 1 and 2 to 2.00 s in Level 3 and 1.61 s in Level 4, leaving participants less time to perform recognisable expressions. The

average number of FER predictions per reenactment remained approximately 8.7 in Levels 1–3 but decreased to 6.63 in Level 4, reducing the opportunities for successful recognition.

Overall, the gameplay results show no consistent improvement in recognition success across repeated reenactments or successive levels. Nevertheless, the increasing difficulty and decreasing recognition opportunities in later levels might have obscured potential learning effects. Furthermore, recognition success does not directly measure changes in facial behaviour. Thus, the observed performance cannot rule out adaptation to the in-game FER model or establish that expression diversity remained unaffected, but it also provides no indication of such adaptation through improved recognition success.

The complete gameplay logs and analyses are available in the thesis repository: https://github.com/thorbenortmann/phd-thesis.

5.7 Collected Data

During the study, all 37 participants completed the four predefined levels of EmojiHeroVR and reenacted a total of 2,590 target expressions: 370 for each of anger, disgust, fear, happiness, neutral, sadness, and surprise. As described in Section 4.4, the central-view camera, the side-view camera, and the Meta Quest Pro recorded images and face-tracking vectors during dedicated recording windows. All stored samples were timestamped and associated with the corresponding session, level, and emoji. Table 5.3 summarises the recording volume. Due to a technical malfunction, no face-tracking data were captured for one participant.

In total, 147,490 PNG files per camera were stored at a resolution of 1280×720 pixels. For Level 1, the mean recording-window duration was 2.14 s, with 64.63 ($\sigma = 0.61$) images per camera on average. As the game became faster in later levels, the mean recording-window duration decreased to 1.61 s in Level 4, while still yielding at least 45 logged frames per reenactment. Across all levels, the

effective frame rate remained close to 30 FPS, with very little variation ($\mu = 30.44$, $\sigma = 0.30$).

The total number of logged face-tracking vectors was 323,846. The Face Tracking API (Meta 2025) of the Meta XR Core SDK (Meta 2023) was queried once per game-loop iteration, resulting in an average logging frequency of 62.80 Hz ($\sigma = 3.48$) and at least 93 associated vectors per reenactment. However, about half of the logged face-tracking vectors were duplicates, resulting in an effective update frequency of about 30 Hz. Each face-tracking vector comprises 63 floating-point values between 0 and 1, each representing the activation strength of a distinct FACS-oriented facial expression.

Overall, the study produced about 194 GB of PNG and CSV data. The outlined metrics show that EmojiHeroVR fulfilled the central data-acquisition requirements derived in Section 3.2.3: it repeatedly elicited posed facial expressions during actual VR use and synchronously logged external-camera recordings and HMD-derived face-tracking signals at high temporal resolution and substantial volume. In total, the study produced 2,590 reenactments, each represented by at least 45 frames per camera and 93 face-tracking vectors.

At the same time, the collected recordings do not yet constitute a benchmark database. The target expression associated with each recording window reflects the game prompt rather than verified ground truth. Thus, the raw recordings must

TABLE 5.3: Overview of the data recorded during the user study ($n = 37$).

Level	**Reenact-ments**	**Duration[a] in s**		**No. of Images[b]**		**No. of FT vectors[c]**	
		Mean	**Min**	**Mean**	**Min**	**Mean**	**Min**
1	333	2.14	2.04	64.63	61	143.14	122
2	444	2.14	2.10	64.74	63	144.87	116
3	777	2.00	1.90	60.43	57	136.83	98
4	1036	1.61	1.57	48.52	45	110.56	93
Total	**2,590**	**85 min 58 s**		**147,490**		**323,846**	

[a] Duration of the recording window (see Section 4.4) per reenactment in seconds.
[b] Per reenactment; values are identical for the central-view and side-view cameras.
[c] Face-tracking vectors per reenactment; only available for 36 instead of 37 participants.

first be labelled, filtered by agreement, temporally aligned across modalities, and structured into benchmark-ready subsets for static and dynamic FER. Chapter 6 addresses these steps and transforms the collected data into the benchmark database EmoHeVRDB.

Chapter 6

Constructing the EmojiHeroVR Database

This chapter presents the construction of the EmojiHeroVR Database (EmoHeVRDB) from the raw multimodal recordings generated in the user study reported in Chapter 5 and detailed in Section 5.7. It thereby realises Artefact 2 as specified by the evidence requirements presented in Section 3.2 and provides the benchmark database for the implementation and evaluation of baseline FER models (Artefact 1) in Chapter 7.

First, Section 6.1 explains why the recorded reenactments require manual annotation, describes the annotation procedure, and establishes a filtered set of defensible benchmark labels. Next, Section 6.2 defines the participant-independent training, validation, and test split used throughout Chapter 7. Finally, Section 6.3 describes the temporal alignment of the recorded modalities and details the construction of EmoHeVRDB's static and dynamic subsets across the external-camera and HMD-derived modalities.

6.1 Data Annotation and Filtering

Although each reenactment recorded in Chapter 5 was associated with a target expression from EmojiHeroVR's game logic, these targets cannot be treated as reliable ground-truth labels for benchmark construction. The expressions were posed by participants who were not trained actors and had to reenact them under the time pressure imposed by the game. In addition, the gameplay itself could elicit spontaneous reactions that differed from the intended target expression, such as stress or frustration when a reenactment failed. Moreover, participants might have adapted their facial expressions to 'please' the in-game FER model and maximise their score, although the analysis in Section 5.6.3 found no consistent improvement in recognition success. To establish defensible ground-truth labels for the recorded expressions, additional manual annotation was required for the construction of EmoHeVRDB.

Accordingly, Subsection 6.1.1 explains how annotation candidates were selected from the raw recordings. Subsection 6.1.2 then describes the manual annotation procedure. Finally, Subsection 6.1.3 presents the annotation results, reports inter-rater agreement, and defines the filtering step that yields the ground-truth labels used for EmoHeVRDB.

6.1.1 Selection of Annotation Candidates

The first step of the annotation process selected one central-view image per reenactment for manual labelling from the collected data reported in Section 5.7. Restricting the annotation to one image per reenactment reduced the effort to a manageable set of 2,590 candidate samples. The central-view modality was chosen because it already served as input to EmojiHeroVR's FER subsystem, and the target expression is more directly visible there than in side-view images or face-tracking vectors.

To identify the annotation candidate from the 45 to 66 stored central-view images per reenactment, the FER model described in Section 4.3 was reused as a selection aid. It predicted a probability vector over the seven target classes for each central-view image within a recording window, except for the first 15 frames, which corresponded to the 0.5 s buffer before an emoji entered the reenactment zone (see Section 4.4). Excluding these first 15 frames also ensured that at least 15 preceding frames remained available as temporal context for the later construction of sequence samples for dynamic FER. From the remaining 30 to 51 frames, the image with the highest probability for the target expression was selected for manual labelling as the most likely representative apex frame.

6.1.2 Manual Annotation Procedure

Each of the 2,590 selected central-view images was manually labelled by three student raters with prior experience in FER. The raters worked independently and assigned each image to exactly one of the seven target classes: anger, disgust, fear, happiness, neutral, sadness, or surprise. For this purpose, a minimalist labelling web service was implemented using the Python Flask framework (Pallets 2023). Its GUI is depicted in Figure 6.1. The web interface presented images

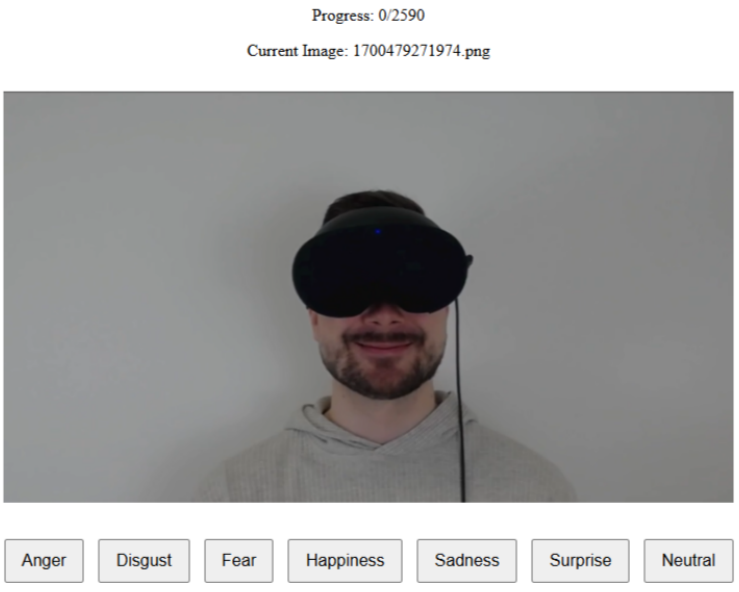


FIGURE 6.1: Screenshot of the labelling service.

one at a time in random order and allowed raters to assign a label by selecting the corresponding class button. No information about the target expression was shown during annotation, so that labelling decisions were solely based on the facial expression visible in the presented image.

6.1.3 Results, Agreement, and Filtering

The three raters produced 1,110 labels per class and agreed with the target expression in 71.87% of cases. The confusion matrix in Figure 6.2 summarises the manual labelling outcomes per class. The most accurately recognised classes were happiness, surprise, and neutral, with F1-scores of 92.44%, 80.37%, and 77.68%, respectively. In contrast, anger, fear, and disgust were recognised less reliably, with recalls of 51.44%, 54.23%, and 57.12%, respectively.

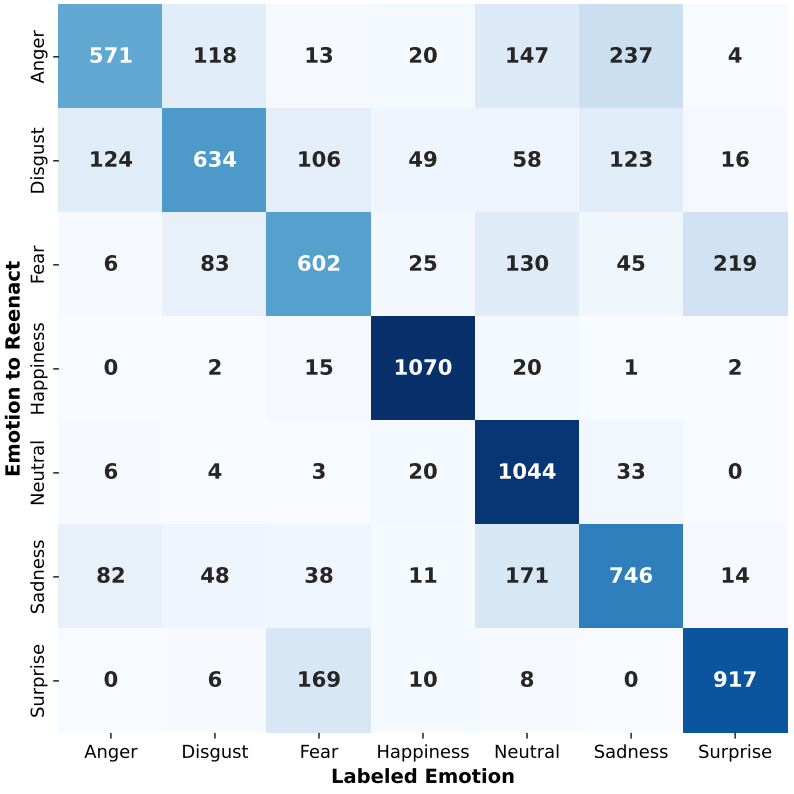


FIGURE 6.2: Confusion matrix: emotion to reenact versus labelled emotion; adapted from (Ortmann, Q. Wang et al. 2024, Figure 4), © 2024 IEEE.

For contextual comparison, Table 6.1 lists per-class recognition rates from validation studies for KDEF (Goeleven et al. 2008) and RaFD (Langner et al. 2010), alongside

the present ratings. Confusion patterns consistent with these validations of in-lab

TABLE 6.1: Recognition rates for KDEF[a], RaFD[b], and EmoHeVRDB's annotation candidates in % and their differences in percentage points.[c]

	Reference Studies		**Present Study**		
Expression	**KDEF**[a]	**RaFD**[b]	**EmoHeVRDB**	**Δ KDEF**	**Δ RaFD**
Anger	78.81	85.00	51.44	−27.37	−33.56
Disgust	72.17	81.00	57.12	−15.05	−23.88
Fear	43.03	83.00	54.23	+11.20	−28.77
Happiness	92.65	98.00	96.40	+ 3.75	− 1.60
Neutral	62.64	84.00	94.05	+31.41	+10.05
Sadness	76.70	80.00	67.21	− 9.49	−12.79
Surprise	77.09	90.00	82.61	+ 5.52	− 7.39
Mean	**71.87**	**85.86**	**71.87**	**0.00**	**−13.99**

[a] KDEF values are the mean biased hit rates reported in (Goeleven et al. 2008, Table 1).
[b] RaFD values are the mean percentage choices shown in (Langner et al. 2010, Figure 4), excluding contempt.
[c] The values are intended as contextual reference only, as the underlying databases and validation procedures differ between studies.

FER databases were observed for anger and disgust (242 cases), as well as for fear and surprise (388 cases). However, the confusion matrix in Figure 6.2 also shows notable differences from these earlier validation studies. Anger samples were most often misrated as sadness (237 of 1,110 cases), and anger, fear, and sadness were frequently labelled as neutral, with 147, 130, and 171 labels, respectively. A plausible explanation for these additional confusions is the occlusion of the upper face induced by the HMD. Anger, fear, and sadness rely strongly on cues from the eye and eyebrow regions (Wegrzyn et al. 2017). Thus, masking these cues likely increased both confusion and the tendency towards neutral assessments.

To quantify inter-rater reliability, Cohen's kappa and Fleiss' kappa were used, as introduced in Section 2.5.3. As listed in Table 6.2, the pairwise Cohen's kappa values ranged from 63.33% to 72.82%, with an average of 67.82%. Across all three raters, Fleiss' kappa was 67.76%. These values indicate substantial agreement and therefore provide confidence in the quality of the manual labels.

Finally, the labelled samples were filtered based on rater agreement to obtain a more reliable ground truth for the construction of EmoHeVRDB. Only samples

TABLE 6.2: Cohen's kappa scores in %; adapted from (Ortmann, Q. Wang et al. 2024, Table II), © 2024 IEEE.

Annotator	**Annotator**	**Cohen's kappa**
1	2	67.32
1	3	63.33
2	3	72.82
Average		67.82

for which at least two of the three raters assigned the expected target expression were retained. This filtering step reduced the sample pool from 2,590 to 1,921 labelled central-view images. For this retained set, the overall rater agreement increased from 71.87% to 93.18% and from 67.76% to 78.60% for Fleiss' kappa, indicating substantially stronger label consistency. At the same time, the filtering step introduced class imbalance in favour of expressions that were easier to reenact and recognise. The resulting class frequencies, ranging from 185 samples for anger to 360 for neutral, are listed in Table 6.3.

TABLE 6.3: Class frequencies of the retained 1,921 samples after filtering.

Anger	**Disgust**	**Fear**	**Happiness**	**Neutral**	**Sadness**	**Surprise**
185	218	219	358	360	259	322

6.2 Data Split

Following the filtering step described in Subsection 6.1.3, the retained labelled central-view images were divided into participant-independent training, validation, and test sets. Of the 37 participants, 21 were assigned to the training set, 8 to the validation set, and 8 to the test set. The split was stratified using the reported age, gender, and ethnicity information. Table 6.4 details the age and gender composition of the three participant groups.

A fixed participant-independent split was adopted to establish a common and reproducible benchmark protocol for static, dynamic, unimodal, and multimodal FER experiments. Keeping the participant allocation fixed supports comparisons

TABLE 6.4: Composition of reported age and gender for the participant-independent training, validation, and test sets.

Statistic	**Training Set**	**Validation Set**	**Test Set**
n	21	8	8
Female	8	4	3
Male	12	4	4
No Disclosure	1	0	1
Min Age	19	19	22
Max Age	50	48	48
Mean Age	25.81	28.75	28.75
SD Age	6.87	8.97	9.02

on corresponding held-out cases and allows subsequent studies to evaluate their models against the same benchmark. The training set supports model fitting, the validation set supports hyperparameter tuning and model selection, and the test set provides the final performance assessment. Participant-independent cross-validation, as discussed in Section 2.4.4, could provide a more robust assessment of performance variability across participants. However, it would require repeated training, model selection, and reconstruction of participant folds, including balancing of the evaluation folds. Thus, the fixed split was prioritised for comparability, reproducibility, and computational feasibility.

After the participant-level split, the validation and test sets were balanced at the class level to make later evaluation results more comparable and easier to interpret. For each expression class, samples from the most represented participant were iteratively removed at random until the class frequency matched the lowest class frequency in that split. In total, 64 of 449 validation samples and 79 of 457 test samples were discarded in this step. In contrast, the training set was left unbalanced to retain as much data as possible for model fitting.

The resulting class distribution of the benchmark split is visualised in Figure 6.3. It comprises an unbalanced training set of 1,015 samples, a balanced validation set of 385 samples, and a balanced test set of 378 samples. These 1,778 manually labelled central-view images, together with their participant-independent split, serve as

the common ground truth for the subsequent construction of EmoHeVRDB's benchmark subsets in Section 6.3.

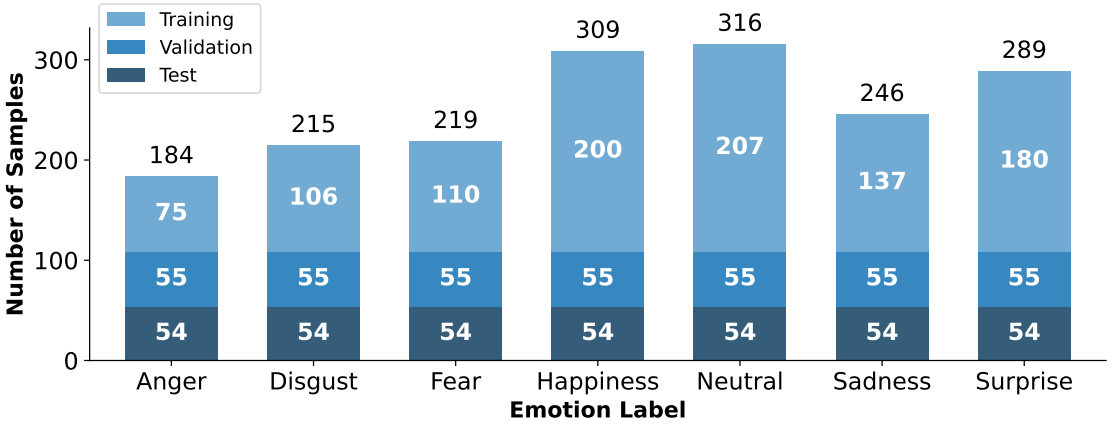


FIGURE 6.3: Label distribution of the training, validation, and test sets; adapted from (Ortmann, Q. Wang et al. 2024, Figure 5), © 2024 IEEE.

6.3 Benchmark Subsets

Building on the 1,778 manually labelled central-view images and their participant-independent split defined in Section 6.2, this section describes the construction of the four benchmark subsets of EmoHeVRDB: EmoHeVRDB-SI (Subsection 6.3.1), EmoHeVRDB-SFEA (Subsection 6.3.2), EmoHeVRDB-DI (Subsection 6.3.3), and EmoHeVRDB-DFEA (Subsection 6.3.4). In line with the requirements for the benchmark database in Section 3.2.2, these subsets were designed to support static and dynamic FER using both external-camera recordings and HMD-derived face-tracking signals under a single consistent protocol. Together, they form the benchmark resources for the implementation and evaluation of baseline FER models in Chapter 7. Because the subsets share common reenactments, labels, participant-independent splits, and timestamps for cross-modal alignment, they also support multimodal benchmarking. An overview of the four subsets is presented in Table 6.5.

TABLE 6.5: Overview of EmoHeVRDB's benchmark subsets.

Subset	Task	Sample Unit	Number of Samples			
			Train.	Val.	Test	Total
SI	Static	224×224 RGB image (central or 45° side view)	2,030	770	756	3,556
SFEA	Static	63-dimensional FEA vector	964[a]	385	378	1,727
DI	Dynamic	Sequence of 30 RGB images (central or 45° side view)	2,030	770	756	3,556
DFEA	Dynamic	Sequence of 30 FEA vectors	964[a]	385	378	1,727

Abbreviations: FEA = Facial Expression Activation, RGB = Red–Green–Blue, Train. = Training, Val. = Validation.
[a] Face-tracking data for one training participant were not recorded. Consequently, the FEA-based training splits contain 20 instead of 21 participants and 964 instead of 1,015 samples.

6.3.1 EmoHeVRDB-SI

EmoHeVRDB-SI was constructed for static image-based FER. For each of the 1,778 manually labelled central-view images, the simultaneously recorded 45-degree side-view image with the same timestamp was added. Starting from the original 1280×720 PNG frames, face detection was applied using the MediaPipe Python package, and a 720×720 region centred on the face was extracted. The images were then cropped more tightly around the detected face while ensuring a minimum crop size of 400×400 pixels. Finally, all images were resized to a uniform resolution of 224×224 pixels using Lanczos interpolation from the OpenCV Python package and converted from PNG to JPEG with a quality setting of 95 to reduce file size. As a result, EmoHeVRDB-SI comprises 3,556 RGB images from central and 45-degree side views at a resolution of 224×224 pixels. In addition to this standard benchmark version, wider 720×720 crops and 400×400 crops with the same face bounding boxes as the 224×224 version are also available. The sample distribution of EmoHeVRDB-SI is visualised in Figure 6.4.

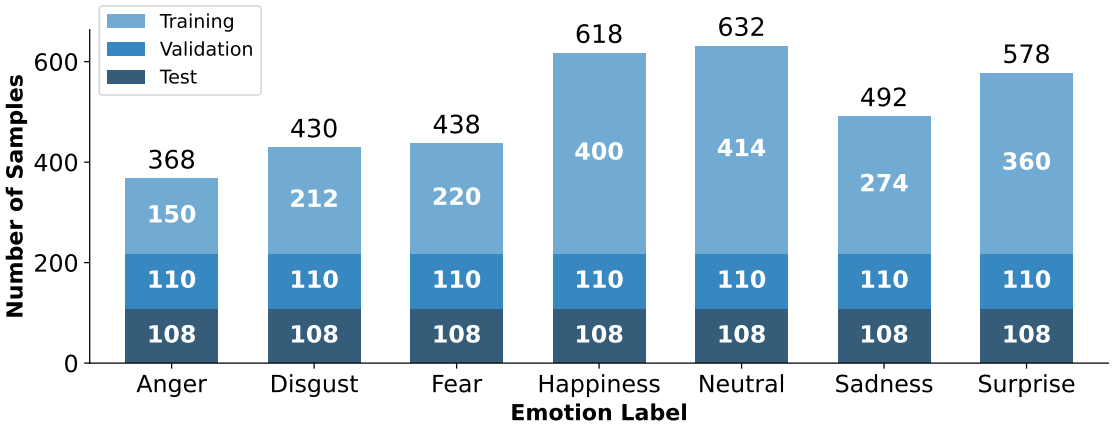


FIGURE 6.4: Sample distribution of EmoHeVRDB-SI and EmoHeVRDB-DI.

6.3.2 EmoHeVRDB-SFEA

EmoHeVRDB-SFEA was constructed for static FER using facial expression activations (FEAs; formerly referred to as face-tracking vectors) from the Meta Quest Pro. For each of the 1,778 manually labelled central-view images, the FEA vector with the smallest temporal distance within the same recording window was selected, using a tolerance of 5 ms. Each FEA vector originates from the Face Tracking API (Meta 2025) of the Meta XR Core SDK (Meta 2023) and contains 63 floating-point values between 0 and 1, each representing the activation strength of a distinct FACS-oriented facial expression. As a result, EmoHeVRDB-SFEA comprises 1,727 FEA vectors. The sample distribution is visualised in Figure 6.5. The overall sample size is smaller than that of EmoHeVRDB-SI because face-tracking data for one participant were not recorded. Consequently, the training set contains 964 instead of 1,015 samples. EmoHeVRDB-SFEA is available as a single CSV file and as one JSON file per FEA vector.

6.3.3 EmoHeVRDB-DI

EmoHeVRDB-DI is a superset of EmoHeVRDB-SI constructed for dynamic image-based FER. Each sample from EmoHeVRDB-SI was extended into a 30-frame sequence by adding 29 additional frames from the same recording window, prioritising preceding frames and appending subsequent frames when necessary. As a

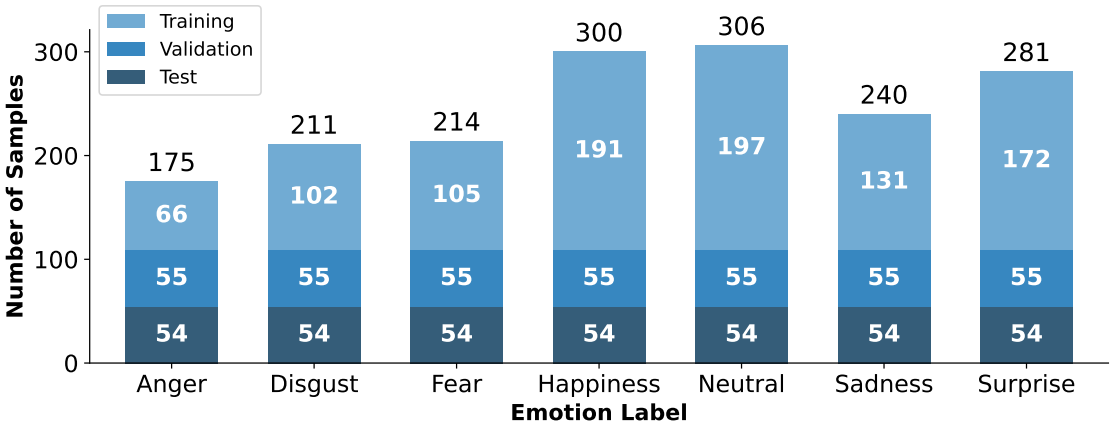


FIGURE 6.5: Sample distribution of EmoHeVRDB-SFEA and EmoHeVRDB-DFEA.

result, EmoHeVRDB-DI comprises 3,556 sequences of 30 RGB frames from central and 45-degree side views. As in EmoHeVRDB-SI, the standard benchmark version uses a resolution of 224×224 pixels, while corresponding 400×400 and 720×720 variants are also available. EmoHeVRDB-DI has the same sample distribution as EmoHeVRDB-SI, as visualised in Figure 6.4.

6.3.4 EmoHeVRDB-DFEA

EmoHeVRDB-DFEA was constructed for dynamic FER using FEA sequences. For each central-view frame in EmoHeVRDB-DI, the FEA vector with the smallest temporal distance within the same recording window was selected, preserving EmoHeVRDB-DI's sequence structure. As a result, EmoHeVRDB-DFEA comprises 1,727 sequences of 30 FEA vectors, with each vector containing 63 values between 0 and 1. Similar to EmoHeVRDB-SFEA, the training split contains 964 instead of 1,015 samples. EmoHeVRDB-DFEA has the same sample distribution as EmoHeVRDB-SFEA, as visualised in Figure 6.5. It is available as a single CSV file and as one JSON file per FEA vector.

Chapter 7

Facial Expression Recognition with EmoHeVRDB

This chapter presents the implementation and evaluation of baseline FER models (Artefact 1) using EmoHeVRDB as constructed in Chapter 6. It reports static and dynamic benchmark results under a common participant-independent protocol and synthesises them across modelling settings and sensing modalities.

Section 7.1 details the static FER experiments on EmoHeVRDB-SI and EmoHeVRDB-SFEA, covering image-based, FEA-based, and multimodal approaches, as well as the comparison between natural and simulated HMD occlusion. Section 7.2 then presents the corresponding dynamic FER experiments on the sequential benchmark subsets EmoHeVRDB-DI and EmoHeVRDB-DFEA for unimodal and multimodal FER. Finally, Section 7.3 consolidates these findings and discusses the resulting benchmark evidence to derive practical trade-offs, limitations, and design implications for FER in VR settings.

7.1 Static Facial Expression Recognition

This section reports the static FER experiments using EmoHeVRDB as the benchmark database to implement and evaluate the static FER baseline models required

for Artefact 1 in Section 3.2.1. All static experiments were executed on an Ubuntu 22.04 machine with CUDA 12.2 and cuDNN 8.9 installed. It had an Intel i9-12900K CPU, 64 GB of DDR4 RAM, and an Nvidia GeForce RTX 3090 GPU. Detailed configurations and hyperparameters for each static baseline model are provided in Appendix B.1.

First, Subsection 7.1.1 presents the image-based baseline on EmoHeVRDB-SI. Then, Subsection 7.1.2 compares static image-based FER under natural HMD occlusion with results obtained under simulated HMD occlusion to examine the extent of the domain gap. Next, Subsection 7.1.3 details the HMD-derived FEA baseline on EmoHeVRDB-SFEA. Building on the unimodal baselines from Subsections 7.1.1 and 7.1.3, Subsection 7.1.4 analyses the complementarity between image predictions and FEA predictions to assess the potential value of combining the two modalities. Finally, Subsection 7.1.5 reports the multimodal static FER experiments with synchronised images and FEAs.

7.1.1 Image-Based Baseline on EmoHeVRDB-SI

This section establishes the static image-based baseline on EmoHeVRDB-SI (see Section 6.3.1). In this FER setting, each input image is classified into one of the seven expression categories: anger, disgust, fear, happiness, neutral, sadness, and surprise. The following subsections first outline the model setup and training procedure (7.1.1.1), then justify the choice of EfficientNet-B0 as the convolutional backbone (7.1.1.2), and finally report the resulting performance on the naturally HMD-occluded images of EmoHeVRDB-SI (7.1.1.3).

7.1.1.1 Setup and Training

The experiments were implemented in TensorFlow 2.15 using the Keras interface. The EfficientNet-B0 architecture (M. Tan and Le 2020) was chosen as the CNN backbone for all static image-based experiments. This choice is justified in detail in Section 7.1.1.2. As in related work reviewed in Section 2.7.1, training relied

on transfer learning followed by fine-tuning. Accordingly, an EfficientNet-B0 with weights pretrained on ImageNet (Russakovsky et al. 2015) was initialised using Keras' standard implementation of the network. Its original classification head was then replaced with a Dropout layer (Srivastava et al. 2014) followed by a Dense layer with 7 units and softmax activation for the 7 target classes. The model architecture is visualised in Figure 7.1.

Training proceeded in two stages. First, only the new classification head was trained with a comparatively high learning rate of about 10^{-3}. Subsequently, progressively larger parts of the network were unfrozen and fine-tuned with lower learning rates down to 10^{-6}. For all experiments, the SparseCategoricalCrossentropy loss function, the Adam optimiser (Kingma and J. Ba 2014), and a batch size of 32 were used. To account for class imbalance in the training set, the loss was weighted with normalised inverse class frequencies. In addition, random flipping, translation, rotation, zoom, contrast, and brightness augmentation were applied during training. Overall, the hyperparameter space was kept small, and tuning focused primarily on the fine-tuning procedure, particularly the combination of layers to unfreeze, the number of epochs, and the learning rate. Finally, the model with the highest validation accuracy was evaluated on the test set. Its configuration and hyperparameters are detailed in Appendix B.1.1.

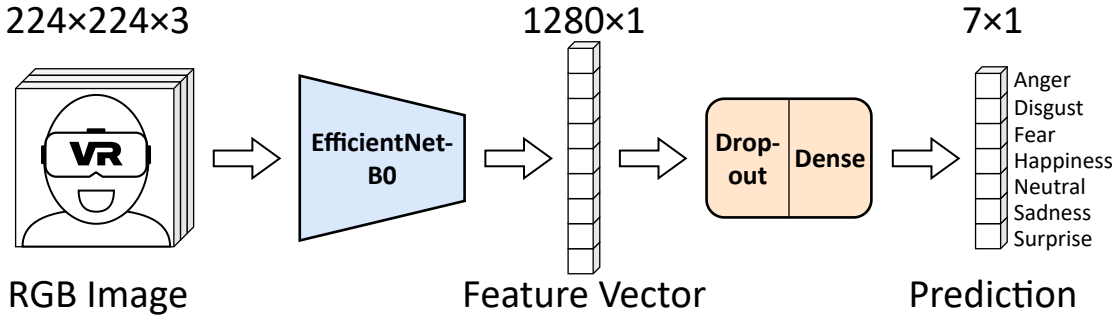


FIGURE 7.1: Architecture of the image-based baseline model.

7.1.1.2 Backbone Justification

For comparability, reproducibility, and moderate training cost, this thesis relies on EfficientNets (M. Tan and Le 2020; M. Tan and Le 2021) for image-based

feature extraction across all experiments reported in Chapter 7. The EfficientNet model family has demonstrated strong performance across several computer-vision tasks while remaining computationally efficient. EfficientNet-B0 is the smallest variant of this family. For the 8-class static FER task on AffectNet (Mollahosseini et al. 2019), Savchenko (2021) reported 61.32% accuracy using an EfficientNet-B0 pre-trained on the VGGFace2 database (Cao et al. 2018), thereby exceeding the 59.03% reported by Georgescu and Ionescu (2019) with a larger VGG-Face-based model. When using ImageNet-pretrained weights, the much larger Inception-v3 network ($\approx 24M$ parameters; Szegedy et al. 2016) exceeded EfficientNet-B0's accuracy ($\approx 5.3M$ parameters; M. Tan and Le 2020) by 2.10 percentage points (pp), whereas EfficientNet-B0 outperformed the slightly smaller MobileNet-v1 ($\approx 4.2M$ parameters; Howard et al. 2017) by 0.67 pp (Savchenko 2021). To further support the choice of EfficientNet-B0 as the CNN backbone empirically, two reference datasets were prepared:

AffectNet-7. First, a 7-class version of AffectNet (Mollahosseini et al. 2019), denoted AffectNet-7, was constructed by removing all images of the contempt class from the original 8-class dataset, resulting in 287,401 available images. Following common practice, the predefined AffectNet validation set, comprising 500 images per class, was used as the test set. In addition, a balanced validation set was created by randomly splitting 380 samples from each class, corresponding to 10% of the lowest class frequency. All images were kept at their standard resolution of 224×224 pixels.

AffectNet-7-occl. Second, an artificially occluded version of this dataset, denoted AffectNet-7-occl, was created to simulate HMD occlusion. Using the occlusion procedure from related work (see Section 2.7.1), a black rectangle was added to each image based on face- and facial-keypoint-detection results. If no face was detected or the detection result was implausible, the masking procedure defaulted to occluding the top 54% of the image at 90% of its width.

Following the training procedure described in Section 7.1.1.1, the best EfficientNet-B0 models achieved 64.31% accuracy on AffectNet-7 and 52.57% on AffectNet-7-occl. These accuracies, as well as the performance drop due to simulated HMD occlusion, are in the same range as the results of related work (Georgescu and Ionescu 2019; Houshmand and Khan 2020; Savchenko 2021), listed in Table 7.1. Although most results are not directly comparable because the number of target classes and the databases used for pre-training differ, they demonstrate that the comparatively lightweight EfficientNet-B0 is well suited for FER, achieving performance comparable to that of related work using significantly larger architectures.

TABLE 7.1: Comparison of classification accuracies on AffectNet; extended from (Ortmann, Q. Wang et al. 2024, Table III), © 2024 IEEE.

Model	**Pre-trained on**	**Simulated HMD Occlusion**	**Accuracy (%)**	
			7 classes	**8 classes**
Inception-v3[a]	ImageNet	No	63.10	59.65
EfficientNet-B0[a]	ImageNet	No	60.80	57.55
EfficientNet-B0[a]	VGGFace2	No	64.57	61.32
VGG-Face[b]	VGGFace	No	–	59.03
VGG-Face[b]	VGGFace	Yes	–	49.23
VGG-Face[c]	VGGFace	Yes	–	50.13
EfficientNet-B0[d]	ImageNet	No	64.31	–
EfficientNet-B0[d]	ImageNet	Yes	52.57	–

Results from: [a]Savchenko (2021), [b]Georgescu and Ionescu (2019), [c]Houshmand and Khan (2020), [d]present study.

7.1.1.3 Results on EmoHeVRDB-SI

Using EmoHeVRDB-SI at 224×224 pixels and the training procedure from Section 7.1.1.1, the best model achieved 69.84% accuracy on the test set, with a corresponding validation accuracy of 70.39%. This result establishes the static image-based benchmark baseline for participant-independent FER under natural HMD occlusion. Table 7.2 lists per-class precision, recall, and F1-scores on the test set, and Figure 7.2 shows the corresponding confusion matrix. Performance differed substantially between classes. Surprise, happiness, and neutral were recognised most reliably, with F1-scores of 86.12%, 77.18%, and 76.42%, respectively.

TABLE 7.2: Performance metrics on EmoHeVRDB-SI in %.

Expression	Precision	Recall	F1-Score
Anger	47.41	50.93	49.11
Disgust	56.06	68.52	61.67
Fear	78.16	62.96	69.74
Happiness	69.92	86.11	77.18
Neutral	77.88	75.00	76.42
Sadness	80.72	62.04	70.16
Surprise	89.11	83.33	86.12
Average	71.33	69.84	70.06

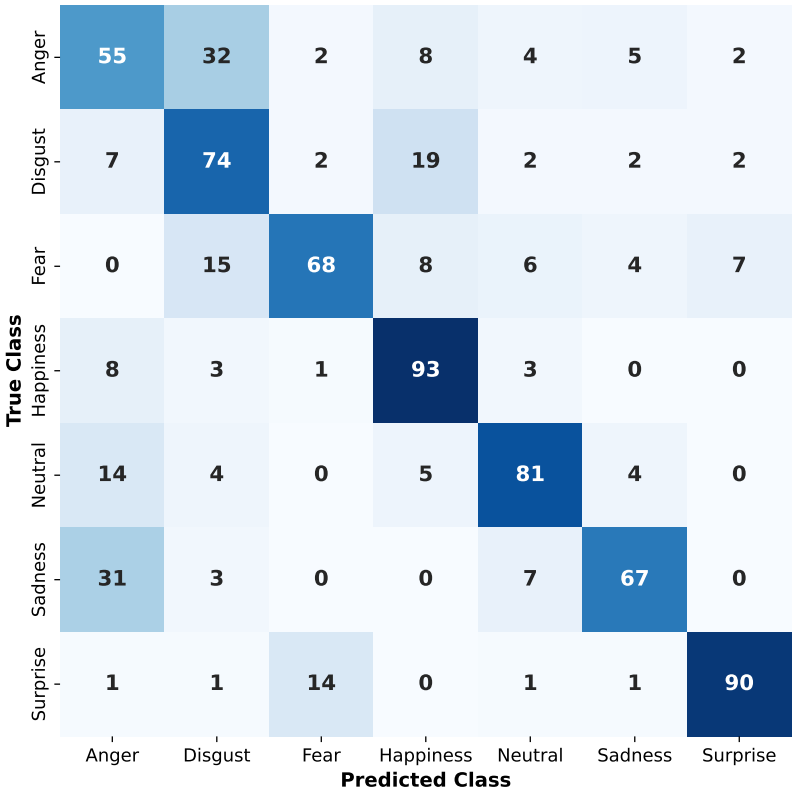


FIGURE 7.2: Confusion matrix on EmoHeVRDB-SI's test set.

In contrast, anger and disgust were the most challenging classes, with F1-scores of 49.11% and 61.67%, respectively. Fear and sadness occupied an intermediate position, with F1-scores of 69.74% and 70.16%. Out of 108 samples per class, the model correctly recognised only 55 for anger, 67 for sadness, and 68 for fear. The dominant confusion patterns were classifying anger as disgust (32 cases) and sadness as anger (31 cases), followed by disgust as happiness (19 cases), fear as disgust (15 cases), neutral as anger (14 cases), and surprise as fear (14 cases). Overall, these results indicate that static FER based on external-camera images remains

feasible despite natural HMD occlusion, but that the task is clearly challenging, particularly for anger and disgust.

7.1.2 Natural versus Simulated HMD Occlusion

This section compares the effects of natural and simulated HMD occlusion on static FER. The following subsections first prepare unoccluded and artificially occluded versions of AffectNet and KDEF (7.1.2.1), and then describe and interpret within-database and cross-database experiments using these databases, as well as EmoHeVRDB-SI (7.1.2.2).

7.1.2.1 Reference Databases and Evaluation Procedure

In addition to the AffectNet-7 and AffectNet-7-occl reference databases introduced in Section 7.1.1.2, a KDEF-based reference pair was prepared.

KDEF-SHR. The KDEF database contains 4,900 facial images of 70 subjects for seven facial-expression categories and five viewing angles (Lundqvist et al. 1998). For the present comparison, only the 1,960 central-view and 45-degree left-side-view images, coded *S (straight)* and *HR (half right profile)*, respectively, were retained to resemble EmoHeVRDB-SI. These images were cropped more tightly around the face and resized to 224×224 pixels. The resulting subset, denoted KDEF-SHR, was then split at the subject level by randomly selecting 14 of the 70 subjects, including seven female and seven male subjects, for the validation set and another 14 for the test set.

KDEF-SHR-occl. KDEF-SHR-occl was created by applying the same landmark-based masking procedure to KDEF-SHR as for the creation of AffectNet-7-occl.

All experiments used the same EfficientNet-B0 architecture and the same transfer-learning and fine-tuning procedure as described in Section 7.1.1.1. First, a separate

model was fine-tuned for each of the five databases: AffectNet-7, AffectNet-7-occl, KDEF-SHR, KDEF-SHR-occl, and EmoHeVRDB-SI. Second, cross-database evaluations were performed by applying the models trained on AffectNet-7 and AffectNet-7-occl to KDEF-SHR, KDEF-SHR-occl, and EmoHeVRDB-SI. Finally, additional experiments combined AffectNet-7-occl with EmoHeVRDB-SI during training to examine whether artificially occluded large-scale data can benefit recognition on naturally HMD-occluded faces.

7.1.2.2 Results and Interpretation

Within-Database. As listed in Table 7.3, FER on KDEF is significantly easier than on AffectNet and EmoHeVRDB-SI. Good recording conditions and very expressive faces enable the model to achieve 86.48% accuracy on KDEF-SHR, and the accuracy decreases by only 6.89 pp to 79.59% on KDEF-SHR-occl. In comparison, the corresponding results on AffectNet-7 and AffectNet-7-occl decrease by 11.74 pp from 64.31% to 52.57%. Accordingly, AffectNet-7 appears more affected by artificial occlusion than KDEF-SHR, likely because its in-the-wild images contain fewer distinctive features, making additional occlusion more likely to mask the crucial information available. The result of 69.84% accuracy on EmoHeVRDB-SI is 16.64 pp below KDEF-SHR, indicating that FER under natural HMD occlusion is substantially harder than regular FER, and still 9.75 pp below KDEF-SHR-occl, which was constructed to resemble EmoHeVRDB-SI. This remaining gap is likely due to the higher data variety in EmoHeVRDB-SI. The subjects in KDEF were trained amateur actors instructed to pose strong, clear facial expressions.

TABLE 7.3: Comparison of classification accuracies between databases without, with simulated, and with natural HMD occlusion; extended from (Ortmann, Q. Wang et al. 2024, Table IV), © 2024 IEEE.

Model	Pre-trained on	Database	Accuracy (%)
EfficientNet-B0	ImageNet	AffectNet-7	64.31
EfficientNet-B0	ImageNet	AffectNet-7-occl	52.57
EfficientNet-B0	ImageNet	KDEF-SHR	86.48
EfficientNet-B0	ImageNet	KDEF-SHR-occl	79.59
EfficientNet-B0	ImageNet	EmoHeVRDB-SI	69.84

They were between 20 and 30 years old and wore no beards, moustaches, earrings, eyeglasses, or visible makeup (Goeleven et al. 2008). In contrast, the participants in the EmojiHeroVR study were between 19 and 50 years old, wore various beards, makeup, earrings, and similar jewellery, and were only briefly trained in posing facial expressions. In addition, the expressions in EmoHeVRDB-SI were reenacted under time pressure while playing a VR game, resulting in greater variation in facial appearance, head pose, and expression intensity.

Cross-Database. The cross-database results are listed in Table 7.4. Applying a model trained on AffectNet-7 to KDEF-SHR achieves an average accuracy of 81.82%, which is only 4.66 pp below the accuracy obtained when training directly on KDEF-SHR. However, evaluating this model on KDEF-SHR-occl drops the accuracy to 55.06%. Training on AffectNet-7-occl improves performance on KDEF-SHR-occl to 64.10%. Accordingly, the transfer from AffectNet-7-occl to KDEF-SHR-occl performs significantly worse (−15.49 pp) than the transfer between the unoccluded versions (−4.66 pp). Nevertheless, using AffectNet-7-occl instead of AffectNet-7 could still improve the performance on KDEF-SHR-occl by 9.04 pp. Most notably, however, the AffectNet-7 and AffectNet-7-occl models perform similarly poorly on EmoHeVRDB-SI, with average accuracies of only 35.45% and 34.73%, respectively. This indicates a substantial difference between AffectNet-7(-occl) and EmoHeVRDB-SI and, more specifically, a pronounced domain gap between simulated and natural HMD occlusion.

Mixed Training. To investigate this difference further, additional experiments combined AffectNet-7-occl and EmoHeVRDB-SI during training while sampling according to the class frequencies of EmoHeVRDB-SI. A 1:1 ratio in the training set resulted in 70.11% accuracy on EmoHeVRDB-SI's test set. Using a 1:4 ratio further improved accuracy to 74.74%, thereby exceeding the 69.84% obtained when training exclusively on EmoHeVRDB-SI by almost 5 pp. This suggests that the underlying expression representations in AffectNet and EmoHeVRDB-SI do not differ fundamentally. Instead, the more diverse samples from AffectNet-7-occl can

TABLE 7.4: Cross-dataset evaluations; adapted from (Ortmann, Q. Wang et al. 2024, Table V), © 2024 IEEE.

Trained	**Evaluated**	**Accuracy (%)**			
		Train.	**Val.**	**Test**	**Avg.**
AffectNet-7	KDEF-SHR	80.68	83.38	83.67	81.82
AffectNet-7	KDEF-SHR-occl	52.68	59.08	58.16	55.06
AffectNet-7-occl	KDEF-SHR-occl	62.72	67.26	65.05	64.10
AffectNet-7	EmoHeVRDB-SI	–	33.38	37.57	35.45
AffectNet-7-occl	EmoHeVRDB-SI	–	32.60	36.90	34.73
AffectNet-7-occl & EmoHeVRDB-SI	EmoHeVRDB-SI	–	–	74.74	74.74

Abbreviations: Avg. = Average, Train. = Training, Val. = Validation.

enrich and regularise the learning process on naturally HMD-occluded samples. Thus, the primary reason for the very low cross-database accuracy appears to be the absence of naturally HMD-occluded faces in AffectNet-7-occl. Accordingly, the artificial occlusion procedure does not simulate HMDs realistically enough to make models robust against natural HMD occlusion on their own. These findings underline the importance of EmoHeVRDB for developing reliable FER models for real VR scenarios.

7.1.3 FEA-Based Baseline on EmoHeVRDB-SFEA

This section establishes the static baseline on EmoHeVRDB-SFEA (see Section 6.3.2) using its HMD-derived facial expression activations (FEAs). In this FER setting, each input sample is a single 63-dimensional FEA vector. The following subsections first describe the experimental setup and training procedure (7.1.3.1), and then report the resulting benchmark performance on EmoHeVRDB-SFEA (7.1.3.2).

7.1.3.1 Setup and Training

To establish the FEA-based baseline, several machine learning models were trained to classify EmoHeVRDB-SFEA's 63-dimensional vectors into one of the seven target

classes. The selected models covered distinct classification approaches to provide a reliable basis for the baseline evaluations. They comprised logistic regression as a linear baseline, support vector machines for margin-based classification, and random forests and gradient boosting as nonlinear tree-based alternatives, with random forests combining independently fitted trees and gradient boosting constructing an ensemble sequentially (Hastie et al. 2001). This selection enabled comparison of different well-established classification approaches on the same FEA representation under a common evaluation protocol. All four models were implemented using the scikit-learn machine learning library (version 1.4.2). For each model type, an automated grid search was performed on the training set, and the resulting candidates were selected based on validation accuracy. The search space included hyperparameters such as the regularisation strength and solver algorithm for logistic regression, the number of trees, maximum depth, and splitting criteria for the tree-based models, as well as optional class weighting to mitigate imbalance in the training set.

To complement these classical machine learning models, a multilayer perceptron (MLP) was implemented in TensorFlow (version 2.15) to learn non-linear feature representations from the FEA vectors. It was manually tuned on the validation set by varying the learning rate, the number and size of Dense layers, and the number and rate of Dropout layers. For all MLP experiments, a batch size of 32, the Adam optimiser, and the SparseCategoricalCrossentropy loss function with normalised inverse class frequencies as class weights were used. The best-performing MLP architecture, illustrated in Figure 7.3, consists sequentially of a Dense layer with 128 units and ReLU activation, a Dropout layer with a rate of 0.2, a Dense layer with 64 units and ReLU activation, a Dropout layer with a rate of 0.2, and a final Dense layer with 7 units and softmax activation.

After hyperparameter optimisation, the configuration with the highest validation accuracy for each model type was evaluated on the test set. The results are listed in Table 7.5 and discussed in the following section. The grid-search spaces, selected hyperparameters, and MLP configuration are detailed in Appendix B.1.2.

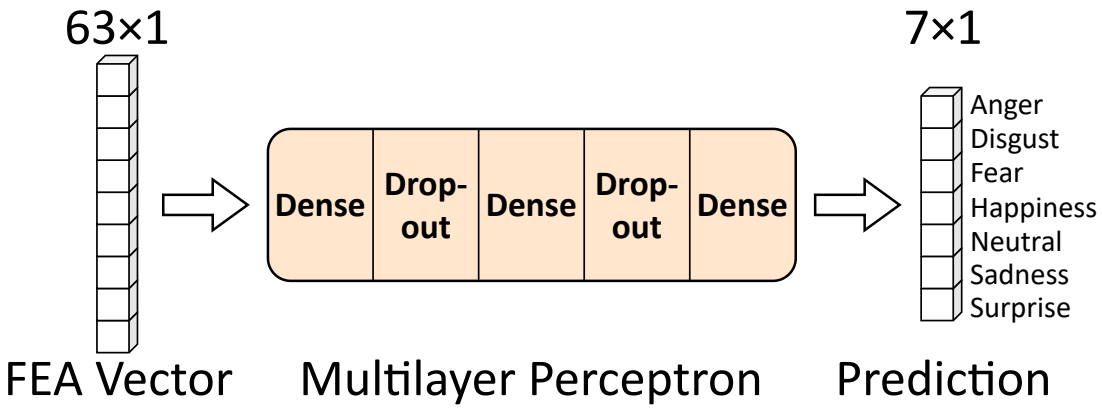


FIGURE 7.3: Architecture of the FEA-based baseline model.

TABLE 7.5: Performance of static FEA-based models on EmoHeVRDB-SFEA in %; adapted from (Ortmann, Q. Wang et al. 2025, Table I), © 2025 IEEE.

Model	**Accuracy**		**F1-Score**	**Min.**	**Min.**
Type	**Val.**	**Test**		**F1-Score**	**Recall**
LR	**82.34**	**73.02**	**72.56**	56.47	44.44
SVM	80.52	71.43	70.90	56.82	46.30
RF	82.08	71.16	69.78	46.58	31.48
GB	80.52	70.90	70.23	49.38	37.04
MLP	81.56	71.69	71.10	**57.14**	**48.15**
EffNet[a]	70.39	69.84	70.06	49.11	50.93

[a] Image-based EfficientNet-B0 baseline on EmoHeVRDB-SI from Section 7.1.1.
Abbreviations: EffNet = EfficientNet-B0, GB = Gradient Boosting, LR = Logistic Regression, Min. = Minimum, MLP = Multilayer Perceptron, RF = Random Forest, SVM = Support Vector Machine, Val. = Validation.

7.1.3.2 Results on EmoHeVRDB-SFEA

Across the evaluated model types, performance was broadly similar. Test accuracies ranged from 70.90% for gradient boosting to 73.02% for logistic regression, while the MLP achieved 71.69% (see Table 7.5). As in the image-based setting, performance differed substantially between expression classes. For all model types, happiness was recognised most reliably, with a recall of 98.15%. In contrast, anger was the least recognised class, with a maximum class recall of only 48.15% and the lowest F1-scores across all model types. Disgust and fear were consistently the next most challenging classes, with F1-scores ranging from 51.67% to 58.41% and from 59.26% to 68.42%, respectively. The most frequent error across all models was

misclassifying anger as disgust, which occurred for 22 to 29 of the 54 anger samples in the test set.

For the remainder of this thesis, the MLP serves as the FEA-based baseline because its learned intermediate feature representations can be combined with those of the image-based baseline model in one neural fusion architecture. This choice prioritises architectural compatibility and comparability over the slightly higher test accuracy of the logistic regression model. The MLP's class-wise performance metrics are reported in Table 7.6. The corresponding confusion matrix is illustrated in Figure 7.4. Besides happiness, surprise, sadness, and neutral were also recognised

TABLE 7.6: Performance metrics on EmoHeVRDB-SFEA in %.

Expression	**Precision**	**Recall**	**F1-Score**
Anger	70.27	48.15	57.14
Disgust	54.10	61.11	57.39
Fear	64.58	57.41	60.78
Happiness	79.10	98.15	87.60
Neutral	69.23	83.33	75.63
Sadness	92.31	66.67	77.42
Surprise	77.05	87.04	81.74
Average	72.38	71.69	71.10

comparatively well, with F1-scores of 81.74%, 77.42%, and 75.63%, respectively. By contrast, anger, disgust, and fear remained the most challenging classes, with F1-scores of 57.14%, 57.39%, and 60.78%, respectively.

Overall, these results demonstrate that headset-derived signals from the Meta Quest Pro are a viable modality for static FER under natural HMD occlusion. With 71.69% overall accuracy, the FEA-based baseline slightly exceeds the 69.84% accuracy of the image-based baseline model, while the class-wise recognition patterns are broadly similar.

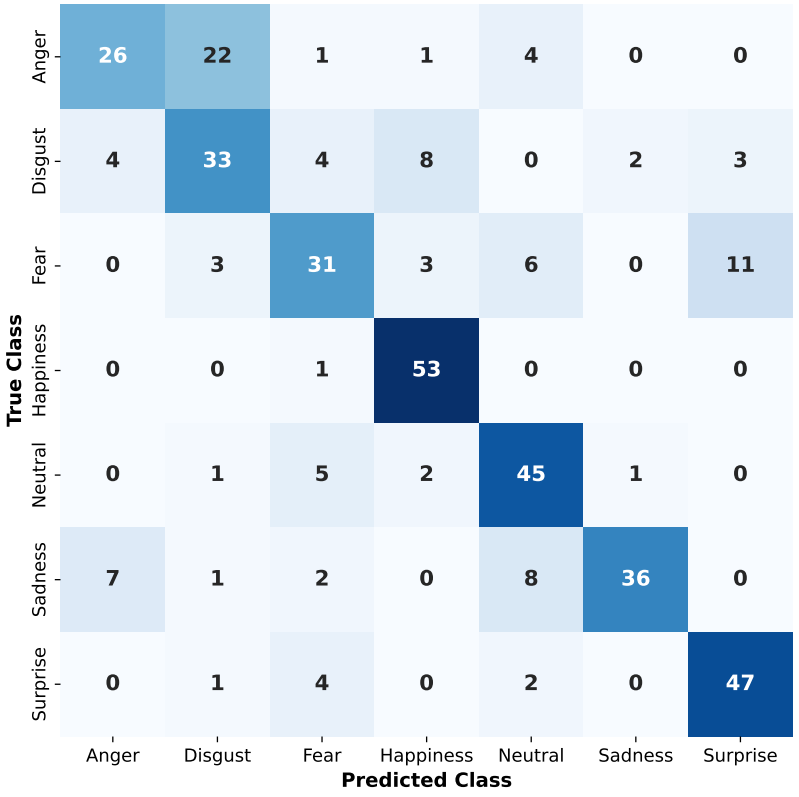


FIGURE 7.4: Confusion matrix on EmoHeVRDB-SFEA's test set.

7.1.4 Complementarity of Both Modalities

To assess the potential value of multimodal fusion in the static setting, this section examines the overlap between the predictions of the image-based EfficientNet-B0 from Section 7.1.1 and the FEA-based MLP from Section 7.1.3. If the two unimodal baselines are correct on different samples, this indicates complementary information and thus potential for multimodal static FER.

Figure 7.5 visualises the agreement between both models on the test data. For each of the 378 test samples in EmoHeVRDB-SFEA, EmoHeVRDB-SI provides one corresponding central-view image and one corresponding 45-degree side-view image. Consequently, the comparison comprises 756 paired cases in total. Each class bar is divided into two sub-bars: the left sub-bar represents comparisons with the central-view images, and the right sub-bar represents comparisons with the side-view images. Green denotes cases in which both models predict the correct class, orange cases in which only the FEA-based model is correct, blue cases in

which only the image-based model is correct, and red cases in which both models are incorrect.

At the class level, the FEA-based MLP performs noticeably better on happiness, neutral, and sadness, whereas the reverse pattern appears only for the central-view images of disgust and fear. Anger remains the most challenging class for both modalities. Across the 108 anger samples in the multimodal test cases, both models misclassify the same 34 samples, corresponding to a joint error rate of 31.48%. The next-highest shared error rate occurs for sadness with 17.59%.

Across all 756 paired test cases, both models predict the correct class in 414 cases and both fail in 100 cases. Of the remaining 242 cases, only the FEA-based MLP is correct in 128 cases, whereas only the image-based EfficientNet-B0 is correct in 114 cases. Thus, nearly one-third of all test cases exhibit non-overlapping errors. If an oracle could decide for each sample which unimodal prediction to trust, it would correctly classify 656 of the 756 cases, corresponding to an accuracy of 86.77%. This large gap between the unimodal baselines and the oracle result indicates substantial potential for fusing external-camera images and HMD-derived FEAs for multimodal FER under natural HMD occlusion.

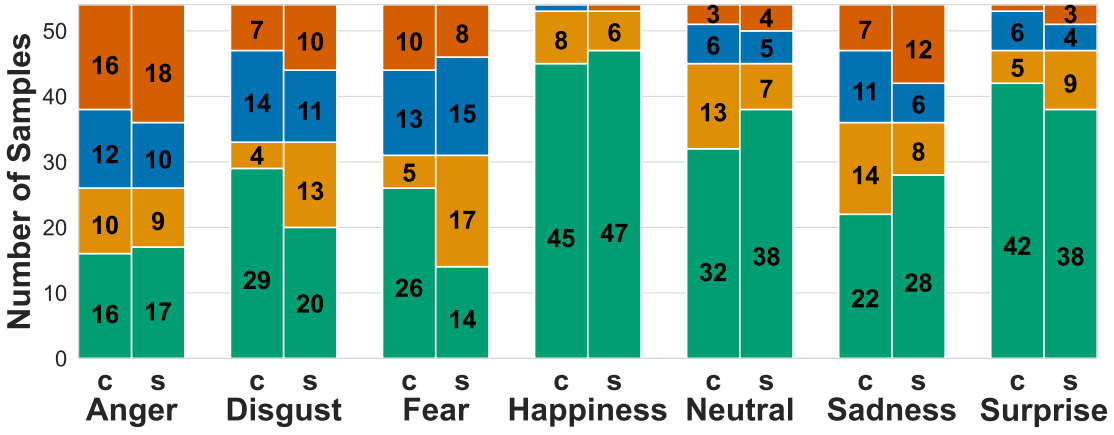


FIGURE 7.5: Agreement between the image-based and FEA-based baseline models; (c=central-view, s=side-view; colours from bottom to top: green=both correct, orange=only FEA model correct, blue=only image model correct, red=both incorrect); adapted from (Ortmann, Q. Wang et al. 2025, Figure 5), © 2025 IEEE.

7.1.5 Multimodal Static Facial Expression Recognition

This section establishes the multimodal static baseline on the combination of EmoHeVRDB-SI and EmoHeVRDB-SFEA, reusing the baseline image model from Section 7.1.1 and the baseline FEA model from Section 7.1.3. First, Subsection 7.1.5.1 explains the logic for constructing multimodal samples from the external-camera images in EmoHeVRDB-SI and the HMD-derived FEAs in EmoHeVRDB-SFEA. Then, Subsections 7.1.5.2 and 7.1.5.3 describe the training procedure and report the results for late fusion and intermediate fusion (see Figure 2.9), respectively. These two fusion levels were selected to compare decision-level integration of the unimodal predictions with feature-level integration that can exploit interactions between the learned representations (see Figure 2.9). Early fusion was not investigated because combining 224×224 RGB images and 63-dimensional FEA vectors would require a separate input-level architecture rather than reusing the established unimodal baselines. All fusion models were implemented in TensorFlow 2.15, using the Adam optimiser and the SparseCategoricalCrossentropy loss function, with balanced inverse class frequencies passed as class weights. Across all multimodal fusion experiments, the unimodal baseline models were reused—either as a whole or in part—and their weights were kept frozen to compare fusion approaches while limiting additional training to the fusion layers.

7.1.5.1 Multimodal Sample Construction

For each of the 1,727 FEA samples in EmoHeVRDB-SFEA, EmoHeVRDB-SI contains two simultaneously recorded images: one central-view image and one 45-degree side-view image. Accordingly, each FEA vector was paired with its two corresponding images, resulting in 3,454 multimodal samples. These multimodal samples preserve the same class distribution and the same participant-independent training, validation, and test split structure as the underlying static benchmark subsets.

7.1.5.2 Late Fusion

For late fusion, the image-based EfficientNet-B0 from Section 7.1.1 and the FEA-based MLP from Section 7.1.3 were treated as fixed models, and only their 7-dimensional softmax output vectors were combined. As a simple baseline, the two probability vectors were averaged, already increasing test accuracy to 76.46%. Subsequently, several learnable late-fusion variants were evaluated, including a weighted sum, concatenation followed by a Dense classification layer, and bilinear fusion. A custom cross-attention layer achieved the best result with 78.97% accuracy. It consisted of two parallel Dense layers with 7 units and softmax activation. Each Dense layer learned to weight one model's prediction vector based on the other model's prediction vector. The resulting weighted vectors were then summed and passed through a final softmax activation. Per-class precision, recall, and F1-scores, as well as F1-score deltas, are reported in Table 7.7.

TABLE 7.7: Late-fusion performance on EmoHeVRDB-SI and EmoHeVRDB-SFEA in % and differences (Δ) to unimodal baselines in pp; extended from (Ortmann, Q. Wang et al. 2025, Table II), © 2025 IEEE.

Expression	**Precision**	**Recall**	**F1-Score**	**Δ SI**[a]	**Δ SFEA**[b]
Anger	67.01	60.19	63.41	+14.30	+ 6.27
Disgust	60.77	73.15	66.39	+ 4.72	+ 9.00
Fear	81.19	75.93	78.47	+ 8.73	+17.69
Happiness	80.30	98.15	88.33	+11.15	+ 0.73
Neutral	86.36	87.96	87.16	+10.74	+11.53
Sadness	93.90	71.30	81.05	+10.89	+ 3.63
Surprise	89.42	86.11	87.74	+ 1.62	+ 6.00
Average	79.85	78.97	78.94	+ 8.88	+ 7.84

[a] Δ SI = F1-score difference to the image baseline on EmoHeVRDB-SI (Table 7.2).
[b] Δ SFEA = F1-score difference to the FEA baseline on EmoHeVRDB-SFEA (Table 7.6).

Compared with the image-based baseline, late fusion improved precision, recall, and F1-scores across all classes. In terms of F1-score, the largest gains were observed for anger (+14.30 pp), happiness (+11.15 pp), sadness (+10.89 pp), and neutral (+10.74 pp). These gains resulted from clear improvements in both precision and recall, most notably a 19.60 pp increase in precision for anger, while recall increased

by 12.97 pp for fear and 12.96 pp for neutral. In contrast, increases for surprise and disgust remained below 5 pp.

Relative to the FEA-based baseline, the strongest F1-score gains occurred for fear (+17.69 pp), neutral (+11.53 pp), and disgust (+9.00 pp). Here, the most pronounced changes were recall increases for fear (+18.52 pp), anger (+12.04 pp), and disgust (+12.04 pp), as well as precision increases of 17.13 pp for neutral, 16.61 pp for fear, and 12.37 pp for surprise. At the same time, happiness improved only marginally in F1-score (+0.73 pp), which is plausible given the already very strong unimodal FEA result for this class (87.60%). The only trade-offs were lower anger precision (−3.26 pp) and slightly lower surprise recall (−0.93 pp).

Overall, these class-wise changes are consistent with the complementarity analysis in Section 7.1.4. With 78.97% accuracy, the late-fusion model achieved gains of 9.13 pp relative to the image-based baseline and 7.28 pp relative to the FEA-based baseline, while still remaining 7.80 pp below the oracle accuracy of 86.77% derived in Section 7.1.4.

7.1.5.3 Intermediate Fusion

For intermediate fusion, the two unimodal networks were used as frozen feature extractors rather than as final classifiers. On the image side, the final GlobalAveragePooling layer of EfficientNet-B0 provided a 1,280-dimensional feature vector. On the FEA side, the first Dense layer of the MLP provided a 128-dimensional feature vector. In contrast to late fusion, the two vectors differed substantially in size, with the image representation being ten times larger than the FEA representation. During the experiments, projecting both feature vectors with additional Dense layers to the same dimensionality, between 128 and 512 units, improved performance and stabilised learning. Moreover, BatchNormalization (Ioffe and Szegedy 2015) layers were added to bring both feature vectors into similar value ranges, and the batch size was increased to 128 to obtain more stable batch statistics. To fuse both batch-normalised feature vectors of equal size, several fusion operations were

evaluated, including concatenation, bilinear fusion, and cross-attention, followed by additional Dense and intermediate Dropout layers.

The best intermediate-fusion model, illustrated in Figure 7.6, first projected both feature vectors to 512 dimensions and batch-normalised them. The normalised vectors were then weighted by a custom cross-attention layer, concatenated, and passed through a Dropout layer with a rate of 0.4 before the final Dense layer with 7 units and softmax activation. Further details of the selected architecture and training hyperparameters are provided in Appendix B.1.3.

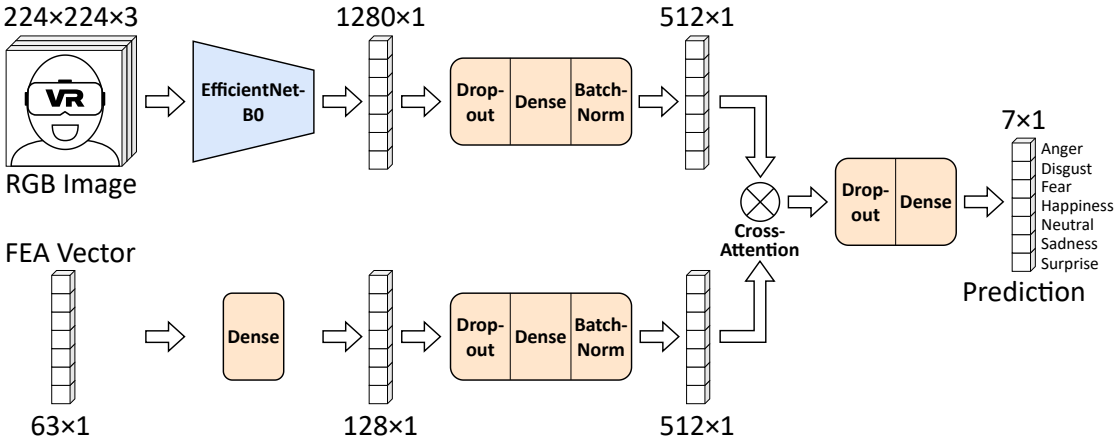


FIGURE 7.6: Architecture of the static intermediate fusion model.

As shown in Table 7.8, this configuration achieved the best overall multimodal result with 80.42% accuracy and an F1-score of 80.22%. Compared with late fusion, intermediate fusion yielded a modest improvement, increasing accuracy by 1.45 pp and the average F1-score by 1.28 pp. The intermediate-fusion model primarily increased precision for anger (+8.27 pp) and disgust (+5.61 pp), as well as recall for fear (+5.55 pp), neutral (+4.63 pp), and sadness (+4.63 pp). All remaining changes were increases or decreases of less than 5 pp. The clearest F1-score gains were observed for anger (+4.61 pp) and disgust (+2.36 pp). In contrast, the F1-score for surprise decreased slightly (−1.89 pp), due to a decrease in recall (−4.63 pp).

In terms of error patterns, the most frequent misclassification was anger predicted as disgust, occurring 29 times (see Figure 7.7). The next most common confusions were surprise predicted as fear (20 cases), disgust predicted as happiness (15 cases), and sadness predicted as anger (13 cases). Thus, the main residual errors remained

TABLE 7.8: Intermediate-fusion performance on EmoHeVRDB-SI and EmoHeVRDB-SFEA in % and differences (Δ) to unimodal baselines and late fusion in pp; extended from (Ortmann, Q. Wang et al. 2025, Table III), © 2025 IEEE.

Expression	**Precision**	**Recall**	**F1-Score**	**Δ SI**[a]	**Δ SFEA**[b]	**Δ LF**[c]
Anger	75.28	62.04	68.02	+18.91	+10.88	+ 4.61
Disgust	66.38	71.30	68.75	+ 7.08	+11.36	+ 2.36
Fear	76.52	81.48	78.92	+ 9.18	+18.14	+ 0.45
Happiness	80.92	98.15	88.70	+11.52	+ 1.10	+ 0.37
Neutral	84.75	92.59	88.50	+12.08	+12.87	+ 1.34
Sadness	91.11	75.93	82.83	+12.67	+ 5.41	+ 1.78
Surprise	90.72	81.48	85.85	− 0.27	+ 4.11	− 1.89
Average	80.81	80.42	80.22	+10.16	+ 9.12	+ 1.28

[a] Δ SI = F1-score difference to the image baseline on EmoHeVRDB-SI (Table 7.2).
[b] Δ SFEA = F1-score difference to the FEA baseline on EmoHeVRDB-SFEA (Table 7.6).
[c] Δ LF = F1-score difference to the best late-fusion model (Table 7.7).

concentrated in the same difficult class pairs already identified in the unimodal and late-fusion settings, particularly anger–disgust and fear–surprise, although at a reduced level overall.

Overall, these results indicate that intermediate fusion can further improve static FER beyond decision-level fusion by leveraging more complex feature-level interdependencies. Although the gain over the late-fusion model is moderate, the achieved accuracy of 80.42% substantially exceeds the unimodal baselines of 69.84% for the image-based model and 71.69% for the FEA-based model, thereby demonstrating the benefit of multimodal fusion.

7.2 Dynamic Facial Expression Recognition

This section reports the dynamic FER experiments using EmoHeVRDB as the benchmark database to implement and evaluate the dynamic FER baseline models required for Artefact 1 in Section 3.2.1. All dynamic experiments were executed on an Ubuntu 24.04 machine, using the Nvidia TensorFlow NGC container *25.02-tf2-py3*, which bundles TensorFlow 2.17 with CUDA 12.8 and cuDNN 9.7. The

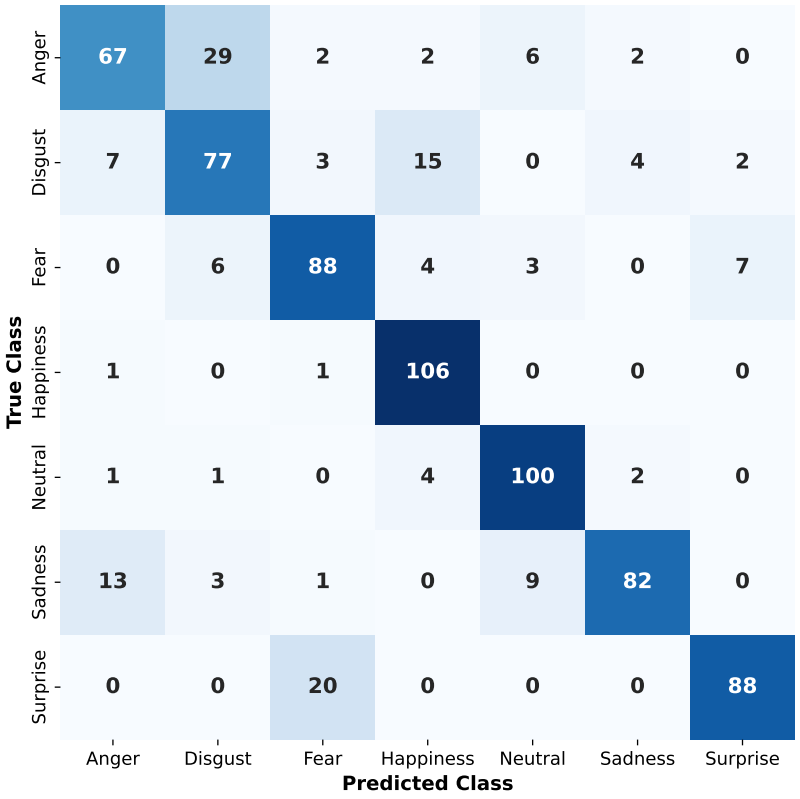


FIGURE 7.7: Confusion matrix for static intermediate fusion.

computer had an Intel Core Ultra 9 285K CPU, 128 GB of DDR5 RAM, and an Nvidia GeForce RTX 5090 GPU. In line with the static baselines, the dynamic experiments relied on established model architectures and kept the hyperparameter space small. All models were implemented in TensorFlow 2.17 with the Keras interface and trained using the Adam optimiser and the SparseCategoricalCrossentropy loss function with normalised inverse class frequencies as class weights. Detailed configurations and hyperparameters for the dynamic baseline models are provided in Appendix B.2.

First, Subsection 7.2.1 presents the image-sequence-based baseline on EmoHeVRDB-DI. Then, Subsection 7.2.2 describes the FEA-sequence-based baseline on EmoHeVRDB-DFEA. Building on these unimodal dynamic baselines, Subsection 7.2.3 analyses the complementarity between the baseline models' predictions to assess the potential of combining both modalities. Finally, Subsection 7.2.4 reports the multimodal dynamic FER experiments with synchronised image and FEA sequences.

7.2.1 Image-Sequence-Based Baseline on EmoHeVRDB-DI

This section establishes the dynamic image-sequence-based baseline on EmoHeVRDB-DI (see Section 6.3.3). In this setting, each sample is a sequence of 30 RGB frames at 224×224 pixels. Together, these frames capture about one second of facial movement during a single facial-expression reenactment under natural HMD occlusion. The following subsections first present the selected model architecture (7.2.1.1), then describe the training procedure (7.2.1.2), and finally report the resulting benchmark performance on EmoHeVRDB-DI (7.2.1.3).

7.2.1.1 Model Architecture

For dynamic image-sequence-based FER, this thesis adopts a hybrid CNN–LSTM architecture, illustrated in Figure 7.8. The model first extracts spatial features from each frame, then models their temporal progression across the sequence, and finally classifies the facial expression. This design follows a well-established pattern in dynamic FER, where convolutional layers encode per-frame appearance and recurrent layers capture temporal dependencies across onset, apex, and offset phases (see Section 2.4.2). Compared with alternative architectures such as 3D CNNs and video Transformers, which jointly learn spatial and temporal features, the CNN–LSTM approach separates spatial feature extraction from temporal modelling. This allows the direct reuse of an ImageNet-pretrained 2D CNN for extracting per-frame features, which is particularly useful given the limited size of

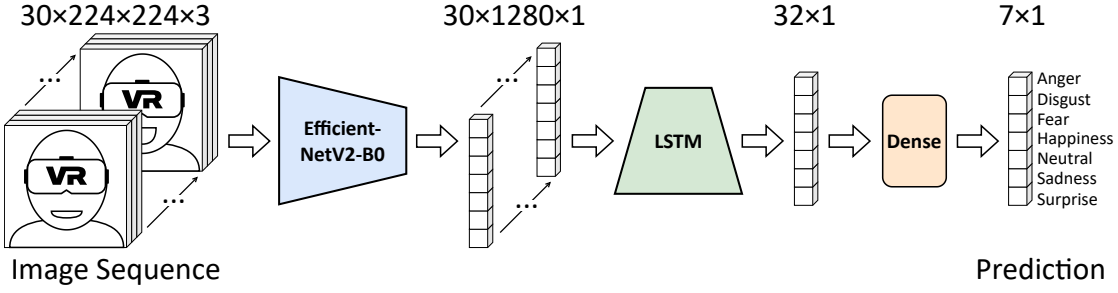


FIGURE 7.8: Architecture of the image-sequence-based baseline model; Dropout layers are omitted for better readability.

EmoHeVRDB-DI, and maintains a similar spatial feature-extraction strategy to the static image-based baseline.

In contrast to the static image-based baseline in Section 7.1.1, which processes individual images, the dynamic setting operates on sequences of 30 RGB frames at 224×224 pixels. Consequently, the computational demand is substantially higher, as the convolutional backbone must process roughly 30 times more data per sample. Therefore, EfficientNetV2-B0 was selected as the CNN backbone instead of EfficientNet-B0. EfficientNetV2 was explicitly introduced as a family of models optimised for faster training and improved parameter efficiency (M. Tan and Le 2021). A throughput benchmark with EfficientNet-B0, EfficientNetV2-B0, and sequences from EmoHeVRDB-DI on the exact setup described at the beginning of Section 7.2 supported this efficiency-oriented choice. Using a batch size of 16, EfficientNetV2-B0 processed a mean of 93.3 rather than 82.8 sequences per second when the backbone was frozen (+12.7%) and 38.0 rather than 25.9 sequences per second when the backbone was unfrozen during fine-tuning (+46.7%). At the same time, the predictive performance of EfficientNetV2-B0 and EfficientNet-B0 is broadly similar. The original publications report 78.7% top-1 accuracy for EfficientNetV2-B0 (M. Tan and Le 2021) and 77.1% for EfficientNet-B0 (M. Tan and Le 2020) on ImageNet. Studies from other domains report mixed downstream performance, with some tasks slightly favouring EfficientNet-B0 and others EfficientNetV2-B0 (Giedra et al. 2025; Sivasubramanian et al. 2025; Venancio et al. 2025; Voon et al. 2022).

The original classification head of EfficientNetV2-B0 was removed so that the CNN backbone produced a 1,280-dimensional feature vector for each frame. Applied in a time-distributed manner, the CNN transformed each 30-frame input sequence into a sequence of 30 feature vectors. A SpatialDropout1D layer (Tompson et al. 2015) then regularised these features before a unidirectional LSTM captured their temporal progression. Finally, an additional Dropout layer preceded the final Dense layer with 7 softmax-activated units, which produced the prediction vector.

7.2.1.2 Training

As in the static image-based baseline, training followed a transfer-learning strategy. EfficientNetV2-B0 was initialised with ImageNet-pretrained weights, and all backbone layers were frozen initially. In this first stage, only the LSTM and classification layers were trained, using a learning rate of 10^{-3} and a batch size of 32. Subsequently, progressively larger parts of the backbone were unfrozen and fine-tuned on EmoHeVRDB-DI. During fine-tuning, the learning rate was reduced to 5×10^{-4} and subsequently 10^{-4}, and the batch size was reduced from 32 to 16.

To reduce overfitting, both spatial and temporal data augmentation were applied. Spatial augmentation comprised random cropping, scaling, rotation, and changes in brightness and contrast, applied consistently across all frames within a sequence. Temporal augmentation randomly shifted frames within a sequence and, with small probability, replaced individual frames with their predecessor or with the per-sequence average frame. In addition, a SpatialDropout layer between the EfficientNetV2-B0 and the LSTM, as well as a Dropout layer between the LSTM and the final Dense layer, provided further regularisation. Hyperparameter tuning on the validation set focused on the dropout rates, the number of LSTM units, and the combination of unfrozen layers, training epochs, and learning-rate schedules. The augmentation settings, model architecture, and five-phase training schedule for the selected model are detailed in Appendix B.2.1.

7.2.1.3 Results

The best-performing configuration used a SpatialDropout1D rate of 0.15, followed by an LSTM layer with 32 units and a Dropout rate of 0.3 before the final Dense layer. This baseline model achieved an accuracy of 72.75% on the 756 image sequences of EmoHeVRDB-DI's balanced test set. It thereby outperformed the static image-based baseline from Section 7.1.1 by 2.91 pp in terms of accuracy. The class-wise precision, recall, and F1-scores are reported in Table 7.9, and the corresponding confusion matrix is shown in Figure 7.9.

TABLE 7.9: Performance metrics on EmoHeVRDB-DI in %.

Expression	Precision	Recall	F1-Score
Anger	54.95	46.30	50.25
Disgust	51.39	68.52	58.73
Fear	73.20	65.74	69.27
Happiness	75.00	91.67	82.50
Neutral	87.10	75.00	80.60
Sadness	84.62	81.48	83.02
Surprise	91.58	80.56	85.71
Average	73.97	72.75	72.87

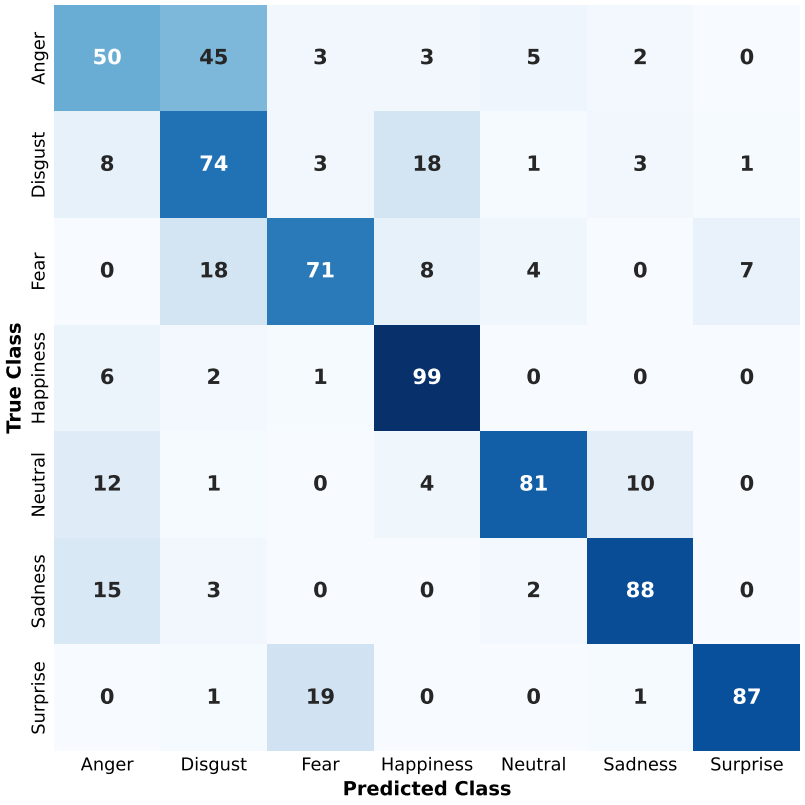


FIGURE 7.9: Confusion matrix on EmoHeVRDB-DI's test set.

Performance differed substantially between classes. Surprise, sadness, happiness, and neutral were recognised most reliably, with F1-scores of 85.71%, 83.02%, 82.50%, and 80.60%, respectively. In contrast, anger, disgust, and fear remained the most challenging classes, with F1-scores of 50.25%, 58.73%, and 69.27%, respectively. Most notably, the model classified only 50 of 108 anger samples in the test set correctly, resulting in a recall of 46.30%. Precision was also lowest for anger and disgust, at 54.95% and 51.39%, respectively.

The dominant confusion patterns were classifying anger as disgust (45 cases), surprise as fear (19 cases), disgust as happiness (18 cases), and fear as disgust (18 cases). Relative to the static image-based baseline, the largest changes were an increase in anger being classified as disgust (+13 cases) and a decrease in sadness being classified as anger (−16 cases).

Overall, these results show that image-sequence-based FER is feasible despite natural HMD occlusion, and that temporal modelling provides a modest improvement over the static image-based baseline. At the same time, substantial class-specific confusions remain, particularly for anger and disgust.

7.2.2 FEA-Sequence-Based Baseline on EmoHeVRDB-DFEA

This section establishes the dynamic baseline on EmoHeVRDB-DFEA (see Section 6.3.4) using its HMD-derived FEA sequences. In this setting, each sample is a 30-element sequence of 63-dimensional FEA vectors. Together, these vectors capture about one second of facial movement during a single facial-expression reenactment. The following subsections first present the selected model architecture (7.2.2.1), then describe the training procedure (7.2.2.2), and finally report the resulting benchmark performance on EmoHeVRDB-DFEA (7.2.2.3).

7.2.2.1 Model Architecture

For dynamic FER based on FEA sequences, this thesis adopts an LSTM-based architecture as illustrated in Figure 7.10. Unlike the image sequences, the FEA vectors already provide compact facial-movement representations, eliminating the need for a separate spatial feature extractor. The LSTM was selected as an established approach for modelling temporal dependencies, consistent with the temporal modelling component of the image-sequence-based baseline in Section 7.2.1. Alternative sequence models, such as GRUs, temporal CNNs, or Transformers,

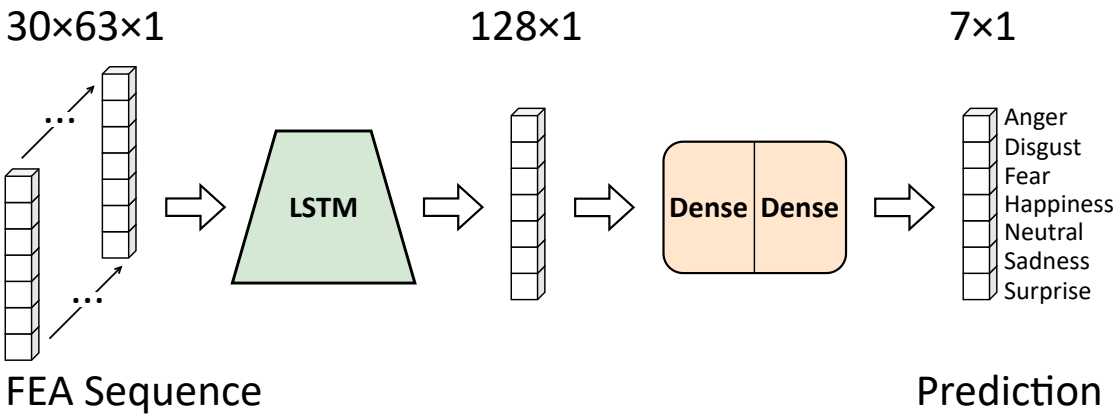


FIGURE 7.10: Architecture of the FEA-sequence-based baseline model; Dropout layers are omitted for better readability.

were not evaluated, as the objective was to establish a comparable baseline rather than exhaustively compare temporal architectures.

The LSTM-based model processes each sample as a sequence of 30 FEA vectors along the temporal axis using a unidirectional LSTM, followed by a Dense classification head. A SpatialDropout1D layer is applied to the input sequence before the LSTM, while further Dropout layers regularise the classification head. The final Dense layer has 7 softmax-activated units and produces the prediction vector for the seven expression categories used throughout EmoHeVRDB.

7.2.2.2 Training

Training focused on establishing a robust baseline while keeping the hyperparameter space compact. Hyperparameter tuning on the validation set varied the learning rate, the number of LSTM units, the number and size of Dense layers in the classification head, and the dropout rates. Finally, the configuration with the highest validation accuracy was selected and evaluated on the test set. Its architecture and training hyperparameters are detailed in Appendix B.2.2.

7.2.2.3 Results

The best-performing configuration used an initial learning rate of 5×10^{-4} and the following architecture: a SpatialDropout layer with a rate of 0.25, followed

by an LSTM layer with 128 units, a Dropout layer with a rate of 0.2, a Dense layer with 128 units, another Dropout layer with a rate of 0.2, and the final Dense classification layer. This baseline model achieved an accuracy of 78.31% on the 378 FEA sequences of EmoHeVRDB-DFEA's balanced test set. It thereby outperformed the static FEA baseline from Section 7.1.3 by 6.61 pp in terms of accuracy. The class-wise precision, recall, and F1-scores are reported in Table 7.10, and the corresponding confusion matrix is shown in Figure 7.11.

TABLE 7.10: Performance metrics on EmoHeVRDB-DFEA in %.

Expression	**Precision**	**Recall**	**F1-Score**
Anger	93.33	51.85	66.67
Disgust	48.39	55.56	51.72
Fear	76.74	61.11	68.04
Happiness	77.14	100.00	87.10
Neutral	83.61	94.44	88.70
Sadness	96.00	88.89	92.31
Surprise	83.87	96.30	89.66
Average	79.87	78.31	77.74

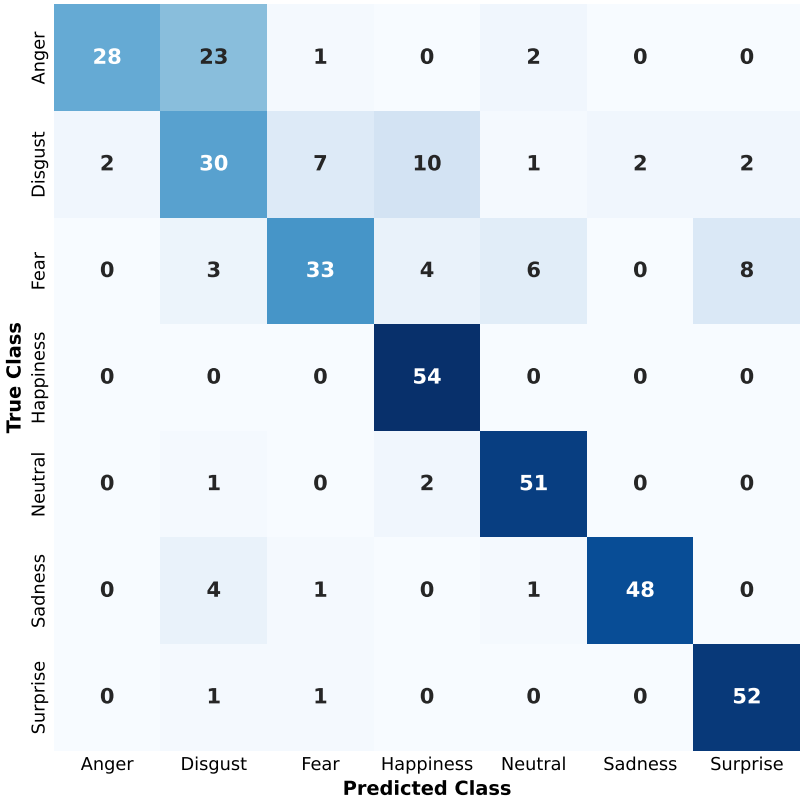


FIGURE 7.11: Confusion matrix on EmoHeVRDB-DFEA's test set.

Performance again differed substantially between classes. Sadness, surprise, neutral, and happiness were recognised most reliably, with F1-scores of 92.31%, 89.66%, 88.70%, and 87.10%, respectively. The baseline recognised nearly all samples of happiness, surprise, and neutral, as reflected in recall values of 100.00%, 96.30%, and 94.44%, respectively. In contrast, disgust, anger, and fear remained the most challenging classes, with F1-scores of 51.72%, 66.67%, and 68.04%, respectively. Precision was lowest for disgust at 48.39%, whereas recall was lowest for anger at 51.85%.

The dominant confusion patterns were anger being classified as disgust (23 cases), disgust as happiness (10 cases), and fear as surprise (8 cases). Compared with the static FEA-based baseline, the largest F1-score gains were observed for sadness (+14.89 pp), neutral (+13.07 pp), anger (+9.53 pp), and surprise (+7.92 pp), whereas disgust deteriorated by 5.67 pp. Overall, these results show that dynamic FER based on HMD-derived FEA sequences is clearly feasible and benefits substantially from temporal modelling. However, the remaining errors indicate that anger and disgust remain difficult to recognise reliably even when temporal information is available.

7.2.3 Complementarity of Both Sequence Modalities

To assess the potential value of multimodal fusion in the dynamic setting, this section examines the overlap between the predictions of the image-sequence baseline from Section 7.2.1 and the FEA-sequence baseline from Section 7.2.2. If the two unimodal baselines are correct on different samples, this indicates complementary information and thus potential for multimodal dynamic FER.

Figure 7.12 visualises the agreement between both models on the test data. For each of the 378 test samples in EmoHeVRDB-DFEA, EmoHeVRDB-DI provides one corresponding central-view sequence and one corresponding 45-degree side-view sequence. Consequently, the comparison comprises 756 paired cases in total. As in Figure 7.5, green denotes cases in which both models predict the correct class,

orange cases in which only the FEA-sequence model is correct, blue cases in which only the image-sequence model is correct, and red cases in which both models are incorrect.

At the class level, the image-sequence baseline performs better on disgust and fear, whereas the FEA-sequence baseline is stronger on happiness, neutral, sadness, and surprise. Anger remains the most challenging class for both modalities. Across the 108 anger test cases, both models misclassify the same 42 samples, corresponding to a joint error rate of 38.89%. Disgust and fear also show substantial shared difficulty, with 25 and 18 jointly misclassified samples, respectively. Compared to the static complementarity analysis in Section 7.1.4, the joint error rates for anger and disgust are both higher by 7.41 pp, whereas the joint error rate for fear is unchanged. In contrast, all happiness and neutral samples are correctly classified by at least one of the two baselines, and only three sadness samples and two surprise samples are misclassified by both.

Across all 756 paired test cases, both models predict the correct class in 476 cases and both fail in 90 cases. Of the remaining 190 cases, only the FEA-sequence baseline is correct in 116 cases, whereas only the image-sequence baseline is correct in 74 cases. Thus, about one quarter of all test cases exhibit non-overlapping errors. If an oracle could decide for each sample which unimodal prediction to

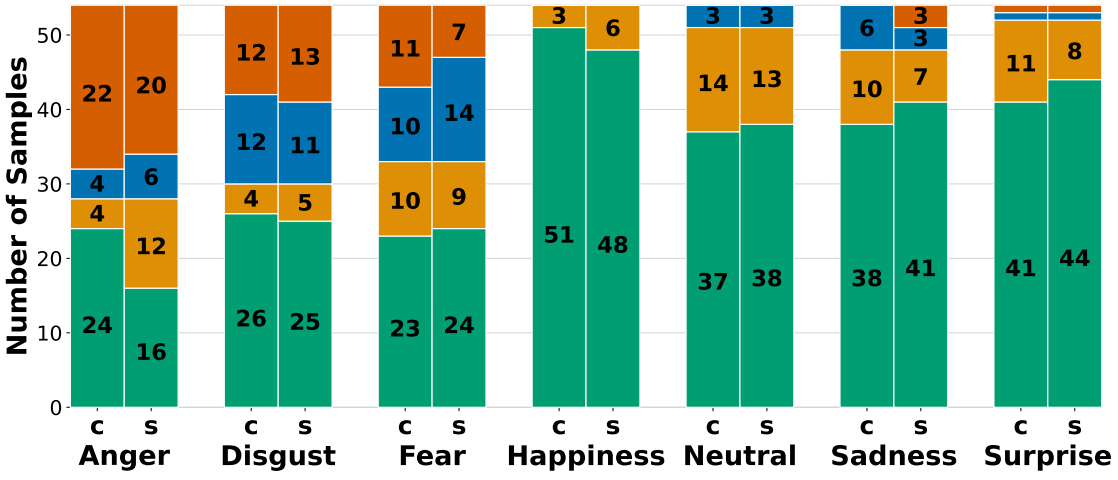


FIGURE 7.12: Agreement between the image-sequence-based and FEA-sequence-based baseline models; (c=central-view, s=side-view; colours from bottom to top: green=both correct, orange=only FEA-sequence model correct, blue=only image-sequence model correct, red=both incorrect).

trust, it would correctly classify 666 of the 756 cases, corresponding to an accuracy of 88.10%. This result remains well above the two unimodal dynamic baselines at 72.75% and 78.31%, indicating clear potential for multimodal fusion. At the same time, the oracle accuracy is only 1.32 pp higher than the corresponding static oracle accuracy of 86.77%. Together with the smaller number of non-overlapping errors, this suggests that temporal modelling already resolves part of the variability that multimodal fusion can later exploit.

7.2.4 Multimodal Dynamic Facial Expression Recognition

This section establishes the multimodal dynamic baseline on the combination of EmoHeVRDB-DI and EmoHeVRDB-DFEA, reusing the baseline image-sequence model from Section 7.2.1 and the baseline FEA-sequence model from Section 7.2.2. First, Subsection 7.2.4.1 explains the logic for constructing multimodal samples from the external-camera image sequences in EmoHeVRDB-DI and the HMD-derived FEA sequences in EmoHeVRDB-DFEA. Then, Subsections 7.2.4.2 and 7.2.4.3 describe the training procedure and report the results for late fusion and intermediate fusion (see Figure 2.9), respectively. Following the static multimodal experiments in Section 7.1.5, late and intermediate fusion were used to compare the combination of final predictions with the integration of learned feature representations. Accordingly, all multimodal fusion experiments reused the unimodal dynamic baseline models—either as a whole or in part—and kept their weights frozen.

7.2.4.1 Multimodal Sample Construction

For each of the 1,727 FEA sequences in EmoHeVRDB-DFEA, EmoHeVRDB-DI contains two simultaneously recorded image sequences: one central-view sequence and one 45-degree side-view sequence. Accordingly, each FEA sequence was paired with its two corresponding image sequences, resulting in 3,454 multimodal samples. Therefore, each multimodal sample comprises one sequence of 30 RGB frames

and one synchronised sequence of 30 FEA vectors. These multimodal samples preserve the same class distribution and the same participant-independent training, validation, and test split structure as the underlying dynamic benchmark subsets.

7.2.4.2 Late Fusion

For late fusion, the image-sequence baseline from Section 7.2.1 and the FEA-sequence baseline from Section 7.2.2 were treated as fixed models, and only their 7-dimensional softmax output vectors were combined. As a simple baseline, the two probability vectors were averaged, increasing test accuracy by 0.13 pp to 78.44% compared to the FEA-sequence baseline. Subsequently, several learnable late-fusion variants were evaluated, including a weighted sum, concatenation followed by a Dense classification layer, and bilinear fusion. A custom cross-attention layer achieved the best result with 80.56% accuracy. It consisted of two parallel Dense layers with 7 units and softmax activation. Each Dense layer learned to weight one model's prediction vector based on the other model's prediction vector. The resulting weighted vectors were then summed and passed through a final softmax activation. Per-class precision, recall, and F1-scores are reported in Table 7.11.

Compared with the image-sequence baseline, late fusion improved the F1-score for all classes except disgust, whose performance remained effectively unchanged

TABLE 7.11: Late-fusion performance on EmoHeVRDB-DI and EmoHeVRDB-DFEA in % and differences (Δ) to unimodal baselines in pp.

Expression	**Precision**	**Recall**	**F1-Score**	**Δ DI**[a]	**Δ DFEA**[b]
Anger	81.43	52.78	64.04	+ 13.79	− 2.63
Disgust	54.33	63.89	58.72	− 0.01	+ 7.00
Fear	81.19	75.93	78.47	+ 9.20	+10.43
Happiness	74.48	100.00	85.38	+ 2.88	− 1.72
Neutral	93.64	95.37	94.50	+13.90	+ 5.80
Sadness	96.94	87.96	92.23	+ 9.21	− 0.08
Surprise	90.48	87.96	89.20	+ 3.49	− 0.46
Average	81.78	80.56	80.36	+ 7.49	+ 2.62

[a] Δ DI = F1-score difference to the image-sequence baseline (Table 7.9).
[b] Δ DFEA = F1-score difference to the FEA-sequence baseline (Table 7.10).

(−0.01 pp). The largest gains were observed for neutral (+13.90 pp), anger (+13.79 pp), sadness (+9.21 pp), and fear (+9.20 pp), which resulted from clear improvements in both precision and recall, most notably a 26.48 pp increase in precision for anger and a 20.37 pp increase in recall for neutral. In contrast, happiness and surprise improved more modestly in F1-score, by 2.88 pp and 3.49 pp, respectively. For disgust, a 2.94 pp increase in precision was offset by a 4.63 pp decrease in recall.

Relative to the FEA-sequence baseline, the strongest F1-score gains occurred for fear (+10.43 pp), disgust (+7.00 pp), and neutral (+5.80 pp). Here, the most pronounced changes were a 14.82 pp increase in recall for fear, a 10.03 pp increase in precision for neutral, and simultaneous gains of 5.94 pp in precision and 8.33 pp in recall for disgust. At the same time, late fusion slightly underperformed the FEA-sequence baseline for anger, happiness, surprise, and sadness with F1-score decreases of 2.63 pp, 1.72 pp, 0.46 pp, and 0.08 pp, respectively. The clearest trade-off was for surprise, where precision increased by 6.61 pp, but recall decreased by 8.34 pp. Likewise, anger recall improved slightly by 0.93 pp, but precision dropped by 11.90 pp.

Overall, these class-wise changes are broadly consistent with the complementarity analysis in Section 7.2.3. With 80.56% accuracy, the late-fusion model achieved gains of 7.81 pp relative to the image-sequence baseline, but only 2.25 pp relative to the FEA-sequence baseline, while still remaining 7.54 pp below the oracle accuracy of 88.10%. Anger and disgust remain the most challenging classes, whereas fear improved substantially relative to both unimodal baselines. Thus, the late-fusion results confirm that combining both sequence modalities is also beneficial in the dynamic setting, but that the performance increase is smaller than in the static setting.

7.2.4.3 Intermediate Fusion

For intermediate fusion, the two unimodal baseline networks were used as frozen feature extractors rather than as final classifiers. The LSTM layer of the image-sequence baseline provided a 32-dimensional feature vector, whereas the Dense layer following the LSTM of the FEA-sequence baseline provided a 128-dimensional feature vector. In contrast to late fusion, the two vectors differed substantially in size. Thus, the experiments used additional Dense layers to project both feature vectors to the same dimensionality. Moreover, LayerNormalization layers (J. L. Ba et al. 2016) were added to bring both feature vectors into similar value ranges. To fuse both normalised feature vectors of equal size, several fusion operations were evaluated, including concatenation, bilinear fusion, and cross-attention, followed by additional Dense and intermediate Dropout layers.

The best intermediate-fusion model, illustrated in Figure 7.13, first applied Dropout with rates of 0.15 and 0.25 to the image and FEA feature vectors, respectively. Both vectors were projected to 16 dimensions via Dense layers with ReLU activation and then layer-normalised. A custom cross-attention layer computed a sigmoid gate for each modality from the other modality's normalised features and applied these gates element-wise. The weighted feature vectors were summed and passed through a sigmoid activation. After fusion, a Dropout layer with a rate of 0.2 preceded the final Dense layer with 7 units and softmax activation. Only the newly introduced fusion head was trained. The checkpoint with the highest validation accuracy was selected for test evaluation. The model configuration and training hyperparameters are detailed in Appendix B.2.3.

As shown in Table 7.12, this configuration achieved the best overall multimodal result in the dynamic setting, with 81.61% accuracy and an F1-score of 81.49%. Compared with late fusion, intermediate fusion increased accuracy by 1.06 pp and the average F1-score by 1.13 pp. F1-scores improved for five of the seven classes, whereas those for neutral and surprise decreased by 0.05 pp and 2.20 pp, respectively. The largest precision gains occurred for anger (+5.53 pp) and surprise (+4.09 pp), while recall increased most for fear (+5.56 pp) and sadness (+4.63 pp).

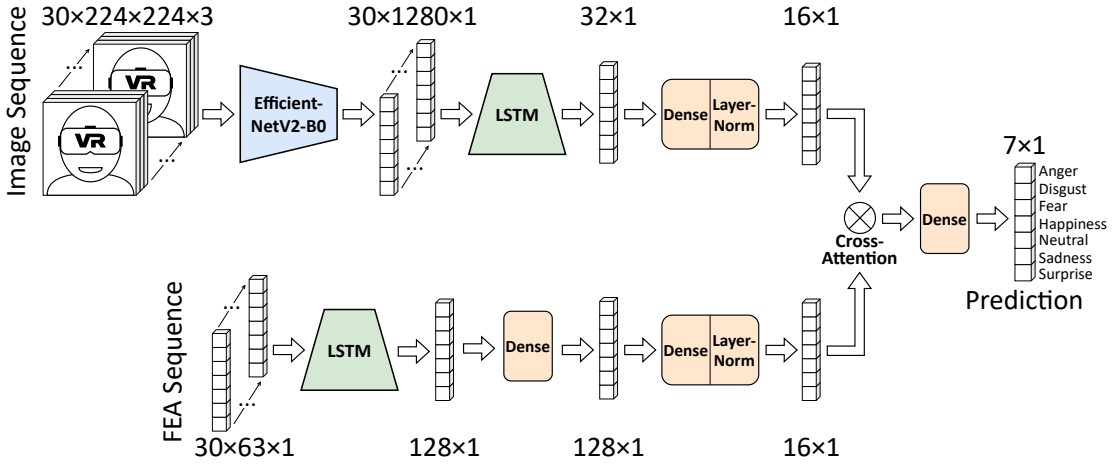


FIGURE 7.13: Architecture of the dynamic intermediate fusion model; Dropout layers are omitted for better readability.

TABLE 7.12: Intermediate-fusion performance on EmoHeVRDB-DI and EmoHeVRDB-DFEA in % and differences (Δ) to unimodal baselines and late fusion in pp.

Expression	**Precision**	**Recall**	**F1-Score**	**Δ DI**[a]	**Δ DFEA**[b]	**Δ LF**[c]
Anger	86.96	55.56	67.80	+17.55	+ 1.13	+ 3.75
Disgust	56.69	66.67	61.28	+ 2.55	+ 9.55	+ 2.55
Fear	77.88	81.48	79.64	+10.37	+11.60	+ 1.17
Happiness	76.06	100.00	86.40	+ 3.90	− 0.70	+ 1.02
Neutral	94.44	94.44	94.44	+13.85	+ 5.75	− 0.05
Sadness	95.24	92.59	93.90	+10.88	+ 1.59	+ 1.66
Surprise	94.57	80.56	87.00	+ 1.29	− 2.66	− 2.20
Average	83.12	81.61	81.49	+ 8.62	+ 3.75	+ 1.13

[a] Δ DI = F1-score difference to the image-sequence baseline (Table 7.9).
[b] Δ DFEA = F1-score difference to the FEA-sequence baseline (Table 7.10).
[c] Δ LF = F1-score difference to the best late-fusion model (Table 7.11).

For surprise, however, recall decreased by 7.41 pp. Anger and disgust remained the most challenging classes, with F1-scores of 67.80% and 61.28%, respectively.

In terms of error patterns, the most frequent misclassification remained anger predicted as disgust, with 44 cases compared with 45 cases in the late-fusion model (see Figure 7.14). The next most common confusions were disgust predicted as happiness and surprise predicted as fear (20 cases each), followed by fear predicted as happiness (10 cases). Thus, the main residual errors remained concentrated in a small number of difficult class pairs.

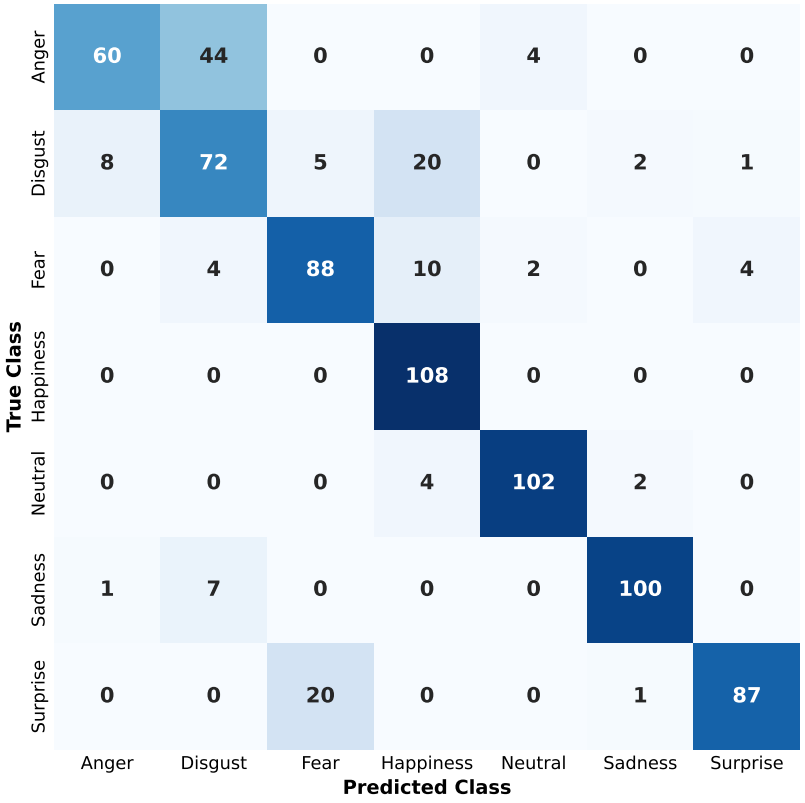


FIGURE 7.14: Confusion matrix for dynamic intermediate fusion.

Overall, these results indicate that intermediate fusion can slightly improve dynamic FER beyond decision-level fusion by exploiting feature-level interactions between the two sequence modalities. The achieved accuracy of 81.61% is a small gain over the late-fusion result, but clearly exceeds both unimodal baselines of 72.75% for the image-sequence model and 78.31% for the FEA-sequence model. At the same time, the remaining gap of 6.48 pp to the oracle accuracy of 88.10% derived in Section 7.2.3 indicates that some complementary potential between both modalities remains unexploited.

7.3 Discussion

This section synthesises the benchmark evidence reported in Section 7.1 and Section 7.2 and interprets it across the modelling settings and sensing modalities. As required for the evaluation of Artefact 1 in Section 3.2.1, this section embeds

the baseline results in broader contexts and derives practical trade-offs and design implications for FER under HMD occlusion.

Table 7.13 summarises the baseline accuracies for the six evaluated FER conditions. Across these six conditions, three general patterns emerged. First, the dynamic baselines outperformed their static counterparts across modalities, indicating that temporal modelling was beneficial overall. Whereas this benefit was modest in the image-based and multimodal conditions, it was substantial in the FEA-based condition, with an increase of 6.61 pp in accuracy. Second, the HMD-derived FEA modality consistently outperformed the external-camera image modality within both the static and dynamic settings. Third, multimodal fusion achieved the strongest overall results, reaching 80.42% accuracy in the static setting and 81.61% in the dynamic setting. At the same time, the gain from fusion was markedly larger in the static setting than in the dynamic one, suggesting that temporal modelling and multimodal fusion address FER under HMD occlusion partly in overlapping ways rather than contributing fully independent information.

TABLE 7.13: Baseline accuracies across FER conditions.

	Accuracy in %		
Modality / Fusion	**Static**	**Dynamic**	**Δ in pp**
Image-based	69.84	72.75	+2.91
FEA-based	71.69	78.31	+6.61
Multimodal Fusion	80.42	81.61	+1.19

The remainder of this section interprets the benchmark evidence from four perspectives. First, Subsection 7.3.1 compares static and dynamic FER under HMD occlusion and discusses the practical trade-offs between the two settings. Then, Subsection 7.3.2 examines the relative strengths and limitations of image data, HMD-derived FEAs, and their fusion across the benchmark. Next, Subsection 7.3.3 interprets the recurring class-wise error patterns and clarifies what they reveal about the benchmark database and results as a whole. Finally, Subsection 7.3.4 defines the interpretive boundaries of the reported benchmark evidence.

7.3.1 Static versus Dynamic FER under Natural HMD Occlusion

Across all three modality conditions, the dynamic baselines outperformed their static counterparts, confirming that temporal modelling is beneficial for FER under natural HMD occlusion (see Table 7.13). The accuracy gain was clearly strongest for HMD-derived FEAs (+6.61 pp), smaller for images (+2.91 pp), and smallest for multimodal fusion (+1.19 pp). Table 7.14 shows that these gains were not uniform across classes. Sadness improved most strongly, followed by neutral, and both improved consistently across all three conditions. Accordingly, temporal modelling particularly reduced confusions involving these classes, particularly sadness predicted as anger and sadness predicted as neutral. In contrast, the most common error of classifying anger as disgust became more pronounced in the dynamic setting. Overall, anger showed mixed results, while disgust deteriorated from the static to the dynamic setting in all three modality conditions. Therefore, the overall advantage of dynamic FER did not arise from a broad lifting of the full class profile, but from clear gains in selected classes alongside persistent hard residual confusions.

TABLE 7.14: Per-class F1-score differences between dynamic and static baselines in pp; + values indicate an advantage for the dynamic setting.

Expression	**Image-Based**	**FEA-Based**	**Multimodal**
Anger	+ 1.14	+ 9.53	− 0.22
Disgust	− 2.94	− 5.67	− 7.47
Fear	− 0.47	+ 7.26	+ 0.72
Happiness	+ 5.32	− 0.50	− 2.30
Neutral	+ 4.18	+13.07	+ 5.94
Sadness	+12.86	+14.89	+11.07
Surprise	− 0.41	+ 7.92	+ 1.15
Average	+ 2.81	+ 6.64	+ 1.27

These improvements are noteworthy because the dynamic setting is more demanding than the static one. Besides the substantially larger amount of data to process in the dynamic setting, the benchmark database is, by construction, more closely aligned with the static task. EmoHeVRDB is built around one manually labelled

central-view image per reenactment, and the static subsets directly reuse that annotated frame or the corresponding FEA vector as the sample representation (see Chapter 6). Thus, the static setting benefits from labels assigned to a selected representative moment close to an apex frame. In contrast, the dynamic subsets use full 30-step sequences, thereby requiring the models to identify and aggregate the most important temporal information without explicit apex supervision.

The large gain for the FEA-based condition further suggests that temporal modelling is particularly valuable for HMD-derived face-tracking signals. One plausible explanation is that the compact 63-dimensional FEA vectors benefit strongly from temporal aggregation, which can smooth frame-wise noise and emphasise more stable activation patterns across a reenactment. For the image-based condition, the smaller gain suggests that temporal modelling can only partly compensate for the permanent loss of upper-face information caused by HMD occlusion. For multimodal fusion, the even smaller gain implies that the strong static fusion model already captures much of the complementary information available across modalities.

From a practical perspective, the dynamic setting is more realistic for VR applications because real systems observe continuous data streams rather than isolated representative frames, and the most expressive moment is not known in advance. At the same time, dynamic FER requires more data, more processing, and more complex models than static FER. Therefore, static FER remains a valuable reference condition. However, the present results show that when the requirements are met, dynamic FER provides a more meaningful operating point and better results for FER under natural HMD occlusion, particularly for HMD-derived face-tracking signals.

7.3.2 Modality Trade-offs and Multimodal Fusion

Across both static and dynamic FER, multimodal fusion achieved the strongest results, HMD-derived FEAs were the stronger unimodal modality, and external-camera images were the weaker one (see Table 7.13). However, the gain from fusion differed markedly across settings: it was substantial for static FER, but much smaller for dynamic FER, where the FEA-sequence baseline was already comparatively strong. Thus, the benchmark evidence shows that fusion performed best overall, but that its benefit depended on how much complementary information remained beyond the strongest unimodal baseline.

In the static setting, exactly one unimodal model was correct in 242 of 756 cases on the multimodal test set (see Figure 7.5), and an oracle that always selected the correct unimodal prediction would have reached 86.77% accuracy, increasing accuracy by 15.08 pp over the stronger FEA baseline. This large residual gap helps explain the strong gain from static multimodal fusion. Class-wise, both unimodal baselines found anger and disgust particularly challenging, followed by fear. Static fusion then primarily improved on these difficult classes, with F1-scores increasing by 10.88 pp for anger, 9.18 pp for fear, 7.08 pp for disgust, and 12.08 pp for neutral over the respective stronger unimodal baseline (see Table 7.8). In the dynamic setting, the unimodal models were individually stronger, but less complementary. Exactly one unimodal model was correct in 190 of 756 cases, yielding a dynamic oracle accuracy of 88.10% (see Figure 7.12), which is only 1.32 pp above the static oracle and 9.79 pp above the stronger FEA-sequence baseline. Class-wise, the image-sequence baseline contributed mainly to disgust and fear, whereas the FEA-sequence baseline was stronger on anger, happiness, neutral, sadness, and surprise (see Table 7.12). Taken together, multimodal fusion was most beneficial where the two modalities made more distinct errors, and less beneficial where both were already either strong or similarly limited. Intermediate fusion outperformed late fusion in both settings, but only modestly, suggesting that decision-level fusion already captures most of the available multimodal benefit.

Despite its weaker benchmark results, the image modality retains important strengths. It can build directly on the mature ecosystem of conventional visual FER, including established datasets, preprocessing pipelines, and model architectures. Section 7.1.2 demonstrated this: combining an artificially occluded version of AffectNet (AffectNet-7-occl) with EmoHeVRDB-SI during training increased test accuracy on EmoHeVRDB-SI from 69.84% to 74.74%. Additionally, image-based FER is not tied to a specific headset-integrated face-tracking system, and external cameras are low-cost and widely available. However, image-based FER remains constrained by HMD occlusion, which removes much of the upper-face information on which conventional FER relies. Furthermore, the EmojiHeroVR study used fixed camera and participant positions with little head or body movement (see Chapter 5). Consequently, the image data in EmoHeVRDB reflect the challenges of real VR use only partially, especially with respect to head motion, body motion, self-occlusion, and unstable face detection.

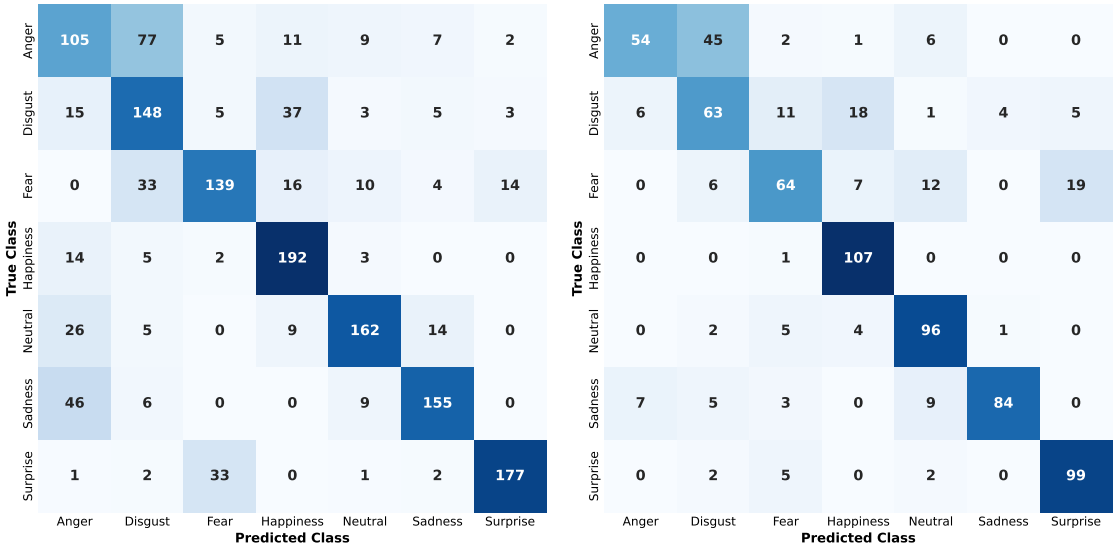
In contrast, the stronger performance of the FEA modality indicates that HMD-derived face tracking is a particularly suitable unimodal signal for FER in the present benchmark. Beyond its stronger results, HMD-integrated sensing offers practical advantages: it requires no additional environmental hardware, remains fixed relative to the wearer's face, and is less sensitive to VR-specific challenges, such as head motion and changing camera pose. In addition, face-tracking outputs provide a much more compact facial representation than images. At the same time, this modality has its own limitations. In this thesis, FEAs are derived from the Meta Quest Pro and exposed through the Face Tracking API of the Meta XR Core SDK (Meta 2023) as 63-dimensional vectors. Although there are industry efforts for standardisation—such as the OpenXR initiative (Group 2026)—the FEA benchmark and baseline models in this thesis still depend on provider-specific hardware and software. Furthermore, the wider ecosystem of models and benchmarks remains far less developed than for the image modality. Accordingly, FEAs appear to be the more effective unimodal route in this thesis, but not the more universally transferable one.

Overall, these results suggest a clear practical trade-off. Image-based FER remains attractive where portability, comparability with the broader FER literature, and reuse of existing image resources are important. FEAs provide the stronger unimodal results, particularly in the dynamic setting, and are preferable where supported HMD hardware is available and efficient, compact sensing is prioritised. They are also well suited to FER in VR because they are compact, headset-integrated, and less exposed to some of the geometric limitations of external-camera capture. Multimodal fusion provides the best overall performance and is therefore the most promising benchmark route where synchronised access to both modalities is available. However, this advantage comes at the cost of additional sensing, synchronisation, and model complexity. Consequently, the present evidence does not support a single universally optimal modality choice, but a hierarchy of options: images as the more established baseline, FEAs as the stronger unimodal solution for this VR setting, and multimodal fusion as the highest-performing but most complex configuration.

7.3.3 Class-wise Error Patterns, Database Difficulty, and Benchmark Interpretation

Across the six benchmark FER conditions evaluated in Sections 7.1 and 7.2 (see Table 7.13), largely similar class hierarchies emerge. In terms of F1-scores, anger and disgust are consistently the two weakest classes, followed by fear. Happiness and surprise are already strong in the static unimodal settings, whereas neutral and sadness benefit particularly from the dynamic setting and multimodal fusion. The confusion matrices show similarly stable patterns. Figure 7.15 summarises the unimodal confusion patterns across static and dynamic settings. In all six benchmark FER conditions, the most frequent error is the misclassification of anger as disgust. Fear and surprise also form a persistent confusion pair, though the dominant direction varies by modality. Additionally, disgust predicted as happiness recurs across the image-based, FEA-based, and fusion conditions as a major error. In contrast, sadness predicted as anger, neutral predicted as anger,

and fear predicted as disgust mainly occur in the image-based conditions. In the FEA-based conditions, there are no comparably frequent modality-specific confusions beyond the shared errors mentioned above. The fusion models reduce the unimodal errors in many cases, but largely preserve the major error patterns.



(A) Image-based conditions. (B) FEA-based conditions.

FIGURE 7.15: Summed confusion matrices across static and dynamic unimodal baselines. The image-based matrix contains twice as many test samples because each reenactment was recorded from two camera views.

The similarity between image-based and FEA-based patterns is noteworthy because both modalities differ substantially in structure and available facial information. The image-based models operate on 224×224×3 facial images in which the HMD occludes most of the upper face, whereas the FEA-based models process 63-dimensional FEA vectors that retain a structured approximation of the full face. Therefore, the persistence of class and error patterns suggests that these patterns do not solely arise from the modelling task or from HMD occlusion, but also reflect underlying properties of EmoHeVRDB itself. In contrast, the confusions that occur mainly in the image-based conditions—particularly those of anger, sadness, and neutral—are plausibly linked more directly to the loss of upper-face information caused by HMD occlusion. This interpretation is also consistent with the annotation results in Section 6.1.3, where anger, sadness, and neutral were also frequently confused during manual labelling of the central-view images. Regarding

the persistence of class and error patterns across conditions, the following three explanations are especially plausible.

First, the training sets in EmoHeVRDB are strongly imbalanced. EmoHeVRDB-SI and EmoHeVRDB-DI contain only 150, 212, and 220 training samples for anger, disgust, and fear, respectively, but 414 for neutral and 400 for happiness (see Figure 6.4). The training sets of EmoHeVRDB-SFEA and EmoHeVRDB-DFEA contain even fewer samples, with only 66 for anger, 102 for disgust, and 105 for fear (see Figure 6.5), because the FEA modality provides no separate camera perspectives and face tracking malfunctioned for one participant. Across conditions, these least represented classes are also consistently the most difficult for the baseline models. To account for class imbalance during training, normalised inverse class frequencies were passed as class weights to the loss function in all baseline experiments. However, while class weighting gives samples from less-represented classes more weight, it cannot compensate for a lack of diversity in the underlying data, which is a likely constraint given the low sample counts for anger, disgust, and fear.

Second, the imbalance in EmoHeVRDB results from the annotation and filtering pipeline. Three raters manually annotated 2,590 central-view images. Subsequently, a majority-vote filter was applied to ensure that the retained samples were representative of the target expression. Because the hit rates for anger (51.44%), disgust (57.12%), and fear (54.23%) were substantially lower than for the other classes, this filtering step produced the skewed class distribution in EmoHeVRDB. Although the majority-vote filter removed the most ambiguous samples, the retained anger, disgust, and fear samples may still be less distinctive or prototypical than the samples of easier classes, particularly happiness and neutral, with hit rates of 96.40% and 94.05%, respectively. This is also plausible because EmoHeVRDB was collected from non-actors who posed expressions under time pressure while playing EmojiHeroVR. Consequently, the reduced reenactment accuracy or distinctiveness of anger, disgust, and fear samples appears to be a limitation of EmoHeVRDB.

Third, some expression categories appear to be inherently more difficult to recognise and distinguish than others, particularly under HMD occlusion. This interpretation is supported by the low annotation hit rates for anger, disgust, and fear, as well as by these classes' consistently low F1-scores across all baseline models. Moreover, the recurring confusions observed during both labelling and model evaluation align with the validation studies for KDEF (Goeleven et al. 2008) and RaFD (Langner et al. 2010), where anger–disgust and fear–surprise were also prominent confusion pairs.

These explanations are not mutually exclusive. The persistent weakness of anger, disgust, and, to a lesser extent, fear is best interpreted as the result of interacting factors. Some expression categories appear less separable than others under the present benchmark conditions. Additionally, expressions were posed by non-actors under time pressure while playing EmojiHeroVR, which may have made them less distinctive or non-prototypical. These factors plausibly contributed to lower annotation hit rates, particularly for anger, disgust, and fear. Subsequently, the majority-vote filter retained fewer samples from these difficult classes, which in turn reduced the amount and diversity of training data available to the baseline models.

Consequently, the persistent class and error patterns should not be interpreted solely as limitations of the baseline models, but also as a benchmark limitation related to the underrepresented expression classes in EmoHeVRDB. The most direct improvement for EmoHeVRDB would therefore be to collect additional samples for anger, disgust, and fear to reduce class imbalance and increase the within-class diversity of the most difficult expression categories. Given the current class distribution and annotation pipeline of EmoHeVRDB, the benchmark results should be read not only as evidence of general model performance, but also as evidence of the class-dependent difficulty of recognising posed expressions in EmoHeVRDB under natural HMD occlusion.

7.3.4 Interpretive Boundaries of the Benchmark Evidence

The preceding subsections of Section 7.3 have already discussed specific interpretive limits of the benchmark results, particularly with respect to temporal modelling, modality choice, multimodal fusion, and class-wise database difficulty. Therefore, this final subsection does not revisit each of these points individually, but defines the broader benchmark-level boundaries of the evidence presented in this chapter.

As defined in the requirements for Artefact 1 in Section 3.2.1, the baseline models evaluated in Sections 7.1 and 7.2 are not intended to establish the maximum achievable performance for FER under HMD occlusion. Instead, they provide comparable reference points across static, dynamic, unimodal, and multimodal conditions under a common benchmark protocol, using the fixed participant-independent split defined in Section 6.2. Accordingly, the training procedures, sampling strategies, and hyperparameter spaces were deliberately limited. For instance, class imbalance was addressed through normalised inverse class frequencies as class weights, but no additional imbalance-handling strategies—such as augmentation-based oversampling or balanced sampling—were applied. Likewise, the dynamic baselines used a fixed 30-step sequence representation for each sample, without systematically exploring alternative temporal sampling strategies, such as shorter sub-sequences or sliding windows. In addition, all neural-network baselines were trained with the Adam optimiser and the SparseCategoricalCrossentropy loss function, without systematically comparing alternative optimisers or loss functions. These methodological choices kept the benchmark pipeline simple, reproducible, and comparable across conditions, but they also mean that the reported accuracies should be interpreted as baseline evidence rather than as final performance ceilings for EmoHeVRDB or for FER in VR more generally.

The evidence is also bounded by the construction of EmoHeVRDB and the FER task considered in this thesis. Compared with prior work based solely on simulated HMD occlusion, EmoHeVRDB provides naturally HMD-occluded, time-synchronised multimodal data and a participant-independent benchmark protocol, thereby enabling a more realistic benchmark for FER in VR settings. At the same time, the

benchmark remains a controlled, game-mediated, seven-class FER setting based on posed facial expressions. The labels represent manually filtered facial-expression categories rather than direct measurements of participants' internal emotional states, and both the static and dynamic subsets inherit the same underlying annotation logic described in Chapter 6. Thus, the benchmark is well suited for comparing FER approaches under natural HMD occlusion, but it does not by itself validate broader claims about spontaneous emotion recognition in open-ended VR interactions.

Finally, the results are tied to the specific study and sensing configuration used in this thesis. The image-based baselines depend on external-camera recordings from the controlled setup described in Chapter 5, while the FEA-based baselines depend on the face-tracking capabilities of the employed HMD and its software interface. Accordingly, the chapter provides evidence for this concrete combination of external-camera images, HMD-derived FEAs, and their fusion, but not yet for all camera placements, HMDs, face-tracking APIs, user populations, or deployment contexts.

Within these boundaries, the benchmark results establish the feasibility of FER under natural HMD occlusion, identify the main comparative trade-offs between static and dynamic FER, image-based and FEA-based sensing, and unimodal and multimodal modelling, and provide a reproducible empirical foundation for further research on FER in VR.

Chapter 8

Conclusion

This thesis set out to enhance emotion recognition for VR users by enabling effective FER despite partial occlusion caused by HMDs. Building on this research aim and the research questions and objectives defined in Section 1.2, Chapter 2 reviewed the related literature and concluded with five research gaps that guided the remainder of the thesis (see Section 2.9.3). Chapter 3 then followed the DSR research strategy to derive requirements for three artefacts and their evaluation to address the research gaps and questions. Accordingly, Chapter 4 presented the design and implementation of EmojiHeroVR (Artefact 3) as the data-acquisition system for capturing multimodal facial-expression data from VR users under natural HMD occlusion. In Chapter 5, EmojiHeroVR was subsequently deployed in a user study to collect the external-camera and HMD-derived data required to build a benchmark database for FER under HMD occlusion. Next, Chapter 6 constructed EmoHeVRDB as a time-synchronised multimodal benchmark database (Artefact 2) from the collected data through manual annotation, majority-vote filtering, participant-independent splitting, and benchmark subset construction. Finally, Chapter 7 implemented and evaluated baseline FER models (Artefact 1) to establish benchmark evidence across static, dynamic, unimodal, and multimodal conditions.

This chapter concludes the thesis by synthesising the evidence generated in the preceding chapters. First, Section 8.1 assesses the developed artefacts and their

evaluations against the evidence requirements defined in Section 3.2 and examines how they contribute to addressing the research gaps. Second, Section 8.2 answers the research questions based on the generated evidence. Third, Section 8.3 consolidates the main limitations and conditions of interpretation that define the scope of the evidence presented in this thesis. Finally, Section 8.4 outlines directions for future work and provides concluding remarks.

8.1 Contributions in Relation to the Research Gaps

This section synthesises the thesis's main artefact-level contributions in relation to the research gaps identified in Section 2.9.3. It follows the evidence chain established across Chapters 4–7 and closes the DSR logic introduced in Section 3.2 by assessing whether the developed artefacts and their evaluations fulfil the defined evidence requirements. First, Subsection 8.1.1 discusses EmojiHeroVR as the data-acquisition system that enabled the controlled collection of multimodal facial-expression data under natural HMD occlusion. Then, Subsection 8.1.2 examines EmoHeVRDB as the resulting time-synchronised multimodal benchmark resource. Finally, Subsection 8.1.3 discusses the benchmark evidence generated from the baseline FER models and explains how it advances the evaluation of static, dynamic, unimodal, and multimodal FER for VR users.

8.1.1 Artefact 3: EmojiHeroVR as a Data-Acquisition System

The required data-acquisition system (Artefact 3) was realised as EmojiHeroVR in Chapter 4 and deployed in the user study reported in Chapter 5. Its design combines a game-based elicitation procedure with multimodal recording and synchronised logging. In the user study, participants wore an HMD and were prompted by EmojiHeroVR to reenact the seven facial-expression categories of anger, disgust,

fear, happiness, sadness, surprise, and neutral, as defined in Section 3.3.1. The system captured two complementary modalities: naturally HMD-occluded facial images from two external-camera perspectives and HMD-derived FEAs from the Meta Quest Pro. These recordings were associated with defined recording windows and logged in a temporally aligned form. EmojiHeroVR therefore provided the elicitation, capture, and synchronisation functions required to generate the raw data for Artefact 2.

EmojiHeroVR's deployment in the user study provided the evaluation evidence for Artefact 3. The system was successfully used with human participants and generated external-camera recordings and HMD-derived data from which EmoHeVRDB, the benchmark database (Artefact 2), was later constructed. This outcome demonstrates its feasibility as a repeated-use VR study instrument and its suitability for producing the raw data required to build a time-synchronised multimodal benchmark database. The questionnaire results further support this assessment: the GEQ indicated an engaging and largely positive user experience, while the VRSQ indicated low VR-related discomfort.

In relation to the research gaps, EmojiHeroVR contributes as both an implemented data-acquisition instrument and a repeatable procedure for collecting FER data under HMD occlusion. By recording participants while they wore an actual HMD, it enabled data collection under natural rather than simulated HMD occlusion (RG1). By embedding the recording in a VR game rather than a non-VR image-capture setup, it increased the VR-specificity and ecological relevance of the evidence base (RG3). By combining external cameras with the Meta Quest Pro's face tracking, it provided external-camera and HMD-derived facial signals under a consistent study protocol (RG2). Finally, by logging both modalities within matched recording windows, it created the temporal and cross-modal structure required for dynamic FER and multimodal fusion (RG4, RG5). To support reuse and further development beyond this thesis, EmojiHeroVR's source code is publicly available[1], together with a demo video showing the third level of EmojiHeroVR used in the user study.[2]

[1] https://github.com/thorbenortmann/EmojiHeroVR
[2] https://youtu.be/TnJrGYOjKJc

Overall, EmojiHeroVR does not directly solve FER under HMD occlusion, but provides a data-acquisition mechanism for subsequent database construction and benchmark evaluations.

However, this contribution remains bounded by the thesis-level design decisions discussed in Section 3.3 and the empirical scope of the user study reported in Chapter 5. EmojiHeroVR elicits posed rather than spontaneous expressions and uses a fixed set of seven facial-expression categories. In addition, the HMD-derived modality is based specifically on the Meta Quest Pro's face-tracking pipeline and therefore cannot be directly generalised to other HMDs. Finally, EmojiHeroVR was evaluated in a controlled laboratory setting with a limited participant sample, which further constrains ecological and population-level generalisability. Nevertheless, within these boundaries, EmojiHeroVR fulfils the evidence requirements for Artefact 3.

8.1.2 Artefact 2: EmoHeVRDB as a Benchmark Database

The required benchmark database (Artefact 2) was realised as EmoHeVRDB in Chapter 6, using the multimodal study data collected with EmojiHeroVR. Its construction began with manual annotation of 2,590 reenactments by three raters, followed by majority-vote filtering, which retained 1,921 samples. The retained data were then organised into participant-independent training, validation, and test splits with 21, 8, and 8 participants, respectively. Subsequently, the validation and test sets were balanced by discarding samples, whereas the training set remained imbalanced. Finally, the data were temporally aligned across modalities and structured into task-specific benchmark subsets. The resulting database supports static image-based FER via EmoHeVRDB-SI, static FEA-based FER via EmoHeVRDB-SFEA, dynamic image-sequence-based FER via EmoHeVRDB-DI, and dynamic FEA-sequence-based FER via EmoHeVRDB-DFEA. Because these subsets share participant-independent splits and preserve temporal alignment across modalities, EmoHeVRDB supports unimodal and multimodal benchmarking under one consistent protocol.

The evaluation evidence for Artefact 2 lies in its transparent ground-truth construction and its demonstrated benchmark utility. Manual annotation yielded substantial inter-rater agreement, with a Fleiss' kappa of 67.76%, while majority-vote filtering increased agreement in the retained set to 78.60%. Together with the participant-independent split, temporal alignment rules, and benchmark subset definitions, this made EmoHeVRDB suitable for reproducible evaluation. Its benchmark utility was demonstrated in Chapter 7, where it supported all static, dynamic, unimodal, and multimodal baseline evaluations.

In relation to the research gaps, this thesis contributes EmoHeVRDB as a benchmark database for systematic FER evaluation under HMD occlusion. It provides naturally HMD-occluded external-camera recordings captured during actual VR use (RG1, RG3), HMD-derived FEAs obtained from unmodified consumer hardware under a consistent study protocol (RG2), sequential recordings for dynamic FER (RG4), and time-synchronised multimodal samples for fusion experiments (RG5). EmoHeVRDB is available to other researchers for non-commercial use through controlled distribution under an end-user licence agreement.[3]

This contribution remains bounded by the data-collection conditions inherited from EmojiHeroVR (see Section 8.1.1) and by the construction choices applied during database creation. Majority-vote filtering improved label consistency but introduced class imbalance, particularly by reducing the number of retained samples for anger, disgust, and fear. This limits the fairness of class-wise benchmark comparisons because class difficulty and training-set imbalance cannot be fully separated. Nevertheless, EmoHeVRDB fulfils the evidence requirements for Artefact 2 by providing a transparent, participant-independent, time-synchronised benchmark resource and the empirical foundation for the baseline FER models discussed in the following Section 8.1.3.

[3] https://github.com/thorbenortmann/emoji-hero-vr-database

8.1.3 Artefact 1: Baseline FER Models and Evaluations

The required baseline FER models (Artefact 1) were implemented and evaluated in Chapter 7 as benchmark references for static, dynamic, unimodal, and multimodal FER on EmoHeVRDB. The unimodal evaluations covered image-based FER on EmoHeVRDB-SI, FEA-based FER on EmoHeVRDB-SFEA, image-sequence-based FER on EmoHeVRDB-DI, and FEA-sequence-based FER on EmoHeVRDB-DFEA. The multimodal evaluations added late and intermediate fusion in the static and dynamic settings. In addition, the static image-based evaluation directly compared simulated and natural HMD occlusion.

The evaluation evidence for Artefact 1 lies in the comparable benchmark results produced under the shared participant-independent protocol. As summarised in Table 7.13, the baseline accuracies varied across conditions, ranging from 69.84% for static image-based FER to 81.61% for dynamic intermediate fusion. FEA-based models outperformed their image-based counterparts in both the static and dynamic settings, while intermediate fusion yielded the strongest result in each setting. Consequently, the benchmark results provide the evidence needed to compare modality choice, temporal modelling, and fusion strategy under one consistent evaluation framework.

In relation to the research gaps, Artefact 1 contributes the baseline evidence required to assess FER for VR users under HMD occlusion. The comparison between simulated and natural HMD occlusion addresses the transferability of findings from artificial occlusion to actual HMD recordings (RG1). The FEA-based baselines evaluate HMD-derived facial signals from unmodified consumer hardware (RG2). By using EmoHeVRDB, the evaluations are grounded in data collected during actual VR interaction (RG3). The sequence-based models examine dynamic FER (RG4), while the late- and intermediate-fusion experiments compare unimodal models with multimodal fusion (RG5). Thus, Artefact 1 provides a structured benchmark comparison across the main design dimensions identified by

the research gaps. The code and trained models for the baseline experiments are publicly available.[4]

This contribution remains bounded by the intended role of the models as baselines. The architectures, training procedures, sampling strategies, and hyperparameter searches were deliberately limited to produce comparable reference points rather than maximise performance. The results also inherit the constraints of EmoHe-VRDB, including posed expressions, seven categorical labels, dependence on the Meta Quest Pro's face-tracking pipeline, and training-set imbalance. The consistent difficulty in recognising anger, disgust, and fear should therefore be interpreted partly in light of their limited representation in the training data. Within this scope, Artefact 1 fulfils the evidence requirements for the baseline FER models and provides the empirical basis for answering RQ3 and RQ4.

8.2 Answers to the Research Questions

This section shifts from the artefact-level contributions to the research-question perspective. It synthesises the evidence generated across the literature review, system design, user study, database construction, and benchmark experiments to provide concise answers to the research questions formulated in Section 1.2.1.

8.2.1 RQ1: What are the key challenges and limitations that hinder effective FER for VR users?

The literature review in Chapter 2 showed that the central challenge for FER in VR settings is HMD-induced occlusion, which hides much of the upper face and thereby removes facial cues that conventional image-based FER systems rely on. As synthesised in Section 2.9.3, this challenge is compounded by several research gaps in the existing evidence base. Existing external-camera studies mainly relied on artificially occluded images from established FER databases, leaving the

[4] https://github.com/thorbenortmann/phd-thesis

transfer to naturally HMD-occluded recordings during actual VR use unclear (see Section 2.7.1 and RG1). In contrast, headset-integrated approaches used heterogeneous hardware, sensing pipelines, target representations, and evaluation protocols, which limited comparability and reproducibility (see Section 2.7.2 and RG2). More broadly, the literature on FER in VR settings provided limited evidence from actual VR use (RG3), rarely examined dynamic FER (RG4), and lacked directly comparable multimodal evaluations combining external-camera and HMD-derived signals (RG5).

In addressing these gaps, the thesis confirmed literature-derived challenges and made additional constraints visible. Chapter 3 showed that central design decisions required for FER in VR settings involve trade-offs: categorical labels support comparability but simplify affective behaviour, posed expressions enable controlled and ethically acceptable data collection but limit ecological validity, and HMD-derived face tracking provides otherwise unavailable facial signals but remains hardware- and software-specific. Chapters 4 and 5 made clear that collecting data suitable for a time-synchronised multimodal benchmark database depends on a complex data-acquisition setup and controlled study procedure. At the same time, the laboratory-based setup limited sample size and participant diversity. Chapter 6 further showed that prompted posed expressions could not simply be used as reliable ground-truth labels. Manual annotation and majority-vote filtering improved label consistency, but reduced the retained sample set and introduced class imbalance. Finally, Chapter 7 confirmed that FER under HMD occlusion is feasible but remains challenging. In particular, Section 7.1.2 showed that findings from artificially occluded databases do not transfer directly to natural HMD occlusion. Sections 7.1–7.3 further showed that performance depends strongly on modality choice, temporal modelling, and fusion strategy, with anger, disgust, and fear remaining especially difficult across the benchmark conditions. Thus, the key challenges hindering effective FER for VR users are not limited to the physical occlusion caused by HMDs, but extend to the transfer from artificial to natural occlusion, benchmark-suitable data acquisition, reliable ground-truth construction, class imbalance, hardware

dependence, and the remaining modality-specific limitations of external-camera recordings, HMD-derived signals, and multimodal fusion.

8.2.2 RQ2: How can facial expressions of VR users be collected to construct a time-synchronised multimodal benchmark database that supports static and dynamic FER under HMD-induced occlusion?

The thesis shows that such a database can be constructed through an integrated data-acquisition and database-construction pipeline, as developed across Chapters 4–6 and evaluated in Chapter 7. To ensure sufficient ecological validity, facial expressions must be elicited during actual VR use rather than only simulated outside VR. In this thesis, EmojiHeroVR's gameplay served as the elicitation method by prompting players to reenact seven facial-expression categories while wearing an HMD (see Chapter 4). This controlled, repeated elicitation enabled the collection of a substantial amount of facial-expression data from comparatively few participants while maintaining an engaging user experience (see Chapter 5).

To construct a time-synchronised multimodal database, the data-acquisition system must record multiple modalities in a temporally alignable form. Accordingly, EmojiHeroVR combined two external cameras with HMD-derived face tracking from the Meta Quest Pro. For each prompted reenactment, the system defined a recording window and logged the external-camera frames and FEA vectors with timestamps and recording-window identifiers (see Section 4.4). These shared temporal markers made it possible to associate both modalities with the same facial-expression event and provided the structure required for static samples, dynamic sequences, and multimodal pairings.

To make the database defensible as a benchmark resource, the collected raw data must then be converted through transparent ground-truth and benchmark-construction procedures. For EmoHeVRDB, this involved manual annotation, majority-vote filtering, participant-independent splitting, temporal alignment, and

task-specific subset construction (see Chapter 6). The resulting benchmark subsets support static image-based FER via EmoHeVRDB-SI, static FEA-based FER via EmoHeVRDB-SFEA, dynamic image-sequence-based FER via EmoHeVRDB-DI, and dynamic FEA-sequence-based FER via EmoHeVRDB-DFEA (see Section 6.3). Because these subsets are based on the same underlying reenactments and share labels, participant-independent splits, and temporal alignment across modalities, they support unimodal and multimodal benchmarking under one consistent protocol.

Thus, this thesis answers RQ2 by demonstrating one viable route for collecting facial expressions of VR users and transforming them into a benchmark database, as described above. This route produced EmoHeVRDB as a time-synchronised multimodal benchmark database for static and dynamic FER under natural HMD occlusion, whose benchmark utility was demonstrated through the baseline evaluations in Chapter 7. At the same time, alternative routes are also possible, including approaches based on spontaneous expressions, other emotion representations, additional modalities, or different sensing hardware.

8.2.3 RQ3: How do unimodal baselines and multimodal fusion perform for static and dynamic FER under HMD-induced occlusion?

This thesis conducted benchmark experiments for static and dynamic FER across image-based, FEA-based, and multimodal conditions (see Sections 7.1 and 7.2). Their goal was not to maximise performance through extensive optimisation, but to provide comparable baseline results. These baseline accuracies are summarised in Table 7.13. In the static setting, the image-based baseline achieved 69.84%, the selected FEA-based MLP baseline reached 71.69%, and intermediate fusion achieved 80.42%. In the dynamic setting, image-sequence-based FER achieved 72.75%, FEA-sequence-based FER reached 78.31%, and intermediate fusion achieved the highest overall accuracy of 81.61%.

These results show several main patterns. First, unimodal FER under HMD occlusion is feasible with both external-camera images and HMD-derived FEAs. Second, the FEA-based models outperformed their image-based counterparts in both the static and dynamic settings. Third, temporal modelling improved both unimodal baselines, but the gain was larger for FEAs than for images. Fourth, multimodal fusion achieved the strongest result in both settings, although the gain over the best unimodal baseline was substantially smaller in the dynamic setting than in the static setting. Fifth, similar class-wise performance and error patterns emerged across conditions. Anger and disgust were consistently the most difficult classes, followed by fear. The most frequent error was anger being misclassified as disgust, while fear and surprise also formed a recurring confusion pair. In contrast to these shared error patterns, misclassifications of sadness as anger, neutral as anger, and fear as disgust mainly occurred in the image-based conditions. Overall, the baseline results show performance advantages for HMD-derived FEAs, temporal modelling, and multimodal fusion, while also revealing persistent class-wise difficulties.

8.2.4 RQ4: What interpretations, trade-offs, and design implications can be derived from these baseline results to advance FER research and applications in VR settings?

Interpretation of the Baseline Evidence. The comparable baseline results, summarised in the previous subsection for answering RQ3, support several broader interpretations. First, the comparison between static and dynamic FER showed that temporal modelling benefited the FEA modality more strongly than the image modality. A plausible explanation is that temporal aggregation makes the compact FEA signal more stable across a reenactment, whereas image-based FER remains constrained by the permanent loss of upper-face information caused by HMD occlusion (see Section 7.3.1).

Second, the reduced fusion gain in the dynamic setting indicates that temporal modelling and multimodal fusion provide partly overlapping benefits. The strong FEA-sequence baseline likely captures much of the information that made image–FEA fusion particularly beneficial in the static setting, leaving less additional benefit for the image modality to add through fusion (see Section 7.3.2).

Third, the persistent weakness of anger, disgust, and, to a lesser extent, fear should not be interpreted solely as model failure but also as evidence of class-dependent benchmark difficulty in EmoHeVRDB. These classes were substantially underrepresented in the imbalanced training data, harder for raters to label consistently, and likely more difficult for non-actors to pose distinctly while playing EmojiHeroVR (see Section 7.3.3). At the same time, the error patterns that occurred mainly in the image-based conditions are plausibly linked more directly to the loss of upper-face information caused by HMD occlusion. This interpretation is consistent with the annotation results in Section 6.1.3, where the raters also frequently confused anger, sadness, and neutral in the central-view images.

Finally, the cross-database experiments in Section 7.1.2 revealed that models trained on artificially occluded images from AffectNet (AffectNet-7-occl) transferred poorly to EmoHeVRDB-SI, indicating a pronounced domain gap between simulated and natural HMD occlusion. However, combining AffectNet-7-occl with EmoHeVRDB-SI during training improved performance beyond training on EmoHeVRDB-SI alone, suggesting that artificially occluded data are useful as supplementary training data, but not sufficient as a standalone substitute for naturally HMD-occluded recordings.

Trade-offs and Design Implications. Together, the baseline results and their interpretations reveal practical trade-offs and design implications for FER in VR settings. Regarding temporality, static FER remains useful as a simpler reference condition because it requires less data, processing, and modelling complexity. In contrast, dynamic FER achieved better overall accuracies across all evaluated conditions and better reflects real-world VR use, where systems observe continuous

data streams rather than isolated static samples. Thus, future work should retain static FER as a controlled reference condition, but prioritise dynamic FER as the stronger and more deployment-oriented setting when sequential data and processing capacity are available.

Regarding modality choice, external-camera FER can benefit from the wider visual FER ecosystem and remains attractive when low-cost sensing, HMD independence, and comparability with the visual FER literature are important. However, it is fundamentally constrained by the loss of facial information caused by HMD occlusion. FER based on HMD-derived FEAs achieved stronger unimodal results and is well suited to VR because the signal is compact and headset-integrated. Nevertheless, it depends on provider-specific face-tracking hardware and software and has a less mature ecosystem than image-based FER. Multimodal fusion achieved the strongest results overall, but requires synchronised access to both modalities and has the highest complexity in sensing, synchronisation, and modelling. Taken together, the baseline evidence does not support a universally optimal modality configuration for FER in VR settings. Instead, it shows that performance advantages should be balanced against hardware and data availability, processing requirements, comparability with existing FER work, and system complexity.

For benchmark design, the results imply that naturally HMD-occluded recordings should form the primary evaluation target, while artificially occluded image data are better understood as supplementary training material. Given the substantial class-wise performance differences and persistent error patterns, future benchmarks should also report class-wise metrics and confusion matrices for detailed analysis. Finally, future extensions of EmoHeVRDB and similar resources should prioritise additional samples for anger, disgust, and fear to reduce imbalance and increase within-class diversity.

Overall, RQ4 is answered by showing that the baseline evidence provides more than reference performance values: it clarifies how FER in VR can be advanced

through careful benchmark interpretation, appropriate temporal modelling, context-dependent modality choices, naturally HMD-occluded evaluation data, and targeted improvement of class-wise bottlenecks.

8.3 Limitations and Conditions of Interpretation

The preceding sections have summarised the thesis contributions and answered the research questions. This section outlines the thesis's main limitations and clarifies the conditions under which its findings should be interpreted.

Observable Expressions versus Internal Emotions. This thesis aims to enhance emotion recognition for VR users by focusing on the narrower task of FER. Accordingly, it is concerned with observable facial expressions rather than internal emotional states directly. Thus, labels and model outputs in this thesis should be interpreted as classifications of facial configurations and movements into one of seven categories, not as definitive evidence that a person experienced the corresponding emotion (see Section 2.4.1).

Categorical Emotion Representation. As described in Section 3.3.1, this thesis adopts a categorical emotion representation with seven classes commonly used in FER research: anger, disgust, fear, happiness, neutral, sadness, and surprise. This choice supports comparability with established FER research and enables a clear benchmark protocol. However, it also limits the complexity of facial expressions and does not incorporate other aspects, such as expression intensity, compound expressions, or dimensional affect. Therefore, the produced evidence is restricted to categorical FER under the class definition used throughout the thesis.

Ecological Validity. As described in Section 3.3.2, this thesis relies on posed facial expressions rather than spontaneous ones. This decision was operationalised in EmojiHeroVR's game concept, which prompts players to reenact seven target

expressions (see Section 4.2). The user study reported in Chapter 5 then recorded participants playing EmojiHeroVR in a seated, stationary position. While this approach enabled the collection of sufficient data for the construction of a benchmark database, it also limits the ecological validity of the collected data. First, spontaneous expressions can differ from posed expressions in terms of AU combinations, onset–apex–offset dynamics, and intensity (see Section 2.5.2). Second, the controlled VR game setting of the user study does not reflect the dynamic, open-ended complexity of VR interactions involving, for example, speech, hand movements, locomotion, or social interaction. Consequently, a limitation of this thesis is that it addresses RG3, which concerns the insufficient ecological validity of current evidence, only in part. The thesis improves ecological validity by enabling the benchmark evaluation of FER on data recorded during actual VR use, but the evidence remains restricted to posed expressions collected under controlled study conditions.

Recognition-Dependent Scoring. EmojiHeroVR rewards participants when the in-game FER model recognises the prompted expression. This feedback may encourage participants to favour facial configurations recognised by that model, potentially leaving more difficult-to-recognise expression variants underrepresented or absent from EmoHeVRDB. Although the gameplay analysis in Section 5.6.3 showed no consistent improvement in recognition success, such adaptation cannot be ruled out. The subsequent manual annotation (see Section 6.1) improved label reliability but could not recover expression variants that participants never produced. Thus, potential adaptation of participants' facial expressions to the in-game FER model remains a limitation of EmoHeVRDB.

Realised Sample. The realised participant sample limits the generalisability of the findings to the population level. As described in Section 5.5, the user study involved 37 participants recruited primarily from the environment of HAW Hamburg, including many students and faculty members. Consequently, the sample does not represent the demographic, cultural, morphological, and behavioural diversity of

the wider VR user population. Moreover, the fixed participant-independent split restricts the reported test performance to 8 of the 37 participants. The reported results should therefore be interpreted as baseline evidence for the realised sample and data split rather than as population-level evidence of demographic robustness.

Sensing Setup. The generalisability of the modality results is bound by the recording setup used in the user study (see Section 5.1). The image-based modalities are tied to the external-camera setup, which used two RGB cameras to record stationary participants at 1280×720 pixels from two fixed angles under consistent illumination in front of a white wall. This controlled setup limits transferability to less controlled visual settings. Nevertheless, the data remain conventional facial images and are therefore relatively comparable with other image-based FER resources. By contrast, the FEA-based modalities are more tightly bound to the study's specific sensing setup. They depend on the Meta Quest Pro and the Face Tracking API of the Meta XR Core SDK, which exposes the headset's face-tracking capabilities as 63-dimensional vectors. Consequently, the FEA-based results should be interpreted as evidence for this specific headset-derived representation rather than as evidence that transfers directly to other HMDs or face-tracking systems.

Data Annotation and Benchmark Construction. The target expressions prompted by EmojiHeroVR could not be used directly as reliable ground-truth labels (see Section 6.1). Therefore, one central-view image was selected per reenactment, manually labelled, and retained only when at least two of three raters agreed on the target expression. This procedure improved label defensibility, but also shaped the resulting benchmark database. The model-assisted selection of annotation candidates aimed to identify representative apex frames and thus selected the frames most recognisable to the selection model. This approach focused the subsequent manual labelling on more likely apex frames than random selection would have, but also biased the annotation candidates towards the selection model's capabilities. Furthermore, the majority-vote filter retained fewer samples for classes that were more difficult to label and excluded ambiguous samples rather than modelling

ambiguity explicitly. As a result, EmoHeVRDB's training sets are imbalanced, with anger, disgust, and fear substantially underrepresented. Consequently, the reported baseline results should not be interpreted solely as evidence of model performance, but also in relation to the limitations of the benchmark database and its construction.

Baseline Scope. The baseline models evaluated in Sections 7.1 and 7.2 were designed to enable consistent comparison across static, dynamic, unimodal, and multimodal FER conditions. Accordingly, the experiments deliberately used a constrained modelling pipeline rather than exhaustive model optimisation (see Section 7.3.4). The reported results should therefore be interpreted as comparable baseline evidence rather than as upper limits of FER performance on EmoHeVRDB or under HMD occlusion more generally.

8.4 Future Work and Concluding Remarks

This thesis aimed to enhance emotion recognition for VR users by enabling effective FER despite partial occlusion caused by HMDs. Within the limitations outlined in Section 8.3, this aim was addressed through the contributions presented in Section 8.1 and the answers to the research questions provided in Section 8.2. Together, the developed artefacts and the generated evidence establish a reproducible foundation for future work on FER in VR settings.

The most immediate direction for future work is to extend the empirical basis established by EmoHeVRDB. Larger and more diverse participant samples are needed to improve generalisability across demographic, cultural, morphological, and behavioural variation. Additional data should particularly target the anger, disgust, and fear classes, which were difficult to elicit, label, and recognise throughout this thesis. Increasing the number and diversity of these samples would reduce class imbalance and support more robust model training. It would also enable fairer

class-wise comparisons by helping to separate model limitations from class-specific difficulty in the benchmark database.

Future work should also further increase ecological validity by moving towards less constrained, more application-realistic VR settings. This thesis collected data during actual VR use, but the expressions were posed, prompted, and recorded under controlled laboratory conditions. Future studies should therefore collect spontaneous facial expressions in VR contexts that include speech, hand movement, locomotion, social interaction, and task-specific emotional demands. Such contexts would make it possible to examine whether the benchmark patterns observed in this thesis also transfer to facial behaviour that is less deliberate, more context-dependent, and embedded in richer VR activity. Practical use cases—such as exposure therapy, emergency response training, and social VR—would provide stronger evidence for FER in the contexts in which it is ultimately intended to operate. In these settings, the relevant question is not only whether facial expressions can be recognised, but whether the recognised signals provide useful, timely, and interpretable information for the system or the user.

This shift towards less controlled and more application-realistic settings also motivates broader expression and emotion representations. The seven categorical labels used in this thesis support comparability with established FER research and are suitable for controlled benchmarking. However, they may not fit all VR applications equally well. For example, exposure therapy may primarily require sensitivity to anxiety or distress, training systems may focus on stress or cognitive overload, and entertainment applications may be more interested in engagement or frustration. Future work should therefore consider representations that better match the intended use, including dimensional affect, action units, expression intensity, soft or multi-label annotations, and application-specific targets.

A further direction concerns the portability of the FEA-based results. These results are tied to the Meta Quest Pro and the Face Tracking API of the Meta XR Core SDK. Future work should therefore examine whether the findings transfer to other HMDs, face-tracking APIs, and sensor configurations. It should also assess whether

expression representations from different devices or APIs can be mapped onto each other to support cross-device data reuse.

The models evaluated in this thesis were designed to provide comparable baselines, not to maximise recognition performance. Future work should therefore move beyond the deliberately constrained modelling choices and investigate more comprehensive modelling and training options, particularly for architectures, temporal modelling, multimodal fusion, class imbalance handling, and hyperparameter optimisation. For deployment-oriented VR systems, model evaluation should also account for computational efficiency, latency, robustness to noisy signals, confidence and interpretability of predictions, and the impact on user experience.

Overall, this thesis moves the study of FER under HMD occlusion beyond evidence based mainly on simulated occlusion, imitation protocols, and custom sensing setups. It contributes EmojiHeroVR as a VR game for eliciting and collecting posed facial expressions, constructs EmoHeVRDB as a benchmark database based on external-camera recordings and HMD-derived face-tracking signals, and provides comparable baseline evaluations across static, dynamic, unimodal, and multimodal FER conditions. The results show that FER in VR settings is feasible despite HMD occlusion, but also that the evidence must be interpreted in relation to the categorical emotion representation, the controlled study setting, the realised sample, the benchmark-construction procedure, the sensing setup, and the baseline scope. Future work can extend this foundation through larger and more diverse benchmark data, more ecologically valid VR studies, broader expression and emotion representations, cross-device validation of HMD-derived signals, and more deployment-oriented model evaluation.

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Appendix A

Questionnaire Response Distributions

A.1 Game Experience Questionnaire (GEQ) Item Response Distributions

TABLE A.1: GEQ core-module item response frequencies ($n = 36$). Response options: 0=not-at-all, 1=slightly, 2=moderately, 3=fairly, 4=extremely.

Component	**Item**	**Frequency**					**Mean**
		0	**1**	**2**	**3**	**4**	
Competence	I felt skilful	3	7	15	10	1	1.97
	I felt competent	1	7	19	8	1	2.03
	I was good at it	4	9	10	11	2	1.94
	I felt successful	2	10	6	16	2	2.17
	I was fast at reaching the game's targets	0	7	10	9	10	2.61
Sensory and Imaginative Immersion	I was interested in the game's story	8	8	12	4	4	1.67

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Table A.1 continued from previous page

Component	**Item**	**Frequency**					**Mean**
		0	**1**	**2**	**3**	**4**	
	It was aesthetically pleasing	3	4	14	4	11	2.44
	I felt imaginative	6	12	11	5	2	1.58
	I felt that I could explore things	13	8	10	4	1	1.22
	I found it impressive	1	7	13	7	8	2.39
	It felt like a rich experience	2	5	13	16	0	2.19
Flow	I was fully occupied with the game	0	1	4	12	19	3.36
	I forgot everything around me	3	6	9	11	7	2.36
	I lost track of time	4	3	6	13	10	2.61
	I was deeply concentrated in the game	0	0	6	14	16	3.28
	I lost connection with the outside world	4	10	7	10	5	2.06
Challenge	I thought it was hard	4	8	14	9	1	1.86
	I felt pressured	8	11	10	6	1	1.47
	I felt challenged	1	1	11	12	11	2.86
	I felt time pressure	8	9	7	7	5	1.78
	I had to put a lot of effort into it	4	5	9	17	1	2.17
Tension / Annoyance	I felt annoyed	17	14	3	2	0	0.72
	I felt irritable	23	7	5	1	0	0.56
	I felt frustrated	9	13	4	9	1	1.44
Negative Affect	It gave me a bad mood	27	7	1	1	0	0.33

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Table A.1 continued from previous page

Component	Item	Frequency					Mean
		0	**1**	**2**	**3**	**4**	
	I thought about other things	23	8	2	3	0	0.58
	I found it tiresome	18	14	3	1	0	0.64
	I felt bored	21	10	3	2	0	0.61
Positive Affect	I felt content	0	3	7	16	10	2.92
	I thought it was fun	0	3	6	13	14	3.06
	I felt happy	0	5	10	13	8	2.67
	I felt good	0	0	12	13	11	2.97
	I enjoyed it	1	1	8	17	9	2.89

A.2 Virtual Reality Sickness Questionnaire (VRSQ) Item Response Distributions

TABLE A.2: VRSQ item response frequencies ($n = 36$). Response options: 0=not-at-all, 1=slightly, 2=moderately, 3=very.

Component	Item	Frequency				Mean
		0	**1**	**2**	**3**	
Oculomotor	General discomfort	33	2	1	0	0.11
	Fatigue	30	6	0	0	0.17
	Eyestrain	18	15	3	0	0.58
	Difficulty focusing	24	10	2	0	0.39
Disorientation	Headache	31	5	0	0	0.14
	Fullness of head	23	9	3	1	0.50
	Blurred vision	19	14	3	0	0.56
	Dizzy (eyes closed)	33	3	0	0	0.08

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Table A.2 continued from previous page

Component	**Item**	**Frequency**				**Mean**
		0	**1**	**2**	**3**	
	Vertigo	33	3	0	0	0.08

Appendix B

Experimental Configurations and Hyperparameters

This appendix documents the configurations and hyperparameters used in the FER experiments presented in Chapter 7. It complements the main text with details of input processing, model architectures, training procedures, and model selection.

B.1 Static FER Experiments

This section documents the static FER configurations. It covers the image-based baseline, the FEA-based models, and the multimodal intermediate-fusion baseline in Sections B.1.1–B.1.3, respectively.

B.1.1 Image-Based Baseline

The following subsections detail the preprocessing and augmentation, architecture, and training configuration of the static image-based baseline on EmoHeVRDB-SI.

B.1.1.1 Preprocessing and Augmentation

Each input comprised one RGB image of 224×224 pixels from either camera view in EmoHeVRDB-SI. Training augmentation followed the order in Table B.1 and was disabled during validation and test evaluation. Parameters were sampled uniformly from the listed ranges.

TABLE B.1: Image training augmentation in execution order. Intensities use the 0–255 scale.

Operation	**Probability / range**	**Application**
Horizontal flip	Probability 0.50	Random left–right flip.
Translation	Horizontal and vertical fractions: -0.05 to 0.05	Shift by up to 5% of image width and height; bilinear interpolation; constant fill 0.
Rotation	Angle: $-36°$ to $36°$	Keras rotation factor 0.1; bilinear interpolation; constant fill 0.
Zoom	Zoom factor: 0.90–1.10	Keras height factor 0.1 and default width factor; preserve aspect ratio; bilinear interpolation; constant fill 0.
Contrast	Factor: 0.90–1.10	Adjust about each image's per-channel mean; clip to $[0, 255]$.
Brightness	Offset: -25.5 to 25.5	Keras brightness factor 0.1 with value range $[0, 255]$; add the offset and clip to $[0, 255]$.

B.1.1.2 Architecture

The model used TensorFlow 2.15 with the Keras interface and its default settings unless otherwise specified. Table B.2 details the architecture of the static image-based baseline model.

TABLE B.2: Architecture of the static image-based baseline. Output dimensions exclude the batch dimension.

Layer / operation	**Output dimensions**	**Configuration**
Input	$224 \times 224 \times 3$	–
EfficientNet-B0 with global average pooling	1280	ImageNet initialisation; original classification head excluded; spatial average of each feature channel.
Dropout	1280	Rate 0.20.
Dense classifier	7	7 units; softmax activation.

B.1.1.3 Training and checkpoint selection

Training used the Adam optimiser (Kingma and J. Ba 2014) with Keras defaults ($\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-7}$). Sparse categorical cross-entropy used normalised inverse class frequencies as class weights. Training counts were 150, 212, 220, 400, 414, 274, and 360 for anger, disgust, fear, happiness, neutral, sadness, and surprise, respectively. Training followed the fixed six-phase schedule in Table B.3, without early stopping or an adaptive learning-rate scheduler.

TABLE B.3: Six-phase training schedule for the static image-based baseline on EmoHeVRDB-SI.

Phase	**Epochs**	**Learning rate**	**Batch size**	**Layers selected for training**[1]
1	1–20	10^{-3}	32	Classification head only (last 3 layers)
2	21–40	10^{-3}	32	Last 19 layers, starting at `block7a_expand_conv`
3	41–60	10^{-3}	32	Last 78 layers, starting at `block6a_expand_conv`
4	61–80	10^{-4}	32	Last 194 layers, starting at `block3a_expand_conv`
5	81–120	10^{-5}	32	Entire model (242 layers)
6	121–130	10^{-6}	32	Entire model (242 layers)

[1] Layer counts include all Keras layer objects, including those without trainable parameters. Batch normalisation layers remained frozen throughout.

Each phase saved the checkpoint with the highest validation accuracy. Subsequent phases restored the preceding phase's selected checkpoint and reinitialised the Adam optimiser. Learning rates remained constant within phases. The checkpoint selected

for test evaluation was obtained at epoch 125 in the final phase and achieved 70.39% validation accuracy. The training dataset was loaded with shuffling enabled and seed 42. Augmentation and initialisation seeds were not explicitly set. Validation used all 770 images after each epoch, and test evaluation used all 756 images. All datasets used batches of 32 and retained incomplete batches.

B.1.2 FEA-Based Models

The static FEA-based experiments evaluated four classical machine-learning models and a multilayer perceptron (MLP). The following subsections document their input preprocessing, the grid-search configurations of the classical models, and the architecture and training procedure of the MLP.

B.1.2.1 Input and Preprocessing

Each input comprised one 63-dimensional FEA vector from EmoHeVRDB-SFEA. The FEA coefficients lie in the range $[0, 1]$. The experiments used 964 training samples, 385 validation samples, and 378 test samples. Timestamps and participant identifiers were excluded from the input features. No data augmentation was applied.

For the classical machine-learning models, the FEA coefficients were standardised using scikit-learn's `StandardScaler`. The mean and standard deviation of each feature were estimated from the training data and then applied to training, validation and test data. In contrast, the TensorFlow-based MLP received the original FEA coefficients without additional standardisation.

B.1.2.2 Classical Machine-Learning Models

Logistic regression (LR), support vector machine (SVM), random forest (RF), and gradient boosting (GB) were implemented using scikit-learn 1.4.2. Hyperparameters were selected through an exhaustive grid search using `GridSearchCV` with

validation accuracy as the selection metric. A `PredefinedSplit` preserved the original training and validation sets, as defined in Section 6.2, rather than performing k-fold cross-validation. Each configuration was trained on the training set and evaluated on the validation set. The configuration with the highest validation accuracy was subsequently fitted to the training set and evaluated on the test set.

Tables B.4–B.7 list the grid-search values and the selected hyperparameters. Parameters not listed in the tables retained their scikit-learn defaults. Random seed 42 was used for data shuffling and estimator random states.

TABLE B.4: Grid-search space and selected configuration for static FEA-based logistic regression.

Hyperparameter	**Search space**	**Selected value**
Regularisation parameter C	0.01, 0.1, 1, 10	0.1
Maximum iterations	10, 100, 1 000, 10 000	1 000
Solver	`lbfgs`, `liblinear`, `newton-cg`, `newton-cholesky`, `sag`, `saga`	`saga`
Class weighting	None, `balanced`	`balanced`

TABLE B.5: Grid-search space and selected configuration for the static FEA-based support vector machine.

Hyperparameter	**Search space**	**Selected value**
Regularisation parameter C	0.01, 0.05, 0.1, 0.2, 0.5, 1, 10	0.05
Kernel	`linear`, `poly`, `rbf`, `sigmoid`	`linear`
Polynomial degree[1]	1, 2, 3, 4, 5	1
Kernel coefficient γ[1]	`scale`, `auto`	`scale`
Class weighting	None, `balanced`	`balanced`
Stopping tolerance	10^{-3}, 10^{-4}	10^{-3}

[1] Degree and γ were included in the search grid but have no effect on the selected linear kernel.

TABLE B.6: Grid-search space and selected configuration for the static FEA-based random forest.

Hyperparameter	**Search space**	**Selected value**
Number of trees	5, 10, 50, 100, 200	100
Maximum tree depth	None, 10, 20	10
Split criterion	`gini`, `entropy`, `log_loss`	`entropy`
Minimum samples to split	2, 4	2
Minimum samples per leaf	1, 2	1
Features per split[1]	`sqrt`, `log2`, None	`log2`
Class weighting	None, `balanced`, `balanced_subsample`	`balanced`

[1] `sqrt` and `log2` select feature counts based on the square root and base-2 logarithm of the input dimensionality. None uses all features. A maximum depth of None leaves tree depth unrestricted. `balanced_subsample` computes class weights separately for each bootstrap sample.

TABLE B.7: Grid-search space and selected configuration for static FEA-based gradient boosting.

Hyperparameter	**Search space**	**Selected value**
Boosting stages	50, 100, 200	100
Learning rate	0.01, 0.1, 0.5	0.1
Maximum tree depth	2, 3, 5, 10	3
Minimum samples to split	2, 4	2
Minimum samples per leaf	1, 2	2
Features per split[1]	`sqrt`, `log2`, None	`log2`
Sample weighting[2]	None, balanced	None

[1] `sqrt` and `log2` select feature counts based on the square root and base-2 logarithm of the input dimensionality. None uses all features.
[2] The same grid was searched separately with and without balanced sample weights derived from training class frequencies.

B.1.2.3 MLP Architecture

The MLP used TensorFlow 2.15 with the Keras interface and its default settings unless otherwise specified. Table B.8 details the final architecture selected through manual hyperparameter optimisation. The model comprised 16,903 trainable parameters.

TABLE B.8: Architecture of the static FEA-based MLP baseline. Output dimensions exclude the batch dimension.

Layer / operation	**Output dimensions**	**Configuration**
Input	63	–
Dense	128	128 units; ReLU activation.
Dropout	128	Rate 0.20.
Dense	64	64 units; ReLU activation.
Dropout	64	Rate 0.20.
Dense classifier	7	7 units; softmax activation.

B.1.2.4 MLP Training and Checkpoint Selection

Training used the Adam optimiser with Keras defaults. Sparse categorical cross-entropy used normalised inverse class frequencies as class weights. Training counts were 66, 102, 105, 191, 197, 131, and 172 for anger, disgust, fear, happiness, neutral, sadness, and surprise, respectively. Table B.9 specifies the training configuration.

TABLE B.9: Training configuration of the static FEA-based MLP baseline.

Setting	**Value**	**Application**
Batch size	32	Training, validation, and test evaluation. Incomplete batches retained.
Maximum epochs	200	Training budget subject to early stopping.
Initial learning rate	10^{-3}	Adam initialisation.
Learning-rate reduction	Factor 0.20	Reduce after 8 epochs without a validation-accuracy improvement exceeding 10^{-4}.
Minimum learning rate	10^{-6}	Lower bound for learning-rate reduction.
Early-stopping patience	20 epochs	Stop after 20 epochs without validation-accuracy improvement and restore the best weights.
Checkpoint selection	Validation accuracy	Save the model with the highest validation accuracy.

Python, NumPy, and TensorFlow random seeds were set to 42. Training samples were reshuffled each epoch, whereas validation and test samples retained a fixed order. Validation used all 385 samples after each epoch. Training stopped after epoch 43. The selected checkpoint was obtained at epoch 23 with a validation accuracy of 81.56% and was reloaded for test evaluation.

B.1.3 Multimodal Intermediate-Fusion Baseline

The multimodal baseline combines features extracted by the pretrained image-based baseline (see Section B.1.1) and FEA-based baseline (see Section B.1.2) models through intermediate fusion with cross-attention. The following subsections specify the input preprocessing, fusion architecture, and training configuration of the selected model.

B.1.3.1 Preprocessing and Augmentation

Each input paired one RGB image of 224×224 pixels from EmoHeVRDB-SI with its corresponding 63-dimensional FEA vector from EmoHeVRDB-SFEA. Each FEA vector was paired separately with one central-view and one side-view image, using the associated timestamps. Images without a corresponding FEA record were excluded, resulting in 1,928 training pairs, 770 validation pairs, and 756 test pairs.

Images were decoded as RGB and resized to 224×224 pixels using bilinear interpolation. The pretrained image-based feature extractor retained the preprocessing and training augmentation described in Section B.1.1. The FEA coefficients were used without additional standardisation or augmentation.

B.1.3.2 Architecture

The model used TensorFlow 2.15 with the Keras interface and its default settings unless otherwise specified. The pretrained image-based and FEA-based models from Sections B.1.1 and B.1.2 were reused as frozen feature extractors. The image branch provided the 1280-dimensional global-average-pooling output of EfficientNet-B0, while the FEA branch provided the 128-dimensional output of the first Dense layer of the MLP. Only the newly introduced fusion layers were trainable. Table B.10 details the architecture of the multimodal intermediate-fusion baseline.

The cross-attention layer computed a 512-dimensional sigmoid gate for each modality from the other modality's projected and normalised features. Each gate

TABLE B.10: Architecture of the static multimodal intermediate-fusion baseline. Output dimensions exclude the batch dimension.

Layer / operation	**Output dimensions**	**Configuration**
Image input	$224 \times 224 \times 3$	RGB image.
FEA input	63	Original FEA coefficients.
Image feature extractor	1280	Frozen pretrained EfficientNet-B0 up to global average pooling.
FEA feature extractor	128	Frozen pretrained MLP up to its first Dense layer.
Image feature Dropout	1280	Rate 0.20.
FEA feature Dropout	128	Rate 0.10.
Dense projections	512 per branch	One Dense layer per modality; 512 units each; ReLU activation.
Batch normalisation	512 per branch	One layer per modality; momentum 0.99; $\epsilon = 10^{-3}$; learned scale and offset.
Cross-attention gating	512 per branch	Two Dense gates with 512 units each; sigmoid activation; Glorot-uniform initialisation; trainable biases.
Concatenation	1024	Concatenate the two attention-weighted feature vectors.
Fusion Dropout	1024	Rate 0.40.
Dense classifier	7	Seven units; softmax activation.

weighted its corresponding modality's features element-wise. The two weighted feature vectors were concatenated and passed through the fusion Dropout layer and final classifier.

B.1.3.3 Training and Checkpoint Selection

Only the 1,256,455 trainable parameters of the fusion head were optimised. Training used the Adam optimiser with Keras defaults. Sparse categorical cross-entropy used normalised inverse class frequencies as class weights. Training FEA record counts were 66, 102, 105, 191, 197, 131, and 172 for anger, disgust, fear, happiness, neutral, sadness, and surprise, respectively. Each record contributed two image–FEA training pairs. Table B.11 specifies the training configuration.

TABLE B.11: Training configuration of the static multimodal intermediate-fusion baseline.

Setting	**Value**	**Application**
Batch size	128	Training, validation, and test evaluation. Incomplete batches retained.
Maximum epochs	400	Training budget subject to early stopping.
Initial learning rate	5×10^{-5}	Adam initialisation.
Learning-rate reduction	Factor 0.20	Reduce after 6 epochs without a validation-loss improvement exceeding 10^{-4}.
Minimum learning rate	10^{-6}	Lower bound for learning-rate reduction.
Early-stopping patience	16 epochs	Stop after 16 epochs without a decrease in validation loss and restore the best weights.
Checkpoint selection	Validation accuracy	Save the model with the highest validation accuracy.
Additional checkpoint	Validation loss	Separately save the model with the lowest validation loss.

Python, NumPy, and TensorFlow random seeds were set to 42. The newly introduced image and FEA feature Dropout layers also used seed 42. Training pairs were reshuffled each epoch. Validation used all 770 pairs after each epoch, and test evaluation used all 756 pairs. Training stopped after epoch 77. The checkpoint selected for test evaluation was obtained at epoch 64 and achieved 82.86% validation accuracy.

B.2 Dynamic FER Experiments

This section documents the dynamic FER configurations. It covers the image-sequence baseline, the FEA-sequence baseline, and the multimodal intermediate-fusion baseline in Sections B.2.1–B.2.3, respectively.

B.2.1 Image-Sequence Baseline

The following subsections detail the preprocessing and augmentation, architecture, and training configuration of the dynamic image-sequence baseline on EmoHeVRDB-DI.

B.2.1.1 Preprocessing and Augmentation

Each input comprised a sequence of 30 RGB frames of 224×224 pixels from one camera view. Training augmentation followed the order in Table B.12. Parameters were sampled uniformly from the listed ranges. Spatial and photometric parameters were shared across frames. Temporal operations preserved sequence length.

TABLE B.12: Image-sequence training augmentation in execution order. Intensities use the 0–255 scale.

Operation	**Probability / range**	**Application**
Crop and resize	Scale s: 0.90–1.00	Scale both crop dimensions by s; random position; bilinear resize to 224×224.
Rotation	Angle: $-30°$ to $30°$	Centre rotation; bilinear interpolation; constant fill 1.
Horizontal flip	Probability 0.50	Flip the complete sequence.
Brightness	Offset: -20 to 20	Add the shared offset to all intensities.
Contrast	Factor: 0.90–1.10	Adjust about each frame's per-channel mean; then clip to $[0, 255]$.
Solarisation	Probability 0.05; threshold τ: 120–160	Replace intensities $x > \tau$ with $255 - x$.
Temporal shift	Integer offset: -2 to 2 frames	Circular shift along the temporal axis.
Frame repetition	Probability 0.10 per frame	Replace with the predecessor before this operation; leave the first frame unchanged.
Temporal masking	Probability 0.10; block length: 1–6 frames	Random contiguous block replaced by the sequence mean frame after shift and repetition.

B.2.1.2 Architecture

The model used Keras 3.12 defaults unless otherwise specified. Table B.13 details the architecture of the dynamic image-sequence baseline.

TABLE B.13: Architecture of the image-sequence baseline. Output dimensions exclude the batch dimension.

Layer / operation	Output dimensions	Configuration
Input	$30 \times 224 \times 224 \times 3$	–
TimeDistributed EfficientNetV2-B0	30×1280	ImageNet initialisation; global average pooling; shared weights across frames.
SpatialDropout1D	30×1280	Rate 0.15; feature-channel dropout shared across time steps.
Unidirectional LSTM	32	32 units; tanh activation; sigmoid gates; input dropout 0.10; recurrent dropout 0; last output.
Dropout	32	Rate 0.30.
Dense classifier	7	7 units; softmax activation; float32 output.

B.2.1.3 Training and Checkpoint Selection

Training used mixed float16 precision and the Adam optimiser with Keras defaults. Sparse categorical cross-entropy used normalised inverse class frequencies as class weights. Training counts were 150, 212, 220, 400, 414, 274, and 360 for anger, disgust, fear, happiness, neutral, sadness, and surprise, respectively. Training followed the fixed five-phase fine-tuning schedule in Table B.14.

TABLE B.14: Five-phase training schedule for the dynamic image-sequence baseline on EmoHeVRDB-DI.

Phase	Epochs	Learning rate	Batch size	Backbone layers selected for training[1]
1	1–4	10^{-3}	32	Backbone frozen
2	5–8	10^{-3}	32	Last 4 layers, starting at `top_conv`
3	9–14	5×10^{-4}	32	Last 195 layers, starting at `block5a_expand_conv`
4	15–20	5×10^{-4}	16	Last 250 layers, starting at `block3a_expand_conv`
5	21–40	10^{-4}	16	Last 265 layers, starting at `block1a_project_conv`

[1] Layer counts refer to the Keras layer objects of the EfficientNetV2-B0 backbone, including those without trainable parameters. Backbone batch normalisation layers remained frozen throughout. The LSTM and Dense classification layers remained trainable in every phase.

Each phase saved the checkpoint with the lowest validation loss. Subsequent phases restored the preceding phase's selected checkpoint and reinitialised the Adam optimiser. Learning rates were constant within phases. The checkpoint selected for test evaluation was obtained at epoch 27 in the final phase and achieved 80.39% validation accuracy. Python, NumPy, TensorFlow, and Keras random seeds were set to 13. Training sequences were reshuffled each epoch using the same seed. Incomplete training batches were dropped. Validation used all 770 sequences after each epoch, and test evaluation used all 756 image sequences.

B.2.2 FEA-Sequence Baseline

The following subsections detail the input preprocessing, architecture, and training configuration of the dynamic FEA-sequence baseline on EmoHeVRDB-DFEA.

B.2.2.1 Input and Preprocessing

Each input comprised 30 chronologically ordered observations of 63 FEA coefficients. Preprocessing consisted of channel-wise standardisation using means and variances estimated across all training sequences and time steps. The same statistics were applied to validation and test inputs. No data augmentation was applied.

B.2.2.2 Architecture

The model used Keras 3.12 defaults unless otherwise specified. Table B.15 details the architecture of the dynamic FEA-sequence baseline.

TABLE B.15: Architecture of the FEA-sequence baseline. Output dimensions exclude the batch dimension.

Layer / operation	**Output dimensions**	**Configuration**
Input	30×63	–
Normalization	30×63	Channel-wise standardisation; statistics estimated on the training split.
SpatialDropout1D	30×63	Rate 0.25; feature-channel dropout shared across time steps.
Unidirectional LSTM	128	128 units; tanh activation; sigmoid gates; input dropout 0.05; recurrent dropout 0; last output.
Dropout	128	Rate 0.20.
Dense	128	128 units; ReLU activation.
Dropout	128	Rate 0.20.
Dense classifier	7	7 units; softmax activation.

B.2.2.3 Training and Checkpoint Selection

Training used the Adam optimiser with Keras defaults. Sparse categorical cross-entropy used normalised inverse class frequencies as class weights. Training counts were 66, 102, 105, 191, 197, 131, and 172 for anger, disgust, fear, happiness, neutral, sadness, and surprise, respectively. Table B.16 specifies the training configuration.

TABLE B.16: Training configuration of the FEA-sequence baseline.

Setting	**Value**	**Application**
Batch size	32	Training, validation, and test evaluation. Incomplete batches retained.
Maximum epochs	200	Training budget subject to early stopping.
Initial learning rate	5×10^{-4}	Adam initialisation.
Learning-rate reduction	Factor 0.50	Reduce after 3 epochs without a validation-loss improvement exceeding 10^{-4}.
Minimum learning rate	10^{-6}	Lower bound for learning-rate reduction.
Early-stopping patience	12 epochs	Stop after 12 epochs without a decrease in validation loss and restore the weights with the lowest validation loss.
Checkpoint selection	Validation loss	Save the complete model with the lowest validation loss.

Python, NumPy, TensorFlow, and Keras random seeds were set to 31. Training sequences were reshuffled each epoch using the same seed. Validation used all 385 sequences after each epoch, and test evaluation used all 378 sequences. Training stopped after epoch 30. The selected checkpoint was obtained at epoch 18 with a validation accuracy of 85.19% and was reloaded for test evaluation.

B.2.3 Multimodal Intermediate-Fusion Baseline

The multimodal baseline combines features extracted by the pretrained image-sequence baseline (see Section B.2.1) and FEA-sequence baseline (see Section B.2.2) through intermediate fusion with cross-attention. The following subsections specify the input preprocessing and augmentation, fusion architecture, and training configuration of the selected model.

B.2.3.1 Preprocessing and Augmentation

Each input paired a sequence of 30 RGB frames of 224×224 pixels with its corresponding sequence of 30 observations of 63 FEA coefficients. Each FEA sequence was paired separately with one central-view and one side-view image sequence. Inputs were matched by reenactment, timestamps, and labels. The pretrained FEA normalisation statistics were retained. Training augmentation was applied only to the image branch in the order listed in Table B.17.

TABLE B.17: Image augmentation for the multimodal intermediate-fusion baseline in execution order. Intensities use the 0–255 scale.

Operation	**Probability / range**	**Application**
Crop and resize	Scale s: 0.95–1.00	Scale both crop dimensions by s; random position; bilinear resize to 224×224.
Rotation	Angle: $-20°$ to $20°$	Centre rotation; bilinear interpolation; constant fill 1.
Horizontal flip	Probability 0.50	Flip the complete sequence.
Brightness	Offset: -20 to 20	Add the shared offset to all intensities.
Contrast	Factor: 0.90–1.10	Adjust about each frame's per-channel mean.
Gamma adjustment	Exponent γ: 0.90–1.10	Apply $255(x/255)^{\gamma}$ after clipping intensities x to $[0, 255]$.
Solarisation	Probability 0.05; threshold τ: 120–160	Replace intensities $x > \tau$ with $255 - x$.
Gaussian noise	Mean 0; standard deviation 4	Add independent noise to each pixel; then clip to $[0, 255]$.
Frame repetition	Probability 0.05 per frame	Replace with the predecessor before this operation; leave the first frame unchanged.
Temporal masking	Probability 0.05; block length: 1–3 frames	Random contiguous block replaced by the sequence mean frame after repetition.

B.2.3.2 Architecture

The pretrained unimodal models from Sections B.2.1 and B.2.2 were reused as frozen feature extractors. The image branch provided the 32-dimensional LSTM output, and the FEA branch provided the 128-dimensional output of the Dense layer following its LSTM. Both extractors were called in inference mode. Only the newly introduced fusion head was trainable. Table B.18 details the architecture of the multimodal intermediate-fusion baseline.

The cross-attention fusion layer computed a 16-dimensional sigmoid gate for each modality from the other modality's projected and normalised features. Each gate weighted its corresponding modality's features element-wise. The weighted image and FEA features were summed and passed through a sigmoid activation to obtain the fused representation.

TABLE B.18: Architecture of the dynamic multimodal intermediate-fusion baseline. Output dimensions exclude the batch dimension.

Layer / operation	**Output dimensions**	**Configuration**
Image input	$30 \times 224 \times 224 \times 3$	RGB image sequence.
FEA input	30×63	FEA sequence.
Image feature extractor	32	Frozen pretrained image-sequence model through its LSTM.
FEA feature extractor	128	Frozen pretrained FEA-sequence model through the Dense layer following its LSTM.
FEA feature Dropout	128	Rate 0.25.
Image feature Dropout	32	Rate 0.15.
Dense projections	16 per branch	One Dense layer per modality; 16 units each; ReLU activation.
Layer normalisation	16 per branch	One layer per modality; feature axis; $\epsilon = 10^{-3}$; learned scale and offset.
Cross-attention fusion	16	Two bias-free Dense gates with 16 units each; sigmoid activation; Glorot-uniform initialisation.
Fusion Dropout	16	Rate 0.20.
Dense classifier	7	7 units; softmax activation; float32 output.

B.2.3.3 Training and Checkpoint Selection

Only the 3,287 trainable parameters of the fusion head were optimised. Training used mixed float16 precision and the Adam optimiser with Keras defaults. Sparse categorical cross-entropy used normalised inverse class frequencies as class weights. Training counts were 132, 204, 210, 382, 394, 262, and 344 for anger, disgust, fear, happiness, neutral, sadness, and surprise, respectively. Table B.19 specifies the training configuration.

Python, NumPy, TensorFlow, and Keras random seeds were set to 13. Training samples were reshuffled each epoch using the same seed. Validation used all 770 pairs after each epoch, and test evaluation used all 756 pairs. Training stopped after epoch 67. The selected checkpoint was obtained at epoch 51 with a validation accuracy of 87.01% and was reloaded for test evaluation.

TABLE B.19: Training configuration of the multimodal intermediate-fusion baseline.

Setting	**Value**	**Application**
Batch size	32	Training, validation, and test evaluation. Incomplete batches retained.
Maximum epochs	80	Training budget subject to early stopping.
Initial learning rate	5×10^{-4}	Adam initialisation.
Learning-rate reduction	Factor 0.50	Reduce after 4 epochs without a validation-loss improvement exceeding 10^{-4}.
Minimum learning rate	10^{-6}	Lower bound for learning-rate reduction.
Early-stopping patience	16 epochs	Stop after 16 epochs without a decrease in validation loss.
Checkpoint selection	Validation loss	Save the complete model with the lowest validation loss.